Initial Environmental Examination

PUBLIC

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India: Chennai Climate Resilient Water Security and Sewerage Project

Providing a Ring Main System in Chennai City (GCC limit) including Operation and Maintenance for 10 years

Part A: Main Report

Prepared by the Chennai Metropolitan Water Supply and Sewerage Board (CMWSSB) for the Asian Development Bank (ADB).

CURRENCY EQUIVALENTS
(as of 26 February 2026)

Currency Unit	–	Indian rupee (₹)
₹1.00	=	$0.011
$1.00	=	₹ 90.88

ABBREVIATIONS

ADB	–	Asian Development Bank
ASI	–	Archaeological Survey of India
BOD	–	biochemical oxygen demand
CBO	–	community-based organization
CCMC	–	Municipal Corporation
CMA	–	Chennai Metropolitan Area
CMWSSB	–	Chennai Metropolitan Water Supply and Sewerage Board
CMSC	–	construction management and supervision consultant
CPCB	–	Central Pollution Control Board
CTE	–	Consent to Establish
CTO	–	Consent to Operate
DSP	–	desalination plant
EAC	–	Expert Appraisal Committee
EHS	–	environmental health and safety
EIA	–	environmental impact assessment
ESS	–	environmental and social safeguard
EMP	–	environmental management plan
GoI	–	Government of India
GoTN	–	Government of Tamil Nadu
GCC	–	Greater Chennai Corporation
IEE	–	initial environmental examination
MOEFCC	–	Ministry of Environment, Forest and Climate Change
NGO	–	non-government organization
NOC	–	No Objection Certificate
PIA	–	project implementing agency
PIU	–	project implementation unit
PMU	–	project management unit
RMS	–	ring main system
SPS	–	Safeguard Policy Statement
STP	–	sewage treatment plant
TNPCB	–	Tamil Nadu Pollution Control Board
TNUIFSL	–	Tamil Nadu Urban Infrastructure Financial Services Limited
WDSs	–	water distribution stations
WHO	–	World Health Organization
WTP	–	water treatment plant

WEIGHTS AND MEASURES

°C	-	degree Celsius
km	-	kilometer
LPCD	-	liters per capita per day
m	-	meter
Mgd	-	million gallons per day
MLD	-	million litres per day
mm	-	millimeter
nos	-	numbers
km^2	-	square kilometer
m^2	-	square meter

NOTE

In this report, "$" refers to United States dollars.

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EXECUTIVE SUMMARY

The proposed Chennai climate resilient water security and sewerage project (CCRWSSP) aims to upgrade water supply and sanitation systems to deliver safe, reliable, and continuous (24/7) services to 4.5 million people across an area of 426 square kilometer (km²)—covering nearly 70% of Chennai, including slum areas. A new citywide ring main, along with strengthened distribution networks and household connections, will ensure equitable and efficient last-mile service. The project integrates climate-smart and digital solutions to enhance system efficiency, resilience, and customer engagement. Enhanced digitalization—through the expansion of supervisory control and data acquisition (SCADA), geographic information system (GIS), direct memory access (DMA), smart metering, and customer service platforms—will enable real-time monitoring, leak detection, transparent billing, and improved financial performance. A structured digital roadmap supported by staff training and cybersecurity measures will institutionalize data-driven operations.

The proposed CCRWSSP, is aligned with the following impacts: urban public health and environmental conditions in Chennai improved (Tamil Nadu Vision 2023) and universal and equitable access to safe and affordable drinking water and sanitation achieved (National Urban Water Supply and Sanitation Policy 2021–2031). The project will have the following outcome: access to safe, reliable, and inclusive water supply and sanitation services in Greater Chennai improved. The outcome will be achieved through the following three interlined outputs: (i) Output 1: Resilient water and wastewater infrastructure system modernized and expanded, (ii) Output 2: Utility capacity and operations strengthened for reliable and inclusive water and sanitation.

Tamil Nadu Urban Infrastructure Financial Services Limited (TNUIFSL) will be the executing agency and Chennai Metro Water Supply and Sewerage Board (CMWSSB) will be the implementing agency for the project. TNUIFSL will manage the project through its existing project management unit (PMU) and CMWSSB will establish a dedicated project implementing unit (PIU) for the project and will manage procurement, financial management, safeguards, and overall coordination.

Proposed subproject. This initial environmental examination (IEE) report is prepared for the Ring Main System (RMS) subproject, one of the three subprojects proposed under the CCRWSSP. Chennai's existing water distribution system suffers from high pressure losses, unequal supply, and frequent outages due to fragmented transmission lines. This subproject will provide a ring main water system connecting all major reservoirs and treatment plants, allowing water to circulate efficiently and will maintain a balanced pressure across service zones. This arrangement will address existing operational issues by improving system reliability, reducing non-revenue water (NRW), and ensuring uninterrupted water supply during maintenance activities or localized failures. Under the proposed system, water from the Ring Main to the distribution network will be routed through the underground storage tanks (UGTs) at water distribution stations (WDSs) facilities or headworks, with no direct supply to the distribution network. The subproject includes the following:

- Ring main of 97.9 kilometer (km) long, 1,200–2,000-millimeter (mm) diameter, mild steel (MS) pipes (91.2 km new pipeline, 6.7 km existing);
- Feeder mains from treatment plants/reservoirs to ring main: 15.87 km, MS pipes 800–2,000 mm diameter (2.65 km new, 13.22 km existing);
- Transmission mains from ring main to WDSs: 114 km (64.2 km new, 49.8 km existing), MS pipes 150–1,900 mm diameter;
- Seven new UGTs;

- Replacement of pumps in seven clear water pumping stations;
- Eight surge tanks; and,
- SCADA and instrumentation: Real-time flow, pressure, pumping, and reservoir monitoring with a master control center.

Screening and Categorization. The subproject was screened using ADB's rapid environmental assessment (REA) checklist for water supply. Potential impacts during pre-construction, construction, and operation phases were identified and assessed. Based on this, the subproject is classified as category B for environment according to the Safeguard Policy Statement (SPS) 2009 of ADB, and an IEE and corresponding environmental management plan (EMP) was prepared accordingly. As per the Government of India's (GoI) Environmental Impact Assessment (EIA) Notification 2006, this project does not require environmental clearance (EC). Sections of the RMS alignment located within coastal regulation zone (CRZ)-II and CRZ-IV areas, including pipe crossings on Adyar river and Buckingham canal, will be implemented only after obtaining prior CRZ clearance from the Ministry of Environment, Forest and Climate Change (MOEFCC) in accordance with the CRZ Notification, 2019.

An **alternatives analysis** was undertaken for the proposed Chennai RMS within the GCC area to identify the most environmentally sound, technically feasible, and economically viable option in accordance with ADB's SPS (2009). The assessment examined the "no project" scenario, alternative alignments, pipe material options, and construction methodologies (open cut versus trenchless). The no-project option was rejected due to continued hydraulic stress, inequitable supply distribution, high non-revenue water, and reduced system resilience. Alignment alternatives were optimized to follow existing road rights-of-way, maintaining a 62 m setback from Pallikaranai Marshland (345 m diversion length), thereby minimizing ecological disturbance and regulatory risk. Mild Steel pipes were selected for the 1,200–2,000 mm diameter high-pressure system (total proposed network: 91.2 km ring main, 64.2 km transmission main, 2.65 km feeder main) due to structural strength and durability under Chennai's coastal conditions. For construction methodology, open-cut trenching was adopted for standard sections based on cost efficiency, while trenchless technology will be used at critical crossings to reduce environmental and traffic impacts. The preferred alternative thus represents a balanced solution integrating technical reliability, environmental safeguards, regulatory compliance, and long-term operational sustainability.

Description of the Environment. Chennai is located on the flat Eastern Coastal Plains of India with an average elevation of approximately 6.7 meter (m) above mean sea level. The proposed RMS is planned predominantly along existing arterial and sub-arterial roads within the urbanized parts of the city to avoid land acquisition and resettlement. The alignment broadly follows a circumferential orientation, interconnecting major water treatment plants (WTPs), distribution zones, and existing service reservoirs across the Greater Chennai Corporation (GCC) area. The existing desalination plants (DSP) and WTPs feeding into the RMS, while new and existing overhead tanks and underground sumps are located within built-up residential and commercial zones. Although the RMS passes in proximity to sensitive ecological features such as Guindy National Park (just outside the parks' eco sensitive zone) and 62 m from the Pallikaranai Marsh (Ramsar Site) in south-central Chennai, all pipe-laying activities are confined to road corridors, and no direct encroachment, diversion, or construction activity is proposed within these protected areas.

The RMS alignment intersects a highly urbanized drainage and infrastructure network including crossings near the Buckingham Canal, Otteri Nullah, and other major stormwater drains that convey runoff from dense residential and commercial catchments. Pipe crossings at these locations are proposed using bridge attachments or trenchless methods to avoid interference with canal hydraulics. Groundwater occurs at shallow depths (typically 4 m–5 m below ground level) which is a critical consideration for trench excavation, dewatering, and utility protection during construction. Subsurface conditions comprise predominantly of clayey and sandy alluvium with occasional shale and sandstone layers. The alignment also passes through areas with significant underground utilities, including water supply lines, sewers, stormwater drains, power cables, telecommunication ducts, and gas pipelines, particularly in older city areas. An Archaeological Survey of India (ASI)-protected monument (i.e., Old Fort Wall) lies approximately 400 m from the nearest proposed alignment; since no construction is proposed near the monument and works are limited to standard road trenches, no physical, visual, or vibration impacts are anticipated.

RMS will be developed entirely within the 426 km^2 jurisdiction limits of GCC, traversing congested residential neighborhoods, major commercial corridors, industrial pockets, and coastal urban stretches. Pipe-laying will be undertaken almost entirely within existing road rights-of-way (ROW), where the dominant environmental and social baseline issues relate to high traffic volumes, limited carriageway widths, roadside commercial activities, pedestrian movement, and dense utility corridors. Traffic congestion, access to residences and shops, and safety of road users are therefore key considerations during construction. Certain sections of the alignment run parallel to or cross CRZ areas, particularly near the Adyar River mouth, Buckingham Canal, and the eastern coastal corridor. The project involves approximately 1,305.73 m of CRZ-II and 1,106.72 m of CRZ-IV areas, including about 24 pipe-carrying bridges, of which 13 are located within notified CRZ limits. These components will require prior CRZ clearance from the MOEFCC under the CRZ Notification 2019 and will be designed to avoid alteration of coastal processes, tidal flow, or existing drainage patterns.

Potential Environmental Impacts and Mitigation Measures. The subproject is expected to result in limited, site-specific, and largely reversible environmental and social impacts, concentrated mainly during the construction phase. Anticipated impacts include temporary traffic congestion, mobility restrictions, and access disruptions due to pipelaying along narrow roads with heavy vehicular movement, including sections requiring night-time construction to manage traffic flow. Construction activities will cause localized increases in dust, noise, and vibration, particularly in mixed residential–commercial areas. High groundwater tables in certain zones may require continuous dewatering during trench excavation, posing short-term risks of waterlogging, sediment runoff, and instability of trench walls if not properly controlled. Works close to stormwater drains, canals, and waterbody corridors may result in temporary siltation and obstruction of drainage if mitigation measures are not implemented. Construction near environmentally sensitive areas, including CRZ stretches, wetlands, outside eco sensitive zone of Guindy National Park, and 62 m from Pallikaranai Marsh land, presents risks of accidental spills, material encroachment, and safety hazards, although no permanent impacts or physical intrusion into these areas are proposed. The alignment passes through dense utility corridors, and construction may cause temporary disruption or safety risks associated with underground utilities, including gas pipelines, high-voltage power cables, telecommunication lines, sewerage systems, and sections adjacent to the Chennai Metro corridor.

In addition, the subproject interfaces with associated facilities—namely the existing water treatment plants (Kilpauk, Chembarambakkam, Veeranam–Vadakuthu, Puzhal), desalination plants [Nemmeli 110 million liters per day (MLD) and 150 MLD], and pumping stations—where construction activities such as tie-ins, interconnections, and operational adjustments may generate localized, temporary impacts, including noise, minor chemical handling risks, and access limitations for facility staff. During operation, the project is expected to have negligible adverse environmental impacts, while delivering long-term environmental and social benefits through improved reliability, efficiency, and resilience of Chennai's urban water supply system. All identified impacts are manageable through standard engineering practices and mitigation measures outlined in the EMP consistent to the requirements of SPS 2009.

Mitigation Measures have been developed to reduce all negative impacts to acceptable levels. These were discussed with specialists responsible for the engineering aspects, and as a result, significant measures have already been included in the design of the infrastructure. Various measures include a buffer around the RMS, imparting necessary training, safety and personal protective equipment (PPE) for workers, etc.

Potential impacts during construction are considered significant but temporary and are common impacts of construction in urban areas, and there are well developed methods to mitigate the same. Except laying the ring main, the remaining construction activities will be confined to the selected sites, and the interference with the public and community around is minimal. In these works, the temporary negative impacts arise mainly from construction dust and noise, hauling of construction material, waste and equipment on local roads (traffic, dust, safety, etc.), and occupational health and safety (OHS) aspects. Pipeline will be laid along the edge of public roads in the urban area congested with settlements, commercial activities and moving traffic. Therefore, laying of water pipeline will have temporary impacts arising mainly from the disturbance of residents, businesses and traffic due to construction work; safety risk to workers, public and nearby buildings due to trench excavations in the road; access impediment to houses and businesses, etc. These are all general impacts of construction in densely populated areas, and there are well-developed methods of mitigation that are described in the EMP.

Environmental Management Plan. An EMP has been developed to provide mitigation measures to reduce all negative impacts to acceptable levels, along with the delegation of responsibility to the appropriate agency. Various design-related measures have been incorporated in the project design. During construction, the EMP includes mitigation measures such as (i) proper planning of pipe laying works to minimize the public inconvenience (ii) barricading, dust suppression and control measures (iii) traffic management measures for works along the roads and for hauling activities; (iv) provision of walkways and planks over trenches to ensure access will not be impeded; and (iv) identifying beneficial use of excavated materials to extent possible to reduce the quantity for disposal. The EMP will guide the environmentally-sound construction of the RMS subproject. EMP includes a monitoring program to measure the effectiveness of EMP implementation and includes observations on- and off-site, document checks, and interviews with workers and beneficiaries.

The EMP will be included in the bid and contract documents to ensure compliance to the conditions set out in this document. The contractor will be required to submit to PIU, for review and approval, a site environmental management plan (SEMP) including the (i) proposed sites/locations for construction worker camps, storage areas, hauling roads, lay down areas, disposal areas for solid and hazardous wastes; (ii) specific mitigation measures following the approved EMP; and (iii) monitoring program as per EMP. No works are allowed to commence prior to the approval of SEMP. A copy of the EMP/approved SEMP will be always kept on site during the construction period.

Project Implementation Arrangements. Environmental safeguard implementation will be overseen by the Environment and Social Safeguard (ESS) Managers in the PMU at TNUIFSL who retains the overall responsibility for compliance with the IEE, EMP, and SPS 2009. The PMU will be strengthened with additional environmental safeguard staff. At CMWSSB, a separate safeguards and safety cell will be established, led by an Executive Engineer and supported by two Assistant Engineers, one responsible for environmental management and one for health and safety. At the project supervision level, the CMSC will field one Chief Safety Engineer with relevant international experience, one Safety Engineer supported by two staff, and one Environment Specialist supported by one environmental staff. The CMSC will also mobilize a Traffic Management Expert and a Scheduling/Planning Engineer. Each contractor will deploy dedicated environment, health and safety (EHS) supervisors for their respective packages. The CMWSSB Safeguards Officer will coordinate day-to-day safeguard activities, including health and safety, contractor liaison, and grievance redress. Adequate budget provisions will be included, and PMU safeguard staff will conduct periodic supervision to ensure full regulatory and SPS 2009 compliance.

Consultation, Disclosure and Grievance Redress Mechanism. Extensive stakeholder engagement was carried out in line with ADB requirements. A consultation meeting was held on 20 July 2023 at the CMWSSB office with 58 participants (41 male and 17 female), including elected representatives and senior officials. Project details, expected benefits, implementation timelines, and potential temporary construction impacts were presented, along with environmental and social safeguard measures. Project related information was shared with participants in English and Tamil.

Another public consultation for the Ring Main Project was conducted on 29 July 2025 to inform citizens and gather feedback. The officials of CMWSSB informed the participants that all concerns related to water quality, sewage mixing, pipeline condition, equal distribution, and construction impacts will be addressed through replacement of old and damaged pipelines, improved monitoring, and implementation of the Ring Main System. The public also requested that EMP measures be strictly followed during construction to minimize disruption and ensure safety. The PIU will continue community outreach before construction, particularly on road works, vendor relocation, safety, and traffic management. Site-specific focus group discussions (FGDs) were held from 25–29 July 2025 at six proposed UGT locations, with 74 participants (54 male, 20 female). Community members raised practical concerns regarding access, dust, traffic diversions, and safety, which have been incorporated into mitigation planning.

Separate FGDs were conducted with mobile and informal livelihoods—street vendors, stall operators, auto drivers, and local commuters—likely to experience temporary disruptions. A total of 118 participants (69 male, 49 female) provided feedback on access challenges, business interruptions, dust, and worksite safety. The consultation process will be continued during project implementation.

A project-based grievance redress mechanism (GRM) will be established to receive, register, and resolve complaints from affected persons in a structured and time-bound manner. Grievances will be documented, tracked, and addressed through a multi-tier system, beginning with field-level resolution by the contractor, the construction management and supervision consultant (CMSC), and PIU staff, escalating to PIU and CMWSSB management as required, and further to a Project-level grievance redress committee (GRC) and finally to the State Steering Committee for matters requiring higher-level decisions. Complainants may approach the judiciary at any stage, and, if concerns remain unresolved through government channels, they may access ADB's Accountability Mechanism. Public outreach will be conducted to ensure awareness of procedures, timelines, and contact points, with all actions and outcomes recorded and reported in periodic monitoring submissions to ADB.

Monitoring and Reporting. The contractor will submit a monthly EMP implementation report to PIU. The PIU will monitor the compliance of the Contractor, prepare a Quarterly Environmental

Monitoring Report and submit it to PMU. The PMU will oversee the implementation and compliance and will submit semi-annual environmental monitoring reports (SEMRs) to ADB during construction and annually during operation until the project completion report has been issued by ADB which is usually within 1-2 years after the physical completion of the project. ADB will review SEMRs and disclose them on its website.

Conclusions and Recommendations. The proposed Chennai Integrated Urban Water Management Project is not expected to generate significant adverse environmental impacts. Anticipated impacts are largely construction-related and can be effectively avoided, minimized, or mitigated through the measures prescribed in the EMP. The primary beneficiaries will be the residents of Chennai, who will gain improved water supply reliability and system resilience.

The project is designed to enhance overall service levels and environmental conditions by strengthening bulk water conveyance and distribution. Key benefits include: (i) improved availability and continuity of safe drinking water for all households, including vulnerable communities; (ii) reduced dependence on tanker supply and other alternative sources; (iii) improved public health outcomes through reduced incidence of waterborne and related diseases; and (iv) reduced risk of contamination from ageing or overloaded infrastructure.

Based on the IEE findings, the Chennai RMS subproject is confirmed as category B in accordance with SPS 2009 of ADB and no further special study or full EIA is required under the EIA Notification (2006) of the GOI. The environmental assessment covers the RMS transmission pipelines, pipe-carrying bridges, valve chambers, and associated facilities, namely the existing water treatment plants (Kilpauk, Chembarambakkam, Veeranam–Vadakuthu, and Puzhal), desalination plants (Nemmeli - 110 MLD and 150 MLD), and pumping stations that will supply treated water to the RMS. No new WTPs, DPs, or production sources are proposed under this subproject; however, the RMS will interface with these existing facilities for conveyance and distribution of treated water. Environmental audits of the associated facilities have identified specific compliance gaps related to CTE/CTO documentation, chemical and sludge management, diesel storage and operation of the diesel generator set, fire and electrical safety, and OHS. These issues will be addressed through a time-bound corrective action plan and integration of required mitigation measures into the EMP. The IEE will be updated during implementation if any changes occur in project scope, alignment, design, or site conditions, and such revisions will supersede this draft IEE version. Sections of the RMS alignment located within CRZ-II and CRZ-IV areas, including pipe crossings over Adyar River and Buckingham Canal, will be implemented only after obtaining prior CRZ clearance from the MOEFCC in accordance with the CRZ Notification, 2019, and all clearance conditions will be incorporated into project design, construction, and operation.

I. INTRODUCTION

A. The Chennai Integrated Urban Water Management Project

1. The Chennai climate resilient water security and sewerage project (CCRWSSP) will modernize and strengthen Chennai's water supply and sanitation systems to deliver reliable, safe, and continuous (24/7) services to about 4.5 million residents across 426 km², including slum communities. The project integrates a new citywide water ring main with upgraded transmission and distribution systems, household connections, and digital platforms to ensure equitable, last-mile service delivery. Through climate-smart design and improved system resilience, the project is positioned to make Chennai a national model for inclusive and climate-resilient urban water management.

2. CMWSSB is responsible for service delivery and faces operational and financial constraints, including tariff stagnation, limited cost recovery, slow inter-agency coordination, outdated systems, and inadequate digital capabilities. Strengthening institutional systems, operational efficiency, and financial sustainability is therefore critical to ensuring long-term reliability. The project aligns with national and state programs such as Jal Jeevan Mission (Urban), AMRUT 2.0, Tamil Nadu Vision 2030, and the State Water Policy 2012, and supports ADB's Country Partnership Strategy (2023–2027) and Strategy 2030 priorities on livable cities and climate resilience.

3. CCRWSSP will have the following outcome: access to safe, reliable, and inclusive water supply and sanitation services in Greater Chennai improved. The outcome will be achieved through the following three interlined outputs:

4. **Output 1: Resilient water and wastewater infrastructure system modernized and expanded.** The project will modernize and expand Chennai's core water and wastewater systems to improve service reliability, pressure, and equitable distribution and collection across all service zones. **On water supply,** the project will finance the (i) construction of a 91.2 kilometer (km) water ring main, 15 km feeder main, and 64 km transmission main; and (ii) upgrading of 7 pumping stations. These upgrades will strengthen the city's ability to provide stable water supply even during droughts or localized outages. New generation SCADA and GIS technologies will be introduced for real-time monitoring, proactive maintenance, and water loss reduction.[1] The initiative will also integrate different water sources with a real-time decision support system easing pressure on freshwater sources and enhancing climate resilience through improved water security and water quality control. **On wastewater,** the project will (i) upgrade 38 major sewer pumping stations; (ii) construct 1 new lift station; and (iii) lay 8 km of pumping main and 177 km of sewer network, along with the renewal of existing assets. As a pilot initiative in high-density areas, existing sewer lines shall be rehabilitated in their current location through real-time condition assessment using robotic crawler-mounted cameras and pipe bursting technology. The modern trenchless construction techniques will minimize excavation, traffic disruption, and inconvenience to residents and citizens while ensuring faster installation, reduced environmental footprint, and improved durability, as applicable to the field condition.

5. **Output 2: Utility capacity and operations strengthened for reliable and inclusive water and sanitation.** This output will strengthen four inter-related areas to optimize the

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[1] CMWSSB plans to introduce a cloud-based sentinel and IoT system to integrate SCADA and all treatment and pumping facilities, enabling real-time monitoring, predictive maintenance and AI-based detections.

infrastructure investments. **On utility strengthening**, a 10-year performance-based O&M contract will be introduced to improve service reliability, operational accountability, and long-term asset performance. The project will also complement CMWSSB's ongoing efforts for institutionalized NRW monitoring and reduction to improve cost recovery and financial sustainability, together with capacity development to strengthen capabilities. The project will also finance the preparation of distribution network development in at least two zones of Metro Chennai to achieve the readiness for future investments. **On digital transformation,** the project will finance new SCADA systems, digital twins applications, and integrated customer service platforms to enable real-time system monitoring, data driven decision making, and improve customer responsiveness. Advanced inspection technologies, including robotic closed circuit television and sonar-based sewer surveys, will be financed to assess pipeline conditions, improve asset management, reduce system failures, and eliminate hazardous manual inspections. **On financial sustainability**, the project will support priority actions under the Revenue Improvement Plan, strengthening tariff management, service standards, and financial sustainability while protecting low-income households, and advancing wastewater reuse and integrated water resource management policies. **On inclusive capacity,** using a city-to-city partnership initiative, Chennai will establish a new partnership with a city in Asia to exchange practical knowledge and operational experience on planning, implementation, and management of water supply and sanitation systems. Gender equality will be reinforced through (i) a career scholarship program for women in STEM roles; (ii) women exclusive paid-internship program; (iii) women leadership program; (iv) development of a gender strategy, and (v) training and outreach events targeting women, girls, and other relevant groups.

6. The project is aligned with ADB's Country Partnership Strategy for India, 2023–2027 and Strategy 2030 and its Midterm Review, particularly advancing strategic focus areas as described in its guidance note.[2] It promotes resilient, inclusive, and digitally enabled urban water and sanitation services, improving the quality of life and environmental health in Chennai. The project also reflects ADB's strategic focus on innovation, gender equality, institutional strengthening, and private sector engagement, supporting India's transition toward sustainable, climate-resilient, and resource-efficient urban development.

7. TNUIFSL, under the Municipal Administration and Water Supply Department, will serve as the EA and will house the PMU. CMWSSB will be the IA, and will establish a PIU for the project. Environmental safeguards will be managed by ESS Managers at the PMU level, with day-to-day implementation by the PIU Safeguards Officer, supported by the CMSC Safeguard Specialist and contractor EHS supervisors. Adequate budget provisions and periodic PMU supervision will ensure compliance with Safeguard Policy Statement (SPS) 2009 of ADB and all regulatory requirements of the GoI.

B. Scope of the Initial Environmental Examination

8. Following the SPS 2009, the subproject was screened using the rapid environmental assessment (REA) checklist (Appendix 1) and classified as environment category B requiring an IEE, as potential impacts are expected to be site-specific, temporary, and primarily construction-related. The Chennai Ring Main System (RMS) Project involves laying large-diameter underground pipelines, constructing valve chambers, and establishing interconnections within existing road ROW and CMWSSB premises, thereby minimizing land acquisition and environmental disturbance. The IEE has been prepared based on the available project design,

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[2] ADB. 2023. *Country Partnership Strategy: India, 2023–2027—Catalyze Robust, Climate-Resilient, and Inclusive Growth*. ADB. Strategy 2030 Mid Term Review: Evolution Approach for the Asian Development Bank.

field reconnaissance surveys, secondary environmental data, and stakeholder consultations. The IEE will be updated during implementation if any changes occur in project scope, alignment, design, or site conditions, and such revisions will supersede this draft IEE version.

C. Report Structure

9. In accordance with the requirements of SPS 2009, this IEE report has been organized into the following sections:

(i) Executive Summary,
(ii) Introduction,
(iii) Description of the Project,
(iv) Analysis of Alternatives,
(v) Policy, Legal, and Administrative Framework,
(vi) Description of the Environment,
(vii) Anticipated Environmental Impacts and Mitigation Measures,
(viii) Consultation, Participation, and Information Disclosure,
(ix) Grievance Redress Mechanism,
(x) Environmental Management Plan, and,
(xi) Conclusion and Recommendations.

II. DESCRIPTION OF THE PROJECT

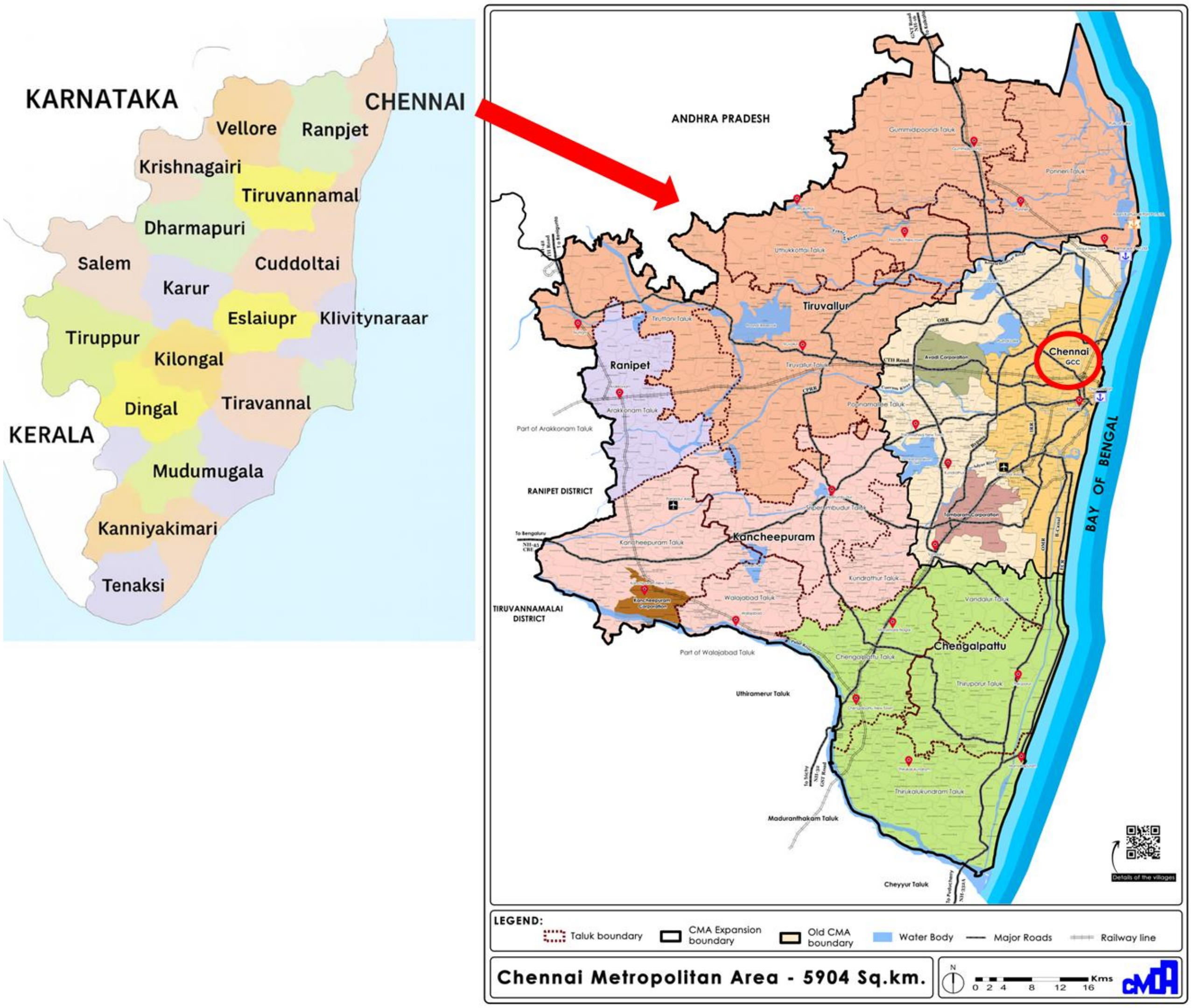
A. Project locations

10. Chennai, the capital of Tamil Nadu state, occupies a strategic position on India's southeastern coast along the Coromandel Coast of the Bay of Bengal. The city lies between latitudes 12°50'N and 13°17'N and longitudes 80°10'E and 80°20'E, with an average elevation of approximately 6 meters above mean sea level (amsl). Location map of Chennai is presented in Figure 1. The city covers 426 square kilometers (km^2) and serves as Tamil Nadu's administrative, cultural, and economic hub. Chennai is bounded by Tiruvallur District (north and northwest), Kanchipuram and Chengalpattu Districts (south and southwest) and by the Bay of Bengal on the east.

11. The coastal topography is predominantly flat, characterized by sandy soil and a network of three major rivers—Cooum, Adyar, and Kosasthalaiyar—which drain into the Bay of Bengal. This geographic setting, combined with coastal accessibility, makes Chennai a major metropolitan center and a hub for trade, industry, and infrastructure development. Chennai's urban extent comprises two administrative and planning regions, Greater Chennai Corporation (GCC) covering an area of 426 km^2 and Chennai Metropolitan Area (CMA), covering a total area of 5,904 km^2 which includes the GCC.

12. This RMS is planned for the GCC entirely within the city’s core urban limits. The project components will be located across major administrative zones of GCC, encompassing diverse land-use areas such as central business districts, dense residential neighborhoods, industrial corridors, and coastal stretches.

Figure 2: Location of Chennai within Tamil Nadu showing CMA area of 5,904 km^2 including the 426 km^2 area of GCC



B. Existing water supply system in Chennai

13. Chennai Metropolitan Water Supply and Sewerage Board, known shortly as CMWSSB, is a statutory board of Government of Tamil Nadu (GoTN) responsible for provision of water supply and sewage services to the city of Chennai and its metropolitan region.

14. Chennai's water supply system has evolved from traditional wells and tanks in the 19th century to a modern, multi-source network serving a population of over 8.5 million. The organized water supply began in 1872 with the construction of the Tamaraipakkam Anicut and conveyance through Cholavaram and Red Hills lakes to the Kilpauk Water Works, marking the city's first structured water system. Over the years, major reservoirs such as Poondi (1944), Cholavaram, Red Hills, and later Chembarambakkam were developed to store and supply surface water. The system expanded significantly between 1946 and 1978, with the introduction of rapid sand filters, new conduits, electrical pumping, and zonal distribution systems. Groundwater sources in the Araniar–Kosasthalaiyar Basin were also developed to supplement surface water.

15. The Krishna Water Supply Project (Telugu Ganga Project), commissioned in 1996, brought water from Andhra Pradesh to Chennai via Poondi and Red Hills reservoirs, substantially increasing supply reliability. Subsequent augmentation included the Veeranam Project (2004) drawing water from the Cauvery River system and two desalination plants (DSP) at Minjur (2010) and Nemmeli (2013), each adding 100 MLD. Currently, Chennai's water is drawn from multiple sources — surface reservoirs, Krishna and Cauvery River systems, desalination, and limited groundwater abstraction.

16. The city's total water treatment capacity at present stands at 1,494 MLD through major plants at Kilpauk, Red Hills, Chembarambakkam, Vadakuthu, Surapet, Minjur, and Nemmeli. A distribution network of 5,247 km provides 100% piped water supply coverage across 16 zones. The city's progressive development of diverse sources and treatment infrastructure has transformed Chennai into a resilient urban water system, capable of meeting growing demands and reducing dependence on monsoon-fed reservoirs.

17. **Current water sources and supply capacities**. Chennai's water supply is derived from a mix of surface water bodies, inter-basin transfer systems, and DPs. The primary surface sources include the Poondi, Puzhal, Cholavaram, Chembarambakkam, and Thervoy Kandigai reservoirs, along with Veeranam Lake and the Krishna River received through the inter-basin transfer scheme. Together, these sources provide 1,467 MLD under normal hydrological conditions. In addition to these, seawater desalination has emerged as a major drought-resilient component of the supply network. The current operational desalination capacity in Chennai includes the Nemmeli plants (100 MLD + 150 MLD), while the Minjur plant (100 MLD) is under rehabilitation and the Perur plant (400 MLD) is in the planning/proposal stage. Once rehabilitation and planned expansions are complete, the combined desalination capacity is expected to reach 750 MLD by 2027, significantly enhancing Chennai's reliable water supply. **Figure 3** below show maps of existing sources of water supply and **Figure 4** showing proposed ring main alignment with connecting WTPs and desalination plants (DSPs).

Figure 3: Map showing sources of water supply in Chennai

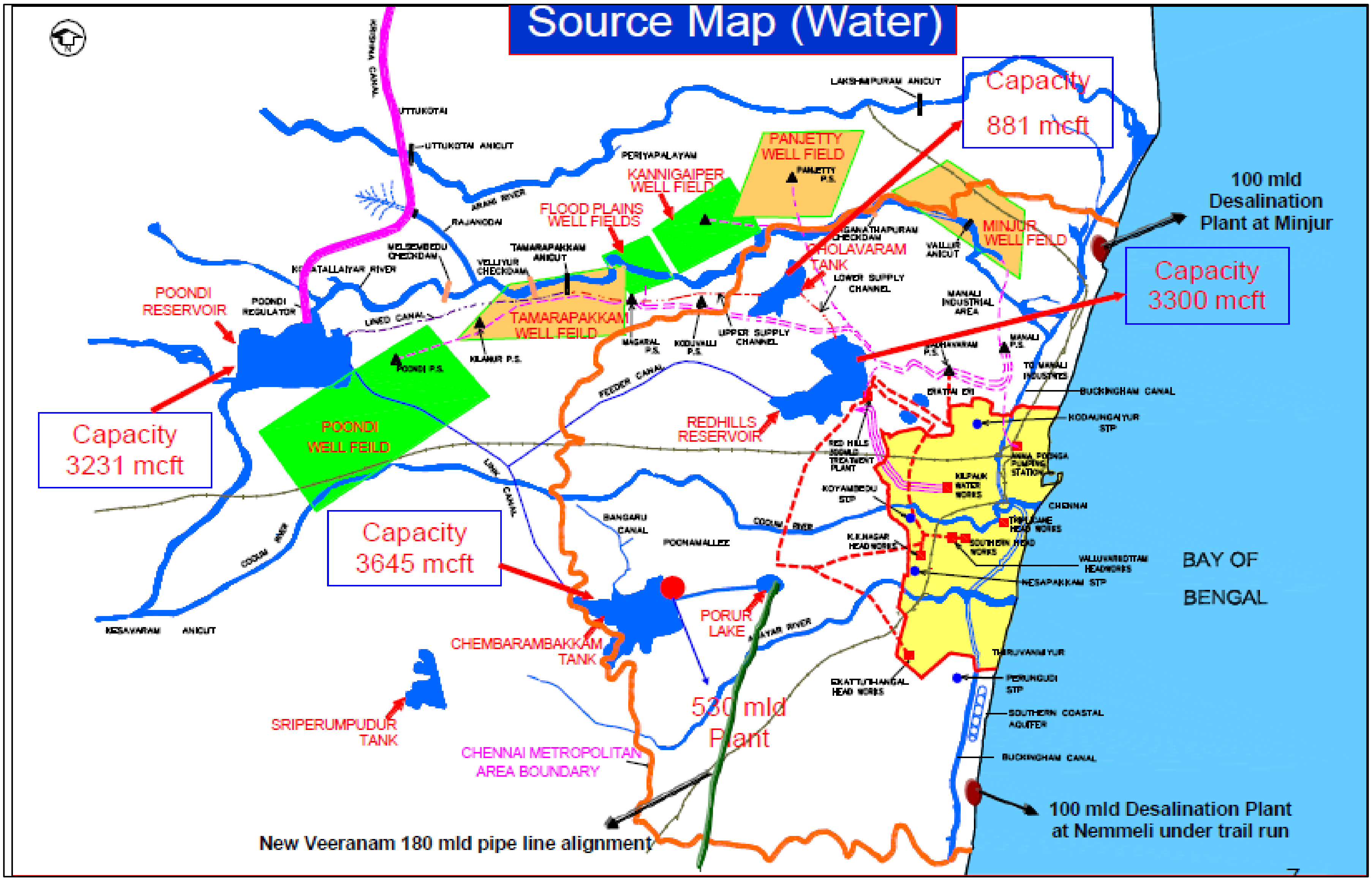
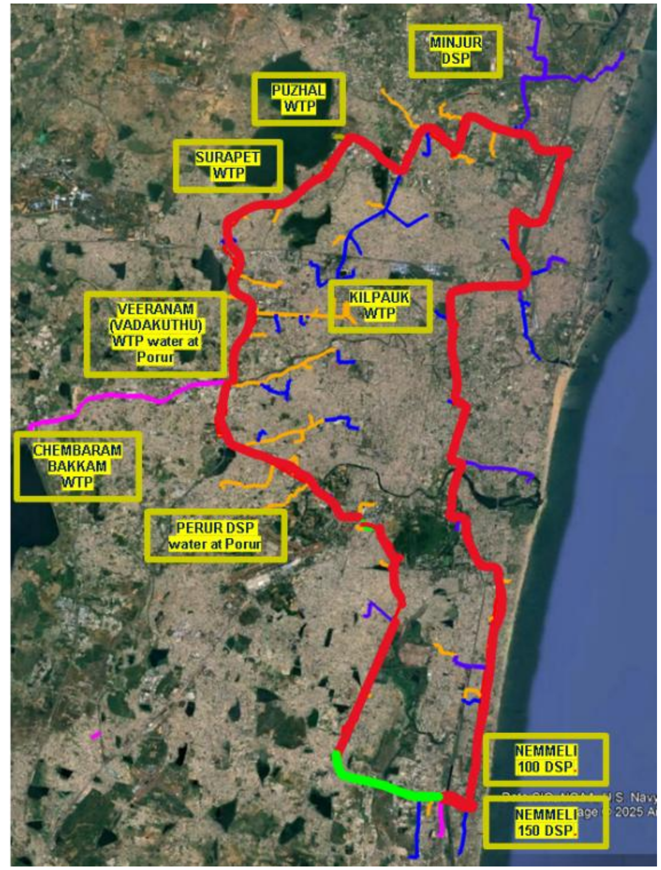


Figure 4: Map showing proposed ring main (Red) alignment and connecting sources

❑ RING MAIN FOR GCC:

Sources to be connected

1. Puzhal WTP
2. Surapet WTP
3. Chembarambakkam WTP
4. Kilpauk WTP
5. Veeranam water at Porur Booster
6. Perur DSP water at Porur Booster (proposed)
7. Minjur water at Madhavaram Booster
8. Nemmeli DSP water at akkarai Booster
9. Nemmeli DSP water at TNHB Booster



18. **Water Treatment Plants (WTPs).** Raw water from the reservoirs and sea water intakes is conveyed to various WTPs through a network of lined canals, open channels, intake wells, and raw water mains. WTPs include Kilpauk (270 MLD), Puzhal/Red Hills (300 MLD), Surapet, currently under rehabilitation (47 MLD), Chembarambakkam (530 MLD), and Vadakuthu (180 MLD). After the treatment, the treated water is conveyed to seven ground level clear water reservoirs (CWRs) and 16 major distribution stations, which form the backbone of the city's transmission and delivery system.

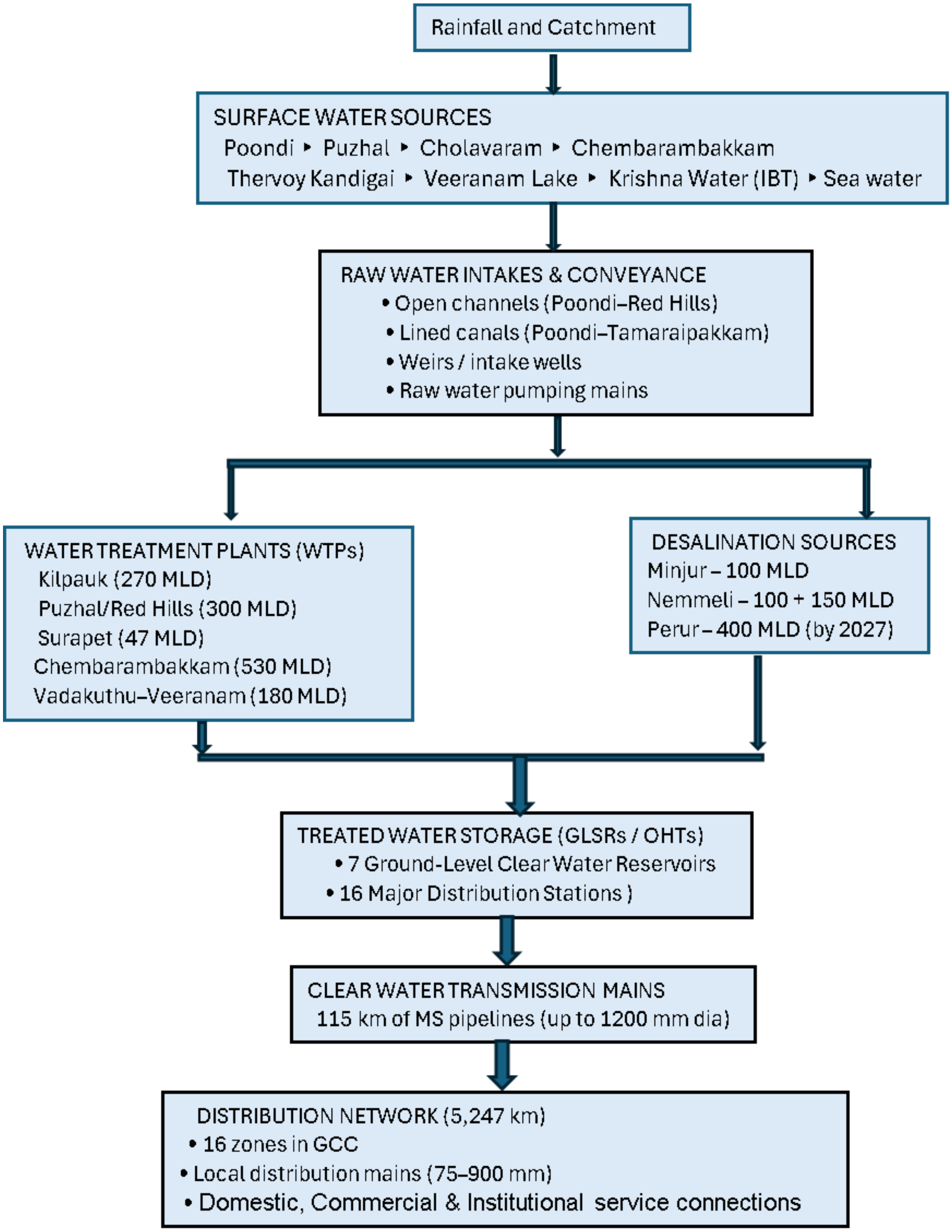
19. The total water demand within GCC area is projected to rise steadily over the planning horizon. In the base year 2027, demand is estimated at 1,230 MLD, intermediate year 2042, 1,496 MLD, by the ultimate year 2057, the total projected demand is 1,766 MLD. Details of existing water supply sources and treatment facilities is given in Table 1. Flow diagram of water supply is presented in **Figure 5**.

Table 1: Details of existing surface sources of water supply

Water source	Capacity (MCM)	Treatment facility	Year of commissioning	Capacity (MLD)	Average water supply (MLD)
Puzhal tank	93.46	Kilpauk WTP	1914/Upgraded 2005	270	584
Poondi reservoir	91.47	Puzhal WTP	1996	310	
Cholavaram Lake	29.12	Surapet WTP	2009	47	530
Chembarambakkam Lake	103.37	Chembarambakkam WTP	2007	530	
Krishna water supply	339.84				
Veeranam Lake	-	Vadakuthu WTP	2004	180	180
Sea Water (North Chennai)	-	Minjur DSP	2010	100	250
Sea Water (South Chennai)	-	Nemmeli DSP	2013	100	
	-	Nemmeli DSP	2024	150	
	-	Perur DSP	To be commissioned in 2027	400	
Orathur & Manimangalam Lake	-	WTP	Proposed by CMA	200	
Total	-	-	**-**	**2,287**	**1,544**
CMA = Chennai Metropolitan Area; MCM = million cubic meters; MLD = million liters per day					

.

Figure 5: Schematic "Flow of Water" Diagram for existing water supply



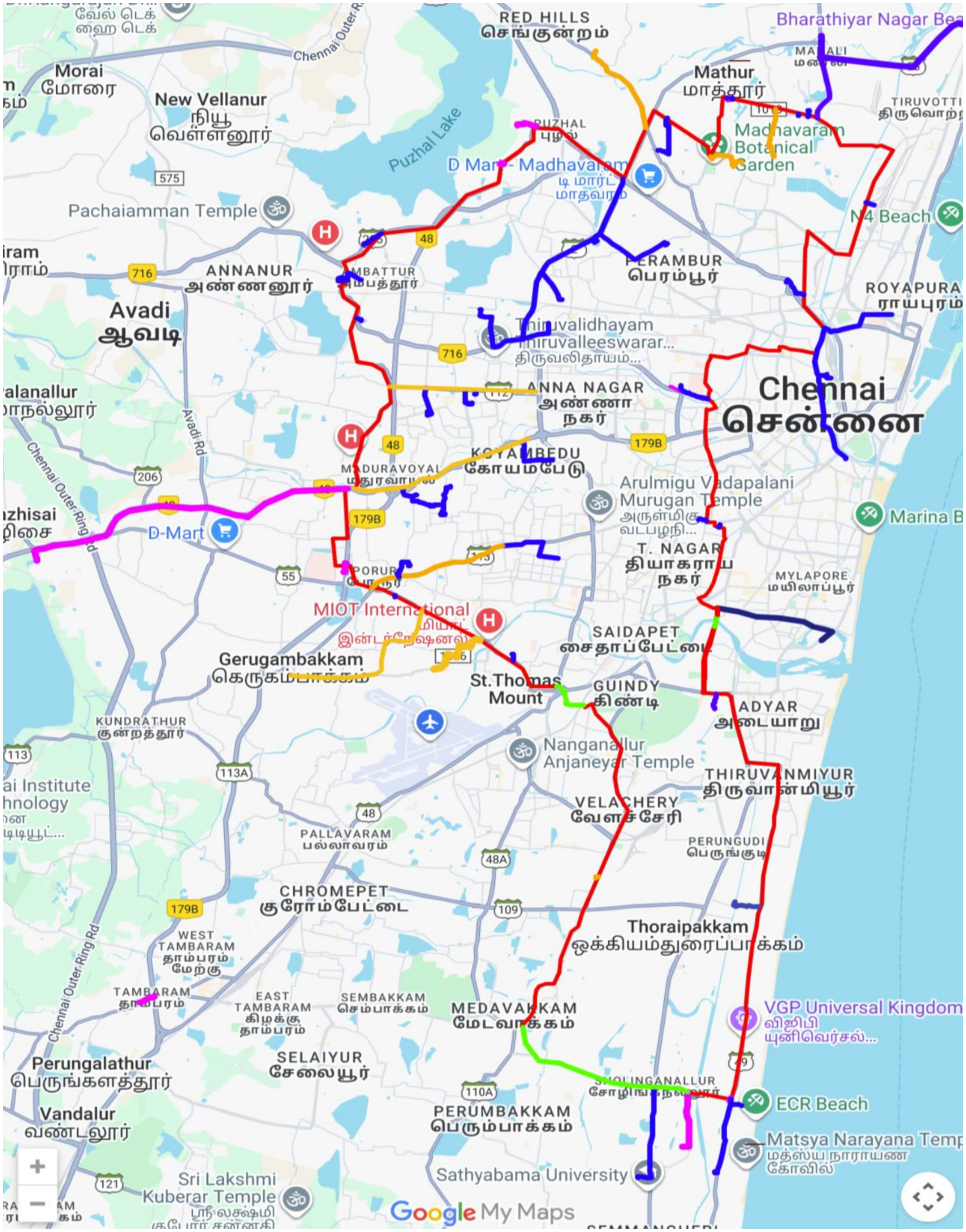
21. **Transmission and Distribution Network.** Clear water transmission is carried out through 115 km of large-diameter (up to 1,200 mm) mild-steel (MS) pipelines, feeding a distribution network of 5,247 km, covering all 16 administrative zones of the GCC. The system supplies over 9.3 lakh (equivalent to 1,000,000) domestic and non-domestic service connections (as of March 2022), ensuring equitable distribution across the urban area.

C. Proposed Ring Main System subproject

22. **Gaps in the Existing Water Supply System.** The current Chennai water supply system functions as separate source-based zones, with each WTP or desalination plant supplying its own independent commanding area. Since there are no interconnections or balancing mechanisms, each zone is fully dependent on the performance of its respective source. Any source failure—such as drought, low reservoir inflow, plant shutdowns, or power issues—directly reduces supply to its area, creating high vulnerability. As a result, water distribution across the city is unequal: zones served by reliable or desalinated sources receive better supply, while peripheral and high-elevation areas face low pressure, intermittent supply, or reliance on tankers. This imbalance affects service quality, economic activity, and future urban growth, and is expected to worsen without system integration.

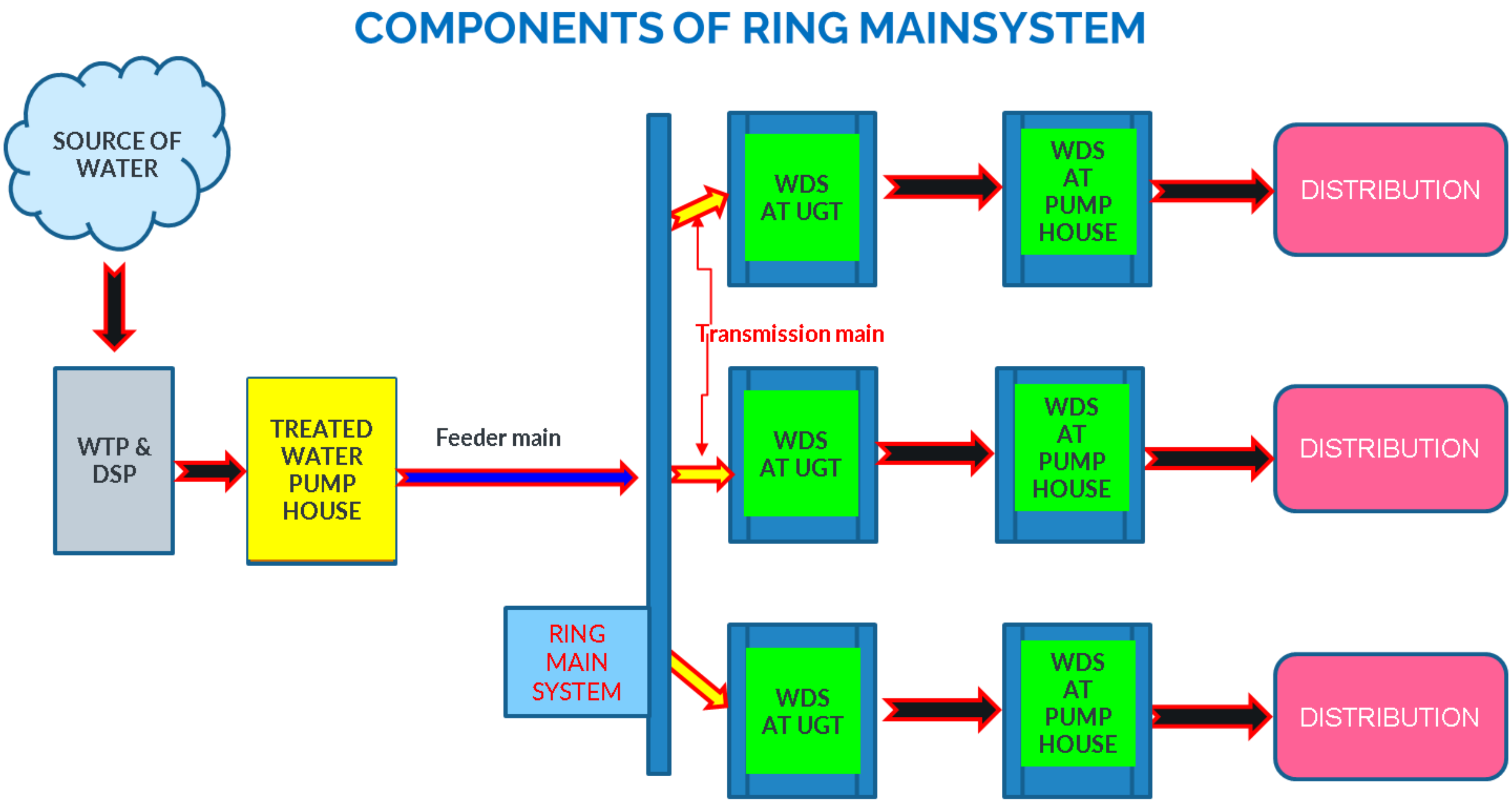
23. **Concept of the Ring Main System (RMS):** The RMS is proposed to address capacity and operational gaps in the existing network by creating a unified, resilient transmission system. It enhances reliability, balances multiple water sources, redistributes supply across zones, and supports equitable, continuous water delivery. All major WTPs and DSPs feed treated water into a continuous circular bulk transmission line forming the city's "ring." From this ring, secondary transmission mains convey water to underground storage tanks (UGTs) at water distribution stations (WDSs) across GCC/CMA. There are no direct links to overhead tanks (OHTs) or consumer networks; distribution continues through existing or upgraded local systems fed from WDS-level UGTs. This structure improves hydraulic stability, flow control, and overall system efficiency. Google map showing proposed ring main for GCC area is presented in **Figure 6** and line diagram showing existing and proposed components is presented in **Figure 7Error! Reference source not found.**.

Figure 6: Proposed RMS for Chennai City



Legend: - Red line - Proposed ring main, Green - existing to be used
Blue line- proposed transmission mains, orange - existing transmission main to be used.
Pink - proposed feeder main

Figure 7: Line diagram showing existing and proposed components



24. **Project Components.** The project comprises a series of interconnected hydraulic, structural, and automation components designed to establish a unified, resilient, and efficient water transmission system for Chennai. Details of proposed project components are presented in Table 2.

Table 2: Details of Proposed Project Components

<table>
<tr><th>Component</th><th>Function</th><th>Description</th><th>Location</th></tr>
<tr><td>Ring main</td><td>To collect water from feeder mains and supply to transmission mains</td><td>
• Total Length - 97.9 km
• New - 91.2 km
• Existing to be used - 6.7 km
• Pipe Material: Mild Steel (MS)
<table>
<tr><th colspan="2">Details of proposed pipelines</th></tr>
<tr><th>Diameter in mm</th><th>Length in m</th></tr>
<tr><td>1,181</td><td>43,744</td></tr>
<tr><td>1,373</td><td>14,770</td></tr>
<tr><td>1,574</td><td>27,046</td></tr>
<tr><td>1,777</td><td>5,654</td></tr>
<tr><td></td><td>91,214</td></tr>
</table>
New – Pipe carrying bridges (PCBs)
• 17 locations including 6 within CRZ area requiring prior CRZ clearance.
• Length – 10 to 150 m
• Span – 12 to 25 m.</td><td>Pipes will be laid within the ROW of major roads and urban corridors. Along highways, the pipelines will be laid within the service roads, while on other roads they will be laid within the carriageway, details of roads are provided in Appendix 9. The exact alignment may be shifted slightly, as required, based on existing utilities, site conditions, and the clearance conditions stipulated by National Highways Authority of India (NHAI), State Highways, and other concerned agencies.

Location of PCBs
• Retteri Lake - 2 locations
• Adyar River - 1 locations
• Buckingham Canal - 6 locations
• Captain Cotton Canal -1 location
• Coovum River -2 locations
• South Boag Road Nalla - 1 location
• Otteri Nalla - 3 locations
• Veerangal Odai, Velachery - 1 location</td></tr>
<tr><td>Feeder mains</td><td>Source to ring main - convey treated water from water sources (WTPs/DSPs) into the RMS.</td><td>
• Total Length - 15.87 km
• New - 2.65 km
• Existing to be used - 13.22 km
• Pipe Material: Mild Steel (MS)
<table>
<tr><th colspan="2">Details of proposed pipelines</th></tr>
<tr><th>Diameter in mm</th><th>Length in m</th></tr>
<tr><td>781</td><td>200</td></tr>
<tr><td>980</td><td>200</td></tr>
<tr><td>1,777</td><td>1,600</td></tr>
</table></td><td>Pipes will be laid within the ROW of major roads and urban corridors. Along highways, the pipelines will be laid within the service roads, while on other roads they will be laid within the carriageway. The exact alignment may be shifted slightly, as required, based on existing utilities, site conditions, and the clearance conditions stipulated by NHAI, State Highways, and other concerned agencies.</td></tr>
</table>

<table>
<tr><th>Component</th><th>Function</th><th>Description</th><th>Location</th></tr>
<tr><td></td><td></td><td><table><tr><td>1,976</td><td>650</td></tr><tr><td>Total</td><td>2,650</td></tr></table></td><td></td></tr>
<tr><td>Transmission mains</td><td>To tap and convey water from ring main to the UGTs of the WDSs</td><td>
Total Length: 114 km
New 64.2 km
Existing to be used - 49.8 km
Pipe Material: Mild Steel (MS)
<table><tr><td colspan="2">Details of proposed pipelines</td></tr><tr><td>Diameter in mm</td><td>Length in m</td></tr><tr><td>150</td><td>428</td></tr><tr><td>200</td><td>223</td></tr><tr><td>300</td><td>6,020</td></tr><tr><td>350</td><td>678</td></tr><tr><td>400</td><td>6,392</td></tr><tr><td>450</td><td>5,675</td></tr><tr><td>500</td><td>12,112</td></tr><tr><td>600</td><td>15,480</td></tr><tr><td>650</td><td>63</td></tr><tr><td>700</td><td>4,211</td></tr><tr><td>781</td><td>3354</td></tr><tr><td>980</td><td>4,349</td></tr><tr><td>1,181</td><td>3,883</td></tr><tr><td>1,574</td><td>1,315</td></tr><tr><td></td><td>64,183</td></tr></table>
New – PCBs
7 locations, all 7 in CRZ area, requiring prior CRZ clearance.
Length – 10 to 60 m
Span – 10 to 25 m.
</td><td>Pipes will be laid within the ROW of major roads and urban corridors. Along highways, the pipelines will be laid within the service roads, while on other roads they will be laid within the carriageway. The exact alignment may be shifted slightly, as required, based on existing utilities, site conditions, and the clearance conditions stipulated by NHAI, State Highways, and other concerned agencies.

Location of PCBs
Adyar River - 1 location
Buckingham Canal - 4 locations
Kosasthalaiyar River -2 locations</td></tr>
<tr><td>Underground storage tanks (UGTs)</td><td>To store clear water for supply</td><td>7 new
Capacity – 1 ML - 6 numbers
0.5 ML - 1 number</td><td>Within the following existing location of water distribution station premises.
Thiruvotriyur
Puzhal
Sholinganallur Neduncheziyan street
Semmenchery
Puzhuthivakkam Okkiyam
Thoraipakkam</td></tr>
</table>

Component	Function	Description	Location
Upgrading existing clear water pumps	Lift and convey treated water from the clear water sump to the distribution system or elevated storage reservoirs at the required pressure and flow.	Existing pumps to be replaced at seven pumping stations	Within the existing pumphouses owned by CMWSSB 1. Chembarambakkam 2. Puzhal 3. Madhavaram 4. Kilpauk 5. Porur 6. Akkarai 7. Surapet
Surge tanks	To protect the transmission system from water hammer and pressure surges	**New** • 8 number - 5 reinforced cement concrete (RCC) + 3 mild steel (MS) • Diameter 1.5 to 10 m • Height – 17.6 to 27.8 m	Within the existing WTP and booster station premises at 1. Chambarambakkam 2. Puzhal 3. Kilpauk 4. Akkarai BS 5. Porur BS Two-way Surge Shaft 6. TNHB BS 7. Surapet WTP 8. Madhavaram BS
SCADA system with instrumentations	To regulate the flow in the RMS to enable equitable water distribution	• **New-** Sensors and insturments for Real-time monitoring of flow, pressure, pumping, and reservoir levels. • Master control center – 1 No.	Within the existing premises owned by CMWSSB

25. **Ring Main.** The Ring Main will function as the backbone of Chennai's bulk water transmission system. It integrates all major water sources and enables flexible, balanced distribution to various transmission corridors. This will ensure redundancy, continuous circulation, and uniform pressure across the network.

26. **Feeder mains.** (Source-to-Ring main Conveyance) will convey treated water from **8 water sources** (WTPs/DSPs) into the RMS. These mains ensure seamless integration of WTPs and DSPs with the Ring main.

27. **Transmission Mains. (**Ring-to-WDS Conveyance**)** will tap water from the Ring Main and deliver it to Underground Tanks (UGTs)

at 85 WDSs. This will enable demand-based water delivery to the localized supply stations.

28. **Existing Clear Water Pumps Upgradation.** The project includes upgradation of the existing clear water pumps. New pumps (centrifugal and vertical turbine) will enhance hydraulic capacity, reduce energy consumption, and improve reliability. Details of pump replacement are provided in Table 3**: - Details of Pumps to be Replaced**.

Table 3: - Details of Pumps to be Replaced

S. No	Name of Pumping Plant	Type of Pumps	No. of Pumps (working +standby)	Capacity	KW
1	Chembarambakkam CWPH	HSC Centrifugal	4W + 2S	3666 m3/hr @ 31m	400
2	Puzhal CWPH	HSC Centrifugal	4W + 2S	3125 m3/hr @ 31m	330
3	Madhavaram Booster Station	HSC Centrifugal	4W + 2S	729 m3/hr @ 38m	100
4	Kilpauk CWPH	Vertical Turbine	4W + 2S	2813 m3/hr @ 37m	350
5	Porur Booster Station	Vertical Turbine	4W + 2S	1667 m3/hr @ 32m	200
6	Akkarai Booster	Vertical Turbine	4W + 2S	1042 m3/hr @ 45m	175
7	Surapet CWPH	Vertical Turbine	2W + 1S	833 m3/hr @ 31m	92
CWPH - clear water pump house, HSC - horizontal split case					

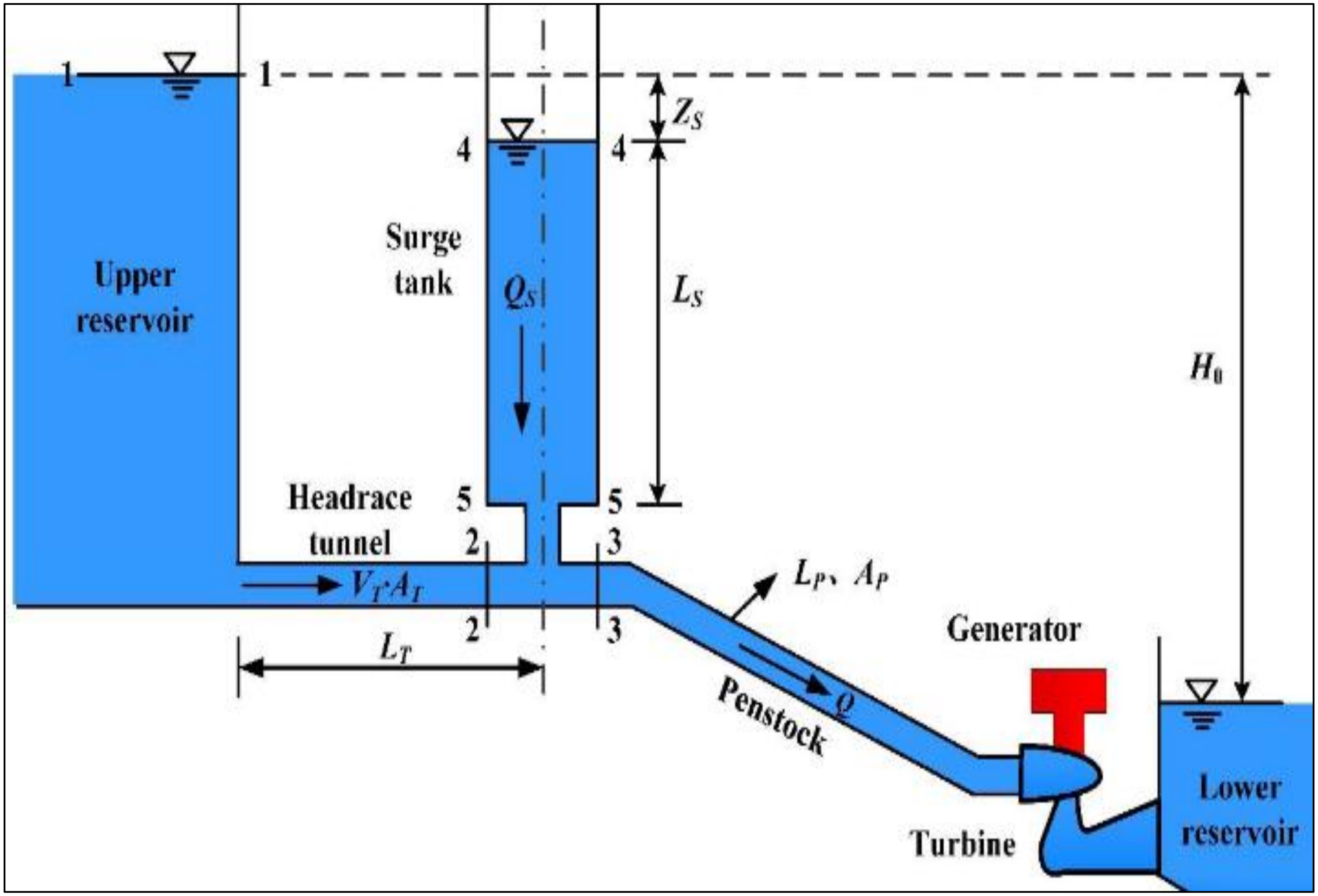
29. **Surge protection system.** Surge tanks will be installed at 8 strategic locations to protect the transmission system from water hammer and pressure surges. Details of proposed surge tanks is presented in

30. . Typical diagram of surge tank is presented in **Figure 8**.

Table 4: **Details of proposed surge tanks**

Sl. No	Description	Type of tank	Diameter (m)	Height (m)
ONE-WAY SURGE TANK				
1	Chembarambakkam WTP	RCC	10	27.8
2	Puzhal WTP	RCC	9	22.38
3	Kilpauk WTP	RCC	7	17.3
4	Akkarai Booster Station	RCC	6	25
5	Porur Booster Station	RCC	5	27
TWO-WAY SURGE SHAFT				
6	TNHB Booster Station	MS	2	22
7	Surapet WTP	MS	2.5	26.6
8	Madhavaram BS	MS	1.5	25.6
RCC = reinforced concrete cement; MS = mild steel; BS = booster station				

Figure 8: Details of Surge Shaft



31. **Pipe Carrying Bridges (PCBs).** Proposed at 24 locations to carry transmission or feeder mains across natural or man-made obstacles. The bridge structures will be designed for adequate carrying pipe diameter, load, and span length, ensuring hydraulic continuity and structural safety. Details are provided in

32. Table **5**. Typical diagram of pipe crossing bridges are presented in **Figure 9** below.

Figure 9: Typical diagram of pipe crossing bridge

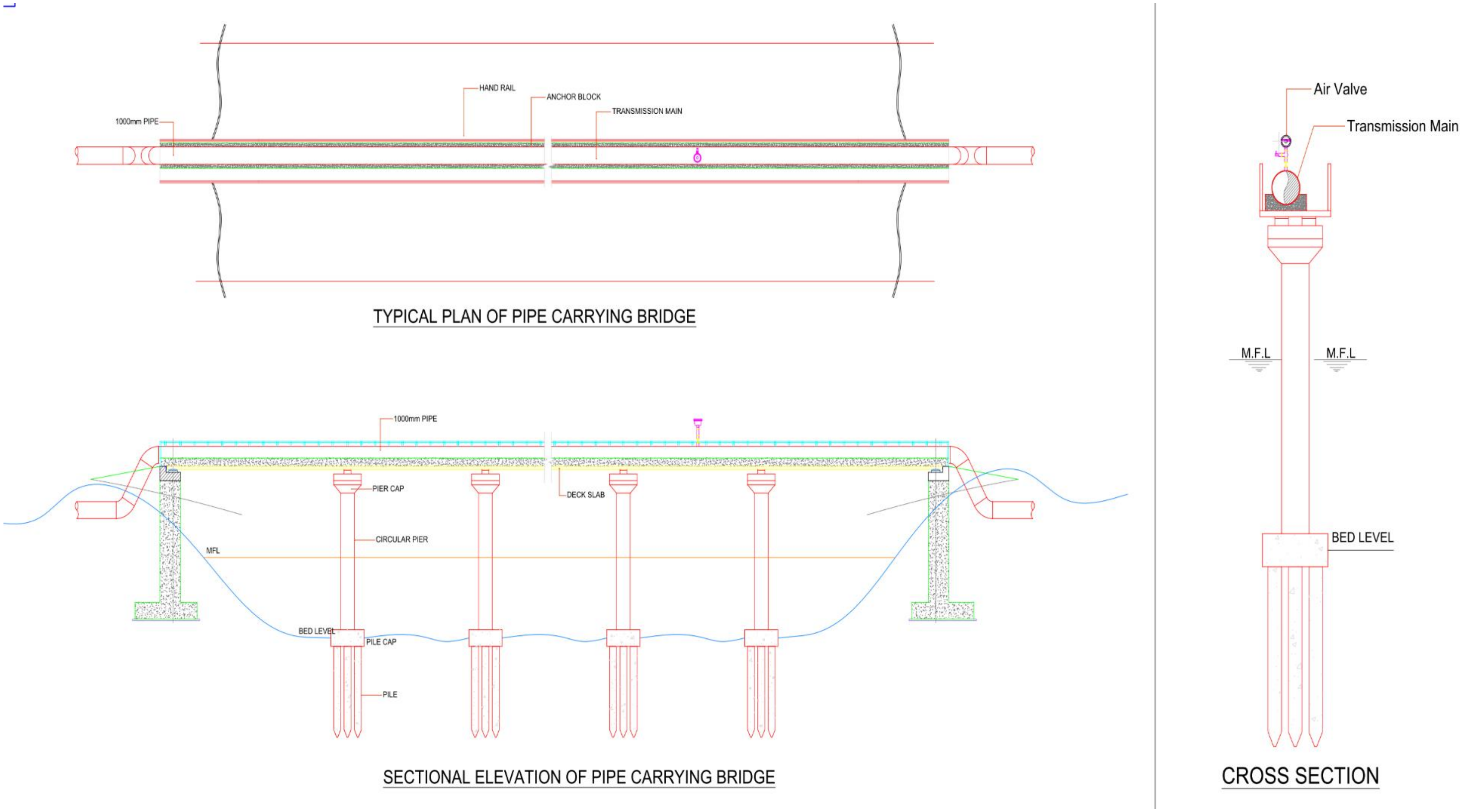


Table 5: Details of proposed pipe carrying bridges

Sl No	Description	Length (m)	Span (m)	Diameter (mm)	Remarks
	Ring Main				
1	Retteri Lake Bridge -1	60	12	1800	
2	Retteri Lake bridge 2	24	12	1800	
3	Adyar River, Madras institute of orthopedics and traumatology (MIOT) Hospital	150	15	1600	
4	Buckingham Canal Manali, Chennai petroleum corporation limited (CPCL) 2	40	20	1600	CRZ *
5	Captain Cotton Canal, Thiru. V. Kalyanasundaram, Link Road	60	20	1600	
6	Coovum River, Vanagaram	60	20	1600	
7	Nalla, South Boag Road	10	10	1200	
8	Otteri Nalla, Basin bridge	36	18	1200	
9	Otteri Nalla, near Kilpauk shaft	20	20	1200	
10	Buckingham canal, Tondiarpet high road 1	30	15	1600	CRZ*
11	Buckingham canal, Tondiarpet high road 2	30	15	1600	
12	Buckingham canal, Akkarai	160	20	1200	CRZ*
13	Coovum River, Nungambakkam bridge	60	20	1200	CRZ*
14	Veerangal Odai, Velachery	40	20	1200	CRZ*
15	Buckingham Canal, Manali CPCL-1	25	25	1600	
16	Otteri Nalla, Otteri	25	25	1200	CRZ*
17	Buckingham Canal, Thiruvanmiyur	25	25	1200	
	Transmission Main				
18	Buckingham canal to Annapark water distribution system (WDS)	60	20	1000	CRZ*
19	Buckingham canal, Thoraipakkam	25	25	600	CRZ*
20	Kosasthalaiyar river	60	20	600	CRZ*
21	Kosasthalaiyar River, Sadayan Kuppam bridge	60	20	300	CRZ*
22	Buckingham canal, Sadayan Kuppam bridge	60	20	300	CRZ*
23	Buckingham canal, Pandian Salai	25	25	500	CRZ*
24	Adyar river (Tholkapiyar park)	10	10	1000	CRZ*
*Project components within CRZ area require CRZ clearance before start of construction works					

33. **Underground Tanks (UGTs).** Six UGTs with capacities of 1.0 MLD and one 0.5 MLD will be constructed at different locations. UGTs receive water from transmission mains and supply to downstream distribution networks. All the proposed tanks are planned to be constructed within the existing premises owned by the CMWSSB.

Figure 9 to Figure 20 below shows location of proposed UGT site on google earth imagery and layout plants of UGTs.

Figure 10: Google map showing proposed area of Thiruvotriyur WDS

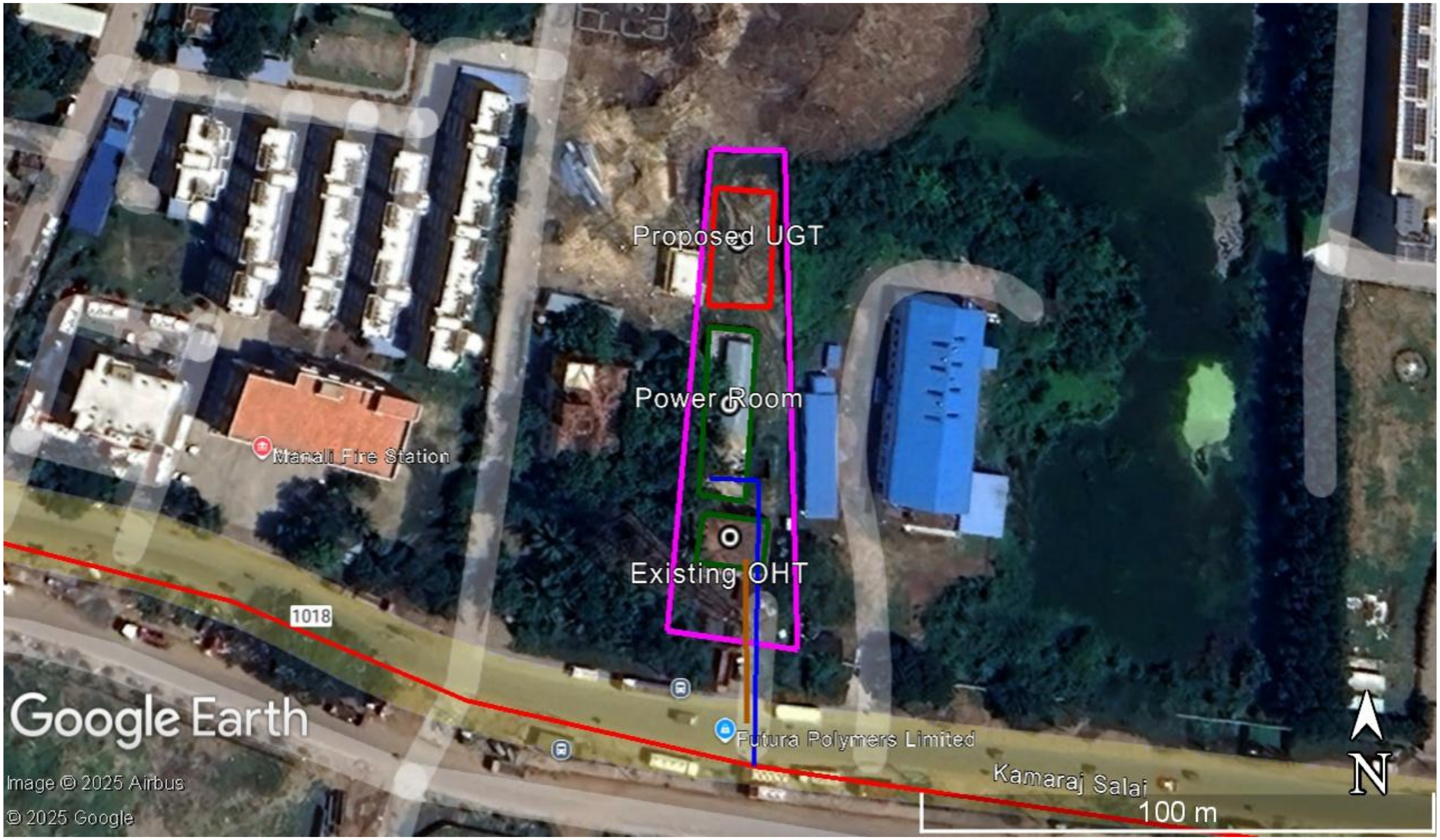


Figure 11: Layout map of proposed Thiruvottriyur Manali UGD (proposed area in pink)

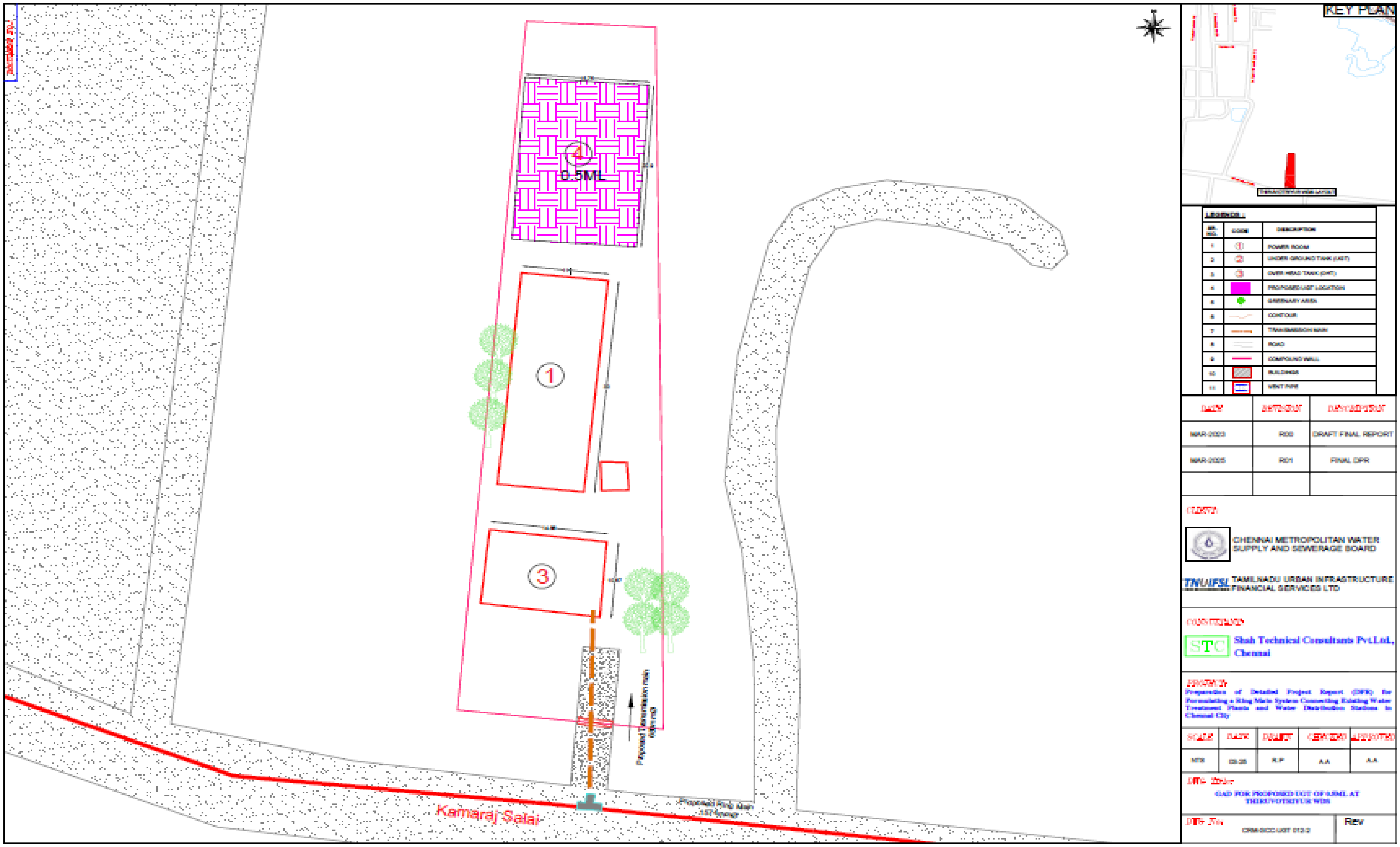


Figure 12: Google map showing proposed area of Puzhal WDS

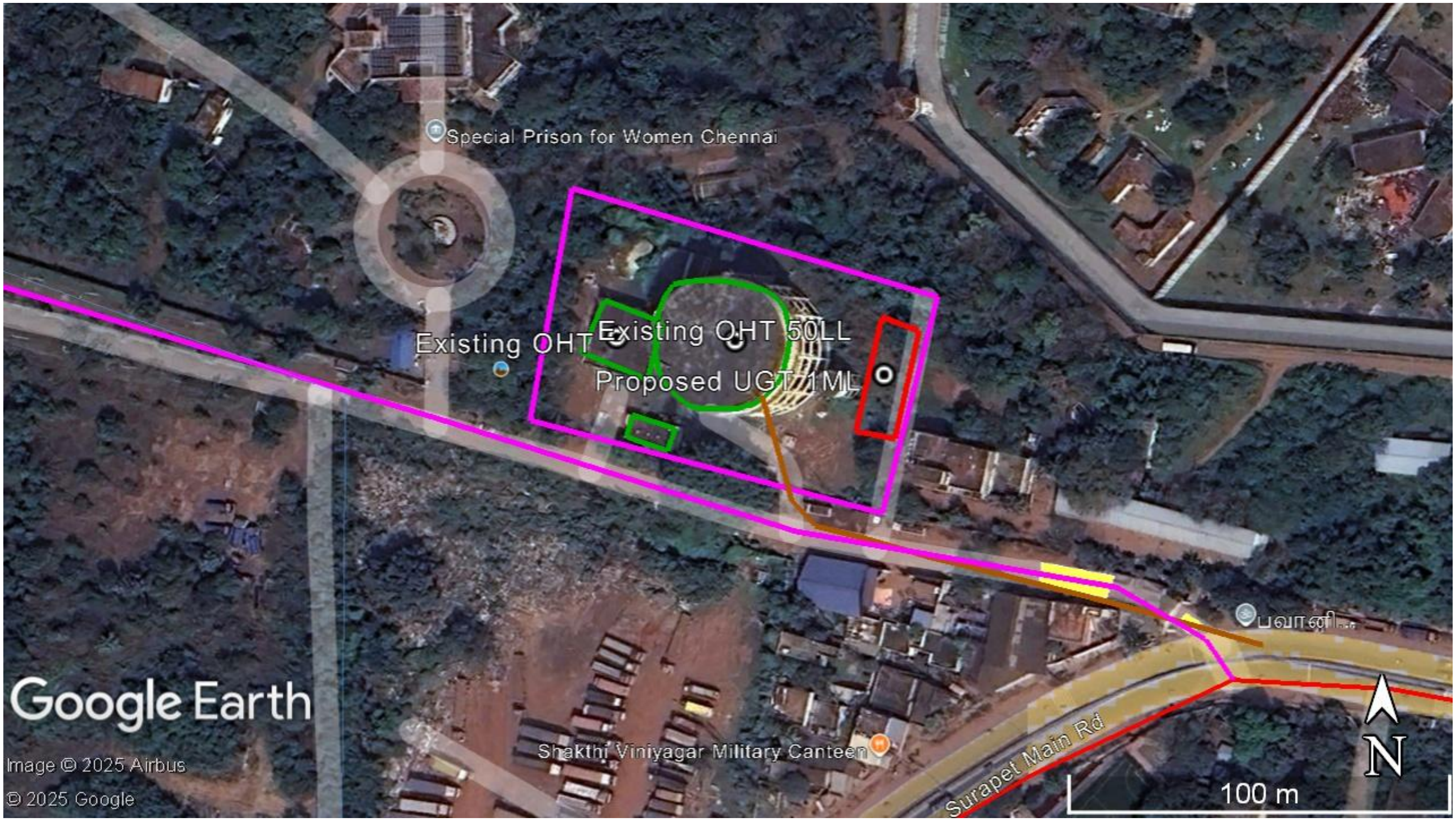


Figure 13: Layout plan of Puzhal WDS Site

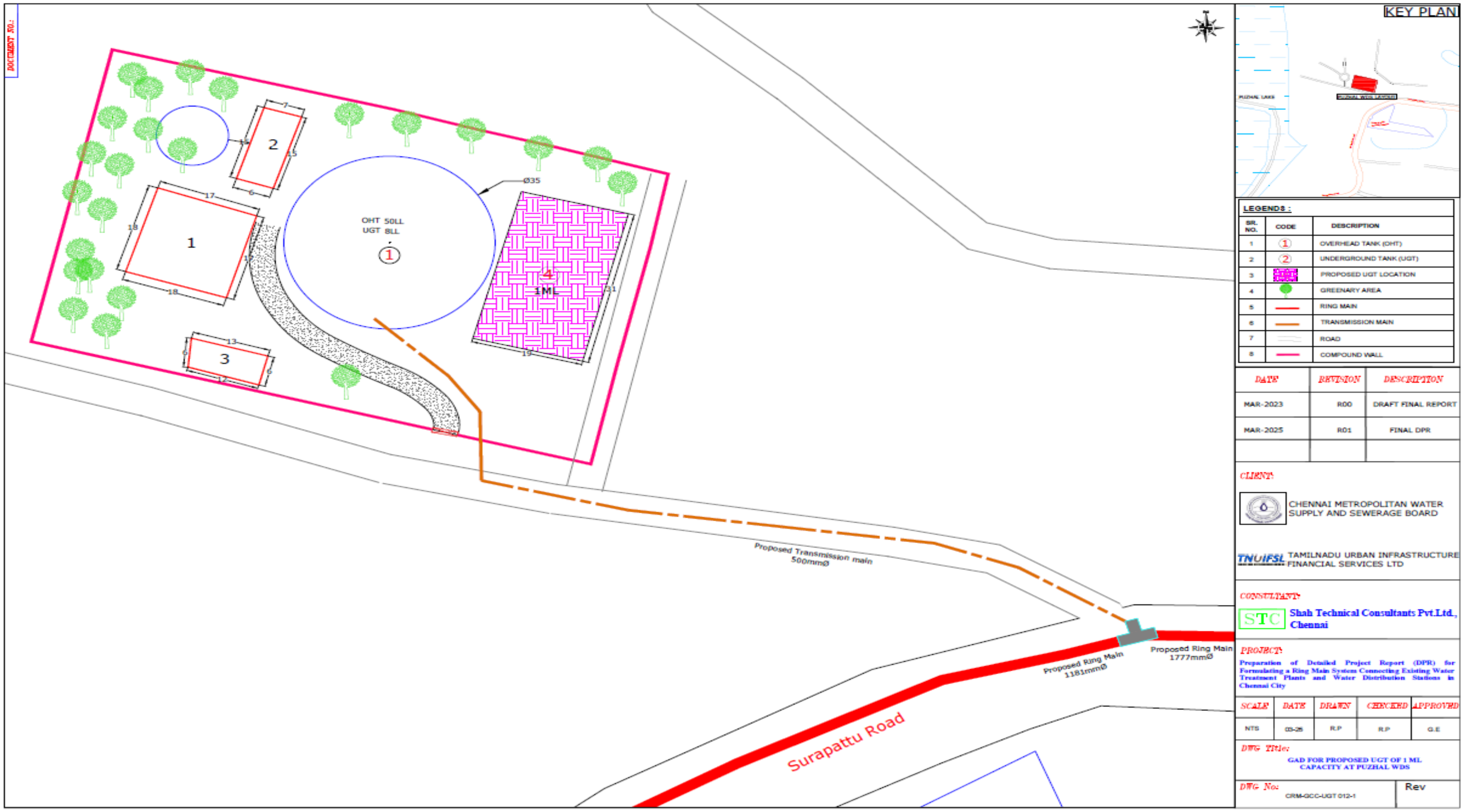


Figure 14: Google map showing proposed area for 2 WDSs in Okkiyam Thoraipakkam area

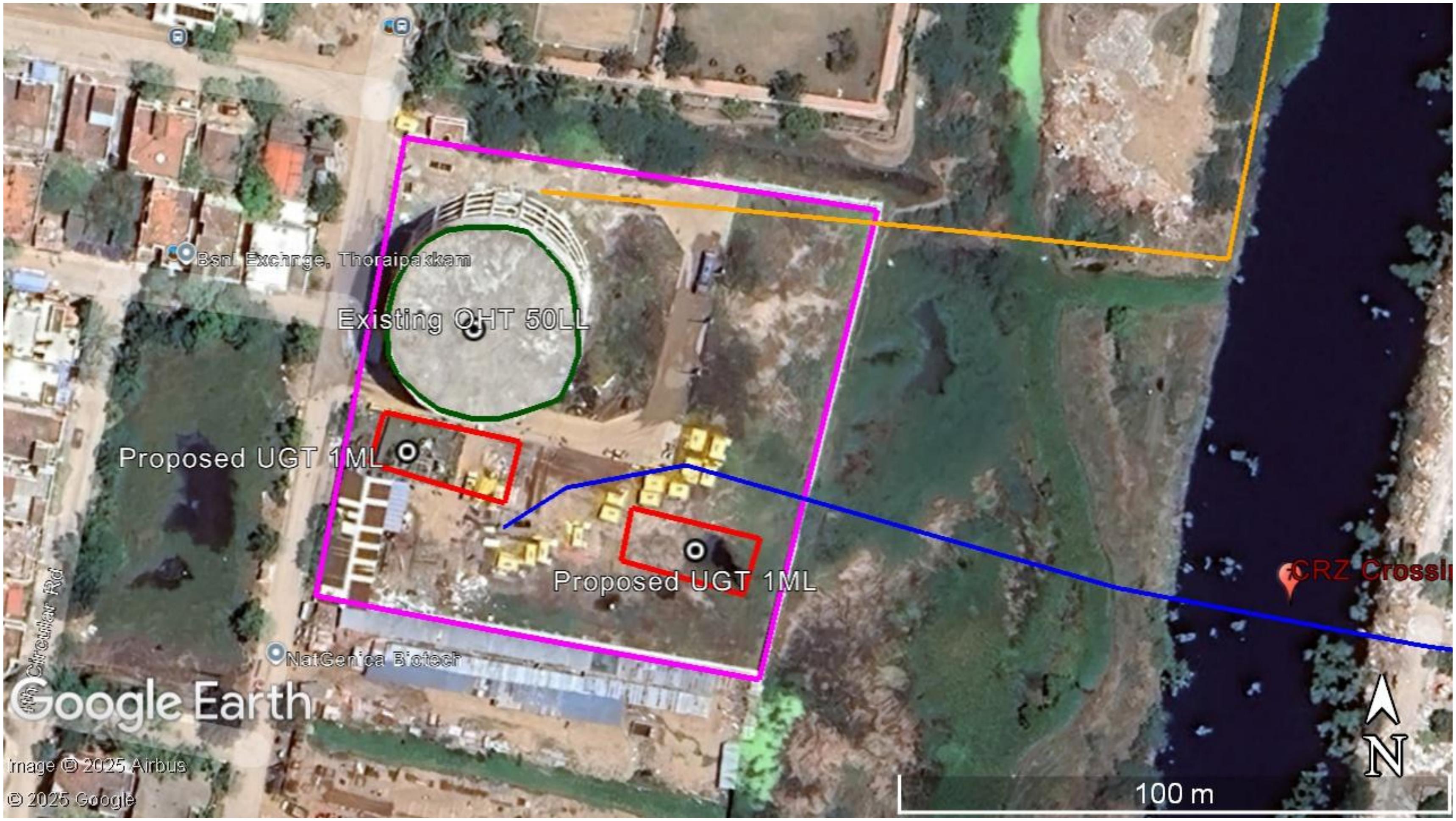


Figure 15: Layout plan for 2 WDSs in Okkiyam Thoraipakkam

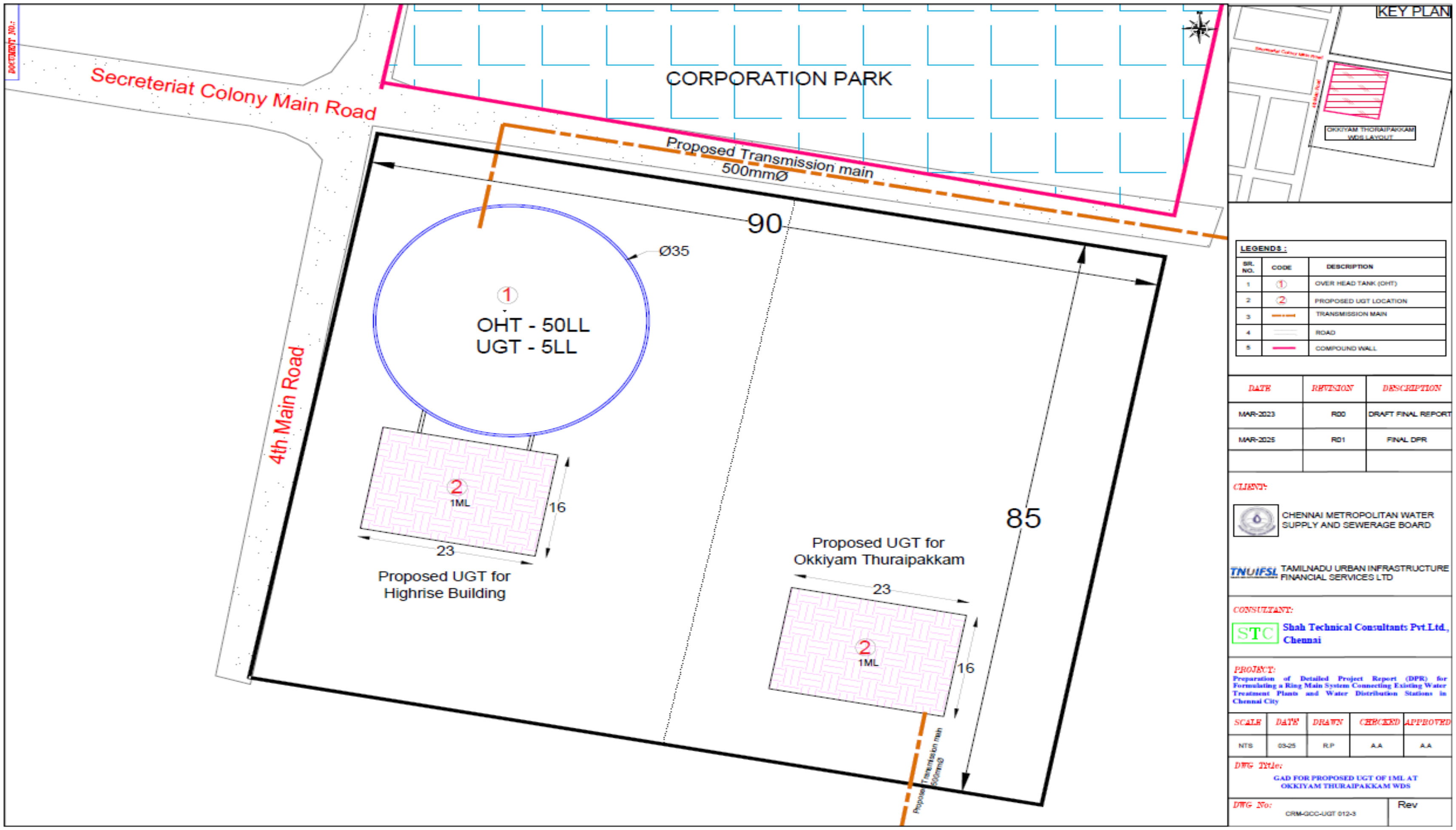


Figure 16: Google map showing proposed area of Sholinganallur Nedunchezhiyan street WDS

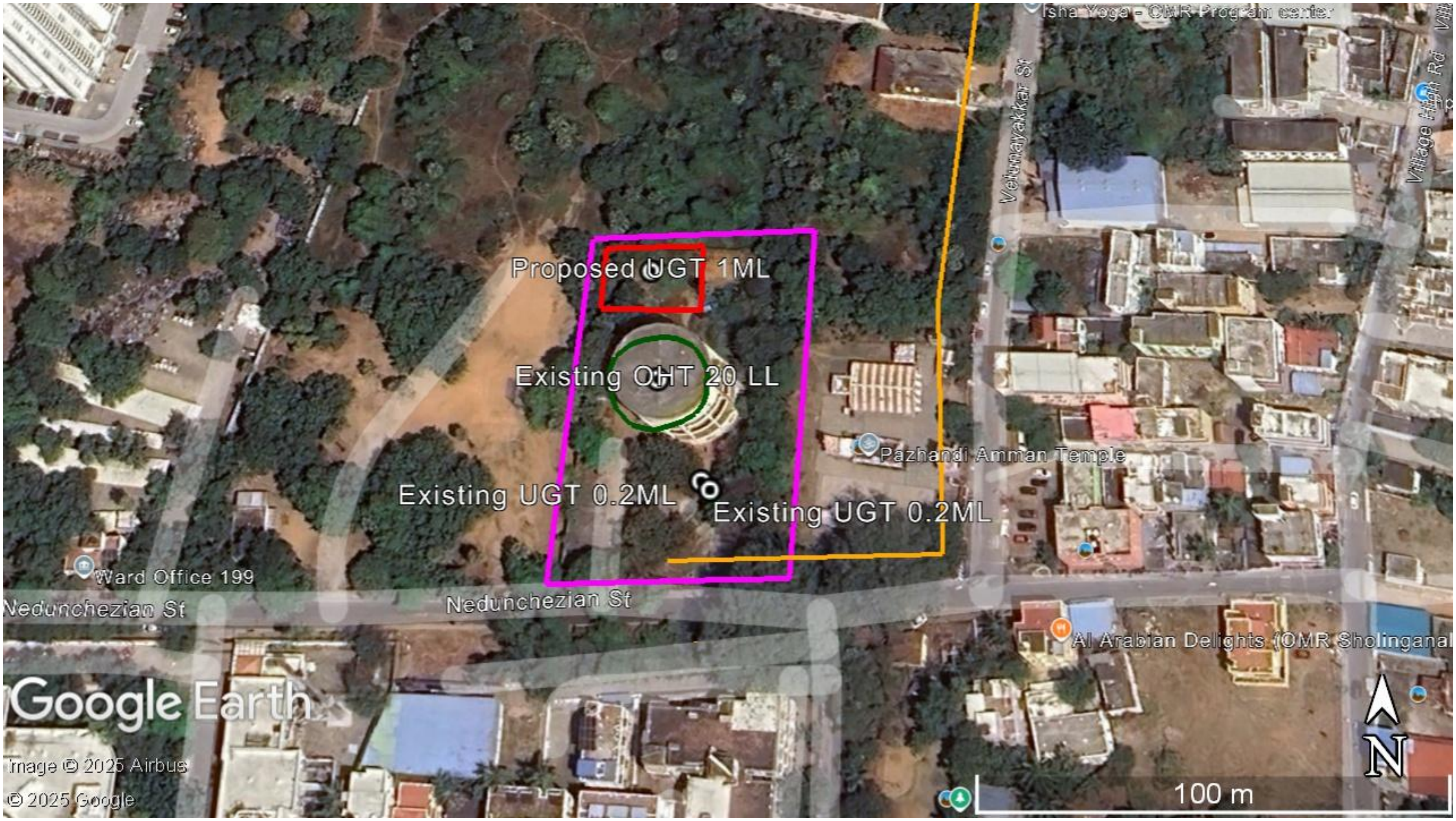


Figure 17: Layout plan of Sholinganallur Neduncheziyan street WDS

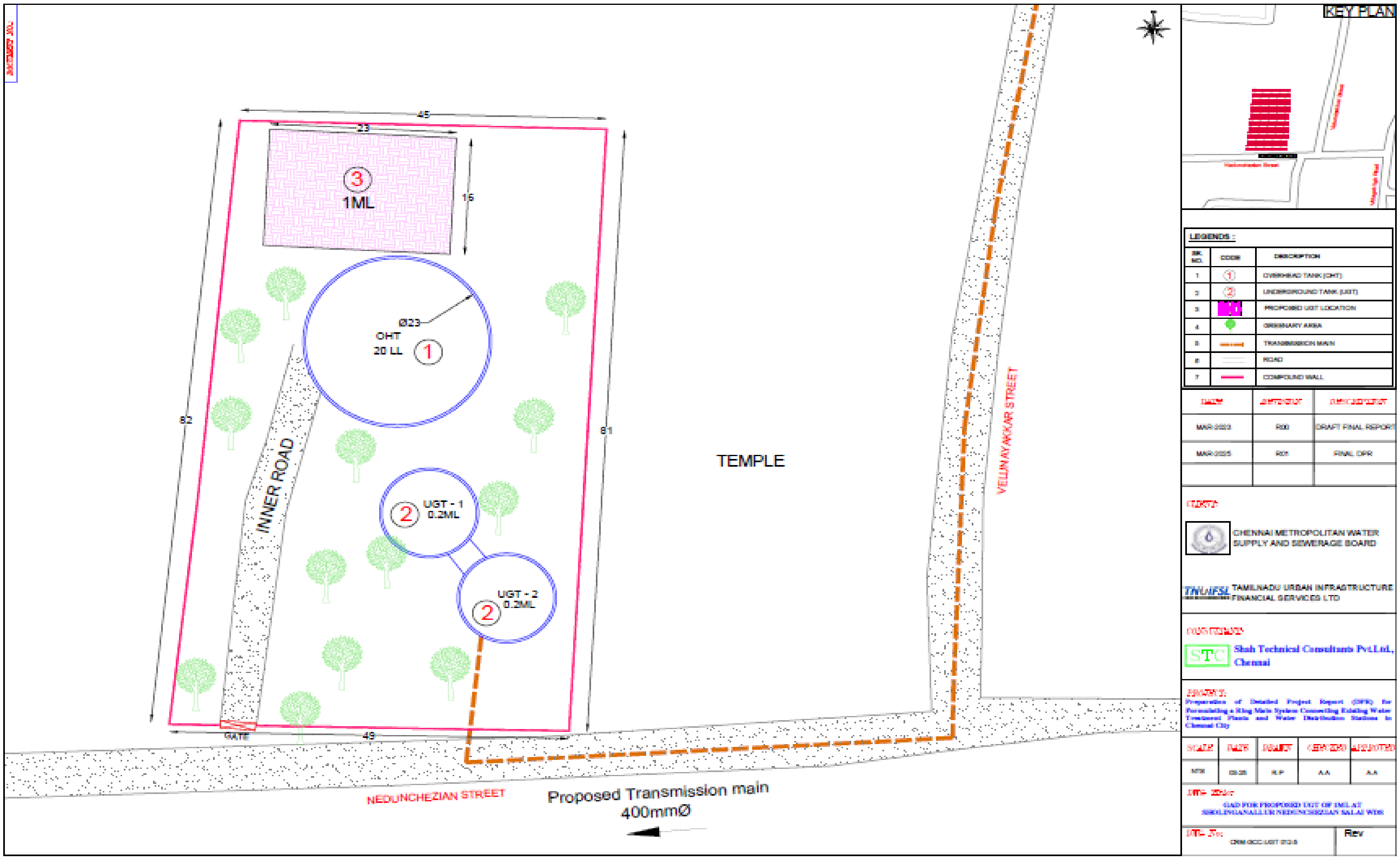


Figure 18: Google map showing proposed area of Semmenchery WDS

Figure 19: Layout plan of Semmenchery WDS

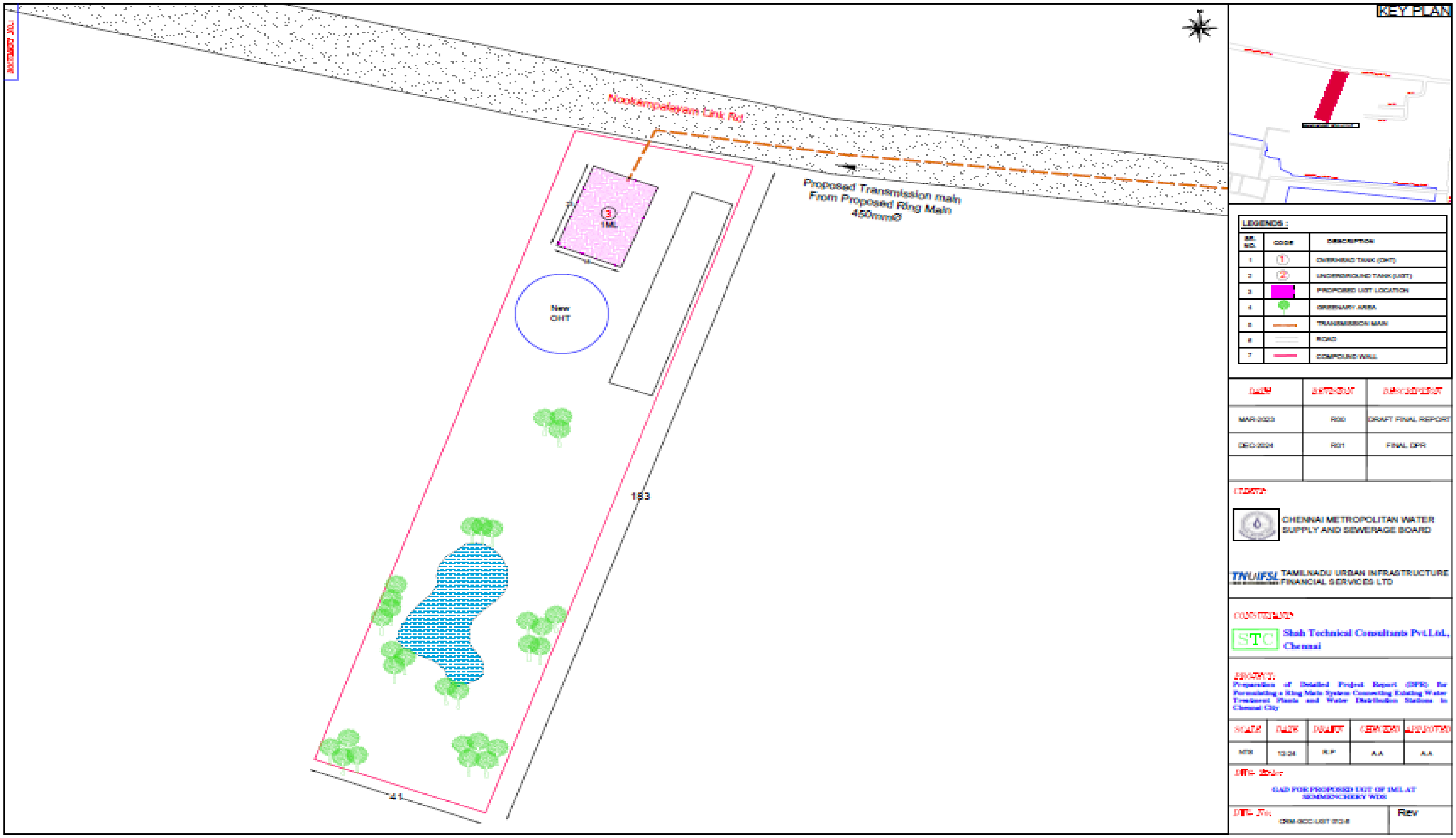


Figure 20: Google map showing proposed area of Puzhuthivakkam WDS

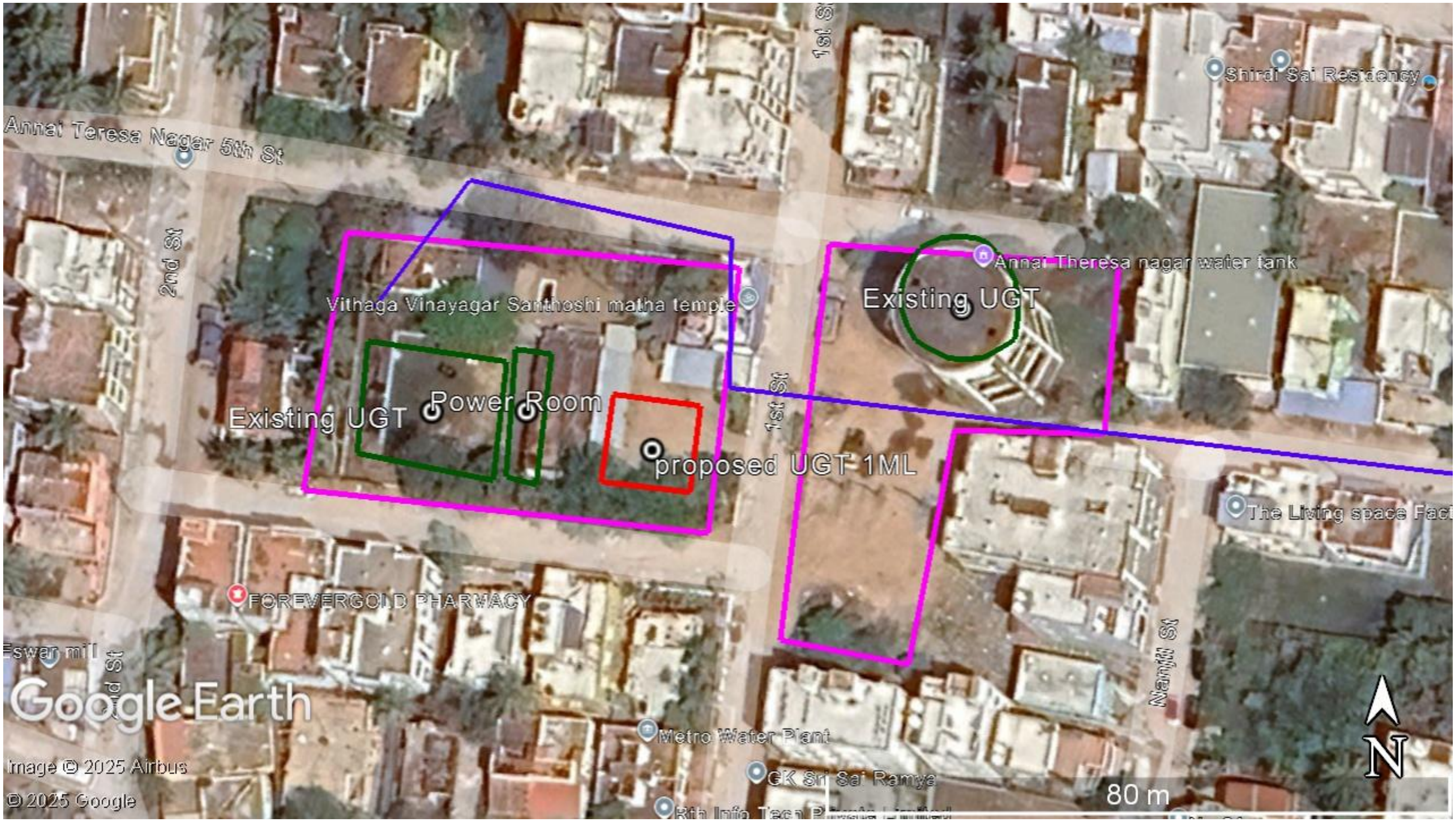
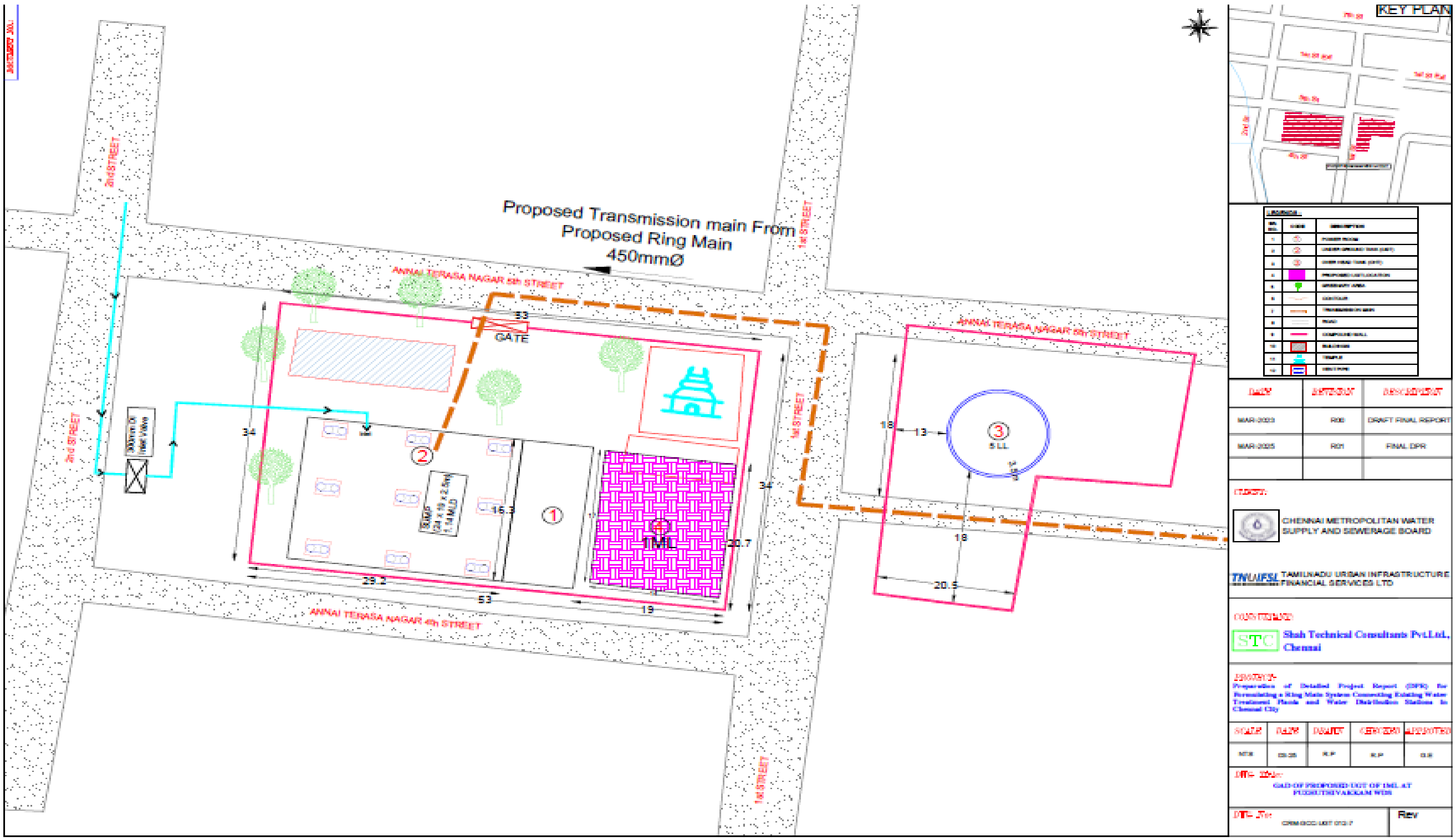


Figure 21: Layout plan of Puzhuthivakkam WDS



34. **SCADA System with Instrumentation.** A comprehensive Supervisory Control and Data Acquisition (SCADA) system will automate and monitor the entire bulk water transmission network. Key system features include: Real-time monitoring of flow, pressure, pumping, and reservoir levels; centralized control at the master control center (MCC) at CMWSSB; sensors and analyzers at all pump houses, injection points, WDSs, and tapping points, and remote operation of flow control valves and electrically operated valves.

D. Construction Methodology

35. All the proposed pipelines, including feeder mains, the Ring Main, and transmission mains—will be laid underground in the existing road corridors to avoid land acquisition and minimize environmental/social impacts. All pipelines will be laid within the ROW of existing roads, typically occupying one traffic lane. Key considerations in selecting the alignment were: maximum use of existing ROW, avoidance of tree cutting and sensitive utilities, coordination with utility departments, adoption of protective measures during excavation and minimizing inconvenience to the public through planned traffic and safety management.

36. **Pipe laying Method.** A total of 158 km of pipelines will be laid, including Ring Main, transmission mains, and feeder mains, along with associated pumps and valves, are proposed under this project. Open-cut method will be used for most stretches. Trenchless technology will be used for about 10 km pipelines, details are provided in Table 6, details of alignment laid using trenchless is presented in Appendix 2. All underground pipelines will be provided with a minimum soil cover of 1.5 m. Figure 21 presents typical strip plan of ring main, figure 22 shows typical strip plan for transmission mains and figure 23 show typical strip plan for feeder main withing existing roads.

Figure 22: **Strip plan showing typical ring main alignment within existing roads**

Figure 23: Strip plan showing typical transmission main alignment within existing roads

Figure 24: Strip plan showing typical feeder main alignment within existing roads

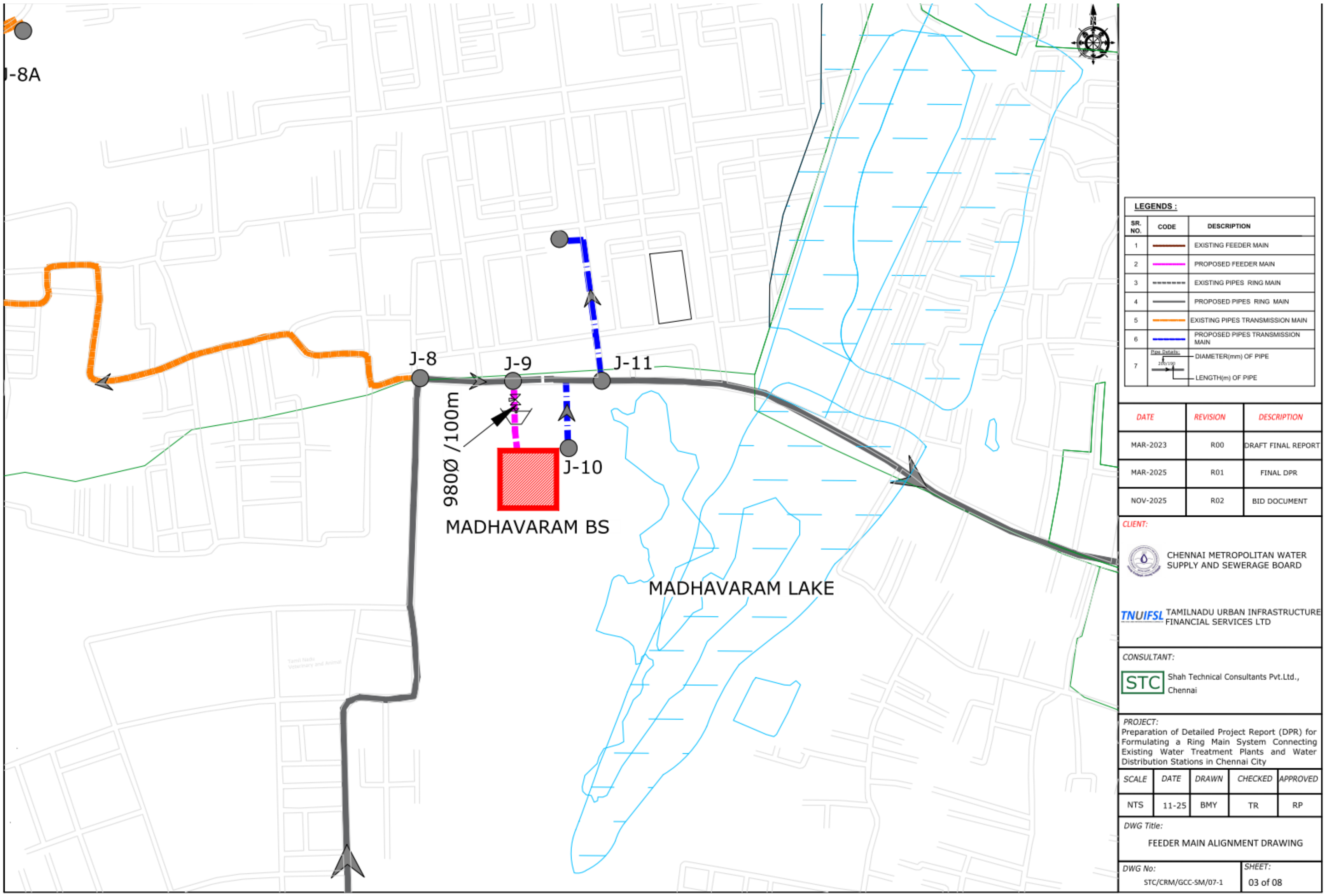


Table 6: Breakup of opencut and trenchless excavation and laying process

Description	Open excavation	Trenchless	Total
Feeder Main	2.65 km		2.65 km
Ring main	82.07 km	9.12 km	91.2 km
Transmission main	63.17 km	1.02 km	64.2 km

37. Trenches will be excavated using backhoes, bulldozers, and pneumatic drills (for concrete roads). Excavated soil will be placed alongside the trench. A bedding layer of sand or gravel (from approved quarries) will be prepared. Pipes transported to site will be lowered using small rigs. Jointing will be done manually or using machinery depending on pipe size. Screened soil will be used for backfilling and compaction. The project will utilize 465,374 cubic meters (m^3) of excavated soil and a surplus soil of 387,023 m^3 (Table 7) will be disposed of as per statutory norms. Excavated spoil will be reused, where suitable, for raising ground levels at pumping stations of CMWSSB above flood levels. Surplus spoil shall be transported in covered vehicles and disposed of only at Government-authorized C&D waste processing facilities, in compliance with the construction and demolition waste management rules, 2016, at Kodungaiyur dumping ground and Perungudi dumping ground.

Table 7: Details of earth work excavation, refilling and surplus earth

Sl. No	Description	Feeder Main	Ring Main	Transmission Main	Total
		m^3			
1	Earthwork excavation - (hard stiff clay, stiff black cotton, hard red earth, shales, murram gravel, stoney earth and earth mixed with small boulders and hard gravelly soil)				
	First depth 2.0 m	15,432	423,152	59,004	497,588
	Second depth 2.0 m to 3.0 m	5,755	153,760	97,006	256,521
	Third depth 3.0 m to 4.0 m	165	5,209	88	5,462
	Total	**21,353**	**582,121**	**156,097**	**759,571**
2	Earthwork excavation (soft disintegrated rock, laterite, soft rock or kankar, not requiring blasting)				
	First depth 2.0 m	2,157	34,529		36,687
	Second depth 2.0 m to 3.0 m	2,397	21,581		23,978
	Third depth 3.0 m to 4.0 m	553			553
	Total	**5,107**	**56,110**		**61,217**
3	Earthwork excavation for Blasting and removing Medium Rock & Dense Medium Rock Measured in solid including stacking.				
b	First depth 0 m to 2.0 m		10,338		10,338
c	Second depth 2.0 m to 3.0 m		298		298
	Third depth 3 m to 4.0 m		20,973		20,973
	Total		**31,609**		**31,609**
	Total excavation	**26,460**	**669,840**	**156,097**	**852,397**

Sl. No	Description	Feeder Main	Ring Main	Transmission Main	Total
		m^3			
3	Refilling with excavated soils	17,285	373,594	74,495	**465,374**
	% of soil excavated soil to be reused	65%	56%	48%	
4	Disposal of surplus excavated earth	9,174	296,246	81,602	387,023
	% of surplus soil to be disposed	35%	44%	52%	

38. **Construction methodology for pipe carrying bridge**.

Stage 1: Pre-Construction Activities
- Obtain soil test report and secure pipe carrying bridges (PCBs) design approval.
- Clear site and develop approach road, labour sheds, and storage facilities.
- Carry out detailed survey and mark pile points (typically four piles per pier).

Stage 2: Pile Foundation
- Bore piles using auger boring with MS casing and bentonite slurry.
- Conduct Standard Penetration Tests (SPT) as per requirement.
- Continue boring up to hard rock and extend MS casing for stability.
- Additional boring is carried out in hard strata for adequate founding depth.
- Lower reinforcement cage and pour concrete.
- Trim the top weak concrete layer to expose sound material.

Stage 3: Substructure Construction
- Tie piles with reinforcement and cast the pile cap.
- Construct pier column in stages with shuttering and concreting.
- Mark pier cap bottom level using a total station and cast the pier cap.
- Provide pedestal beds for elastomeric bearings.

Stage 4: Superstructure Construction
- Install girder beams between pier caps.
- Provide intermediate diaphragms for stiffness.
- Place deck slab reinforcement and complete concreting.

Stage 5: Final Works
- Cast pipe-support concrete beds on deck slab.
- Install handrails and maintenance footpath for safe operation

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39. **Sources of Materials.** The Table 8 provides indicative locations of construction material sources. Based on availability and commercial considerations, the contractor may procure materials from existing or newly identified mines/quarries.

40. However, prior to procurement, the contractor shall obtain and submit copies of all valid statutory permits and clearances for the selected source. It shall be ensured that the mines/quarries hold valid environmental clearance, consent to operate, mining lease, and other applicable licenses from the competent authorities before any material is sourced for the project.

Table 8: Source of Construction Materials

Name of material	Notes and description of the item
Bricks & Tile Products	Sources of supply – Nerkundram, capacity for one lorry taken as 1500 Nos. stock bricks, 2300 Nos. country, 2000 Nos. platform molded bricks, 6000 Nos. tiles, terrace bricks, etc., and brick jelly 2.83 m^3, the number of country bricks of other sizes per lorry load and other than the sizes mentioned in the above conveyance table should be calculated in ratio to the volume content of these bricks
River sand and Sand for filling	Source of supply from Vaduganthangal (Vellore), Arasur (Villupuram)
Crushed stone sand and sand for filling (M-sand)	Source of supply from Nallambakkam (Kanchipuram), Pulipakkam (Kanchipuram), Erumaiyur (Kanchipuram), Thiruneermalai (Chennai).
Lime	Source of supply – Korrukkupet – Capacity for one lorry is taken as 2.83 cu.m.
Gravel	Source of supply from any source beyond 7 km outside the city limits now fixed in Chennai – Nellore Road – capacity for one lorry is taken as 2.83 m^3.
Blue Granite stone jelly	Source of supply – Pallavaram quarries – Capacity for one lorry is taken as 2.83 m^3
Bond stone and rough stone	Source of supply – Nallambakkam quarry in Nallambakkam village in Chengalpattu taluk – capacity of one lorry is taken on 2.83 m^3
Laterite	Source of supply – Thanipoondi – Capacity for one lorry load is taken as 2.83 m^3. The actual lead of individual works to be ascertained and adopted by the concerned Departmental Officer.
Earth	For conveyance of earth, wherever applicable, earth is to be treated like sand, lime, broken stone, gravel.

E. Project's Associated facilities

41. As per the SPS 2009, associated facilities are defined as those facilities that are not financed by ADB, but are directly and significantly related to the project, and are essential for its successful operation. For the proposed Ring Main project, the associated facilities include existing or planned infrastructure that will enable the functioning of the new transmission and distribution system. In the context of the Ring Main project, the associated facilities may include:

42. **Water Treatment Plants (WTPs) and Desalination Plants (DSPs)**: The WTPs that will supply treated water to the RMS are considered associated facilities, as RMS will distribute water drawn from these sources. Similarly, the DSPs form a crucial part of the water supply source for the RMS. These DSPs treat seawater to produce potable water, which is conveyed to the clear water reservoirs and subsequently distributed through the RMS. Although these plants may be financed and operated under separate schemes, their reliable functioning is fundamental to the success of the RMS project. Therefore, they are considered associated facilities, and as per SPS 2009 requirements, an environmental audit of associated facilities was conducted. This involved checking their regulatory compliance and if the treated water is potable and meets the drinking water standards. List of WTPs and DSPs are as follows:

(i) Puzhal WTP (300 MLD)
(ii) Surapet (14 MLD) (Currently under rehabilitation)
(iii) Kilpauk WTP (270 MLD)
(iv) Chembarambakkam WTP (530 MLD)

(v) Vadakkuthu WTP (180 MLD)
(vi) Minjur DSP 100 MLD capacity (Currently under rehabilitation)
(vii) Nemmeli DSP- I 100 MLD capacity
(viii) Nemmeli DSP- II 150 MLD capacity
(ix) Perur DSP 400 MLD capacity (Under construction)

43. **Clear Water Reservoirs and Pumping Stations:** These facilities serve as the source points for the Ring Main and are essential for maintaining required pressure and flow in the system. Any rehabilitation or expansion works at these locations, even if not financed under the current project, are considered associated facility.

44. **Feeder Mains and Transmission Mains** (Outside the Project Scope): Portions of feeder or transmission mains that connect the Ring Main to the WTPs or service reservoirs but are not included in this project, are associated facilities as they are critical for the continuous water supply through the system.

45. **Water Distribution Stations (WDSs) and Ground-Level Storage Reservoirs**: The Ring Main supplies water to multiple WDSs, which in turn distribute water to local networks. The functioning of these WDSs and reservoirs is integral to achieving the overall project objectives and hence they are considered associated facilities.

46. **Power Supply Infrastructure**: Dedicated power supply lines or substations that provide energy for operation of pumping stations and SCADA systems, even if implemented by other agencies, are associated facilities necessary for uninterrupted system performance.

F. Operation and Maintenance

47. Operation and maintenance (O&M) of the above components for 10 years after commissioning will be the responsibility of the contractor, and after which it will be carried out by the CMWSSB directly or through an O&M contractor. The implementation of the RMS in Chennai offers several operational, management, and service-related benefits, enhancing the overall reliability and efficiency of the water supply network: flow regulation through SCADA-operated valves; continuity of supply even if a water source fails; equitable distribution of available water, operational flexibility and reliability, and centralized real-time monitoring for efficiency and leak detection. Key performance indicators (KPI) during the O&M stage are:

(i) Non-revenue water (NRW): The difference between the sum of inflow water into the RMS and the sum of outflows from transmission mains into WDS is the NRW and it should not be more than 5%.
(ii) Pressure: Minimum pressure at the exit point of any of the transmission mains should not be less than 5 m.
(iii) Power Factor: It should be greater than 0.95.
(iv) Flow to the WDS: The flow in each of the transmission main, as percentage of the design flow for the transmission main should be equal to the percentage of actual flow over the design inflow into the system.
(v) Water quality: Quality of water in terms of turbidity, total dissolved solids and free residual chlorine, in the water injected into the ring main and supplied to all the WDSs should be less than the acceptable drinking water standards (turbidity < 1 NTU; total dissolved solids < 500 mg/L; and free residual chlorine shall be maintained within an operational range of 0.2–0.5 mg/L at consumer end points.

The minimum value of 0.2 mg/L is required to ensure effective disinfection, while the upper limit of 0.5 mg/L is adopted as the operational maximum to avoid potential issues related to taste, odor, and consumer acceptability, consistent with the acceptable limits prescribed under bureau of Indian standards IS 10500:2012 (drinking water specification).

(vi) **Expenditure on repairs and replacement:** The expenditure on repairs and replacement of infrastructure in the RMS should be minimum, not more than the specified amount.

G. Implementation Schedule

48. The RMS Project will be implemented through a single, integrated work package considering the scale, interdependency, and complexity of the activities. The package covers the entire length of the ring main, including installation of isolation valves, air valves, scour arrangements, and tees with sluice valves for connecting nine feeder mains and 55 transmission mains. It also includes the supply, installation, testing, and commissioning of seven new sets of pumping machineries (along with two existing sets), nine feeder mains, all transmission mains, surge protection systems at nine pumping/booster stations, and the full instrumentation and SCADA system. The package further encompasses the operation and maintenance of the complete RMS—including the two existing pumping machinery sets—for a period of ten years after completion of construction. The project will deploy approximately 250 workers during peak construction, of which about 70% will be engaged locally, while the remaining 30% will come from outside and will require suitable worker accommodation.

49. The implementation of the RMS Project under the single consolidated package is planned over a total duration of four and half years (54 months). The construction and execution activities will be carried out over a period of 48 months, commencing in June 2026 and concluding in May 2030. This will be followed by a 6-month trial run, scheduled from June 2030 to November 2030, during which system performance, stability, and operational integration will be thoroughly tested before full-scale commissioning and the start of the O&M phase.

III. ANALYSIS OF ALTERNATIVE

A. Introduction

50. The alternative analysis aims to identify and compare feasible options for the proposed Chennai RMS, considering environmental, technical, economic, and social aspects. The analysis evaluates alternatives related to:

(i) the "no project" scenario,
(ii) alignment options (including avoidance of sensitive areas such as Ramsar sites),
(iii) selection of pipe material, and
(iv) construction methodology (open-cut versus trenchless).

51. The objective is to ensure that the selected site and design represents the most environmentally sound and cost-effective solution for sustainable water supply augmentation within the GCC area.

B. No Project Scenario

52. Under the *no project* alternative, the proposed Chennai RMS would not be implemented. Consequently:

(i) Existing transmission and distribution networks would continue operating under severe hydraulic stress and highly fragmented source-based zoning.
(ii) The inequitable water supply across Chennai—particularly in peripheral and low-pressure zones—would persist, worsening during summer months or drought periods.
(iii) Dependence on aging infrastructure would increase the frequency of leakages, bursts, and NRW losses.
(iv) Emergency transfer of water between treatment plants would remain impossible, reducing system resilience during droughts, plant shutdowns, or climate-related disruptions.
(v) Urban growth in GCC would continue facing supply constraints, affecting planned development corridors.

53. Although the no-project scenario avoids temporary construction-related impacts (such as dust, traffic disturbance, and noise), the overall social and public health implications of unreliable water supply outweigh the minimal environmental benefits. Therefore, the *no-project scenario* is not a preferred option.

C. Alignment Alternatives

54. During the preliminary and detailed design, various alignment options were evaluated to minimize environmental, social, and regulatory impacts, details of two broad alignment options were considered is presented in
55.
56.
57. **Table 9**. Special consideration was given to avoiding: • Pallikaranai Marshland (Ramsar Site), Guindy National Park, Adyar River CRZ stretches, heritage structures and ASI-protected monuments, and congested built-up urban pockets wherever possible.

Table 9: Details of two broad alignment options were considered

Option	Description	Environmental Implications	Feasibility/Remarks
Option 1 – Alignment passing near Ramsar Site and Ecologically Sensitive Areas	Section crossing close to Pallikaranai Marshland (Ramsar Site), and Adyar River in CRZ areas.	Risk of disturbing wetland ecology, bird habitats, and potential non-compliance with CRZ regulations.	Not preferred; may require additional clearances under CRZ Notification and TN Forest Department.
Option 2 – Alignment away from Ramsar Site and Sensitive Ecology (Preferred)	Alignment shifted to follow existing road corridors, public ROW, and away from protected or notified areas.	Minimal ecological disturbance; limited tree cutting and easier access for maintenance.	**Preferred option** due to compliance with ADB SPS 2009 and local environmental regulations.

58. Exact chainage of diversion from Pallikaranai Marshland crossing = 345 m length, with the nearest point at 62 m from the Pallikaranai Marshland compound wall. The selected alignment has the following advantages:

(i) Routing along existing urban corridors minimizes new land disturbance.
(ii) Avoiding hydrologically sensitive marsh buffers crucial for flood management.
(iii) Reduces risk of pipeline settlement in soft soil zones typical around marshlands.
(iv) Prevents potential future encroachment concerns inside protected ecological zones.
(v) Ensures continuous access for maintenance without entering restricted areas.

D. Pipe Material Alternatives

59. Various pipe materials were examined for suitability, durability, and environmental implications under Chennai's urban and coastal conditions. Alternate analysis of different material for pipes is presented in Table 10.

Table 10: Alternate analysis of different material for pipes

Pipe Material	Advantages	Disadvantages	Environmental Aspects
Mild Steel (MS)	High strength, suitable for high-pressure main, durable with anti-corrosive coating, flexible for large diameters.	Requires regular maintenance and protection against corrosion.	Recyclable; moderate embodied energy; can be coated to prevent soil contamination.
Ductile Iron (DI)	Easy jointing, corrosion-resistant, suitable for moderate pressures.	Heavy weight, higher transportation and installation cost.	Good reuse potential; larger carbon footprint due to weight.
HDPE	Corrosion-free, lightweight, easy installation.	Limited diameter (not suitable above 1,200 mm), lower pressure tolerance.	Environmentally favorable for smaller diameters.

60. Given the large-diameter requirements (1,200–2,000 mm) and the high transmission pressures in the system, mild steel (MS) pipes were selected as the most appropriate material for the RMS project. The proposed network includes 91.2 km of new ring main, 64.2 km of new transmission main, and 2.65 km of new feeder main. MS pipes provide strength, durability, and

suitability for high-pressure conveyance. All MS pipelines will be protected with appropriate internal and external anti-corrosion systems such as epoxy or cement-mortar lining.

E. Construction Methodology Alternatives

61. Two construction methods were assessed for pipeline laying: open excavation and trenchless methods. Trenchless method is 155–175% more expensive than open excavation. Hence, open excavation was adopted for most of the alignment except at road, river, and railway crossings, where trenchless technology should be adopted to minimize environmental and social disturbance.

Open Excavation Method	**Trenchless Method (Jack and Push)**
Advantages: • Conventional and well-established technique. • Allows for longer pipe lengths (up to 12 m) with fewer joints. • Facilitates visual inspection and better quality control. • Economically viable. • Simplified maintenance in future. **Disadvantages:** • Temporary disturbance to traffic, utilities, and businesses. • Requires road restoration. • Higher occupational safety risks in deeper trenches.	**Advantages:** • Avoids road surface damage and disruption to utilities. • Minimizes traffic disturbance and noise during execution. • Ideal for crossings under roads, rivers, and railways. **Disadvantages:** • Costly due to specialized machinery and more jointing. • Shorter pipe sections (3 m–6 m). • Requires launching and receiving shafts.

F. Preferred Alternative

62. The preferred combination, based on technical feasibility, cost efficiency, and environmental sustainability, is summarized below:

Component	Preferred Option	Key Rationale
Project Scenario	Implement Ring Main Project	To ensure equitable and reliable water supply across GCC.
Alignment	Along public roads, away from Ramsar site and CRZ-sensitive zones	Minimizes ecological disturbance and avoids regulatory complications.
Pipe Material	Mild Steel (MS)	Technically viable for high-pressure, large-diameter pipeline; recyclable.
Construction Method	Open-cut trenching for regular sections; trenchless for crossings	Balances cost efficiency and environmental safety.

IV. POLICY, LEGAL AND ADMINISTRATIVE FRAMEWORKS

A. ADB Safeguard Policy Statement, 2009

63. ADB's Safeguard Policy Statement (SPS) 2009 requires that during the design, construction, and operation of the project, all applicable national and state laws, regulations, and international conventions/treaties are complied with. SPS 2009 further mandates the application of pollution prevention and control technologies and good environmental management practices consistent with international standards and good industry practices.

64. **Screening and Categorization with SPS 2009.** ADB uses a classification system to reflect the significance of a project's potential environmental impacts. A project's category is determined by the category of its most environmentally sensitive component, including direct, indirect, cumulative, and induced impacts in the project's area of influence. Each proposed project is scrutinized as to its type, location, scale, and sensitivity and the magnitude of its potential environmental impacts. Projects are assigned to one of the following four categories:

- (i) **Category A.** A proposed project is classified as category A if it is likely to have significant adverse environmental impacts that are irreversible, diverse, or unprecedented. These impacts may affect an area larger than the sites or facilities subject to physical works. An Environmental Impact Assessment (EIA) is required.
- (ii) **Category B.** A proposed project is classified as category B if its potential adverse environmental impacts are less adverse than those of category A projects. These impacts are site-specific, few if any of them are irreversible, and in most cases mitigation measures can be designed more readily than for category A projects. An initial environmental examination (IEE) is required.
- (iii) **Category C.** A proposed project is classified as category C if it is likely to have minimal or no adverse environmental impacts. No environmental assessment is required although environmental implications need to be reviewed.
- (iv) **Category FI.** A proposed project is classified as category FI if it involves investment of ADB funds to or through a FI.

65. The environmental impacts of the subproject have been systematically identified and assessed during the planning and design stages. An environmental assessment using ADB's REA checklist for water supply (Appendix 1) confirms that the project is unlikely to cause any significant adverse environmental impacts. The proposed works are confined almost entirely to existing utility corridors, public road ROW, and government-owned premises, which substantially minimizes environmental risk and avoids any encroachment into environmentally sensitive areas, protected zones, ASI-protected monuments, or critical natural habitats. All anticipated impacts are site-specific, predictable, temporary, and fully manageable, consisting mainly of routine construction-related disturbances such as dust, noise, vibration, traffic disruption, and trenching works. These impacts are well understood and can be effectively addressed through established engineering practices, occupational health and safety (OHS) measures, and a robust environmental management plan (EMP). As the project does not involve any irreversible, unprecedented, or long-term environmental impacts, and with mitigation measures readily implementable, it firmly qualifies as category B on environment under SPS 2009 requiring an IEE. Accordingly, this IEE has been prepared to meet the requirements of SPS 2009 for an environment category B project.

66. **Environmental Management Plan.** An EMP which addresses the potential impacts and risks identified by the environmental assessment shall be prepared. The level of detail and

complexity of the EMP and the priority of the identified measures and actions will be commensurate with the Project's impact and risks.

67. **Environmental Audit of Existing Facilities.** SPS 2009 requires an environmental audit, if a subproject involves facilities and/or business activities that already exist or are under construction, including an on-site assessment to identify past or present concerns related to impacts on the environment. The objective of this compliance audit is to determine whether actions were in accordance with ADB's safeguard principles and requirements for borrowers, and to identify and plan appropriate measures to address outstanding compliance issues.

68. **Public Disclosure.** The IEE will be put in an accessible place (e.g., local government offices, libraries, community centers, etc.), and a summary translated into local language for the project affected people and other stakeholders. The following safeguard documents will be put up on ADB's website so that the affected people, other stakeholders, and the public can provide meaningful inputs into the project design and implementation:

- Draft, final or updated IEE upon receipt; and
- Semi-annual Environmental monitoring reports (SEMRs) submitted by the PMU during project implementation upon receipt.

69. **Consultation and Participation.** SPS 2009 requires borrower to conduct meaningful consultation[3] with affected people and other concerned stakeholders, including civil society, and facilitate their informed participation. The consultation process and its results are to be documented and reflected in the environmental assessment report.

70. **Grievance Redress Mechanism.** SPS 2009 requires borrowers to establish a mechanism to receive and facilitate resolution of affected people's concerns, complaints, and grievances about the subproject's performance. The grievance mechanism shall be scaled for the risks and adverse impacts of the subproject.

71. **Monitoring and Reporting.** The borrower shall monitor, measure and document the implementation progress of the EMP. If necessary, the borrower shall identify the necessary corrective actions and reflect them in a corrective action plan. The borrower shall prepare and submit to ADB semi-annual EMRs that describe progress with implementation of the EMP and compliance issues and corrective actions, if any. For subprojects likely to have significant adverse environmental impacts during operation, reporting will continue at the minimum on an annual basis until ADB issues a project completion report.

72. **Unanticipated Environmental Impacts.** Where unanticipated environmental impacts become apparent during subproject implementation, SPS 2009 requires the borrower to update the environmental assessment and EMP or prepare a new environmental assessment and EMP to assess the potential impacts, evaluate the alternatives, and outline mitigation measures and resources to address those impacts.

[3] Per ADB SPS 2009, meaningful consultation means a process that (i) begins early in the project preparation stage and is carried out on an ongoing basis throughout the project cycle 1; (ii) provides timely disclosure of relevant and adequate information that is understandable and readily accessible to affected people; (iii) is undertaken in an atmosphere free of intimidation or coercion; (iv) is gender inclusive and responsive, and tailored to the needs of disadvantaged and vulnerable groups; and (v) enables the incorporation of all relevant views of affected people and other stakeholders into decision making, such as project design, mitigation measures, the sharing of development benefits and opportunities, and implementation issues.

73. **Occupational Health and Safety.** SPS 2009 requires the borrower to ensure that workers are provided with a safe and healthy working environment, taking into account risks inherent to the sector and specific classes of hazards in the subproject work areas, including physical, chemical, biological, and radiological hazards. The borrower shall take steps to prevent accidents, injury, and disease arising from, associated with, or occurring during the course of work, including: (i) identifying and minimizing, so far as reasonably practicable, the causes of potential hazards to workers; (ii) providing preventive and protective measures, including modification, substitution, or elimination of hazardous conditions or substances; (iii) providing appropriate equipment to minimize risks and requiring and enforcing its use; (iv) training workers and providing them with appropriate incentives to use and comply with health and safety procedures and protective equipment; (v) documenting and reporting occupational accidents, diseases, and incidents; and (vi) having emergency prevention, preparedness, and response arrangements in place.

74. **Community Health and Safety.** SPS 2009 requires the borrower to identify and assess risks to, and potential impacts on the safety of affected communities during the design, construction, operation, and decommissioning of the subproject, and shall establish preventive measures and plans to address them in a manner commensurate with the identified risks and impacts. The borrower shall ensure to apply preventive and protective measures for both occupational and community health and safety consistent with good international practice, as reflected in internationally recognized standards such as the World Bank Group's Environmental, Health and Safety (EHS) Guidelines. PMU shall also adhere to necessary protocols in response to benefits and opportunities, and implementation issues. In case where responsibility is delegated to subproject contractors during construction phase, borrower shall ensure that the responsibilities on occupational health and safety (OHS) are included in the contract documents including non-employee workers engaged by the borrower/client through contractors or other intermediaries to work on project sites or perform work directly related to the project's core functions. Emerging infectious diseases management should be consistent with the guidelines of relevant government healthcare agencies and the World Health Organization (WHO).

75. **Physical Cultural Resources.** Borrower is responsible for siting and designing the subproject to avoid significant damage to physical cultural resources. SPS 2009 requires that such resources likely to be affected by the subproject are identified and qualified, and experienced experts assess the subproject's potential impacts on these resources using field-based surveys as an integral part of the environmental assessment process. When the proposed location of a subproject component is in areas where physical cultural resources are expected to be found as determined during the environmental assessment process, chance finds procedures shall be included in the EMP.

76. **Pollution Prevention and Control Technologies.** During the design, construction, and operation of the project, PMU, shall apply pollution prevention and control technologies and practices consistent with international good practice, as reflected in internationally recognized standards such as the World Bank Group's EHS Guidelines. These standards contain performance levels and measures that are normally acceptable and applicable to the project infrastructures. SPS 2009 provides that when the government's regulations differ from these levels and measures, the project shall achieve whichever is more stringent. If less stringent levels or measures are appropriate in view of specific project circumstances, the PMU will provide full and detailed justification for any proposed alternatives that are consistent with the requirements presented in SPS 2009.

77. **Adoption of International Reference Guidelines.** To ensure that the subprojects meet high standards of environmental, health, and safety performance, the implementation will adopt

the World Bank Group–IFC General EHS Guidelines. These guidelines provide internationally recognized good practices for managing environmental impacts, OHS risks, community safety, pollution prevention, and resource efficiency across construction and operational stages.

78. The subprojects will also comply with relevant WHO Guidelines, including those pertaining to Air Emissions and Ambient Air Quality, Noise Management, and Wastewater Quality. These guidelines will support the project in achieving and maintaining safe and acceptable levels of environmental quality, especially in dense urban settings where public exposure levels need to be carefully controlled.

79. Given the potential presence or accidental encounter of asbestos-containing materials (ACM) in older utility corridors or structures, the project will adhere to the ADB Good Practice Guidance for the Management and Control of Asbestos: Protecting Workplaces and Communities from Asbestos Exposure Risks (March 2022). This guidance ensures that any ACM-related risks are systematically identified, safely handled, and properly disposed of, thereby protecting workers, nearby communities, and the environment.

80. Furthermore, the project will follow the Guidance Note on Workers' Accommodation: Processes and Standards (August 2009)[4] to ensure that all worker housing and welfare facilities meet adequate standards for health, safety, hygiene, and dignity. This includes requirements for accommodation design, sanitation, food services, security, and grievance redress, helping to safeguard worker well-being throughout the project lifecycle.

81. **Bidding and Contract Documents.** This IEE report, which contains the EMP, shall be included in bidding and contract documents and verified by PMU. The PMU shall also ensure that bidding and contract documents include specific provisions requiring contractors to (i) comply with all other conditions required by ADB (ii) to submit to PMU, for review and approval, a site specific environmental management plan (SEMP), including (i) proposed sites/locations for construction work camps, storage areas, hauling roads, lay down areas, disposal areas for solid and hazardous wastes; (ii) specific mitigation measures following the approved EMP; (iii) monitoring program as per EMP; and (iv) budget for SEMP implementation, among others as may be required. No works can commence prior to approval of SEMP. A copy of the EMP and/or approved SEMP will be always kept on site during the construction period. Non-compliance with, or any deviation from, the conditions set out in the EMP and/or SEMP constitute a failure in compliance and shall require corrective actions.

82. **Conditions for Award of Contract and Commencement of Work.** PMU shall not award any works contract under the subproject until (i) relevant provisions from the EMP are incorporated into the works contract; (ii) this IEE report is updated to reflect subproject's final detailed design and PMU has obtained ADB's clearance of such updated IEE report; and (iii) other necessary permits from relevant government agencies have been obtained.

B. National and State Laws

83. The implementation of the subprojects will be governed by the GoI and Government of Tamil Nadu (GoTN) and other applicable environmental acts, rules, regulations, and standards. These regulations impose restrictions on activities to minimize or mitigate likely impacts on the environment. It is the responsibility of the project executing and implementing agencies to ensure

[4] https://documents.worldbank.org/en/publication/documents-reports/documentdetail/604561468170043490/workers-accommodation-processes-and-standards-a-guidance-note-by-ifc-and-the-ebrd.

subprojects are consistent with the legal framework, whether applicable international, national, state or municipal or local. Key standards include those related to drinking water quality, air quality, noise levels, effluent discharge, and protected areas. Compliance is required in all stages of the subprojects including design, construction, operation and maintenance.

84. **Environmental Assessment.** The EIA Notification of 2006 sets out the requirement for environmental assessment in India. This states that environmental clearance is required for specified activities/projects, and this must be obtained before any construction work or land preparation (except land acquisition) may commence. Projects are categorized as A or B depending on the scale of the project and the nature of its impacts.

(i) **Category A** projects require environmental clearance from the central Ministry of Environment, Forests and Climate Change (MOEFCC). The proponent is required to provide preliminary details of the project in the prescribed manner with all requisite details, after which an Expert Appraisal Committee (EAC) of the MOEFCC prepares comprehensive terms of reference (TOR) for the EIA study. On completion of the study and review of the report by the EAC, MOEFCC considers the recommendation of the EAC and provides the environmental clearance if appropriate.

(ii) **Category B** projects require environmental clearance from the State Environment Impact Assessment Authority (SEIAA). The State level EAC categorizes the project as either B1 (requiring EIA study) or B2 (no EIA study) and prepares TOR for B1 projects within 60 days. On completion of the study and review of the report by the EAC, the SEIAA issues the environmental clearance based on the EAC recommendation. The Notification also provides that any project or activity classified as category B will be treated as category A, if it is in whole or in part within 10 km from the boundary of protected areas, notified areas or inter-state or international boundaries.

85. **Applicable Environmental Regulations.** Besides EIA Notification 2006, there are various other acts, rules, policies and regulations currently in force in India that deal with environmental issues that could apply to infrastructure development. The specific regulatory compliance requirements of the subproject are shown in

86. Table **11**.

Table 11: Applicable environmental regulations

Law	Description	Requirement
National Environment Policy (NEP), 2006	NEP is a comprehensive guiding document in India for all environmental conservation programs and legislations by Central, State and Local Government. The dominant theme of this policy is to promote betterment of livelihoods without compromising or degrading the environmental resources. The policy also advocates collaboration method of different stakeholders to harness potential resources and strengthen environmental management.	Chennai climate resilient water security and sewerage project (CCRWSSP) should adhere to NEP conservation of environmental resources and abatement of pollution.
Environmental Impact Assessment (EIA) Notification, 2006, and its subsequent amendments to date	The EIA Notification of 2006 set out the requirement for environmental assessment in India. Environmental Clearance is required for certain defined activities/projects, and this must be obtained before any construction work or land preparation (except land acquisition) may commence.	The proposed Project is not a listed activity in Schedule I of this notification and hence prior environmental clearance is not required. However, if mining activities are required for sourcing construction materials such as sand, aggregates, or stone, these will require Environmental Clearance (EC) under the EIA Notification, 2006. The Department of Geology and Mining, Government of Tamil Nadu maintains a list of approved sand and aggregate mines within the state. The PIU/PMU shall ensure that the contractor procures construction materials only from legally-approved and environmentally-cleared sources or obtains the necessary EC for any new quarry or borrow area proposed for the project.

Law	Description	Requirement
Water (Prevention and Control of Pollution) Act of 1974, Rules of 1975, and amendments (1987)	Act was enacted to provide for the prevention and control of water pollution and the maintaining or restoring of wholesomeness of water, by the Central and State Pollution Control Boards and for conferring on and assigning to CPCB/SPCBs powers and functions relating to water pollution control. Control of water pollution is achieved through administering conditions imposed in consent issued under provision of the Water (Prevention and Control of Pollution) Act of 1974. These conditions regulate the quantity and quantity of effluent, the location of discharge and the frequency of monitoring of effluents. Any component of the subproject having the potential to generate sewage or trade effluent will come under its purview. Such projects have to obtain Consent to establish (CTE) under Section 25 of the Act from Tamil Nadu Pollution Control Board (TNPCB) before starting implementation and Consent to Operate (CTO) before commissioning.	Not applicable – None of the project components require consent under the Water (Prevention and Control of Pollution) Act, 1974, the Water (Prevention and Control of Pollution) Rules, 1975, and subsequent amendments (1987).
Air (Prevention and Control of Pollution) Act of 1981, Rules of 1982 and amendments.	This Act was enacted to achieve prevention, control and abatement of air pollution activities by assigning regulatory powers to Central and State boards for all such functions. The Act also establishes ambient air quality standards. The projects having potential to emit air pollutants into the atmosphere have to obtain Consent to Establish (CTE) and Consent to Operation (CTO) under Section 21 of the Act from TNPCB. The occupier of the project/facility has the responsibility to adopt necessary air pollution control measures for abating air pollution.	The following will require CTE and CTO from TNPCB: (i) Diesel generators); (ii) Batching Plant hot mix plants; and (iii) stone crushers, if installed for construction.

Law	Description	Requirement
Biodiversity Act of 2002	This Act primarily addresses access to genetic resources and associated knowledge by foreign individuals, institutions or companies, to ensure equitable sharing of benefits arising out of the use of these resources and knowledge to the country and the people.	Not applicable – None of the project components involve access to biological resources or associated traditional knowledge requiring compliance under the Biological Diversity Act, 2002 and associated Rules, 2004.
Wildlife Protection Act, 1972 and amendment	This overarching Act provides protection to wild animals, birds, plants and matters connected with habitat protection, processes to declare protected areas, regulation of wildlife trade, constitution of state and national board for wildlife, zoo authority, tiger conservation authority, penalty clauses and other important regulations.	The provisions of the Wildlife (Protection) Act, 1972 and its Amendment (1991) are applicable to this project. The proposed alignment runs close to the Guindy National Park at two locations, though within existing road ROW. Although the alignment does not fall within the park's core or buffer area, it is located within the ESZ boundary. while another segment crosses the Coastal Regulation Zone (CRZ) area of the Adyar River, forming part of the Tholkappiar Ecological Park. Therefore, No Objection Certificates (NOCs) and necessary clearances will be obtained from the Tamil Nadu Forest Department and the Wildlife Warden, as required under the provisions of the Act.
The Forest (Conservation) Act, 1980	The Forest (Conservation) Act prohibits the use of forest land for non-forest purposes without the approval of MOEFCC.	Not Applicable .
Tamil Nadu Timber Transit Rules, 1968 (as amended)	Regulates the felling, transport, and use of timber and tree produce within Tamil Nadu.	Applicable – If tree cutting or removal is required along the pipeline alignment, prior permission shall be obtained from the Forest Department or competent authority. All timber shall be transported and disposed of as per the Rules. Tree falling will be minimized, and compensatory plantation will be undertaken as directed by the Forest Department. About 75 trees on the shoulder of the road will be affected during the execution of work and compensatory trees 1:10 budget provided in the EMP

<table>
<tr><th>Law</th><th>Description</th><th>Requirement</th></tr>
<tr><td>CRZ Notification 2019</td><td>The Coastal Regulation Zone (CRZ) Notification, 2019, issued under the Environment (Protection) Act, 1986, regulates development activities within coastal stretches influenced by tidal action, including rivers, creeks, and canals. It aims to protect coastal ecosystems while permitting sustainable development in coastal areas. Any construction or activity within the notified CRZ areas requires prior clearance from the Tamil Nadu State Coastal Zone Management Authority (TNSCZMA) and approval from the MOEFCC.</td><td>Applicable –
Following length of pipeline is passing through CRZ area.<table><tr><th>CRZ Classification</th><th>Length in Metres</th></tr><tr><td>CRZ II</td><td>1,305.73</td></tr><tr><td>CRZ IV</td><td>1,106.72</td></tr></table>The alignment includes approximately 24 pipe-carrying bridges, including major crossings over the Buckingham Canal, Adyar River, and Coovum River, 13 of which fall within the notified CRZ area. Accordingly, CRZ clearance will be required from the TNSCZMA under the provisions of the CRZ Notification 2019.

As of December 2025, draft CRZ report is prepared and procurement of CRZ maps from Anna University is under process. Detailed step for clearance with timeframe is presented in Appendix 20.</td></tr>
<tr><td>Environmental (Protection) Act, 1986 amended in 1991 and the following rules/ notifications:</td><td>This is an “umbrella” legislation that empowers the Central Government to take all necessary measures to protect and improve the quality of the environment and prevent, control and abate environmental pollution.
Empowers central government to enact various rules to regulate environmental pollution, including standards for quality of air, water, noise, soil; discharge standards or allowable concentration limits for environmental pollutants, handling of hazardous substances, locating/prohibiting industries, etc.</td><td>There are rules/notifications that have been brought out under this Act, which are relevant to project, and are listed below</td></tr>
<tr><td>Indian Drinking Water Standards IS 10500</td><td>Gives details of the permissible and desirable limits of various parameters in drinking water as per the Bureau of Indian Standards (BIS).</td><td>Appendix 3 provides the drinking water standards</td></tr>
<tr><td>Environmental Standards (ambient and discharge)</td><td>Emissions and discharges from the facilities to be created or refurbished or augmented shall comply with the notified standards</td><td>Appendix 4 provides ambient air quality standards, emission limits for vehicle exhaust, emission limits of diesel generator (DG) sets and emission stack height requirements for DG. Appendix 5 provides the Vehicle Exhaust Emission Norms.</td></tr>
</table>

Law	Description	Requirement
Noise Pollution (Regulation and Control) Rules, 2000 amended up to 2010	Rule 3 of the Act specifies ambient air quality standards in respect of noise for different areas/zones.	Appendix 6 provides applicable noise standards, and noise limits for diesel generators
Solid Waste Management Rules 2016	These rules provide for the requirements of managing solid waste by Urban Local Bodies (ULBs) and Waste Generators to be followed for collection, storage and transport for onward treatment and disposal. The compliance of requirements of these Rules ensures safer EHS conditions	Municipal solid waste (for example, food waste) will be generated during the construction phase. The contractor shall provide separate bins for wet and dry waste in accordance with GCC requirements and shall tie up with the GCC-appointed solid waste collection agency to ensure regular collection and disposal. Construction and demolition waste must not be mixed with municipal solid waste.
Construction and Demolition Waste Management Rules 2016	The provision of this rule is applicable to the project during its construction phase. It sets out requirements to be followed for management of C&D wastes by waste generators depending on quantity of C&D wastes generated by them.	C&D wastes such as concrete cubes, broken concrete, stones, and aggregates may arise from the project. If quantities exceed 20 tons per day or 300 tons in total, the contractor must comply with the C&D Rules. This includes segregating waste into specified streams (concrete; soil; steel, wood, and plastics; bricks and mortar), preparing a waste management plan, and obtaining approvals from the local authority before works commence. C&D waste must not be mixed with municipal or biomedical waste. The contractor must establish a tie-up with the GCC-appointed C&D waste processing agency and ensure all recoverable material is transported to the designated processing facility in Chennai.
Hazardous and Other Wastes (Management and Trans boundary Movement) Rules, 2016,	Generators of hazardous and other wastes covered in Schedules of this Rule are required to obtain an Authorization from TNPCB, maintain monthly and annual records (as per formats specified in these rules), provide a safe storage area to store such wastes, and dispose of the waste in an environmentally sound and legally acceptable manner.	The Rule applies if the project generates waste oil or oil-contaminated cloth from DG set use or equipment and vehicle maintenance. The contractor must obtain TNPCB authorization during construction and dispose these wastes only to TNPCB-authorized recyclers. Forms, fees, and procedures are available on the TNPCB website. Once authorized, the contractor must maintain records of hazardous waste generated, submit annual returns to TNPCB, and retain manifests for all hazardous waste transported for disposal.

Law	Description	Requirement
Biomedical Waste Management Rules, 2016	These rules are applicable to all persons who generate, collect, receive, store, transport, treat, dispose, or handle bio medical waste in any form including hospitals, nursing homes, clinics, dispensaries, veterinary institutions, animal houses, pathological laboratories, blood banks, Ayush hospitals, clinical establishments, research or educational institutions, health camps, medical or surgical camps, vaccination camps, blood donation camps, first aid rooms of schools, forensic laboratories and research labs. The rules outline methods to store bio medical wastes generated by various generators.	The Construction Contractor is expected to provide first aid in case of injuries to workers/staff during project construction phase. The Construction Contractor needs to store any biomedical waste generated at the site and ensure its safe disposal by tying up with the nearest authorized common biomedical waste disposal facilities.
Plastic Waste Management Rules, 2016 (as amended in 2018, 2021 and 2022)	These rules shall apply to every waste generator, local body, manufacturer, Importers and producer.	The contractor shall comply with these Rules to ensure the environmentally-sound management of all plastic waste generated during construction and operation. Requirements include minimizing plastic use, avoiding single-use plastics at worksites, segregating plastic waste at source, safely collecting and storing it, and handing it over only to authorized agencies. The contractor shall maintain a plastic waste register, conduct periodic awareness and training for workers, implement housekeeping and preventive measures, and strictly prohibit burning, burial, or mixing of plastic waste with construction debris.

Law	Description	Requirement
Wetlands (Conservation and Management) Rules, 2017	The Rules specify activities which are harmful and prohibited in the wetlands such as industrialization, construction, dumping of untreated waste and effluents, and reclamation. The Central Government may permit any of the prohibited activities on the recommendation of Central Wetlands Regulatory Authority.	Pallikaranai Marshland, located in south Chennai, is a notified Ramsar Wetland of International Importance (designated in 2022) and recognized as a protected wetland under the Wetlands (Conservation and Management) Rules, 2017. It is one of the last remaining natural wetlands of the city and supports high biodiversity, including migratory birds and threatened species. For the Chennai RMS Project, it is confirmed that no pipeline or construction activity falls within the core area or notified boundary of the Pallikaranai Ramsar site. However, works located in the broader influence zone must follow precautionary measures to avoid impacts such as silt runoff, waste disposal, or hydrological obstruction. PMU and the Contractor must ensure strict compliance with wetland protection norms when working in areas hydrologically connected to the marshland.
Ancient Monuments and Archaeological Sites and Remains Act, 1958 and Ancient Monuments and Archaeological Sites and Remains (Amendment and Validation) Act, 2010.	The Act designates areas within 100 m of the “protected monument/area” as “prohibited area” and beyond that up to 200 m as “regulated area” respectively. No “construction” is permitted in the “prohibited area” and any construction activity in the “regulated area” requires prior permission of the Archaeological Survey of India (ASI).	The proposed pipeline is about 400 m from the nearest ASI protected monument (old fort walls). Hence, ASI permission is not required.
Labor Laws	The contractor shall not make employment decisions based upon personal characteristics unrelated to job requirements. The contractor shall base the employment relationship upon equal opportunity and fair treatment and shall not discriminate with respect to aspects of the employment relationship, including recruitment and hiring, compensation (including wages and benefits), working conditions and terms of employment or retirement, and discipline. The contractor shall provide equal wages and benefits to men and women for work of equal value or type.	Applicable - Appendix 7 provides applicable labor laws including amendments issued from time to time applicable to establishments engaged in construction of civil works.

Law	Description	Requirement
The provisions of the Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996 and the Tamil Nadu Building and Other Construction Workers Rules, 2006	Labour Department, Government. of Tamil Nadu adopted proactive approach and initiated necessary steps to implement the provisions of the BOCW Act through the Tamil Nadu Building and Other Construction Workers (Regulation of Employment And Conditions of Service) Rules, 1996. As per the provision of the BOCW Act, Cess Act and Tamil Nadu Rules, establishments which had employed on any day of the preceding twelve months, 10 or more building workers in any building or other construction work are required to pay cess at the rate of 1% of the total cost of construction incurred by an employer. The cess so collected is required to be spent for the welfare of building and other construction workers.	Contractors are required to follow all the provisions of BOCW Act. Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996 (BOCW Act) and Building and other construction workers' welfare cess act, 1996 (Cess Act) was passed to address the concerns regarding safety, health and welfare of larger number of labor force employed in the building and other constructions sector.
IS 11768: 1986/2005: Recommendations for disposal of asbestos waste material	The standard emphasis that every employer who undertakes work which is liable to generate asbestos containing waste, shall undertake adequate steps to prevent and /or reduce the generation of airborne dust during handling, storing and disposal	Applicable – Some existing underground pipelines made of asbestos-cement (AC) may be encountered during construction. Handling, removal, and disposal of such materials will follow the recommendations of IS 11768:1986/2005, ensuring minimal dust generation, safe removal, and environmentally sound disposal. Waste avoidance and worker safety measures will be strictly implemented.
E-waste (Management) Rules, 2016	Rules apply to manufacturer, producer, consumer, bulk consumer, collection centers, dealers, e-retailer, refurbished, dismantler and recycler involved in manufacture, sale, transfer, purchase, collection, storage and processing of e-waste or electrical and electronic equipment listed in Schedule I, including their components, consumables, parts and spares which make the product operational but shall not apply to batteries, radioactive waste. Schedule 1 specifies electrical and electronic equipment in the following categories: Information technology and telecommunication equipment, consumer electrical and electronics	Applicable - For activities involving the replacement of existing pumps and other machinery, the contractor and PIU must ensure proper segregation, safe storage, authorized disposal of e-waste, and maintenance of e-waste records.

Law	Description	Requirement
International Conventions and Treaties		
Ramsar Convention, 1971	The Ramsar Convention is an intergovernmental treaty that provides the framework for national action and international cooperation for the conservation and wise use of wetlands and their resources. India is one of the signatories to the treaty. The Ramsar convention made it mandatory for the signatory countries to include wetland conservation in their national land use plans.	Pallikaranai Marshland, located in south Chennai, is a notified Ramsar Wetland of International Importance (designated in 2022) and recognized as a protected wetland under the Wetlands (Conservation and Management) Rules, 2017. It is one of the last remaining natural wetlands of the city and supports high biodiversity, including migratory birds and threatened species. For the Chennai RMS Project, it is confirmed that no pipeline or construction activity falls within the core area or notified boundary of the Pallikaranai Ramsar site. However, works located in the broader influence zone must follow precautionary measures to avoid impacts such as silt runoff, waste disposal, or hydrological obstruction. PMU and the Contractor must ensure strict compliance with wetland protection norms when working in areas hydrologically connected to the marshland.
Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), 1973	India is a signatory of this convention which aims to control international commercial trade in endangered species	Not applicable in this project as no endangered species of wild fauna and flora will be impacted on account of project citing or operation.
Montreal Protocol 1992	India is a signatory of this convention which aims to reduction in the consumption and production of ozone-depleting substances (ODS), while recognizing differences in a nation's responsibilities. Ozone depleting substances are divided in two groups Chlorofluorocarbons (CFCs) and Hydro chlorofluorocarbon carbons (HCFCs)	Not applicable in this project as no ODS are involved in construction works.
Basel Convention on Trans-boundary Movement of Hazardous Wastes, 1989	India is a signatory of this convention which aims to reduce trans-boundary movement and creation of hazardous wastes	Contractor to follow the provisions of Hazardous Waste Rules 2016 of the GoI for storage, handling, transport and disposal of any hazardous waste emerged during construction works. Under this Convention, asbestos or asbestos waste in the form of dust and fibers is classified as hazardous waste. There is no usage/disposal of asbestos in the project.

Law	Description	Requirement
Convention on Migratory Species of Wild Animals (CMS), 1979 (Bonn convention)	CMS, also known as Bonn convention, was adopted in 1979 and entered into force on 1 November 1983, which recognizes that states must be the protectors of migratory species that live within or pass through their national jurisdictions,	Not applicable to this project as no migratory species of wild animals are reported in the project areas.

87. Clearances and permissions will be secured prior to commencement of construction. The list provided is indicative, and the contractor is responsible for verifying applicable requirements, obtaining all approvals before work begins, and maintaining valid authorizations throughout implementation. The status of required permissions is presented in Appendix 20.

88. PMU will be the overall responsible for supervision in getting all clearances and provide details to ADB through semi-annual report. PMU will ensure all necessary regulatory clearances and approvals are obtained prior to commencement of works. Respective PIUs, with support of project consultants and contractors, are responsible for obtaining the clearances/permits and ensuring conditions/specifications/provisions are incorporated in the subproject design, costs, and implementation. The PIUs shall report to PMU the status of compliance with clearances/permits as part of the regular progress reporting.

V. DESCRIPTION OF THE ENVIRONMENT

A. Methodology Used for Baseline Study – Chennai Project

89. **Data collection and stakeholder consultations.** Baseline information for the Chennai subproject was collected through a combination of primary surveys and secondary data review. Primary data was gathered for one complete season during February 2023 – March 2023 within the subproject influence area. Secondary information was obtained through a comprehensive literature survey, review of project-related documents, and consultations with stakeholder agencies and relevant government departments.

90. **The literature survey broadly covered the following:**

(i) Project details, reports, maps, and technical documents prepared by CMWSSB and other relevant agencies;
(ii) Consultations and discussions with technical experts, local communities, and concerned government departments;
(iii) Review of previous project reports, published articles, and technical studies relevant to the Chennai water supply sector; and
(iv) Literature relating to land use, soil, geology, hydrology, climate, socio-economic profiles, and other planning documents sourced from government agencies, statutory bodies, and official websites.

91. **Importance of baseline information.** To accurately predict the potential impacts of project implementation and construction activities, it is essential to establish baseline environmental conditions. The interaction between existing environmental conditions and anticipated project impacts forms the basis for preparing the EMP.

92. This chapter presents the existing environmental conditions for various components of the study area. Baseline data collection for each environmental component was carried out considering the project location and the anticipated spatial extent of impacts. The baseline environmental quality—covering **air, noise, water, land, biological, and socio-economic aspects**—was assessed through field surveys conducted in the study area. The study area for each component was defined based on the sensitivity and potential vulnerability of that component to project-related activities.

93. **Ocular inspection.** Multiple site visits were undertaken during the IEE preparation period in 2023 to 2025 to assess the prevailing physical, chemical, biological, and socio-economic conditions at the proposed project locations. These field visits also helped verify ground conditions, land use, existing utilities, and site-specific environmental sensitivities. A separate socio-economic assessment was conducted to document demographic features, current service levels, community needs, and stakeholder priorities relevant to the Chennai project.

B. Physical Features

94. **Geographical location.** Chennai, the capital city of Tamil Nadu, is situated on the southeastern coast of India, along the Coromandel Coast of the Bay of Bengal. It lies at 13.0827° N latitude and 80.2707° E longitude, with the Bay of Bengal forming its eastern boundary for nearly 19 km. To the west and north, it is surrounded by Tiruvallur district, while the Chengalpattu district lies to its south. Its coastal geography and flat terrain give the city both strategic advantages and environmental challenges, including susceptibility to floods during monsoons.

95. Chennai is one of India’s most well-connected metropolitan cities, supported by strong air, rail, road, and port networks. The Chennai International Airport serves as a major aviation hub with extensive domestic links to major Indian cities and international connectivity to Southeast Asia, the Middle East, Europe, and the USA. The city is equally well linked by rail, with Chennai Central and Chennai Egmore functioning as key terminals that connect the region to almost every part of the country, supported by suburban rail, MRTS, and an expanding metro system for seamless intra-city movement.

96. Road connectivity is robust, with major national highways NH 16, NH 48, NH 32, and NH 716 radiating from the city, complemented by a dense public bus network and major infrastructure corridors. Chennai is also a major maritime hub, hosting Chennai Port, Ennore (Kamarajar) Port, and the Kattupalli Shipyard, which together handle diverse cargo and cruise operations, reinforcing the city’s role as a major center for logistics, trade, and regional economic integration.

97. **Topography.** Chennai is characterized by a low-lying, flat terrain typical of coastal plains. It is situated on the Eastern Coastal Plains, a broad and level landform along the southeastern coast of India. The city's average elevation is approximately 6.7 meters (22 feet) above mean sea level, while its highest point reaches around 60 m (200 feet). This gentle elevation profile contributes to the city's expansive urban spread and influences its drainage and land use patterns.

98. **Geology and Geomorphology.** The geology of the district is predominantly composed of recent alluvial deposits, with limited exposure of crystalline rocks belonging to the Archaean age. At greater depths, Tertiary and Gondwana formations are encountered. The Adyar alluvium varies in thickness from 10–20 m, with granular zones occurring at variable depths, while the Cooum alluvium extends up to 28 m in certain areas, particularly showing more granular characteristics in the Kilpauk–Perambur region. Along the coast, beach ridges and sand dunes in the Adyar–Besant Nagar stretch form productive freshwater aquifers.

99. The coastal belt also displays a variety of geomorphic features, including fluvial, marine, and fluvio-marine landforms. Sand bars are scattered along the course of the River Adyar, indicating dynamic sedimentary processes. Although alluvial formations dominate, borehole investigations in the western fringes of the city (Kulathur, Villivakkam, Anna Nagar, etc.) reveal the presence of Tertiary and Gondwana deposits. Due to their clayey nature, however, these deeper formations are reported to have low groundwater yield. Crystalline charnockites occur in localized areas such as Guindy and Velachery, while old lagoonal deposits are found in Tharamani, marked by thick shell beds beneath clay layers.

100. **Soil Type**. The soils of Chennai district are predominantly alluvial in nature, consisting of sand, silt, and clay. The thickness of alluvium varies across the region, reaching up to 28 m in North Chennai near Perambur and about 24 m in the Kilpauk area. In parts of Velachery, a sheet of black clay overlies crystalline rocks, while Gondwana sediments and crystalline formations are also present in localized areas. Overall, the district shows a mix of alluvial, clayey, and crystalline-derived soils, with variations depending on topography and geological setting.

101. **Soil Quality.** Soil quality results from eight locations show generally acceptable conditions with pH ranging from slightly acidic to moderately alkaline (6.9–8.5) and low organic matter, typical of urban soils. Nutrient levels (sodium, potassium, calcium, magnesium, nitrogen, and phosphorus) vary moderately across sites, with higher concentrations near dense urban and industrial areas such as Nandanam and Manali Chennai Petroleum Corporation Limited (CPCL). Electrical conductivity and chloride/sulphate values are within normal ranges, with slightly elevated salinity at Manali CPCL. Importantly, chromium and hexavalent chromium were below

detectable limits at all sites, indicating no heavy-metal contamination. Overall, the soils are within baseline environmental conditions and suitable for project-related works.

102. Table 12 presents results of soil quality testing while Figure 25 shows the location of sampling.

Figure 25: Google earth map showing location of soil sampling

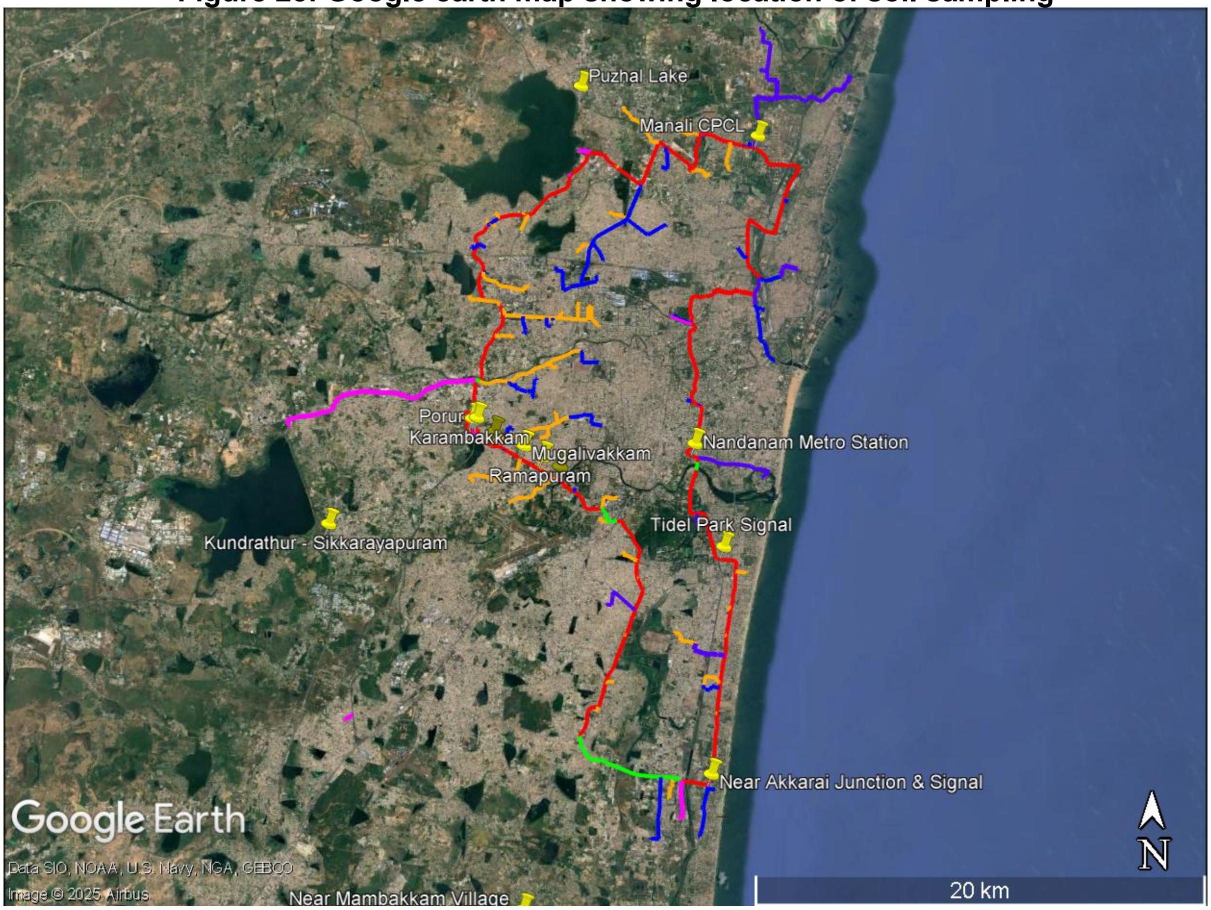


Table 12: Soil quality testing results

Location	Near Akkarai Junction & Signal	Near Mambakkam Village	Porur	Kundrathur - Sikkarayapuram	Tidel Park Signal	Nandanam Metro Station	Manali CPCL	Puzhal Lake
pH	8.1	8.5	7.9	7.4	6.9	7.6	7.6	7.9
Moisture (%)	10.52	21.43	0.51	1.77	2.45	16.47	7.5	13.66
Organic Matter (%)	0.67	0.58	0.74	0.64	0.85	0.77	0.75	0.75
Soluble Sodium as Na (mg/kg)	116	154	126	171	184	275	357	96

Location	Near Akkarai Junction & Signal	Near Mambakkam Village	Porur	Kundrathur - Sikkarayapuram	Tidel Park Signal	Nandanam Metro Station	Manali CPCL	Puzhal Lake
Soluble Potassium as K (mg/kg)	25	71	34	29	152	116	112	29
Soluble Calcium as Ca (mg/kg)	53	49	69	83	102	132	205	74
Soluble Magnesium as Mg (mg/kg)	21	18	25	20	43	52	74	22
Total Phosphorus as P (mg/kg)	7.8	11	21	12	21	35	32	13
Total Kjeldahl Nitrogen (mg/kg)	226	314	211	194	756	546	477	546
Total Soluble Sulphate as SO_4 (mg/kg)	15	76	44	49	29	39	179	52
Chloride as Cl (mg/kg)	67	153	127	153	120	93	580	70
Electrical Conductivity (µS/cm)	170	195	261	151	206	400	715	106
Soluble Chromium as Cr (mg/kg)	BDL (DL:5.0)	BDL (DL:5.0)	BDL (DL:5.0)	BDL (DL:5.0)	BDL (DL:5.0)	BDL (DL:5.0)	BDL (DL:5.0)	BDL (DL:5.0)
Soluble Hexavalent Chromium as Cr^{6+} (mg/kg)	BDL (DL:1.0)	BDL (DL:1.0)	BDL (DL:1.0)	BDL (DL:1.0)	BDL (DL:1.0)	BDL (DL:1.0)	BDL (DL:1.0)	BDL (DL:1.0)
Calcium Carbonate as $CaCO_3$ (%)	14.8	15.7	12.7	11.4	10.45	9.41	13.52	13.7
Date of sample collection and testing – (27.2.2023 to 4.3.2023) Laboratory - Chennai testing laboratory Pvt Ltd, Chennai								

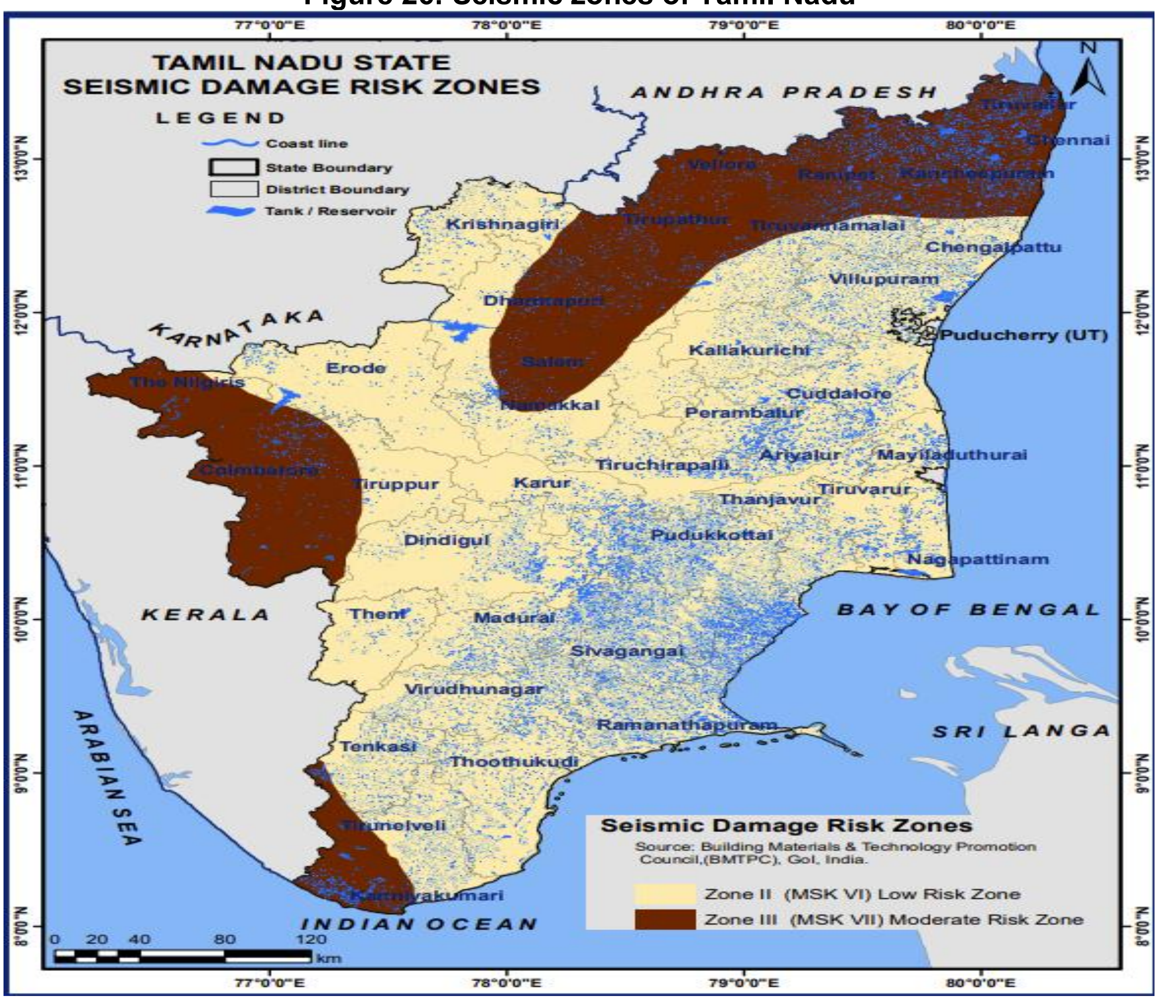
103. **Hydrogeology.** Chennai district is underlain by a wide range of geological formations, spanning from the ancient Archaean crystalline rocks to the Recent alluvium. Broadly, these formations can be grouped into three units: (i) Archaean crystallines, (ii) consolidated Gondwana and Tertiary sediments, and (iii) Recent alluvium. The Archaean rocks, mainly charnockites and gneisses with basic intrusives, are weathered and fractured, with the thickness of the weathered mantle varying from less than 1 m to about 12 m. Productive bore wells in areas such as Saidapet, Adyar, and Ashok Nagar indicate groundwater occurrence in fractured zones down to depths of 60 m.

104. The Gondwana formations, comprising jointed and fractured shales, have been encountered in several boreholes, with thickness varying from about 20 m in Ashok Nagar to more than 130 m in Koyambedu. The tertiary sandstones, though less well defined, are observed beneath alluvium in areas such as Binny Road, Anna Nagar, and Poes Garden. Tube wells tapping granular zones within these sandstones yield about 2–3 liters per second (lps).

105. The recent alluvium, covering a major part of the district, consists of sand, silt, and clay. Thickness varies from place to place, with a maximum of 28 m recorded near Perambur and about 24 m in Kilpauk. These alluvial aquifers serve as the most important groundwater reservoirs of the city. Groundwater occurs in all formations—crystalline, Gondwana, Tertiary, and alluvial—and is developed through dug wells, ring wells, bore wells, tube wells, and filter point wells. Yields range from less than 1 lps in clay-rich zones to about 12 lps in productive alluvial aquifers.

106. **Seismology.** Chennai lies within Seismic Zone III, indicating moderate seismic risk with potential for earthquakes up to roughly magnitude 6–7. While the overall seismic designation applies broadly to the city, localized microzonation reveals higher risk pockets in parts of western-central Chennai, with lower-risk areas in the south and east. These findings support the need for site-specific risk assessments and earthquake-resilient construction practices, particularly in newly developing zones. Figure 26 shows the seismic zone of Tamil Nadu.

Figure 26: Seismic zones of Tamil Nadu



Source : https://tnsdma.tn.gov.in/app/webroot/img/document/map/2021/tn_seismic_damage_zones.pdf

107. **Climate.** Chennai has a tropical wet and dry climate, marked by hot summers, high humidity, and significant seasonal rainfall. The city receives most of its rain from the northeast monsoon (October–December), while summers (April–June) are extremely hot, often crossing 40°C. Being a coastal city along the Bay of Bengal, Chennai is highly vulnerable to climate hazards such as cyclones, storm surges, coastal flooding, and sea-level rise. In recent years, the city has also experienced extreme rainfall events leading to urban floods, as well as heatwaves and water scarcity, making it highly sensitive to the impacts of climate change.

108. **Temperature.** Chennai has a tropical wet and dry climate. The city lies on the thermal equator and is also on the coast, which prevents extreme variation in seasonal temperature. Meteorological data like monthly total rainfall, maximum and minimum temperature, wind rose and relative humidity of the Chennai for a period of Jan 2014 to Dec 2023 collected from Indian Meteorological Department (IMD). Table 13 depicts that the hottest part of the year is in May with maximum temperature ranging from 41.0°C to 43.0°C. The coolest part of the year is in January, with minimum temperature ranging from 18.7°C to 20.6°C.

Table 13: Monthly Highest Maximum Temperature (Degree C)

YEAR	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2014	30.6	32.3	36.6	38.6	42.8	41.8	39.2	38.5	36.7	36.2	32.5	31.8
2015	31.3	33.1	35.1	36.8	42.2	39.6	41.0	37.6	36.9	35.7	32.6	32.4
2016	33.0	34.0	39.0	41.0	41.0	39.0	37.0	38.0	37.0	37.0	34.0	31.0
2017	31.0	36.0	36.0	41.0	43.0	41.0	39.0	37.0	36.0	36.0	34.0	33.0
2018	31.1	32.8	35.6	36.6	39.1	39.8	38.1	37.8	37.3	36.4	32.7	31.3
2019	30.8	34.0	35.6	36.8	41.5	41.5	40.4	38.3	36.8	34.4	35.1	31.0
2020	32.2	33.0	34.4	36.2	41.8	40.6	38.3	36.5	36.6	37.1	33.3	31.2
2021	32.0	33.5	38.3	41.2	40.3	39.9	37.2	36.8	36.6	35.9	32.2	32.4
2022	32.2	33.5	37.6	36.0	39.8	40.1	36.6	37.2	37.5	35.1	31.8	31.9
2023	31.1	34.1	34	38.1	41.8	42.3	37.6	38.2	36.4	36.5	32.6	32.2

Source: IMD Chennai (all the measurements are in degree Celsius)

109. **Rainfall.** Chennai gets most of its seasonal rainfall from the North–East monsoon, from October to December. South-West monsoon prevails from June to September. Cyclones in the Bay of Bengal sometimes hit the city. The highest annual rainfall recorded is 1,049.3 mm in November 2015, the highest recorded since November 1918 when 1,088 mm of rainfall was recorded. The monthly rainfall is given in Table 14.

Table 14: Monthly Rainfall (mm)

Monthly Rainfall (mm) YEAR	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2014	0.1	9.9	0.0	0.0	13.5	96.2	69.7	222.6	130.8	405.5	196.9	149.9
2015	2.8	0.0	0.0	12.3	7.9	20.3	205.9	106.5	75.0	159.9	1049.3	454.7
2016	0.4	0.0	0.0	0.0	216.8	133.1	41.3	24.5	264.7	16.4	73.8	219.9
2017	0.0	5.0	2.5	0.0	0.5	60.0	55.0	90.0	65.0	160.0	155.0	9.0
2018	1.9	1.0	2.9	0.0	0.0	63.1	117.0	191.5	60.7	162.2	190.7	35.8
2019	0.2	4.0	0.0	0.0	0.0	44.7	142.9	120.9	184.1	318.2	108.2	178.8
2020	67.8	0.8	0.0	25.6	0.0	41.8	69.2	69.4	113.0	318.5	525.9	189.1
2021	166.2	8.5	0.0	24.4	16.8	54.7	242.8	168.2	92.1	216.4	1044.3	224.1
2022	90.3	0.0	0.0	0.0	39.5	167.1	107.2	102.2	121.2	171.0	526.2	263.1
2023	4.1	0.8	52.7	1.5	44.9	210.8	56.3	56.3	200.6	109.5	564.7	594.6

Source: IMD Chennai

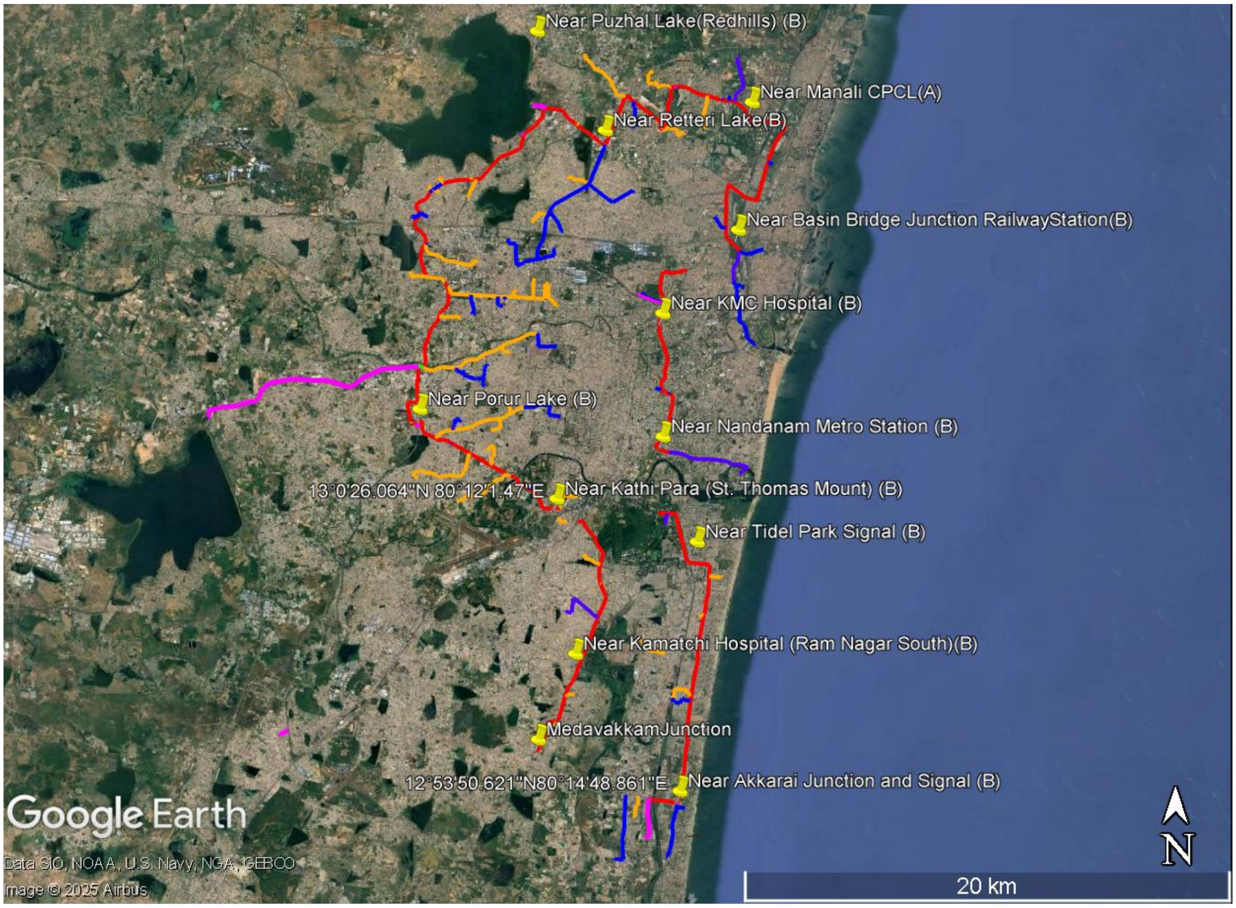
110. **Air Quality.** Ambient air quality monitoring was conducted as part of this study at twelve locations across Chennai covering traffic junctions, residential–commercial areas, and sensitive zones near lakes and hospitals. The results indicate that $PM_{2.5}$ and PM_{10} concentrations exceeded both Central Pollution Control Board (CPCB) 24-hour National Ambient Air Quality Standards (NAAQS) and WHO guidelines at all locations, with the highest $PM_{2.5}$ (85.5 µg/m³) observed near Nandanam Metro Station and the highest PM_{10} (163 µg/m³) at Medavakkam Junction, reflecting significant vehicular emissions and congested traffic corridors. In contrast, SO_2, NO_2, and CO levels remained well within CPCB standards, though NO_2 exceeded the more stringent WHO guideline at all sites except near basin bridge junction railway station, near Manali CPCL and near Porur lake. CO consistently below detectable limits (BDL, DL: 1.15 mg/m³), indicating lower contributions from industrial and combustion sources for these pollutants. Overall, particulate pollution is the key concern across the monitored corridor, requiring focused mitigation measures such as dust suppression, traffic management, and emission control interventions.

111. Table **15** provides the ambient air quality monitoring results while Figure 27 shows the location of sampling.

Table 15: Monitoring results of ambient air quality

S. No.	Location Name	Results				
		$PM_{2.5}$	**PM_{10}**	**SO_2**	**NO_2**	**CO**
1	Near Akkarai Junction and Signal (B)	**73.2**	**139**	14.5	**27.8**	BDL(DL:1.15)
2	Near Kamatchi Hospital (Ram Nagar South) (B)	**76.2**	**142**	19.1	**34.1**	BDL(DL:1.15)
3	Near Medavakkam Junction	**81.0**	**163.0**	20.0	**37.5**	BDL(DL:1.15)
4	Near Porur Lake (B)	**52.4**	**97.0**	11.5	24.0	BDL(DL:1.15)
5	Near Tidel Park Signal (B)	**41.0**	**87.0**	16.2	**31.5**	BDL(DL:1.15)
6	Near Nandanam Metro Station (B)	**85.5**	**151.0**	19.3	**33.0**	BDL(DL:1.15)
7	Near KMC Hospital (B)	**78.0**	**137.0**	18.2	**29.0**	BDL(DL:1.15)
8	Near Basin Bridge Junction Railway Station (B)	**37.5**	**80.0**	11.5	24.9	BDL(DL:1.15)
9	Near Manali CPCL (A)	**41.1**	**86.2**	14.7	23.5	BDL(DL:1.15)
10	Near Retteri Lake (B)	**62.0**	**115.0**	16.1	**28.5**	BDL(DL:1.15)
11	Near Puzhal Lake (Redhills) (B)	**78.5**	**128.0**	15.7	**29.5**	BDL(DL:1.15)
12	Near Kathi Para (St. Thomas Mount) (B)	**64.0**	**122.0**	16.7	**30.0**	BDL(DL:1.15)
Prescribed Limit (CPCB 2009) for NAAQS - 24 hours		60	100	80	80	4.00
WHO Air Quality Guidelines 2021 (µg/m³) [a] mg/m³ 24-hour averaging time		15	45	40	25	4[a]
Date of sample collection and testing -24.2.2023 to 2.3.2023 Laboratory - Chennai testing laboratory Pvt Ltd, Chennai *(A)Industrial area (B) Commercial area (C) Residential area (D) Silence Zone						

Figure 27: Google map showing ambient air and Noise quality monitoring locations



112. **Noise Level.** Daytime and nighttime noise monitoring was carried out at twelve locations representing traffic intersections, commercial corridors, residential–commercial mixed zones, and sensitive areas near hospitals, lakes, and transport hubs. The results show that daytime noise levels ranged from 61.5 to 73.9 dB(A), with several busy junctions such as Medavakkam, Tidel Park, KMC Hospital, Akkarai Junction, and Retteri Lake exceeding the CPCB permissible limits for residential (55 dB) and commercial (65 dB) zones. Nighttime noise levels varied between 48.9 and 64.9 dB(A), again surpassing the 55 dB commercial and 45 dB residential limits at most urban and traffic-influenced locations. Comparatively, the Basin Bridge Railway Station area recorded the lowest noise levels for both day and night.

113. Overall, the monitored corridor exhibits elevated noise levels primarily due to heavy traffic flow, honking, commercial activities, and continuous movement near metro and railway stations. Noise levels also exceed WHO guideline values. These findings indicate the need for noise mitigation measures such as traffic regulation, enforcement of honking restrictions, installation of noise barriers near sensitive locations, and public awareness initiatives.

114.

115.

116. **Table 16** provides ambient noise quality monitoring results.

Table 16: Results of noise quality monitoring

S. No	Location Name	latitude and longitude	Daytime leq dB(a)	Nighttime leq dB(a)
1	Near Akkarai Junction and Signal (B)	12°53'50.582"N 80°14'49.288"E	72.9	57.8
2	Near Kamatchi Hospital (Ram Nagar South) (B)	12°56'56.239"N 80°12'26.471"E	66.3	54.2
3	Near Medavakkam Junction (B)	12°55'2.15"N 80°11'41.333"E	73	64.1
4	Near Porur Lake (C)	13°2'27.354"N 80°8'55.556"E	66.2	51.2
5	Near Tidel Park Signal (B)	12°59'17.849"N 80°15'12.392"E	70.5	64.1
6	Near Nandanam Metro Station (B)	13°1'49.852"N 80°14'26.061"E	67.7	55.2
7	Near KMC Hospital (B)	13°4'38.317"N 80°14'26.369"E	72.2	64.4
8	Near Basin Bridge Junction Railway Station (B)	13°6'29.242"N 80°16'8.609"E	61.5	48.9
9	Near Manali CPCL (A)	13°9'22.499"N 80°16'27.355"E	63.3	52.9
10	Near Retteri Lake (B)	13°8'45.111"N 80°13'7.571"E	73.9	64.9
11	Near Puzhal Lake, Redhills) (B)	13°10'58.57"N 80°11'34.866"E	64.3	54.6
12	Near Kathi Para (St. Thomas Mount) (B)	13°0'26.166"N 80°12'1.254"E	64.2	54.4

Noise Level Standards

Receptor/ Source	India National Noise Level Standards[a] (dBA)		WHO Guidelines Value For Noise Levels Measured Out Doors[b] (One Hour LA_q in dBA)	
	Day	**Night**	**07:00 – 22:00**	**22:00 – 07:00**
Industrial area	75	70	70	70
Commercial area	65	55	70	70
Residential Area	55	45	55	45
Silent Zone	50	40	55	45

Date of sample collection and testing - 25.2.2023 to 27.2.2023 –
Laboratory - Chennai testing laboratory Pvt Ltd, Chennai
Note-
[a] Noise Pollution (Regulation and Control) Rules, 2002 as amended up to 2010.
[b] Guidelines for Community Noise. WHO. 1999 (From WB-IFC EHS Guidelines 2007)

117. **Groundwater.** Chennai district is underlain by a complex sequence of formations ranging from ancient Archaean rocks to Recent Alluvium. The main hydrogeological units are (i) Archaean crystalline rocks composed of charnockites, gneisses and basic–ultrabasic intrusives, (ii) consolidated Gondwana and Tertiary sediments, and (iii) Recent alluvium of sand, silt and clay. The crystalline rocks are weathered and fractured to depths of about 60 m, with variable

weathered mantle thickness from less than 1 m to around 12 m, providing localised fractured aquifers tapped by bore wells in areas such as Saidapet, Adyar, Kasturba Nagar, Gandhi Nagar and Ashok Nagar. Gondwana shales, black to dark grey and jointed, occur with thicknesses ranging from 20–24 m in Kilpauk and Ashok Nagar to more than 130 m in Koyambedu, while Tertiary sandstones are reported beneath the alluvium in parts of Binny Road, Poes Garden, Anna Nagar and Royapuram. Groundwater occurs in all these formations—Archaean crystalline, Gondwana, Tertiary and alluvium—and is developed through dug wells, ring wells, filter-point wells, tube wells and bore wells. Alluvial aquifers, up to about 28 m thick in north Chennai near Perambur and 24 m at Kilpauk, can yield up to about 25 lps in the more productive granular zones, whereas Gondwana and crystalline aquifers generally show moderate yields of about 1–4 lps depending on the degree of fracturing.

118. Water-level behavior reflects both seasonal recharge and long-term stress on the aquifer system. In shallow aquifers, pre-monsoon (May 2006) depths to water ranged from about 2.2–7.6 m below ground level (bgl), with shallower levels (<5 m bgl) in areas such as Alwarpet, Besant Nagar, K.K. Nagar, Tondiarpet and Velachery, and deeper levels (>5 m bgl) in Aminjikarai, T. Nagar, Thirumangalam and Vepery. Post-monsoon (January 2007) water levels rose to 0.45–5.3 m bgl, with very shallow levels (<2 m bgl) in pockets at K.K. Nagar, Villivakkam and Velachery; long-term records for 1998–2007 indicate both rises (up to about 0.93 m/year) and declines (up to about 0.8 m/year). Exploratory wells show transmissivity values between about 6 and 872 m^2/day, with storativity of 2.9×10^{-4} to 4.5×10^{-3}, indicating generally moderate aquifer potential. Groundwater is typically colourless, odourless and slightly alkaline, but often mineralised: specific conductance commonly exceeds 2,250 µS/cm, total hardness is above permissible limits in more than half of the samples, and nitrate exceeds 45 mg/l in around 10% of the samples, largely due to waste disposal practices. Groundwater is developed through dug wells (10–15 m bgl) and tube/bore wells (60–100 m bgl), which can generally sustain 2–4 hours and 6–10 hours of pumping per day respectively, making prudent management and protection of these aquifers essential for Chennai's water security.

119. **Surface water**. Chennai district lies partly in the Korattalaiyar–Arniyar river basin. The major rivers include the Adyar, Cooum, and Otteri Nulla.

- Adyar River (42 km, 800 km^2 catchment) originates near Thiruneermalai and flows through the southern part of the district. It carries perennial flow, with seasonal flooding during monsoon and tidal backwater intrusion up to 3–4 km inland.
- Cooum River, flowing through central Chennai, originates from Cooum tank in Tiruvallur. It mainly carries drainage water, highly polluted, with significant flow only during flood season.
- Otteri Nulla is a small northern stream, while the Buckingham Canal, once built for navigation, now largely serves as a wastewater carrier.

120. **Lakes of Chennai.** The capital city of Tamil Nadu is endowed with a rich network of lakes (aeri in Tamil), which form an integral part of its natural water architecture. These lakes act as rain-fed reservoirs, supporting urban water supply, ecological balance, and biodiversity. Many of them are interconnected with rivers and canals, enabling the seamless flow of water into the Bay of Bengal. With growing demand for water and the ecological significance of lakes, their conservation has become vital for Chennai's sustainable future.

- Chembarambakkam Lake: Spread over 15 km^2, this large rain-fed reservoir is the origin of the Adyar River and one of the primary sources of drinking water for Chennai. Historically, it was known as *Puliyar Kottam* during the Chola period.

- Puzhal Lake (Red Hills Lake): Constructed during the colonial period, it remains one of the largest drinking water reservoirs for the city, holding a capacity of 93 MCM.
- Porur Lake: Covering 200 acres with a storage capacity of 46 MCM, this lake in southwest Chennai is an important water source and also a scenic spot visible from the Chennai Expressway.
- Sholavaram Lake: Spread over 24 km^2, it has historical significance as a World War II airstrip site. Today, it connects with Puzhal Lake through a canal and supports nearby communities.
- Narayanapuram Lake: Located near Pallikaranai Marshland, this lake historically supported irrigation for local farmers. It plays a strategic role in water absorption and flood moderation in the area.
- Velachery Lake: Situated in the heart of the Velachery locality, this marshy lake has both cultural and ecological significance. It continues to serve as a local water source.
- Perumbakkam Lake: Known for its rich biodiversity and birdlife, including storks, flamingos, and pelicans, this freshwater lake is a prime spot for birdwatching and ecological conservation.
- Pulicat Lake: Located about 60 km from Chennai, it is the second-largest brackish water lake in India. Rich in biodiversity, it is home to migratory birds like flamingos and kingfishers and has historical roots dating back to the Chola dynasty.

121. **Surface Water Quality.** Surface and groundwater samples were collected from 10 locations. Most parameters were within the permissible limits of IS 10500 and CPCB Class B standards. Slight exceedance in turbidity and microbial content was recorded at few locations. Total dissolved solids (TDS), chloride, and pH levels were within acceptable limits. Table 17 provides the results of the surface water quality monitoring while

122. Figure **28** shows the sampling location.

Figure 28: Google map showing surface water quality sampling locations

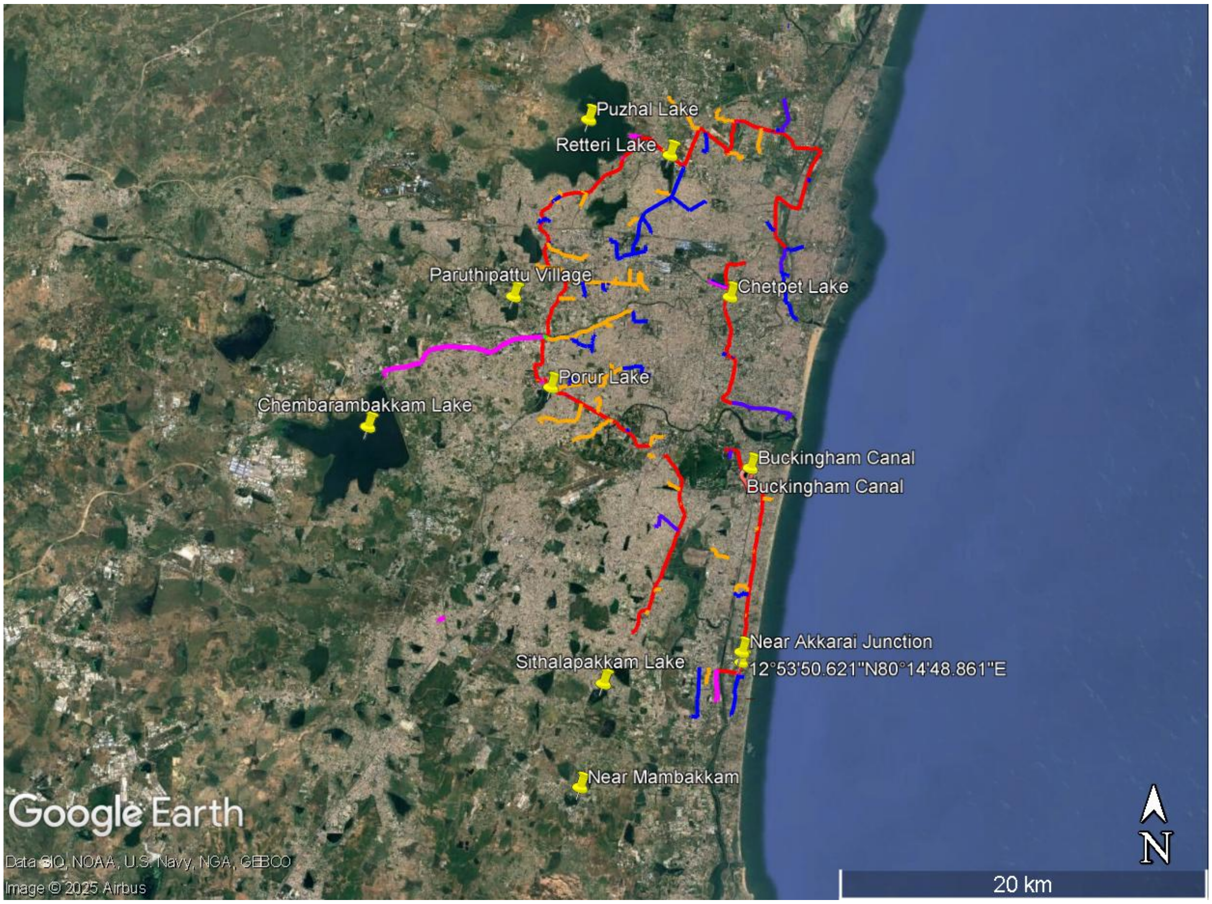


Table 17: Surface water quality test results

Location name	National Standards for Drinking Water [a,b]	WHO Guidelines for Drinking-Water Quality, 4th Edition, 2011[c]	Near Akkarai Junction	Near Mambakkam	Paruthipattu Village	Sithalapakkam Lake	Chetpet Lake	Porur Lake	Chembarambakkam Lake	Buckingham Canal	Puzhal Lake	Retteri Lake
Appearance			Turbid	Turbid liquid	Clear liquid	Slightly turbid liquid	Clear liquid	Turbid liquid	Slightly turbid liquid	Turbid liquid	Slightly turbid liquid	Turbid liquid
Colour (HU)	5 (15)	none	10	10	2	5	2	5	5	10	5	5
pH @ 25°C	6.5 – 8.5	none	7.9	7.9	8	7.9	7.8	8.4	7.9	7.9	7.6	7.7
Odour	Agreeable	-	Disagreeable	Disagreeable	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable	Disagreeable	Agreeable	Agreeable
Turbidity (NTU)	1 (5)	-	21	21	< 1	4	< 1	5	3	17	3	6
Total Hardness as $CaCO_3$ (mg/l)	200 (600)	-	384	384	135	219	337	224	133	384	127	192
Calcium as Ca (mg/l)	75 (200)	-	101	101	36	58	112	70	32	121	26	49
Magnesium as Mg (mg/l)			32	32	11	18	14	12	13	20	15	17
p-Alkalinity as $CaCO_3$ (mg/l)			21	21	8	8	8	12	8	Nil	Nil	Nil
Total Alkalinity as $CaCO_3$ (mg/l)			384	515	136	190	358	260	132	700	124	148
Chloride as Cl^- (mg/l)			101	451	44	105	141	192	68	333	74	152

Location name	National Standards for Drinking Water [(a,b)]	WHO Guidelines for Drinking-Water Quality, 4th Edition, 2011[c]	Near Akkarai Junction	Near Mambakkam	Paruthipattu Village	Sithalapakkam Lake	Chetpet Lake	Porur Lake	Chembarambakkam Lake	Buckingham Canal	Puzhal Lake	Retteri Lake
Sulphate as SO_4 (mg/l)			32	92	106	18	265	58	30	141	28	73
Total Dissolved Solids (mg/l)	500 (2,000)	-	1,378	1,378	370	416	946	654	302	1,330	292	516
Iron as Fe (mg/l)			1.56	1.56	0.04	0.65	0.08	0.89	0.52	1.18	0.49	0.81
Silica as SiO_2 (mg/l)			32	32	45	6.3	70	3.8	11	20	2.1	8.4
Carbonate Hardness (mg/l)			384	384	135	190	337	224	132	384	124	148
Non-Carbonate Hardness (mg/l)			Nil	Nil	Nil	29	Nil	Nil	1	Nil	3	44
Total Coliform (per 100ml)	Must not be detectable in any 100 ml sample		Present	Present	Absent	Absent	Absent	Absent	Present	Present	Present	Present
E. coli (per 100ml)			Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent	Absent
Date of sample collection and testing - 27.2.2023 to 4.3.2023 – Laboratory - Chennai Testing Laboratory Pvt Ltd, Chennai												

[a] Bureau of India Standard 10200: 2012.[b] Figures in parenthesis are maximum limits allowed in the absence of alternate source.
[c] Health-based guideline values

C. Biological Environment

123. Chennai's geographical location supports a wide range of ecological landscapes, including estuaries (Adyar and Cooum), wetlands (Pallikaranai and Singanallur), forested hillocks such as the Pallavaram Hills, and an extensive sandy coastline. In addition, the city contains numerous man-made reservoirs, irrigation tanks, and residual agricultural lands, all of which contribute to its ecological diversity.

124. **Fauna.** Chennai supports remarkable faunal diversity within its protected areas and wetlands. Guindy National Park is home to 14 species of mammals including blackbuck, spotted deer, and jackals, along with over 150 bird species, reptiles, insects, and butterflies. The IIT Madras campus, once part of the park, supports similar fauna, including mongoose, wild cats, toddy cats, and more than 100 bird species.

125. **Protected areas.** The city includes several ecologically sensitive areas such as Reserve Forests, Guindy National Park, Pallikaranai Marsh, and areas classified under Coastal Regulation Zones (CRZ-I, CRZ-II, and CRZ-III). These areas play a critical role in maintaining hydrological balance, biodiversity conservation, and climate resilience within the Chennai Metropolitan Region.

126. Chennai is ecologically connected to larger regional systems, including Pulicat Lake to the north, the Nagari hill range to the northwest, and the Muttukadu estuary to the south. Sacred groves, which once formed an integral part of the regional landscape, now survive only in limited pockets such as within the IIT Madras campus and Guindy National Park, serving as important cultural and ecological heritage sites.

127. **Coastal Regulation Zone (CRZ) area.** The coastal strip of the city is divided into different zones according to ecological sensitivity and development patterns. CRZ-I areas are marked near mangrove and estuarine stretches, particularly around the Adyar and Cooum estuaries. These zones have strict buffer limits to protect sensitive ecology and prevent any harmful activity in such fragile regions.

128. CRZ-II is identified in already developed coastal areas, especially along stretches with urban infrastructure. Places such as Marina Beach, coastal promenades, and parts of the shoreline with established buildings fall under this category. Here, regulated development is permitted since the areas are already urbanized, but construction is subject to CRZ rules.

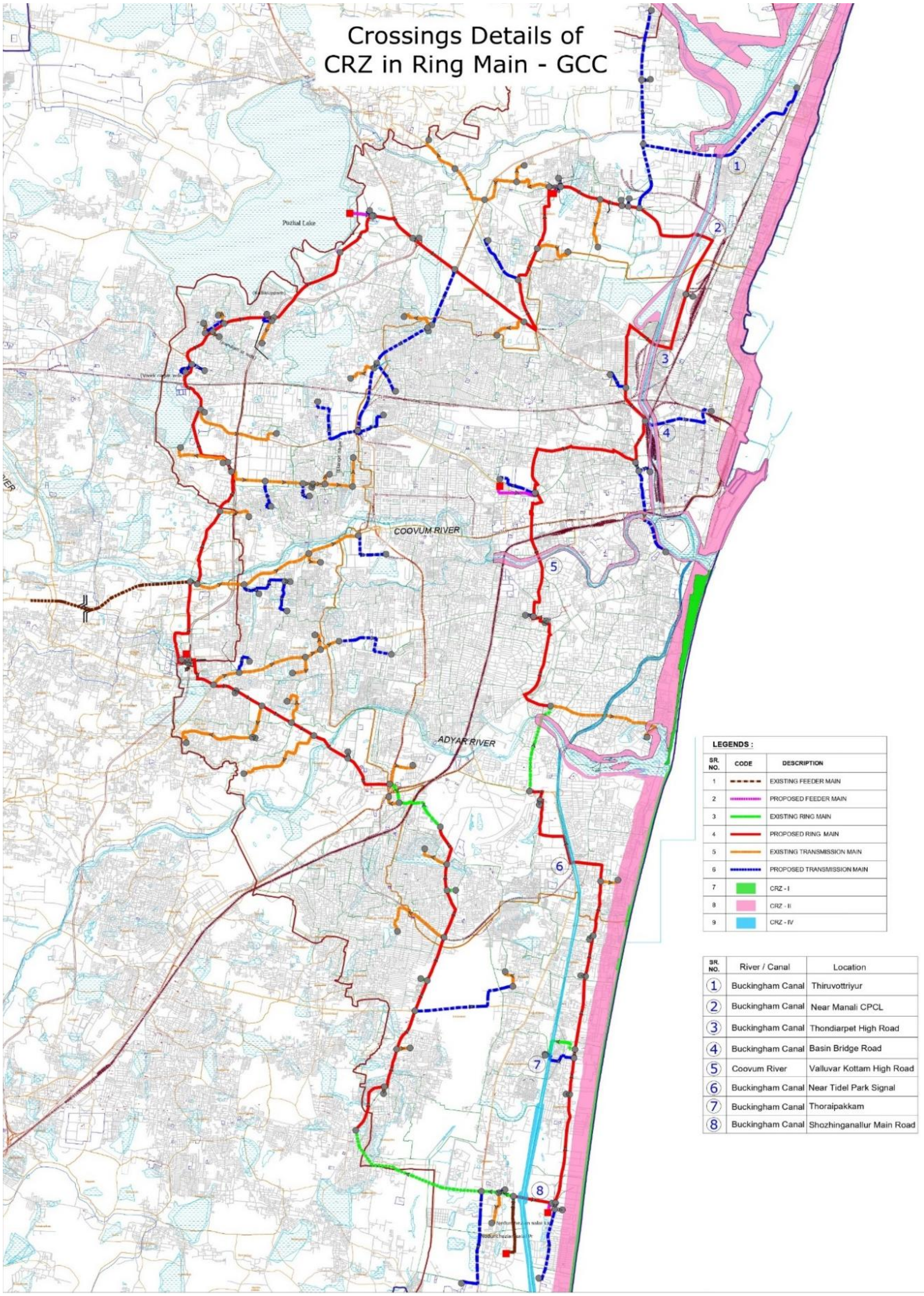
129. In contrast, CRZ-III covers less-developed or rural stretches of the coastline. These areas are generally behind the developed front and are subject to more restrictions, with limited scope for new construction. The emphasis here is on regulating growth to ensure that natural ecosystems are not disturbed.

130. CRZ IV A a) The water area from the Low Tide Line to twelve nautical miles on the seaward side; b) Shall include the water area of the tidal influenced water body from the mouth of the Water body at the sea up to the influence of tide which is measured as five parts per thousand during the driest season of the year.

131. The CRZ map has been prepared by IRS, Anna University, Chennai—an approved agency for demarcation of the High Tide Line (HTL) and Low Tide Line (LTL). The CRZ report for the proposed RMS has been prepared by superimposing the project alignment on the approved Coastal Zone Management Plan (CZMP) as per CRZ Notification, 2011.

132. Based on the CRZ map, the proposed RMS—comprising of existing and new transmission mains interconnecting water treatment plants and water distribution stations across Chennai, Tiruvallur, Kanchipuram, and Chengalpattu districts—passes through both developed and water-area CRZ zones. Approximately 1,306 m of the pipeline alignment falls within CRZ-II areas, while about 1,107 m lies within CRZ-IV areas. Figure 29 shows the proposed pipelines alignment overlaid on the CRZ map.

Figure 29: Map Showing CRZ area and pipe crossing



133. **Protected Areas and Urban Green Spaces:** Within the city limits, two major protected areas exist: Guindy National Park and the Pallikaranai Marsh Reserve Forest, the latter designated as a Ramsar site in 2022. Other areas with remnant natural vegetation include the IIT Madras campus, Madras Christian College (MCC) campus, Theosophical Society campus, and the Kattupalli Island. Along with urban parks and green spaces, these areas contribute significantly to maintaining ecological balance within the metropolitan region.

134. **Guindy National Park** (GNP), located in Mambalam–Guindy Taluk in the south-western part of Chennai city, is one of India's smallest national parks situated within an urban setting. The area was originally part of the Guindy Reserve Forest under the Raj Bhavan and was formally transferred to the Forest Department in 1958. A portion of the land was developed as a Children's Park in 1959, aligning with the vision of providing natural learning spaces for children. Today, the park functions both as a protected area and as a recognized zoo under the Central Zoo Authority (CZA), upgraded to a Medium Category Zoo in 2019.

135. GNP is a unique biodiversity hotspot comprising dry evergreen scrub, thorn forest, grasslands and water bodies, supporting over 350 species of plants, including shrubs, climbers, herbs and grasses. Tree species include *Azadirachta indica*, *Limonia acidissima*, *Borassus flabellifer*, *Acacia chundra*, *Randia spp.*, and native dryland scrub vegetation. The grassland area is particularly important as it represents one of the last remaining grassland habitats in Chennai and supports the near-threatened blackbuck (*Antilope cervicapra*).

136. The park supports rich faunal diversity, including:

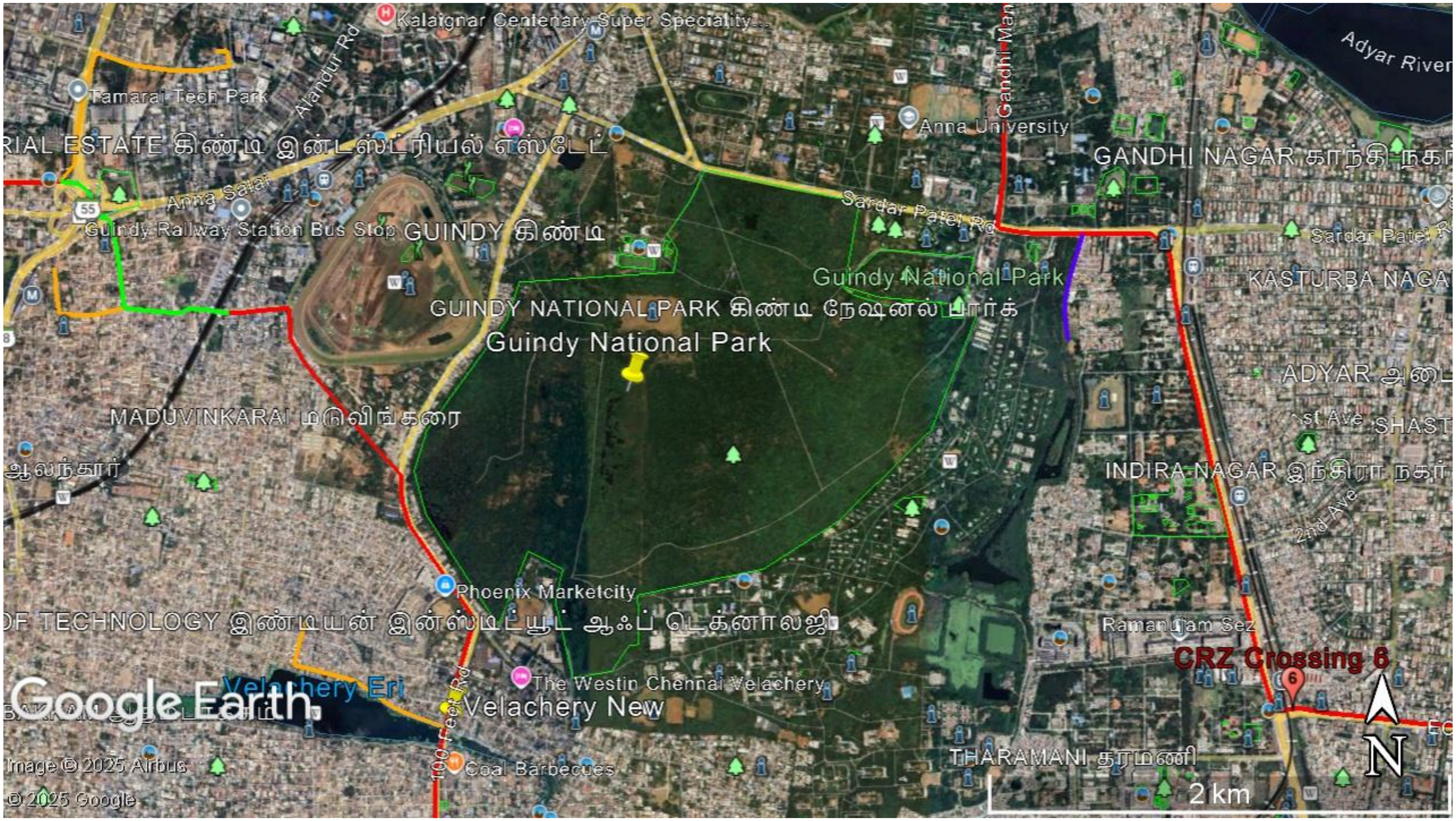
(i) **14 species of mammals** such as blackbuck, chital, jackal, small Indian civet, bonnet macaque, Indian hare, and pangolin.
(ii) **150+ species of resident and migratory birds**, including partridge, lapwings, flycatchers, woodpeckers, drongos and seasonal waterbirds.
(iii) **Reptiles** such as the Indian monitor lizard, fan-throated lizard, saw-scaled viper, turtles and the endangered Indian star tortoise.
(iv) **Amphibians and invertebrates**, including over 60 species of butterflies and spiders.

137. The **Guindy Children's Park Zoo**, located within the park, serves as an ex-situ conservation centre for representative fauna of the Eastern and Western Ghats. It focuses on:
ex-situ conservation of primates, birds and reptiles, public education and conservation awareness for the metropolitan population, strengthening linkages between ex-situ and in-situ conservation, and scientific veterinary care and temporary rehabilitation of urban wildlife.

138. The zoo functions as a Temporary Rescue Centre, receiving injured or rescued wild animals from Chennai and surrounding areas for treatment and release as per veterinary advice. Figure 29 shows Guindy national park and surrounding proposed pipelines.

139. Adjacent to the national park is the Guindy Snake Park, recognized as a medium zoo by CZA since 1995. Spread over 22 acres, it houses snakes, reptiles, small mammals, birds, and includes visitor education facilities. The Children's Park features environmental interpretation facilities, a fossilized tree exhibit, and digital educational show. Figure 29 shows the Guindy National Park and the proposed RMS.

Figure 30: Google image showing Guindy National Park and proposed ring main (red)



140. **Pallikaranai Wetland** is a vast freshwater marsh located about 20 km south of central Chennai, stretching across 1247.54 ha along the Coromandel Coast. Surrounded by rapidly urbanising areas such as Perungudi, Velachery, Taramani, Pallikaranai, Madipakkam, and Sholinganallur, the wetland acts as a crucial aquatic buffer for flood-prone regions of Chennai and Chengalpattu. Its natural sponge-like capacity helps absorb stormwater runoff and recharge the adjoining South Chennai aquifer, making it vital for the city's hydrology.

141. Ecologically, Pallikaranai is one of the most diverse natural habitats in southern Chennai, harbouring over 380 species of flora and fauna. The marsh supports freshwater reservoirs, scrublands, aquatic grasses, irrigated farmlands, and water-filled depressions—creating ideal nesting, foraging, and breeding conditions for birds, fishes, reptiles, amphibians, and invertebrates. It is home to species of global conservation significance, including the endangered Russell's viper, glossy ibis, and pheasant-tailed jacana. The wetland is also a critical habitat for migratory waterbirds and sustains several fish species that use the marsh for feeding, breeding, and migration.

142. The flora of the Pallikaranai wetlands consists primarily of aquatic plants, reeds, liana, and sedges. Acanthus, water lettuce, water hyacinth, joyweed, swamp morning glory, snowbush, and paspalum are featured across marsh areas, as well as transition areas. Among the sedges are umbrella papyrus, purple nutsedge, and slender cyperus. Large tracts of marsh grasses and pastures adjoin the wetlands areas. Figure 31 shows Pallikaranai wetland and nearby proposed pipelines.

Table 18: Detailed of fauna of Pallikaranai wetland

Sl. No.	Faunal Group	No. of Species	Key Details / Examples
1	Birds	165 species (115 commonly observed)	Migratory / Globally Significant: Glossy Ibis, Grey-headed Lapwing, Pheasant-tailed Jacana, *Calidris subminuta, Charadrius alexandrinus, Dendrocygna bicolor, Limicola falcinellus, Himantopus himantopus, Limosa limosa, Mycteria leucocephala, Tringa erythropus*. Common Waterbirds: Cormorants, Darters, Herons, Egrets, Open-billed storks, Spoonbills, White ibis, Little grebe, Indian moorhen, Black-winged stilt, Purple moorhen, Coots, Warblers, Dabchicks. Breeding/Nesting in Large Numbers: *Aquila clanga, Aythya ferina, Calidris tenuirostris, Rynchops albicollis, Sterna acuticauda*
2	Mammals	10 species	Small rodents, bats, Indian civet, mongoose species
3	Reptiles	21 species	Threatened/Notable: Russell's viper (locally highly endangered). Others: *Lissemys punctata* (Indian flapshell turtle – nests in large numbers), snakes, skinks, geckos, marsh-dwelling lizards
4	Amphibians	10 species	Frogs and toads typical of freshwater marsh ecosystems
5	Fishes	46–50 species	*Channa orientalis, Trichogaster chuna* (Dwarf gourami), *Etroplus* spp. (Chromides), *Anguilla bengalensis*, *Anguilla bicolor*
6	Molluscs	9 species	Freshwater snails and clams
7	Crustaceans	5 species	Freshwater prawns and marsh-associated crabs
8	Butterflies	7 species	Common species of the *Indomalayan realm* (specific names not listed)

Figure 31: Google earth image showing Pallaikaranai masrh land, ring main(red) and transmission main (blue)

143. **The Adyar creek and estuary.** Located in the east of the Thiru. Vi. Ka. Bridge, it covers approximately 358 acres. This ecologically significant wetland system historically supported diverse flora and fauna but experienced severe degradation due to urbanisation, siltation, sewage inflow, and solid waste dumping. The Government of Tamil Nadu (GoTN) initiated an eco-restoration programme, designating 58 acres (Survey Nos. 4309/1, 2 & 8) for Phase I development as an Eco Park (Tholkappiar Poonga) and the remaining area for Phase II restoration.

144. *Phase I* Area *Characteristics:* Phase I (Tholkappiar Poonga) covers the inland waterway branching from the Adyar Estuary and curving along Quibble Island up to Greenways Road. Historically, this zone functioned as:

- A stormwater outflow and retention system for the Mandaveli watershed,
- A tidal reservoir during high tides, and
- A natural infiltration zone.

145. In later decades, the creek became a site for sewage discharge, waste dumping, and encroachments, causing ecological decline.

146. *Planning and Institutional Framework:* Since 1997, the Government has pursued the protection and ecological restoration of Adyar Creek. Tamil Nadu Urban Infrastructure Financial Services Ltd. (TNUIFSL) was appointed as the nodal agency to develop the Master Plan, engage experts, and coordinate implementation. Specialists in wetland ecology, hydrology, water management, landscape design, and biodiversity conservation contributed to the project's planning and design.

147. ***Site Condition Before Restoration****:* Before 2008, the Adyar Creek–Estuary area had no natural vegetation. The wetland was heavily degraded due to:

- Dumping of construction debris and municipal waste,
- Siltation and blockage of tidal flow,
- Loss of mangroves and native wetland vegetation, and
- Weak hydrological circulation.

148. To restore *natural* water flow, approximately 2.80 lakh (equivalent to 300,000) m^3 of accumulated sludge and debris were excavated during restoration works.

149. ***Ecological Restoration – Plantation and*** **Mangrove** ***Recovery****:* A large-scale eco-restoration programme was implemented in Phases I and II, involving the planting of 1.4 lakh seedlings from 172 indigenous species. Mangrove recovery formed a major component:

Phase I

- 35,000 mangrove seedlings were planted, covering 9 species.
- About 30,000 well-established true mangrove plants, including:
- *Acanthus ilicifolius, Avicennia marina, Aegiceras corniculatum, Bruguiera cylindrica, Rhizophora mucronata,* and *Rhizophora apiculata*.

Phase II

- 75,000 mangrove seedlings, again representing 9 species.
- 8 species showed successful establishment.
- Approximately 60,000 healthy mangrove plants matured during this phase.

150. **Overall Impact:** The eco-restoration transformed a previously barren, polluted creek into a functional mangrove and estuarine wetland, enhancing:

- Biodiversity and wildlife habitats,
- Water quality and tidal exchange,
- Flood mitigation capacity,
- Climate resilience and carbon sequestration, and
- Environmental awareness among the public through the development of Tholkappiar Poonga.

151. Figure 31 shows **Adyar creek and estuary** I (*Tholkappiar Poonga*) and nearby proposed pipelines.

Figure 32: Google earth image showing Tholkappiyar/Adyar eco park area along with proposed transmission line (violet) and ring main (red)

152. The Integrated Biodiversity Assessment Tool (IBAT) screening was conducted within a 50-km radius around the project site. The screening indicates the presence of areas of biodiversity importance, including one Key Biodiversity Area (KBA) and 80 IUCN Red List threatened species. Table 19 presents list of IUCN listed species around the project area. Within the 50-km radius of the project site, one KBA has been identified, the Pulicat Lake KBA – an ecologically-sensitive wetland ecosystem known for migratory birds and diverse aquatic habitats is about 25 km from nearest proposed project component. A total of 80 threatened species (CR, EN, VU, NT categories) were recorded within the 50-km radius of project area of influence. The taxon-wise distribution of IUCN listed species is given in Table 19.

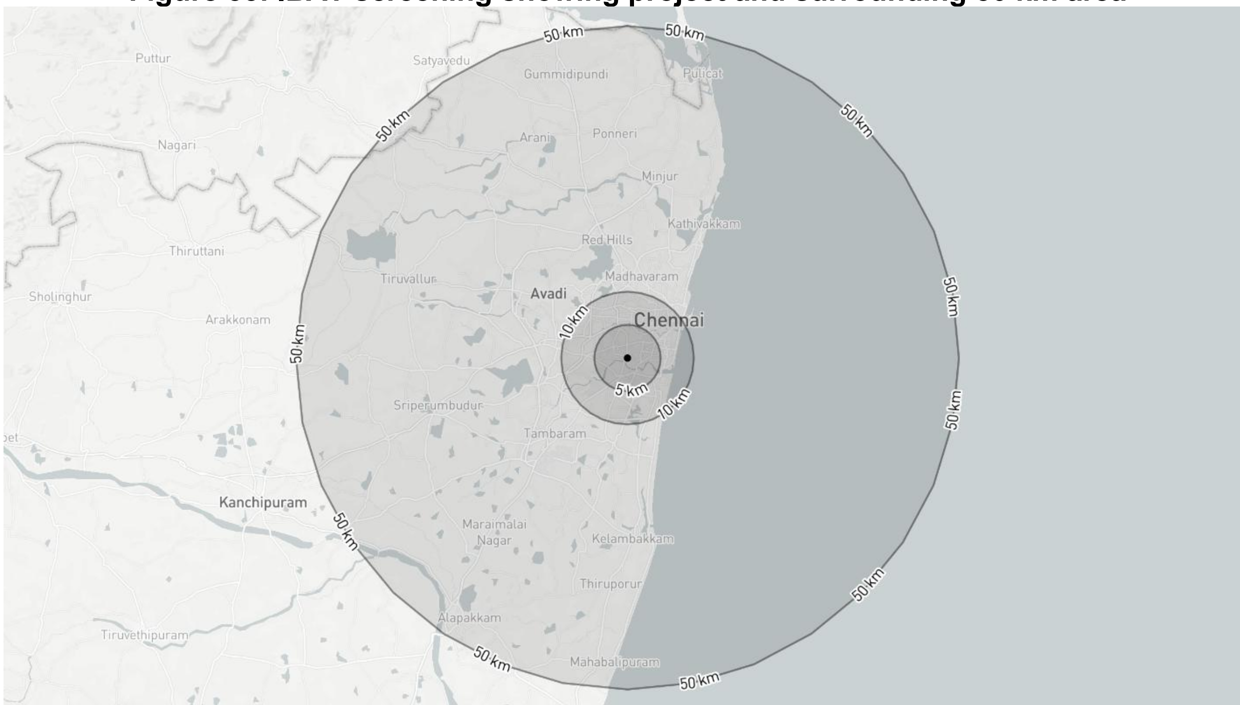
Table 19: Summary of IUCN listed species as per IBAT checklist

Taxonomic Group	CR	EN	NT	VU	Total
Reptilia	2	2	5	6	15
Chondrichthyes	2	–	–	1	3
Aves	3	4	20	8	35
Actinopterygii	–	1	5	3	9
Mammalia	–	2	5	6	13
Magnoliopsida	–	2	1	1	4
Liliopsida	–	–	–	1	1
Total	**7**	**11**	**36**	**26**	**80**

CR =Critically Endangered; EN= Endangered; VU= Vulnerable; NT= Near Threatened

153. However, as per information from the Forest Department, Guindy National Park and the Pallikaranai Marsh (a designated Ramsar Site) are recognized as important protected and ecologically-sensitive biodiversity areas within the broader project influence region. Birds represent the highest proportion of recorded species, with 35 avifaunal species, including several CR and EN species associated with the wetland and coastal habitats near the Pulicat Lake. In addition to birds, the area supports notable diversity across other taxa, including reptiles (15 species), mammals (13 species), and fish species such as Actinopterygii (9 species) and Chondrichthyes (3 species). The presence of threatened plant groups, including Magnoliopsida (4 species) and Liliopsida (1 species) further emphasizes the ecological sensitivity of the landscape. Figure 33 shows IBAT screening in project and surrounding 50 km area.

Figure 33: IBAT screening showing project and surrounding 50 km area



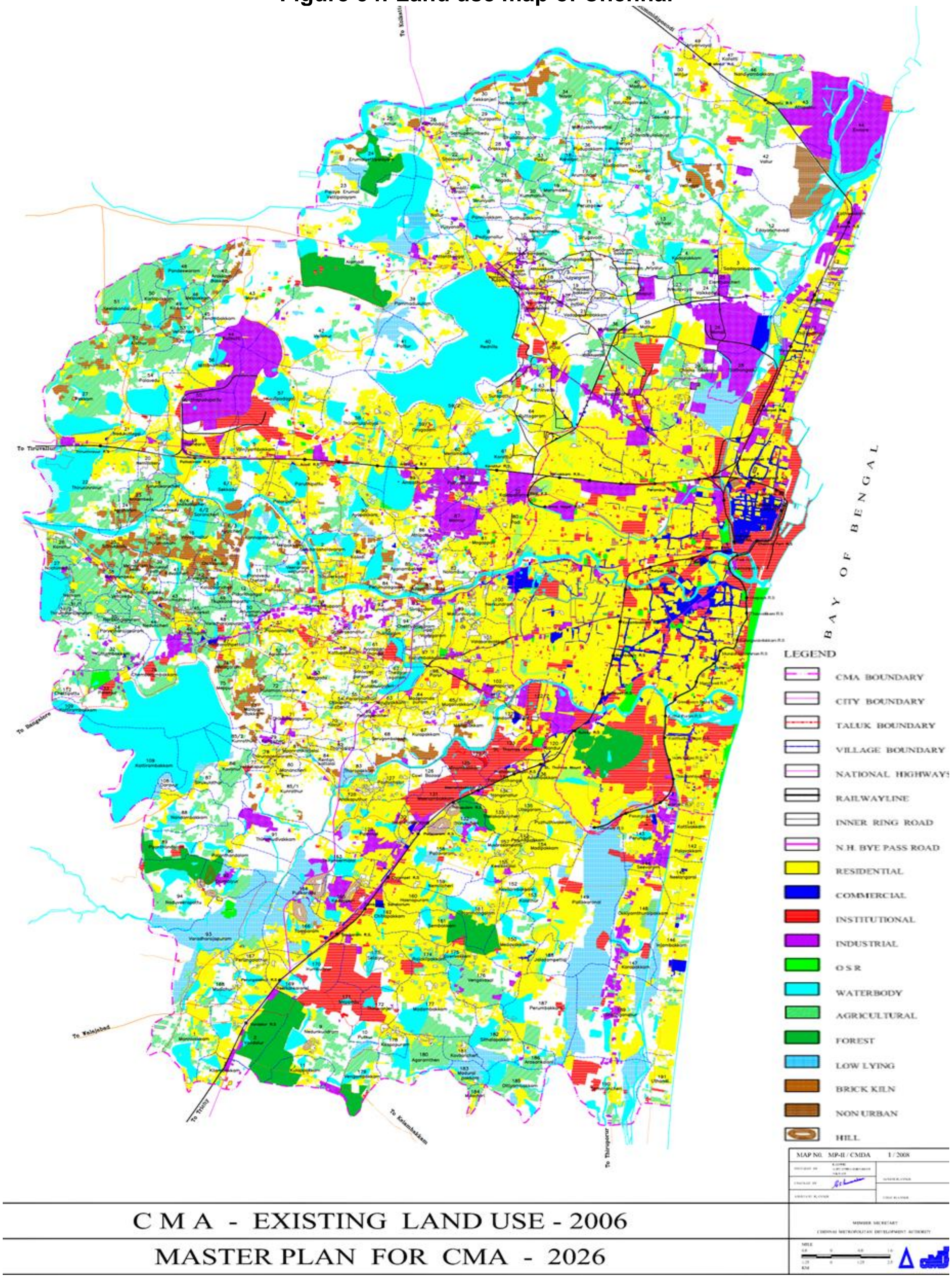
154. Chennai's ecological history is rooted in its coastal and wetland landscapes, represented by the Neithal and Paalai cultural regions. Historically, its coastal plains were dominated by Palmyra and Phoenix palms, along with native trees such as neem, wood-apple, and banyans. In estuarine and beach areas, species like Avicennia, *Calophyllum inophyllum*, and *Thespesia populnea* thrive, while invasive species such as *Prosopis juliflora* and *Lantana camara* have become widespread.

155. The city's urban green spaces—including Guindy National Park, IIT Madras campus, Theosophical Society campus, and Pallikaranai marshland—harbor diverse vegetation. Guindy National Park supports dry evergreen scrub and thorn forests with over 350 plant species, while the Theosophical Society campus hosts a mix of native and exotic trees, shrubs, herbs, and climbers. Pallikaranai marshland, a unique estuarine ecosystem, sustains salt-tolerant sedges, grasses, and over 140 plant species, making it a critical wetland habitat.

D. Overall Land Use

156. **Land Use**. The core city is heavily residential, with significant institutional presence and a balanced integration of open space. The CMA shows more diversified land use: expanded residential zones, increased industrial areas, significant agriculture, and a prominent share of natural land features. The plan clearly promotes compact, sustainable growth—centralizing core functions while channeling industries and infrastructure toward suburban and peri-urban areas. Figure 34 shows the land use of Chennai.

Figure 34: Land use map of Chennai



157. **Other Infrastructures.** GCC maintains 1,160 roads to a length of 370 km and stormwater drain to a length of 962 km. Total number of streetlights in Chennai city under the maintenance of GCC is 213,045 and using 19 megawatts per day and spending 2 lakhs for electric consumption per day. GCC has 260 parks and has constructed 113 community halls for public purposes. GCC is removing 5,000-5,200 metric ton (MT) of solid waste per day through 966 conservancy vehicles and maintaining the Kodungaiyur and Perungudi dumping grounds as disposal sites for solid waste.

158. **Sewerage system.** As of 2024–2025, Chennai's sewerage infrastructure, managed by CMWSSB, has been modernized to serve approximately 98% of the core city and is rapidly expanding into added areas through projects like AMRUT 2.0. The city-wide network now spans over 4,100 km of sewer lines (including 247 km of recently replaced high-capacity pipes and 213 km of new extensions) supported by a robust system of 356 sewage pumping stations, which includes 38 newly commissioned high-capacity units and 23 strategically placed roadside lift stations. Wastewater is processed through a network of sewage treatment plants (STPs), providing a total installed treatment capacity of 769 MLD. Current operational data indicates that the daily quantity of sewage collected and treated has increased to an average of 675 MLD, ensuring improved pollution abatement for the Adyar and Cooum rivers while facilitating water reuse through advanced tertiary treatment facilities.

159. To meet the revised effluent disposal standards prescribed by the Central Pollution Control Board (CPCB) and pursuant to the directions issued by the National Green Tribunal (NGT) vide order dated **30 April 2019**, improvement, modification, and rehabilitation works have been undertaken at all existing STPs in Chennai City. These upgrades cover major facilities, including the 110 MLD plant at Kodungaiyur; 60 MLD and 120 MLD plants at Koyambedu; 40 MLD and 54 MLD plants at Nesapakkam; and 54 MLD and 60 MLD plants at Perungudi. Overall, Chennai’s 13 STPs, commissioned between 1974 and 2015 and located at Kodungaiyur (Zones I and II), Koyambedu (Zone III), Nesapakkam (Zone IV), Perungudi, Alandur (Zone V), and Sholinganallur, together provide an installed treatment capacity of about 745 MLD.

160. **Roads and transport system.** Chennai has a well-developed multi-modal transport system comprising roads, railways, metro, air, and port infrastructure, supporting both intra-city and regional connectivity.

161. **Air Connectivity.** Chennai International Airport (MAA) is among India’s busiest airports, serving both domestic and international routes. It connects Chennai to all major Indian cities, as well as key global destinations in Southeast Asia, the Middle East, Europe, and the USA. The airport has two terminals—Kamaraj (domestic) and Anna (international)—and acts as an important gateway to South India.

162. **Rail Connectivity.** Chennai is a major railway hub in Southern India. It has two main terminals—Chennai Central (Puratchi Thalaivar Dr. M.G. Ramachandran Central Railway Station) and Chennai Egmore—that link the city to almost every part of the country. Suburban train services connect the city with its rapidly growing metropolitan region. The Mass Rapid Transit System (MRTS) and the expanding Chennai Metro Rail further strengthen intra-city rail travel.

163. **Road Connectivity.** Chennai is well connected by national highways radiating in all directions:

- **NH 16** (part of Golden Quadrilateral) connects it to Kolkata via Andhra Pradesh.
- **NH 48** links it with Bengaluru and Mumbai.

- **NH 32** connects it with Puducherry and southern Tamil Nadu.
- **NH 716** leads towards Tirupati.

164. The city has a road network of about 2,780 km, developed in a radial and ring pattern. Major radial corridors include Anna Salai (NH 45), Periyar E.V.R. Salai (NH 4), Chennai–Kolkata Salai (NH 5), and Chennai–Thiruvallur Salai (NH 205), along with other important roads such as Kamarajar Salai, East Coast Road, Rajiv Gandhi Salai (OMR), Arcot Road, and Thiruvottiyur High Road. The Inner Ring Road (Jawaharlal Nehru Road) and Chennai Bypass form the orbital network, easing congestion on core city roads. The Outer Ring Road (ORR) and proposed Peripheral Ring Road (PRR) further improve regional and freight movement. Public transport is supported by an extensive bus network operated by MTC, along with para-transit and ride-hailing services for last-mile connectivity.

165. **Port Connectivity** Chennai has two major ports: Chennai Port (one of the oldest in India) and Ennore Port (Kamarajar Port), along with Kattupalli Shipyard. These handle cargo, automobiles, petroleum, containers, and passenger cruise services. This makes Chennai one of India's largest logistics and maritime trade hubs.

166. **Metro Connectivity.** The Chennai Metro Rail, operational since 2015, connects key parts of the city including the airport, Central railway station, Koyambedu, Alandur, and Anna Nagar. With Phase II under construction, the network will cover more than 100 km, improving last-mile connectivity and easing congestion.

167. **Parks.** Chennai, the fourth largest metropolitan city, is witnessing a green transformation to counter pollution and global warming from rapid industrial and vehicular growth. The GCC has spearheaded initiatives to expand parks, traffic islands, medians, and green corridors, turning the city from dry to greener and more livable place.

168. Some of these major parks that serve as lung spaces spread around the city are Anna Park in Royapuram, Secretariate Park in Rajaji Salai, Peramabur Flyover Park in Perambur, My ladies Park in Periamet, Dr.Visweshwariah Tower Park in Anna Nagar, Jai Nagar Park in Koyambedu,May Day Park in Chintadripet, Independence Day Park in Nungambakkam, Mayor sundararao Park in Egmore, Panagal Park, Natesan Park, Jeeva park in T.Nagar, Erikarai park in Nesapakkam, Sivan Park (Dr.MGR Park) in K.K.Nagar, Dr.Nageshwara Rao Park in Mylapore and Indira Nagar park in Adyar. The parks which require development are rejuvenated as when required. Of these, Independence Day Park, Dr. Visweshwariah Tower Park, Panagal Park, Natesan Park, Anna Park, Sivan Park, Indira Nagar Park have been redeveloped by redesigning with modern landscape designs by architects having all features of parks attracting a larger public to use them. The salient features of parks are lush green manicured lawns, beautiful long walkways, colorful shrubberies, ornamental hedges and edges, avenue palm trees, colorful fountains and water cascades, yoga platforms and meditation centers, sitting gallery, children's playfields with play equipment, Lilly Pond, sitting benches, arches and pergolas and ramp pathways (for differently abled persons). Parks are provided with see-through compound walls to enjoy the beauty of surrounding city while walking or taking rest. Plants such as BougainVilliae, *Caesalpinia, Pisonia, Acalypha, Ixora hybrid, Clerodendron, Tabernamontana, Dracena cordiline, Eranthemum, Allamanda, variegated, Graphtophyllum, Aralia varigated, Mussaenda, Thuja, Callistemon, Juniferous, Prichardia Palm, Areca Palm, Song of India, Tecomacapensis Furcrea,Galphemia, Haemelia patterns, celerodenron, Nerium oleander, Tecoma, Euphorbia splendens*, and Ficus are selected for planting.

E. Socio-economic Environment

169. The demographic profile of the project area, covering the GCC, is summarized in Table 20 based on the data from Census of India 2011, with a comparison to all-India urban indicators to provide regional context.

Table 20: Demographic Profile of Project Area – Greater Chennai Corporation (Census 2011)

Indicator	Greater Chennai Corporation	All-India Urban Average
Administrative Area	Greater Chennai Corporation	Urban Areas of India
Number of Wards	155	—
Total Population	4,646,732	—
Male Population	2,335,844	—
Female Population	2,310,888	—
Child Population (0–6 years)	459,324 (9.88%)	10.9%
Sex Ratio (Females/1000 Males)	989	926
Child Sex Ratio (0–6 years)	950	902
Literacy Rate (Overall)	90.18%	84.10%
Male Literacy Rate	93.70%	88.76%
Female Literacy Rate	86.64%	79.11%
Scheduled Caste (SC) Population	16.78%	12.6%
Scheduled Tribe (ST) Population	0.22%	2.8%
Total Working Population	1,817,297	—
Share of Working Population	39.1%	35.3%
Total Households	1,154,982	—
Source	Census of India, 2011	Census of India, 2011

- GCC exhibits higher literacy levels and better sex ratio compared to the national urban average, indicating relatively strong social development indicators.
- The child population share is slightly lower than the national urban average, reflecting demographic transition typical of large metropolitan cities.
- SC population proportion is higher than the national urban average, while ST presence is minimal, consistent with coastal urban settlement patterns.
- Workforce participation in GCC is higher than the urban national average, reflecting the city's diversified economic base.

F. Socio-cultural Resources

170. **Religious and cultural places in Chennai.** Chennai is dotted with temples, churches, and heritage shrines that highlight its cultural depth.

- Kapaleeshwarar Temple, Vadapalani Murugan Temple, Arulmigu Thiruvallikeni Parthasarathy Temple, Marundeeswarar Temple, Kundrathur Murugan Temple (100 steps to climb), Padi Sivan Temple (1,000 years old, Guru sthalam), Ananthaa Padmanabha Swamy Temple, Siruvapuri Sri Balamurugan Temple, Jagannath Temple, Kanathur (a replica of the Puri shrine), and the Arupadai Veedu Murugan Temple represent Chennai's rich Hindu tradition.
- Christian landmarks include the San Thome Church and St. Thomas Mount National Shrine.
- Other highlights include Shirdi Sai Baba Temple, Kalikambal Temple, Kamakshi Amman

Temple, Ashtalakshmi Temple, and Viswaroopa Anjaneyaswami Temple.

171. **Beaches and Coastal Attractions.** Chennai's coastline offers both leisure and adventure.

- Marina Beach (one of the world's longest), Edward Elliot's Beach (Besant Nagar), Kovalam Beach (best in the mornings), Muttukadu Beach (speed boating at Muttukadu Lake), and VGP Golden Beach are popular for locals and tourists alike.
- The Chennai Lighthouse at Thiruvanmiyur offers panoramic views of the Marina and skyline.

172. **Museums, Libraries, and Historic Sites.** For history and culture enthusiasts, Chennai offers several iconic spaces.

- The Government Museum (Egmore) with 50 galleries spanning art, science, and history.
- Chennai Rail Museum, showcasing India's railway heritage.
- The Anna Centenary Library, South Asia's largest library, is a haven for book lovers.
- The Chennai Central Railway Station itself stands as a colonial architectural landmark.

173. **Parks, Zoos, and Eco Attractions.** Nature and wildlife experiences are easily accessible in the city.

- Arignar Anna Zoological Park, Guindy Snake Park, Guindy National Park (with blackbucks, jackals, and reptiles), and Chetpet Eco Park are family favorites.
- Semmozhi Poonga is a large botanical garden in Teynampet, popular for walks and greenery.
- The Huddleston Gardens at Theosophical Society provide a tranquil 260-acre retreat.
- The VGP Marine Kingdom aquarium and VGP Universal Kingdom amusement park are major attractions.
- For snow-filled fun, Snow Kingdom offers an icy indoor adventure.

174. **Protected monuments of Chennai.** The Archaeological Survey of India (ASI), under the Ancient Monuments and Archaeological Sites and Remains Act, protects and maintains many historically and architecturally significant sites in Chennai. These monuments reflect the city's colonial, religious, and indigenous heritage, with structures dating back several centuries. Chennai has 15 ASI-protected monuments. The proposed pipeline alignment maintains a distance of about 400 m from nearest ASI protected monuments (Old fort walls) surrounded by already built-up houses and other structures all other monuments are more than 1 km away from the proposed project components.

175.

176.

177. **Table 21** lists a few notable ones officially recognized in the Chennai ASI circle while Figure 35 presents a Google Map indicating the distances between the nearest ASI monument and the project components.

Table 21: List of Centrally Protected Monuments – in Chennai

S. No.	Name of the Monument	Locality
1	"Arsenal" between Wellesley house and Clive's House with shells and cannons piled together near the Gateway Block IV/1-12 and 14-18.	Fort St. George, Chennai-9
2	Big Warehouse, south of the Church Library (in Block No. II/7).	Fort St. George, Chennai-9
3	Chaplain's house including portion which is the northern side of the old Wall II/1.	Fort St. George, Chennai-9
4	Clive's house was built in 1753	Fort St. George, Chennai-9
5	Garrison Engineer's Depot, Block No. IV	Fort St. George, Chennai-9
6	Guard room Block No. V	Fort St. George, Chennai-9
7	King's Barracks Block No. XXV	Fort St. George, Chennai-9
8	Last house on the left of 'Snobs Allay' (Oldest house in the Fort with carved staircase) – Block No. I/1	Fort St. George, Chennai-9
9	Nursing Sister's House (Block No. I/3)	Fort St. George, Chennai-9
10	Old British Infantry Officer's Mess (now housing the Fort Museum) Block XXXVI/2	Fort St. George, Chennai-9
11	Ramparts, gates, bastions, Ravelins with vaulted chambers and water cisterns underneath; moat and defense walls all round with glacis to the extent of the existing barbed wire fence	Fort St. George, Chennai-9
12	St. Mary's Church with tablets laid on the ground and enclosed by a compound and a buried wall	Fort St. George, Chennai-9
13	Wellesley house (Built in 1798), Block No. IV/13	Fort St. George, Chennai-9
14	Tomb of David Yale and Joseph Hymners in the compound of Law College, George Town	Muthialpet (Law College/ High Court Compound), Chennai
15	Old town Wall	Tondiarpet, Chennai

Source : https://asichennai.gov.in/monuments_full_list.html.

Figure 35: Old fort wall the nearest ASI protected monument near proposed transmission main (violet)

G. Environmental features of selected sites

178. The subproject components of the Chennai RMS are located entirely within the GCC area, which has long been developed for urban use. Proposed facilities and pipeline alignments fall within existing road ROW, CMWSSB land, and utility corridors (Table 22 and Table 23). These locations contain only sparse vegetation such as shrubs and isolated trees, with no notable wildlife presence. There are no protected areas, ecologically-sensitive zones, wetlands, mangroves, estuaries, or natural habitats within the project footprint. As the project scope is confined to an urbanised environment, impacts on ecological resources are minimal and manageable.

179. Pipeline installation for the RMS will be undertaken along existing major and minor roads within the city. In wide roads, transmission mains will be placed within the road shoulder or utility zone without disturbing traffic lanes, while in narrow urban roads, temporary cutting of the roadway may be required to install pipes within the carriageway. Some densely built pockets contain narrow streets with open drains on either side, and pipe-laying in these stretches may temporarily affect traffic movements. However, the alignment has been selected to minimise disruptions and avoid tree cutting; adequate space is available in most locations to accommodate the pipeline without disturbing mature trees. The specific educational facilities and healthcare centers located in the immediate vicinity of the project corridor are detailed in **Table 24.** Overall, the project relies on existing public land and established transport corridors, ensuring that environmental and social impacts remain limited, predictable, and confined to short-term construction-related disturbances.

Table 22: Environmental features of proposed sites

Proposed components	Environmental features of sites
Ring main and transmission mains	The proposed ring main and transmission main pipeline corridors traverse high-, medium-, and low-density urban areas of Chennai, following exclusively the ROW of existing roads. The alignments pass in the vicinity of important ecological zones such as Guindy National Park, the Pallikaranai Marshland (Ramsar Site), stretches of CRZ near Buckingham Canal and the coast, and several urban waterbodies; however, no works extend beyond the existing carriageway, and no encroachment into protected or notified ecological areas is anticipated. The pipeline routes also lie near select heritage structures in Chennai—such as the Government Museum complex, certain old temples, and historic public buildings—but remain confined to established road corridors without affecting their integrity or requiring land acquisition. In some sections, the pipeline passes close to petroleum refinery zones and other major industrial areas, but all construction activities will be undertaken within the permitted ROW with adherence to safety protocols, traffic management plans, and environmental safeguards. Overall, the routes are located entirely within previously developed urban infrastructure corridors, ensuring minimal disturbance to sensitive ecological habitats, heritage sites, and surrounding communities. The proposed ring main pipeline will be laid largely within the existing ROW of city roads; however, tree impacts may occur at locations requiring open-cut trenching. A preliminary tree inventory along the alignment indicates that 75 trees may be affected due to excavation and pipe-laying works.

<table>
<tr><th>Proposed components</th><th>Environmental features of sites</th></tr>
<tr><td></td><td>These include individual specimens of local, native, and avenue species such as Almond (1), Bodhi (3), Coconut (3), Deerberry (1), Jamun (1), Mahogany (13), Mango (1), Monkeypod (12), Neem (15), Peepal (2), River Tamarind (14), and Vasoka (9).
<table>
<tr><td></td><td></td></tr>
<tr><td>Heavy Traffic areas</td><td>Road with metro rail pillars</td></tr>
<tr><td></td><td></td></tr>
<tr><td>Along with petroleum refineries</td><td>Low traffic area</td></tr>
</table>
</td></tr>
<tr><td>Underground Tank (Annai Theresa Pumping Station, Puzhuthivakkam)</td><td>Site Address: Annai Theresa Pumping Station, Zone 14, Puzhuthivakkam
Site Owner/Occupant: CMWSSB
Land Area: 23.4 m × 16.25 m
The proposed underground tank is located within the Annai Theresa Pumping Station premises in Puzhuthivakkam, on land owned by CMWSSB. The 23.4 m × 16.25 m site lies on plain clay soil with good road connectivity through a BT road and is surrounded on all sides by residential buildings. A small temple is located on the western side, but no other cultural or heritage features, encroachments, or industrial structures are present. The site is clear of trees and shows no evidence of wildlife, nesting, or sensitive on-site receptors. The nearest environmentally sensitive feature is the Pallikaranai Marshland, located approximately 5 km away, while Puzhuthivakkam Lake lies nearby but outside the project footprint. Overall, the site is already part of an operational CMWSSB facility and does not present any significant environmental constraints. Required land area 0.51 acre and available area is 279.2 acres.
</td></tr>
<tr><td>Two Underground Tanks (Okkiyam Thuraipakkam</td><td>Site Address: Okkiyam Thuraipakkam (High-rise building area)
Site Owner/Occupant: CMWSSB
Land Area: 23.4 m × 16.25 m</td></tr>
</table>

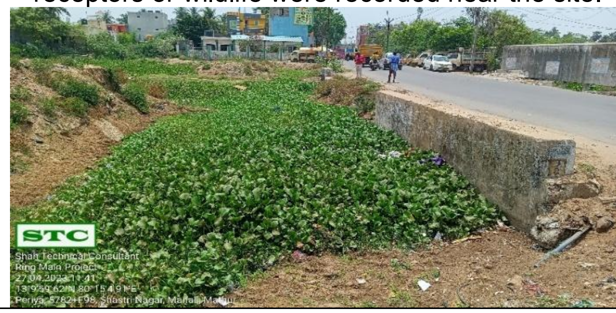
Proposed components	Environmental features of sites
	The proposed underground tank at Okkiyam Thuraipakkam is located within the existing OHT premises owned by CMWSSB, in a plain terrain area measuring approximately 23.4 m × 16.25 m. The site is well connected through a BT road and contains clay soil **Nearest sensitive receptor:** Pallikaranai marshland (~2 km) Required land area 2.55 acres and available area is 1,007 acres. 
Underground Tank (Puzhal, Area III)	**Site Address:** Puzhal, CMWSSB Water Distribution System Area III **Site Owner/Occupant:** CMWSSB **Land Area:** 23.4 m × 16.25 m The proposed facility at Puzhal is located within the existing CMWSSB water tank premises on a 23.4 m × 16.25 m plot comprising plain clay soil and well-connected through a BT road. The surroundings include the Puzhal Prison to the north, road access to the south and east, and the Puzhal Water Treatment Plant to the west. A single neem tree is present but will not be disturbed, and no encroachment, cultural features, or industrial activities are present on or near the site. Puzhal Lake lies about 1 km away, though the site itself has no sensitive receptors, wildlife, or nesting activity. Being within an operational utility compound, the environmental sensitivity of the site is low. Required land area 1.39 acre and available area is 801 acres. 
Underground Tank (Semmancherry)	**Site Address:** Semmancherry **Site Owner/Occupant:** CMWSSB **Land Area:** 23.4 m × 16.25 m The Semmancherry site lies within a CMWSSB compound where OHT construction activities are ongoing, covering an area of 23.4 m × 16.25 m on plain sandy terrain. Road access is available via a BT road, and the surroundings include vacant land to the north and west, a residential building to the east, and a public road to the south. Two small trees (*Prosopis juliflora* and Vasaka species) are present, and some construction material is stored on site, with C&D waste to be managed as per the C&D Waste Management Rules, 2016. No encroachments, industrial structures, cultural features, or sensitive receptors are present, and the nearest water body is the Buckingham Canal, located approximately 5 km away. House

Proposed components	Environmental features of sites
	crow and common Mayna were seen during site visit but no wildlife or nesting activity was observed, making the site suitable for development with minimal environmental risk. Required land area 2 acres and available area is 300 acres. 
Underground Tank (Sholinganallur)	**Site Address:** Sholinganallur **Site Owner/Occupant:** CMWSSB **Land Area:** 23.4 m × 16.25 m The proposed underground tank sites at Thiruvottriyur are located within residential zones and are well connected through existing bitumen (BT) roads. Both locations lie on plain sandy terrain and are free from encroachments or industrial/commercial activities. Only a few local and neem trees are present along the peripheries, none of which will be impacted by construction. Surrounding land uses primarily include vacant areas and residential buildings, with one temple situated to the south of the second site. No sensitive receptors, waste storage areas, or significant wildlife are present in the vicinity. The areas have adequate natural drainage and are situated at safe distances from major water bodies—Kosasthaliyar River is approximately 4 km away from the first site, while Buckingham Canal lies about 2 km from the second. Both locations exhibit stable environmental conditions suitable for construction, with no nearby cultural or heritage sites except the small local temple. Overall, the environmental setting indicates minimal risk of ecological or social disturbance from the proposed works. Required land area 0.83 acre and available area is 249 acres. 
Underground Tank (Thiruvottiyur)	**Site Address:** Thiruvottiyur The proposed Underground Tank (UGT) site is in a residential area of Thiruvottriyur and is accessible through a black-topped road. The land parcel, measuring approximately 16.9 m × 11.85 m, is owned by CMWSSB and remains free from any encroachment. The terrain is plain with sandy

Proposed components	Environmental features of sites
	soil conditions, and the surrounding land use consists primarily of vacant plots on all sides. Three local trees are present along the periphery but will not be impacted by the proposed works. No sensitive receptors or significant wildlife activity is observed in the vicinity, and the nearest major water body, the Kosasthaliyar River, lies at a distance of about 4 km. The environmental setting indicates that the site is suitable for construction with minimal anticipated impacts. Required land area 0.50 acre and available area is 430 acres.  .

Table 23: Environmental features of Pipe Carrying bridges

Site Name	Environmental feature with photos
Pipe Carrying Bridge (Retteri 1, Madhavaram)	**Site Owner/Occupant:** Water Resources Department (WRD) **Dimensions:** Length 60 m; Diameter 1,777 mm (Ch. 2010–2070 m) The site is owned and occupied by the WRD and contains a proposed pipeline section of 60 m length and 1,777 mm diameter between chainage 2010 m–2070 m. The location lies in the northern part of the city with good access through an existing BT road. The terrain is flat with clay soil, and there are two Jamaica cherry trees, both of which will remain unaffected. No encroachments or drainage features are present on the site. The adjoining land use includes residential areas to the north, vacant land to the south, the national highway on the west, and Retteri Lake on the east. The lake is the major environmental feature nearby, although no significant wildlife or bird activity is observed. The nearest habitation is Retteri, and there are no sensitive receptors or cultural/heritage sites in proximity to the proposed alignment. The site is environmentally suitable for construction activities with minimal anticipated impacts. 
Pipe Carrying Bridge (Retteri 2, Madhavaram)	The proposed pipe-carrying bridge involves a 24-m pipeline section of 1,777-mm diameter (Ch. 10020 m–10050 m) located in the northern part of the town. The site has good access through an existing BT road and lies on plain terrain with clay soil conditions. There are no cultural or heritage sites, no trees, and no encroachments observed in the area, and drainage facilities are absent. The surrounding land use is predominantly residential on the northern and southern sides, with a vacant plot to the

Site Name	Environmental feature with photos
	west and a road corridor to the east. Retteri Lake is the nearest waterbody, where water hyacinth growth is present, but no sensitive receptors or wildlife were recorded near the site. 
Ring Main Pipe Carrying Bridge (Buckingham Canal – Manali CPCL)	**Site Name:** No. 3 – Ring Main – PCB **Site Address:** Buckingham Canal near Manali CPCL **Site Owner:** WRD **Site Occupant(s):** WRD **Land Area:** 40 m pipe length; 1,574 mm diameter, Ch. 14240–14280 m This site lies across the Buckingham Canal near Manali, falling under WRD ownership. The bridge section is 40 m in length (pipe diameter 1,574 mm) located between Ch.14240–14280 m. The area consists of plain clay soil terrain with no drainage system and no trees or encroachments. Surrounding land uses include CPCL industrial areas to the west and petrochemical establishments to the east, with road access to the north and south. The nearest habitation is Ennore. No cultural, heritage, or sensitive receptors are present, and the canal (Kosasthalaiyar River) forms the adjacent water body.
Ring Main Pipe Carrying Bridge Location: Buckingham Canal 1 at TH Road	**Site Owner/Occupant:** WRD/WRD **Dimensions** - Length: 30 m - Pipe Diameter: 1,574 mm - Chainage: 18600–18630 m The site is located across Buckingham Canal along TH Road, with a 30 m bridge section (pipe diameter 1,574 mm; Ch.18600–18630 m). It has sandy soil and plain terrain with BT road access and no encroachments or trees. Adjoining land uses include a temple to the north, residential areas to the south, a road to the east, and the river to the west. No sensitive receptors or industrial establishments are present. Only one nearby cultural structure (temple) occurs but will not be impacted by the works. South: Residential area 
Ring Main Pipe Carrying Bridge Location: Buckingham Canal 2 at TH Road	**Site Owner/Occupant:** WRD/WRD **Dimensions** - Length: 30 m - Pipe Diameter: 1,574 mm - Chainage: 19110–19140 m

Site Name	Environmental feature with photos
	This WRD-owned location comprises a 30 m bridge span (pipe diameter 1,574 mm; Ch.19110–19140 m) over Buckingham Canal. The surroundings include residential areas on the north and south, with a road to the east and canal to the west. The site is flat terrain with BT road access, no trees or encroachments, and no heritage structures. Water hyacinth is present in the canal, but no wildlife or nesting activity observed. 
Ring Main Pipe Carrying Bridge Location: Captain Cotton Canal at TVK Link Road	**Site Owner/Occupant:** WRD/WRD **Dimensions** - Length: 60 m - Pipe Diameter: 1,574 mm - Chainage: 19830–19890 m Crossing the Captain Cotton Canal, this 60 m pipe bridge (diameter 1,574 mm; Ch.19830–19890 m) lies in a sandy soil area with available drainage and good road connectivity. The site is surrounded by river stretches on both west and east, vacant land to the north, and residential development to the south. No cultural, industrial, or sensitive receptors are present. 
Ring Main Pipe Carrying Bridge Location: Otteri Nallah at Basin Bridge	**Site Owner/Occupant:** WRD/WRD **Land Requirement:** - Length: 35 m - Pipe Diameter: 1,181 mm - Chainage: 22530–22565 m The site comprises a 35 m bridge section (diameter 1,181 mm; Ch.22530–22565 m) located in a plain clay soil setting with drainage availability. Surroundings are predominantly vacant land, with the nearest habitation being Basin Bridge. A small temple exists nearby but will remain unaffected. Water presence and minor vegetation exist along the Nallah,

Site Name	Environmental feature with photos
	with no wildlife or nesting activity. 
Ring Main Pipe Carrying Bridge Location: Otteri Nallah at Otteri	**Site Owner/Occupant:** WRD/WRD **Land Requirement:** - Length: 25 m - Pipe Diameter: 1,181 mm - Chainage: 25120–25145 m This 25 m pipe bridge (diameter 1,181 mm; Ch.25120–25145 m) traverses Otteri Nalla and is surrounded by vacant lands with BT road access. The area is flat clay soil terrain with no cultural or sensitive receptors. Otteri is the nearest habitation. No encroachments, trees, wildlife, or nesting are found at the site. 
Ring Main – PCB (Otteri Nalla – Kilpauk Shaft)	**Land Area:** 20 m pipe length; 1,181 mm diameter, Ch. 28200–28220 m **Nearby Heritage Site:** One religious gable—will not be affected; trenchless method proposed **Town Location:** South **Road Access:** Yes, BT road This 20-m bridge section (pipe diameter 1,181 mm; Ch. 28200–28220 m) crosses Otteri Nalla near the Kilpauk shaft. The WRD-owned site is surrounded by residential areas on the east, south, and west, while the north side is vacant. The terrain is plain clay soil with available drainage and road connectivity. A small religious gable structure exists nearby but will not be affected since trenchless technology is proposed. Vegetation is limited, and no wildlife, nesting, or sensitive receptors are observed. 

Site Name	Environmental feature with photos
Ring Main – PCB (Coovum River – Nungambakkam Bridge)	**Pipe Length:** 60 m; Diameter 1,181 mm; Ch. 30460–30510 m The proposed pipe-carrying bridge involves laying a 60-m pipeline of 1,181-mm diameter along chainage 30460–30510 m at Nungambakkam, crossing the Coovum River. The site is located on the southern side of the town and can be accessed easily through the existing BT road. There are no cultural or heritage structures within or near the alignment, and the area contains no trees. The terrain is plain with typical urban soil characteristics, and drainage facilities are currently absent. The adjoining land use comprises residential areas on the north and south, a road on the west, and the Coovum River on the east. The nearest habitation is Nungambakkam, situated close to the alignment. The river stretch along the site shows the presence of seasonal shrubs and herbs, representing minor local vegetation. No sensitive receptors, wildlife, nesting sites, or pollution sources were observed in the vicinity. Overall, the site represents a typical urban river-crossing location suitable for establishing the proposed pipe-carrying bridge with minimal environmental constraints. 
Ring Main – PCB (South Boag Road Nallah)	**Pipe Length:** 10 m; Diameter 1,181 mm; Ch. 35370–35380 This 10 m bridge (pipe diameter 1,181 mm; Ch.35370–35380 m) crosses a drainage channel near South Boag Road. The area consists of plain clay soil terrain with residential developments on all sides and the Coovum River flowing nearby. No cultural sites, sensitive receptors, wildlife, or encroachments are present. 
Ring Main – PCB (Veerangal Canal – Velachery Station)	**Pipe Length:** 40 m; Diameter 1,181 mm The site features a 40 m bridge (pipe diameter 1,181 mm; Ch.65370–65410 m) over Veerangal Canal in a clay soil area with available drainage. Surrounding land uses include residential developments and proximity to the Velachery Lake. The terrain is plain with road access and no encroachments or sensitive receptors.

Site Name	Environmental feature with photos
	
Ring Main – PCB (Adyar River – MIOT Hospital)	**Pipe Length:** 150 m; Diameter 1,574 mm; Ch. 73680–73770 A 150 m bridge (pipe diameter 1,574 mm; Ch.73680–73770 m) spans the Adyar River near MIOT Hospital. The terrain is plain clay soil without drainage infrastructure. Surroundings include river stretches to the north and south, residential properties to the east, and the hospital to the west. No cultural or heritage sites, sensitive receptors, or ecological concerns are observed. 
Ring Main – PCB (Coovum – Vanagaram Junction)	**Pipe Length:** 60 m; Diameter: 1,574 mm Located at Coovum River near Nungambakkam, this 60 m section (pipe diameter 1,181 mm; Ch.30460–30510 m) lies in a clay soil area with BT road access. The site is bordered by residential areas and road networks, with the river forming the eastern boundary. Vegetation exists along the riverbanks, but no nesting or wildlife activity is noted. No cultural or sensitive receptors occur in proximity. 
Ring Main – PCB (Thiruvanmiyur MRTS – Buckingham Canal)	**Pipe Length:** 25 m; Diameter: 1,181 mm This 25 m pipe bridge (pipe diameter 1,181 mm; Ch.41460–41485 m) spans Buckingham Canal in a sandy clay region near Thiruvanmiyur MRTS. IT parks lie to the south, residential areas to the north, and the canal borders the east and west. Water hyacinth is present in the canal. No cultural or sensitive receptors exist.

Site Name	Environmental feature with photos
	
Ring Main – PCB (Buckingham Canal – Akkarai)	**Pipe Length:** 155 m; Diameter: 1,181 mm A 155 m bridge (pipe diameter 1,181 mm; Ch.53280–53125 m) spans the Buckingham Canal at Akkarai. The location features sandy clay soil, plain terrain, and road connectivity. Surrounding land uses include the canal to the north and south and roads on both east and west. No cultural sites or sensitive receptors are present. 
Transmission Main – PCB (Buckingham Canal to Anna Park WDS)	**Pipe Length:** 50 m; Diameter: 980 mm This 50 m bridge (pipe diameter 980 mm) lies across the Buckingham Canal in a sandy clay area. The surroundings are largely linear canal and road corridors, with Kondithope as the nearest habitation. No cultural sites, sensitive receptors, or ecological features occur at the site. Nearby Vegetation is castor oil seed plants and water hyacinth. 
Transmission Main – PCB (Kosasthaliyar River)	**Pipe Length:** 70 m; Dia: 600 mm A 70 m bridge (pipe diameter 600 mm) spans the Kosasthalaiyar River in a sandy clay plain terrain. Roads lie on both the northern and southern sides, with the river on the east and west. The area lacks cultural or sensitive features and has no wildlife or nesting activity.

Site Name	Environmental feature with photos
	
Transmission Main – PCB (Buckingham Canal – Thoraipakkam)	**Pipe Length:** 35 m; Diameter: 600 mm This 35 m bridge (pipe diameter 600 mm) crosses the Buckingham Canal near Thoraipakkam. Surrounded by road corridors and riverbanks, the flat clay terrain has no cultural structures or sensitive receptors. Water hyacinth is observed in the canal but no wildlife or vegetation of concern. 
Transmission Main - Pipe Carrying Bridge	A 65 m span (pipe diameter 500 mm) crosses the Buckingham Canal in a sandy soil area with no encroachments, trees, or sensitive receptors. Road corridors flank the site on both sides, and Thiruvotriyur is the nearest habitation. 
Transmission Main Pipe Carrying Bridge (Buckingham Canal near Okkiyam Thuraipakkam)	Pipe length (m) :35 Pipe diameter (mm) : 600 This 35 m bridge (pipe dia 600 mm) runs across the Buckingham Canal near Okkiyam Thuraipakkam. The clay soil site is surrounded by road networks and canal stretches, with no cultural sites or sensitive receptors. The area is free from ecological concerns such as nesting or wildlife presence.

Site Name	Environmental feature with photos
	

Table 24: List of educational institutions and hospitals along the proposed alignment

Sl. No	Name of institution	Chainage	Approximate Distance from Location (m)	Pipe Diameter (mm)	Side of proposed alignment	Road Name
1.	Vellammal global school	0-90	50	1,777	left	Ambattur Redhills road
2.	Tamil Nadu Vertinary and animal sciences university	9270-9360	80	1,574	left	Kamaraj Salai
3.	Chennai Port and Dock Educational Trust Higher Secondary School & Primary School	17280-17370	10	1,574	left	Sathyamoorthy nagar main road
4.	RAASHIDH matriculation higher secondary school	20520-20610	20	1,574	left	Sathyamoorthy nagar main road
5.	Seventh day school	21240-21330	45	1,574	left	Sathyamoorthy nagar main road
6.	Corporation higher secondary school	21600-21690	65	1,181	left	Chennai-Srikulam highway
7.	Don Bosco Higher secondary school	21600-21690	65	1,181	left	Chennai-Srikulam highway
8.	Balaji primary nursery school	22500-22590	90	1,181	left	Chennai-Srikulam highway
9.	Government tuberculous hospital	26370-26460	10	1,181	left	Medavakkam tank road
10.	Physiatrist outpatient Hospital	27090-27180	40	1,181	left	Medavakkam tank road
11.	Kilpack Institute of Mental Health	27270-27900	10	1,181	left	Medavakkam tank road
12.	Ranjith Hospital	27630-27720	25	1,181	left	Medavakkam tank road
13.	K.M. Hospital	27630-27720	25	1,181	left	Medavakkam tank road
14.	Deepam eye hospital	27720-27810	5 to 10	1,181	left	Medavakkam tank road
15.	Divya dental Clinic	29160-29250	5 to 10	1,181	left	Haringston road
16.	KMC hospital	29250-29340	5 to 10	1,181	left	Haringston road
17.	IPC Education, Chennai	30330-	15	1,181	left	Haringston road

Sl. No	Name of institution	Chainage	Approximate Distance from Location (m)	Pipe Diameter (mm)	Side of proposed alignment	Road Name
		30420				
18.	B-Tech Distance Education Chennai	31410-31500	~15	1,181	left	Haringston road
19.	Bridge academy	32220-32310	~35	1,181	left	Haringston road
20.	Valluvar Kottam diagnostic centre	32490-32580	~25	1,181	left	Lake area main road
21.	Koncept hospitals and wellness	32760-32850	~50	1,181	left	Lake area main road
22.	Sathyamoorthi Higher secondary school	33210-33300	~55	1,181	left	Prakasam street
23.	Anjuman Matriculation higher secondary school	33750-33840	~75	1,181	left	Prakasam street
24.	Be well Hospital	33840-33930	~85	1,181	left	Soth Boag Road
25.	Global Rabbe play school	34830-34920	~35	1,181	left	Soth Boag Road
26.	Kavitha clinic	34920-35010	~15	1,181	left	Soth Boag Road
27.	Anna university	37350-37890	5 to 10	1,181	Right	Gandhi mandapam road
28.	University of madras	37980-38070	5 to 10	1,181	Right	Gandhi mandapam road
29.	National Centre for nanoscience and nanotechnology, university of madras	37980-38070	~10	1,181	Right	Gandhi mandapam road
30.	Adyar Cancer institute	38340-38430	~20	1,181	Right	Gandhi mandapam road
31.	school of architecture and planning campus	38430-38520	~25	1,181	Right	Gandhi mandapam road
32.	IIT Madras	38790-38880	~10	1,181	left	Sardhar Patel road
33.	Voluntary Health Services	39690-39780	~10	1,181	left	Sri Ram nagar main road
34.	IPT-Institute of printing Technology	40140-40230	~65	1,181	left	Sri Ram nagar main road
35.	Roja Muthiah research library	40230-40320	~25	1,181	left	Sri Ram nagar main road
36.	Institute of textile technology	40230-40320	~65	1,181	left	Sri Ram nagar main road
37.	Dr.Dharmambal government polytechnic college for women	40590-40680	~40	1,181	left	Sri Ram nagar main road
38.	Primrose school	50850-20940	~55	1,373	Right	East coast road (ecr)
39.	S.I Government higher secondary school, Pallikaranai	60120-60210	~30	1,181	left	Velacherry road / Medavakkam fly over
40.	KKT Clinic Velachery	65790-65880	~40	1,181	left	Velachery - Tambaram main road

Sl. No	Name of institution	Chainage	Approximate Distance from Location (m)	Pipe Diameter (mm)	Side of proposed alignment	Road Name
41.	Igene school of VFX Technology	66240-66330	~30	1,181	left	Velachery - Tambaram Main Road
42.	Excellent Care hospital	67500-67590	~30	1,181	left	100 feet road
43.	Government Higher secondary school	67770-67860	~30	1,181	left	100 feet road
44.	Venkateswara homeopathic medical college	78030-78120	~30	1,574	left	Mount Ponamalle road / MGR flyover
45.	Omayal Achi college of nursing	97380-97470	~20	1,181	Right	Ambattur Redhills Road

VI. ANTICIPATED ENVIRONMENTAL IMPACTS AND MITIGATION MEASURES

A. Introduction

180. Potential environmental impacts of the proposed infrastructure components are presented in this section. Mitigation measures to minimize/mitigate negative impacts, if any, are recommended along with the agency responsible for implementation. Monitoring actions to be conducted during the implementation phase is also recommended to reduce the impact.

181. Screening of potential environmental impacts are categorized into the following subproject phases: location and design (pre-construction phase), construction phase, and operations and maintenance (O&M) phase.

(i) **Location impacts** are associated with site selection and include loss of on-site biophysical array and encroachment either directly or indirectly on adjacent environments. It also includes impacts on people who will lose their livelihood or any other structures by the development of that site.

(ii) **Design impacts** include those arising from project design, including technology used, scale of operation/throughput, waste production, discharge specifications, pollution sources and ancillary services.

(iii) Pre-construction impacts are anticipated during construction works but planning is required for proposed mitigation measures before start of construction works, i.e. during SIP period such as taking consents from various departments, planning for construction and workers camps, deployment of safety officer, arrangement of required barricades and caution boards, etc.

(iv) **Construction impacts** are caused by site clearing, earthworks, machinery, vehicles and workers. Construction site impacts include erosion, dust, noise, traffic congestion and waste production.

(v) **O&M impacts** arise from the operation and maintenance activities of the infrastructure facility. These include routine management of operational waste streams, and occupational health and safety issues.

182. Screening of environmental impacts has been based on the impact magnitude (negligible/moderate/severe - in the order of increasing degree) and impact duration

(temporary/permanent).

183. This section of the IEE reviews possible project-related impacts, to identify issues requiring further attention and screen out issues of no relevance. SPS 2009 require that impacts and risks will be analysed during pre-construction, construction, and operational stages in the context of the project's area of influence. The rapid environmental assessment checklist (Appendix 1) was used to screen the project for environmental impacts and to determine the scope of the IEE.

184. **Methodology for Impact Assessment and Mitigation – Ring Main Project:** The methodology for assessing and mitigating environmental and social impacts followed a systematic, phased approach to ensure that all potential adverse effects are identified, prioritized, and addressed through effective mitigation measures.

185. **Identification of Environmental Components.** The project first identified the environmental components that could be affected during construction, operation, and post-construction phases:

- Physical Environment:
  - ➢ Air quality (dust, emissions)
  - ➢ Noise and vibration levels
  - ➢ Land and soil (erosion, spoil disposal)
  - ➢ Surface water quality and stormwater flow
  - ➢ Groundwater quality and quantity
- Biological Environment:
  - ➢ Terrestrial and aquatic vegetation
  - ➢ Fauna (mammals, birds, aquatic species)
  - ➢ Ecologically sensitive areas (wetlands, marshes, river corridors)
- Social Environment:
  - ➢ Private land, residential buildings, and commercial establishments
  - ➢ Public infrastructure and utilities (roads, drains, pipelines)
  - ➢ Occupational health and safety of workers
  - ➢ Community health and safety, including emergency services

186. **Screening of Impacts.** Impacts were screened by project phase and significant impacts were prioritized for targeted mitigation.

- Design/Pre-construction:
  - ➢ Land use changes
  - ➢ Tree removal
  - ➢ Utility shifting
  - ➢ Access planning
- Construction:
  - ➢ Access disruption to residents, shops, and institutions
  - ➢ Safety risks to road users and workers
  - ➢ Soil erosion and sedimentation
  - ➢ Water quality impacts in rivers, drains, canals, and bridge-crossing areas
  - ➢ Air pollution from dust and machinery emissions
  - ➢ Noise and vibration affecting nearby communities
  - ➢ Occupational health and safety of workers
  - ➢ Community health and safety risks from traffic and construction activities
- Operation:

- Leakage, pipe bursts, or water loss
- Traffic disruption during maintenance
- Energy consumption and carbon footprint of pumping stations
- Water quality deterioration in pipelines
- Occupational health and safety during operation and maintenance
- Community health and safety, including access to water infrastructure

187. **Development of Mitigation Measures.** Mitigation measures were developed based on the type and severity of each impact and included:

(i) Access Management: Temporary pedestrian and vehicular arrangements, advance notice, ramps for vulnerable users, coordination with institutions and emergency services
(ii) Traffic and Worker Safety: Barricades, PPE, speed limits, trained flag persons, hazard identification, emergency response planning
(iii) Soil and Water Management: Sediment control, backfilling, containment of hazardous materials, proper debris disposal, water quality monitoring, bridge-crossing protocols
(iv) Air, Noise, and Vibration Control: Dust suppression, machinery maintenance, noise barriers, restricted working hours, vibration monitoring near sensitive receptors
(v) Occupational and Community Health: OHS plans, hygiene and sanitation facilities, disease prevention, training, emergency preparedness, worker rights, public safety signage, and community communication
(vi) Post-Construction Restoration: Removal of debris, stabilization of access roads, vegetation replanting, slope protection, conditional acceptance after inspection
(vii) Operation Phase Measures: SCADA monitoring, preventive maintenance, emergency response for leaks, water quality monitoring, energy efficiency measures, and restricted access to water facilities

188. **Monitoring and Adaptive Management**

(i) Continuous monitoring of air, water, noise, vibration, and health & safety indicators
(ii) Corrective actions implemented in case of deviations
(iii) Reporting to the Project staff to ensure compliance and adaptive mitigation

B. Location Impacts

189. **Sensitive areas (Guindy national park, Pallikaranai marshland, CRZ zones) in project area** - The transmission and ring main pipelines are largely confined to built-up arterial corridors, so intrinsic ecological impacts are limited. However, the alignment interfaces with multiple regulated environmental receptors, making spatial accuracy and compliance critical. The pipeline remains within a 10-km radius of Guindy National Park and passes along existing roads near the Pallikaranai Marsh, which is both a Reserved Forest and a Ramsar site. In these locations, even minor deviations from approved corridors—arising from inconsistencies between GIS data, CRZ maps, LIDAR drawings, and printed alignment plans—could unintentionally shift work fronts, construction lay-down, or vegetation removal into areas where activities are restricted and subject to prior clearance. This elevates regulatory risk despite low direct habitat loss.

190. Sensitivity increases at the Pallikaranai Marsh, where the risk is not land acquisition but degradation of wetland hydrology and water quality through silt, wastewater, fuel, and concrete

contamination. This carries international obligations associated with Ramsar designation. Along the Adyar River, Cooum River, Buckingham Canal, and other canals and lakes, the alignment includes 24 waterbody crossings, 13 of which fall within CRZ-influenced tidal stretches—1 across the Adyar, 1 across the Cooum, and 11 across Buckingham Canal. Here, only linear works within existing rights-of-way are permitted, and construction of pipe-carrying bridges may temporarily disturb aquatic habitats, embankments, turbidity, and tidal flow. These interactions heighten compliance requirements, given that violations trigger CRZ enforcement.

191. Accordingly, potential impacts relate to disturbance of sensitive ecosystems, obstruction of drainage, deterioration of wetland and coastal water quality, and legal non-compliance with Eco-sensitive zone (ESZ) and CRZ provisions. While the nearest ASI-regulated heritage structure is approximately 400 m away and Pulicat Lake is 24 km from the project—indicating negligible heritage and regional coastal impacts—the urban ESZ, Ramsar, and CRZ interfaces remain high-sensitivity elements. Minutes of consultation with Forest Department officials are included in Appendix 10 and the required mitigation and safeguard actions.

192. Inaccurate alignment may trigger legal non-compliance, ecosystem disturbance, and delays during statutory review. Following mitigation measures / actions shall be implemented:

(i) **Alignment Verification and Regulatory Compliance**
- Conduct GIS-based alignment validation and ground-truth surveys prior to construction.
- Mark Guindy ESZ, Pallikaranai Marsh boundaries, CRZ zones, and ASI buffers clearly on alignment drawings.
- Update IEE/EMP, CRZ maps, and statutory submissions with verified coordinates before initiating works.
- Avoid any deviation near sensitive locations unless re-screened and cleared by relevant authorities.
- Maintain coordination with the Forest Department, Department of Environment and Climate Change, Greater Chennai Corporation, and ASI for location verification and compliance.
- Obtain No Objection from the Forest Department for works within 10 km of Guindy National Park and secure CRZ clearance in accordance with CRZ Notification 2019, complying with issued conditions.

(ii) **Protection of Ramsar, ESZ, and CRZ Areas**
- Prohibit construction camps, equipment parking, material storage, and waste disposal inside CRZ and Ramsar boundaries.
- Enforce strict adherence to CRZ authority conditions during execution.

(iii) **Construction Management in Environmentally Sensitive Zones**
- Restrict works in sensitive stretches to non-monsoon months.
- Prefer trenchless construction (HDD, micro-tunneling) at water crossings to minimize disturbance.
- Install silt curtains, oil traps, and spill-control systems in wetland and CRZ work zones.

(iv) **Environmental Oversight and Monitoring**
- Engage a qualified ecological expert to supervise works in ecologically sensitive sections and ensure implementation of safeguards.

193. **River and canal crossings (Adyar river, Cooum river, Buckingham canal).** The project involves 24 pipeline crossings over rivers and canals, including the Adyar, Cooum, and several

locations on the Buckingham Canal. These works can generate short-term but notable environmental risks.

194. Construction of bridge foundations and supports may increase turbidity by disturbing bed sediments, reducing water clarity and affecting aquatic organisms. Use of concrete, fuels, and equipment wash water creates a pollution risk if spills or runoff enter the channel. Temporary works can alter tidal hydraulics in sensitive stretches of the Adyar, Cooum, and Buckingham Canal, affecting flushing and local aquatic habitats. Access and excavation near embankments may cause bank instability, soil erosion, and small-scale loss of riparian habitat.

195. Following mitigation measures/actions shall be implemented:

(i) Regulatory compliance
- CRZ Clearance from DoE&CC, GoTN shall be obtained prior to any work within the influence zone.
- All designs shall comply with CRZ clearance conditions and incorporate the approved EMP and method statements.

(ii) Environmentally safe construction design
- No heavy machinery, fuel, or material storage shall be planned within 50 m of the CRZ boundary.
- All temporary structures (ramps, platforms, staging) will be designed outside the 100 m buffer from tidal waterbodies.
- Designs will incorporate silt fencing, controlled excavation, and sediment barriers to prevent water contamination.
- Temporary bunds/cofferdams, where required, will be designed to avoid altering tidal flow and removed immediately post-work.
- Use pre-cast bridge components to minimize in-stream work and avoid bed disturbance.
- Ensure zero discharge of spoil, debris, or wastewater into water bodies.
- Conduct hydraulic/tidal flow studies to determine bridge elevation and required freeboard.
- Design abutments to avoid constricting natural river/canal width.
- Install silt traps, barriers, and slope protection using native vegetation.
- Restrict all works near water bodies to non-monsoon periods.
- Inspect bridge supports regularly during operation to prevent leakage or structural settlement.

196. **Cultural heritage areas (fort walls and ASI-protected monuments)**. The pipeline corridor does not directly intersect any notified heritage assets; the nearest ASI-protected remains (old fort walls) are approximately 400 m away and are already surrounded by built-up structures. In this context, major cultural-heritage impacts are unlikely. The key risk arises from indirect effects—namely uncontrolled excavation, excessive vibration, or movement of heavy equipment that could destabilize aging masonry, accelerate surface deterioration, or encroach into regulated buffer zones. Any inadvertent approach toward the 300-m ASI control zone without clearance could also constitute a compliance violation even in the absence of physical damage. Accordingly, impacts are low in magnitude but high in regulatory sensitivity and require preventive controls.

197. Following mitigation measures/actions shall be implemented:

(i) Superimpose the finalized alignment on ASI-published heritage maps and verify horizontal distances before construction mobilization.

(ii) Employ low-vibration excavation and compaction techniques when working in the vicinity of heritage structures.

(iii) Conduct pre- and post-construction structural condition assessments of nearby protected structures to confirm no deterioration attributable to project activity.

(iv) Establish and implement a formal chance find procedure to address any archaeological materials unexpectedly encountered during excavation.

198. **Impacts due to utility congestion near metro corridors and underground services**. Underground water lines, sewers, storm drains, gas mains, optical fiber cable (OFC) networks, and power cables are positioned close to the proposed trench line. Incorrect utility mapping or uncontrolled excavation may rupture or cut these services, causing water supply interruption, sewage overflow, stormwater blockage, gas leak hazards, communication outages, or electrical faults.

199. Where the alignment runs parallel to or crosses near Metro tunnels, stations, or piers, mechanical excavation and compaction can transmit vibration. In sensitive zones, this may threaten tunnel waterproofing, induce micro-fractures, or compromise settlement tolerances around piers, triggering metro-operator review and potential work suspensions.

200. Open trenching along narrow urban corridors can constrain traffic and pedestrian access. Combined with dust emissions from excavation, spoil loading, and vehicle movements, this may reduce visibility, increase respiratory nuisance, and heighten accident risks in high-traffic zones.

201. If utilities are damaged or Metro operators impose stoppages, unplanned corrective works are required. These generate programme delays, cost increases, and schedule uncertainty due to emergency restoration, stakeholder clearances, and temporary service reinstatement.

202. Following mitigation measures/actions shall be implemented:

(i) Pre-Construction Planning
- Conduct ground penetrating radar (GPR) surveys and collect utility details from Metro Rail, GCC, telecom agencies, and CMWSSB.
- Prepare a detailed Utility Relocation Plan with ownership and depth.
- Identify all Metro-related crossings and secure written NOCs.

(ii) Coordination
- Hold joint coordination meetings with each utility agency.
- All works within 25 m of Metro tunnels/pillars require prior approval and supervision from Chennai Metro Rail Limited (CMRL).

(iii) Construction Controls
- Use manual or low-vibration excavation near sensitive utilities.
- Maintain minimum offsets from foundations and chambers.
- Ensure proper barricading, signage, and public safety arrangements.

(iv) Traffic, Noise, Dust Management
- Schedule work in off-peak hours where feasible.
- Use water sprinkling and cover spoil heaps.
- Coordinate closely with Traffic Police for diversions.

(v) Monitoring
- Maintain daily documentation of utility exposure and relocation.
- Conduct condition surveys of Metro structures before/after work.
- Immediately report and rectify any damage at Contractor's cost.

203. **Loss of trees and impacts on urban green cover.** A total of 75 trees may be impacted due to the proximity of the proposed alignment. All identified trees are common local species and fall under the "Least Concern (LC)" category of the IUCN Red List. The species and their counts are as follows: Almond – 1, Bodhi – 3, Coconut – 3, Deerberry – 1, Jamun – 1, Mahogany – 13, Mango – 1, Monkeypod – 12, Neem – 15, Peepal – 2, River Tamarind – 14, and Vasoka – 9, with girths varying from 0.3 m to 4 m, indicating a mix of young, medium, and mature trees. These impacts will be minimized through careful micro-alignment adjustments loss, trees in shoulder areas may be affected due to trenching, excavation, and equipment movement. The alignment follows major arterial and sub-arterial roads, service corridors, and existing utility alignments, thereby avoiding any diversion of forest land or natural habitats. However, in certain narrow sections, selective removal or pruning of roadside trees may be necessary to ensure safe construction, manoeuvring of heavy machinery, and pipe-laying operations.

204. A total of 75 trees were identified along the proposed alignment from Puzhal WTP to T. Nagar and from Ramapuram to Ambattur (near Rakki Cinemas), predominantly located on the left-hand side (LHS) of the corridor at distances generally ranging between 2–23 m from the centre line. Tree clusters are more prominent around chainages 9.4–10.1 km, 30.5–34.7 km, and 82.3–82.6 km, with notable concentrations of Monkey Pod, Vasaka, Mahogany, and River Tamarind in these segments. Coordinates for each tree were recorded for precise mapping and to facilitate site-specific mitigation planning under the tree management protocol. List of trees may get impacted due to ring main and transmission line alignment are presented in Appendix 8.

(i) Alignment and design optimization
- Keep alignment strictly within roadway ROW; apply micro-adjustments to avoid mature trees.
- Avoid any tree removal in CRZ or marsh buffer zones; use trenchless methods where required.

(ii) Controlled construction practices
- Minimize working width to essential needs only.
- No pruning/felling without formal GCC/Horticulture approval.
- Use dust suppression and cover spoil to protect vegetation.
- Restrict equipment movement near tree root zones.

(iii) Compensatory plantation
- Implement compensatory plantation at a minimum ratio of 1:10 using native species.
- Prefer planting along same road corridors, waterbody buffers, or notified public lands.
- Restore all disturbed verges, medians, and margins post-construction.

(iv) Awareness & supervision
- Train site staff on tree protection protocols.
- Mark and protect tree zones before excavation.
- Maintain a Tree Impact Register and submit quarterly compliance reports.
- No works permitted within 10 km of Guindy NP without statutory ESZ/NOC compliance.

205. **Flood-prone and waterlogging zones.** Several stretches of the alignment fall within

chronically waterlogged or historically flood-prone areas such as Velachery, Madipakkam, Sholinganallur, and Saidapet. In these locations, shallow groundwater, reduced surface gradients, and dependence on storm-drain networks increase construction sensitivity. If trenching is not carefully sequenced and drained, excavations can impede natural runoff, leading to localized ponding or aggravation of monsoon flooding. Saturated soils may also lose bearing capacity, creating instability for pipe bedding, increasing the risk of differential settlement, joint failure, or the need for over-excavation and replacement with engineered fill. Prolonged inundation can accelerate pavement deterioration and expose adjacent utilities to washout or collapse. For the public, standing water and weakened pavement elevate accident risks, impede access, and prolong construction durations due to repeated pumping, stabilization, and reinstatement requirements.

206. Following mitigation measures/actions shall be implemented:
- Conduct localized hydrology and drainage assessments along vulnerable sections to understand flow paths, storm-drain capacity, and groundwater conditions.
- Avoid trenching during monsoon periods and provide temporary culverts or cross-drainage channels to maintain uninterrupted runoff during construction.
- Deploy dewatering pumps equipped with silt-control systems to prevent sediment discharge into nearby drains or waterbodies.
- Strengthen pipeline bedding and lateral support with suitable granular sub-base material to minimize settlement and ensure long-term stability under saturated conditions.
- Restore full road-surface drainage immediately after pipe installation, including re-establishing cross-fall, shoulder drains, and stormwater connectivity.

C. Design phase impacts

207. Design of project components follows the Central Public Health and Environmental Engineering Organisation (CPHEEO) Manual 2019, adopting durable MS pipes (1200–2000 mm) with suitable internal linings and external corrosion-protection coatings. The design incorporates trenchless construction (such as HDD or micro-tunneling) in environmentally sensitive locations, near heritage structures, waterbodies, and high-traffic corridors to reduce excavation. Alignment planning ensures no direct impact on Guindy National Park, Pallikaranai Marshland, areas, waterways, or heritage structures. Where the pipeline passes near the Ashok Pillar monument or other protected structures, appropriate buffer distances, vibration-controlled equipment, and permissions from the competent authorities (e.g., ASI/state archaeology) will be obtained**.**

208. Climate change resilience and extreme weather adaptation: Chennai is highly exposed to cyclones, extreme rainfall, storm surges, and recurrent urban flooding, which can impose physical and operational stresses on linear water-supply assets. In severe inundation, buried pipelines may experience upward buoyant forces leading to flotation if not adequately anchored; embankments supporting above-ground or bridge-mounted segments may erode under high-velocity flow; and prolonged submergence can damage electrical systems at pumping stations. High wind loading and debris impact can stress exposed pipeline sections, while power failures and communication outages can disrupt SCADA-based automation and monitoring.

209. Following mitigation measures/actions shall be implemented:

(i) Engineering design adaptation
- Use corrosion-resistant materials with thermal expansion allowances for large-diameter mains.

- Design pumping station foundations and pipe-bridges for scouring, liquefaction, and flood loads
- Use elevated or encased pipe designs at river crossings and low-lying flood-prone zones.

(ii) Infrastructure protection and flood-proofing
- Elevate SCADA rooms, electrical panels, and essential machinery above the highest flood level.
- Provide flood-proof enclosures, surface drainage, and waterproofing systems.
- Include backup power systems during grid failures.

(iii) Smart operational management
- SCADA shall include predictive analytics for rainfall and pressure surges.
- Automatic shut-off/isolation valves will prevent contamination during inundation.
- Real-time communication protocols with GCC and State Disaster Response Fund (SDRF) will be integrated for emergency events.
- Post-flood inspection and restoration protocols shall be included.

210. **Impacts on water resources and risks to existing supply systems.** Although the project does not add new abstractions from lakes or reservoirs, certain secondary impacts may still occur during construction and commissioning. Tie-ins with existing distribution mains may require temporary shutdowns or diversion of flows, creating short-term service interruptions in connected zones. During flushing, pressure testing, or disinfection of new pipelines, treated water may be discharged to waste if not recovered or reused, resulting in avoidable water loss. Localized dewatering to maintain workable trench conditions can temporarily depress groundwater levels, affecting nearby wells or inducing settlement in loose soils. In addition, if chlorinated test water, sediment-laden trench water, or other wastewater streams are released without proper control, they may enter urban drains, contributing to pollutant loading and downstream water-quality issues.

211. Following mitigation measures/actions shall be implemented:

(i) No abstraction policy
- No new surface/groundwater extraction will be included in the design.
- All supply is routed through existing CMWSSB systems without altering allocations.

(ii) Protection of existing supply mains
- Detailed utility mapping and pre-construction surveys will be integral to design.
- Tie-ins will be scheduled during controlled shutdowns under CMWSSB supervision.

(iii) Groundwater and dewatering management
- Designs shall include provision for collection, filtration, and safe discharge of dewatered water.
- No groundwater will be used for construction or curing.

(iv) Construction wastewater control
- All testing/commissioning water will be directed to drains via silt traps.
- No chemical additives will be discharged into stormwater networks.

(v) Coordination with source operators
- Designs will ensure uninterrupted reservoir operations, intake flows, and distribution of hydraulics.

212. **Environmental audit of existing infrastructure – Chennai ring main project:** The Chennai RMS Project aims to strengthen the city's water supply by interlinking major treatment and storage facilities to ensure equitable and reliable distribution. The project leverages existing infrastructure, including:

(i) Puzhal WTP (300 MLD)
(ii) Surapet (14 MLD) (under rehabilitation)
(iii) Kilpauk WTP (270 MLD)
(iv) Chembarambakkam WTP (530 MLD)
(v) Vadakkuthu WTP (180 MLD)
(vi) Minjur DSP 100 MLD capacity (under rehabilitation)
(vii) Nemmeli DSP- I 100 MLD capacity
(viii) Nemmeli DSP- II 150 MLD capacity

213. SPS 2009 requires an environmental audit, if a subproject involves facilities and/or business activities that already exist or are under construction, including an on-site assessment to identify past or present concerns related to impacts on the environment. The objective of this compliance audit is to determine whether actions were in accordance with ADB's safeguard principles and requirements for borrowers, and to identify and plan appropriate measures to address outstanding compliance issues.

214. A environmental audit was conducted to:
- Assess compliance of existing infrastructure with national environmental laws and SPS 2009.
- Identify operational risks, data gaps, and improvement measures.
- Develop corrective actions to ensure sustainable performance.

215. Audit Methodology
- Document Review: Technical designs, consent documents, operation manuals, and laboratory records.
- Site Inspection: Visual verification of plant operations, sludge handling, and safety measures.
- Interviews: Discussions with CMWSSB operation staff and site engineers.
- Compliance Check: Against relevant Acts and Rules – Water (Prevention and Control of Pollution) Act 1974, Air (Prevention and Control of Pollution) Act 1981, Hazardous Waste Rules 2016, sludge and water quality monitoring
- Data Gaps Identification: Missing or incomplete records, unmonitored parameters, and areas requiring further study during stage.

216. The Chennai RMS subproject involves only the interconnection and utilization of existing water treatment plants (WTPs), desalination plants (DSPs), and pumping stations, with no proposal for construction of new treatment facilities.

217. The treated water supply to the Chennai Ring Main system will be sourced from existing facilities, including the Kilpauk (270 MLD), Chembarambakkam (530 MLD), Veeranam–Vadakuthu (180 MLD), and Puzhal (300 MLD) water treatment plants, as well as the Nemmeli 110 MLD and 150 MLD desalination plants.

218. As no new WTPs or water production sources are proposed and the existing facilities will continue their current operations, no Consent to Establish (CTE) or Consent to Operate (CTO) requirements are applicable for WTPs in Tamil Nadu. For the desalination plants, the CTOs and environmental clearances are reported to be valid. The Surapet WTP and Minjur DSP are currently under rehabilitation and are not operational at present.

Table 25: Summary of Environmental audit of existing WTPs and DSPs

Items	Preliminary audit observations	Issues of concern/non-compliances	Proposed corrective actions*
WTP – 530 MLD - Chembarambakkam			
Compliance with environmental regulatory framework	The WTP has obtained the required TNPCB CTE and CTO for DG set operation and chlorine storage exceeding 10 tonnes and holds a valid Petroleum and Explosives Safety Organisation (PESO) license for storage of chlorine gas cylinders.	The sludge testing report was not available Air emission data for DG sets stack monitoring was not available	Sludge characterization testing and DG set stack emission monitoring shall be conducted periodically as per applicable standards, and the reports shall be maintained at the facility for compliance verification
Treatment efficiency – treated water to meet Indian drinking water standards (10500-2012)	As per the external laboratory report dated 21 October 2025, treated water meets all drinking water standards except aluminum, which was recorded at 0.4 mg/L against the acceptable limit of 0.3 mg/L.	The aluminum concentration in treated water (0.4 mg/L) exceeds the acceptable limit of 0.3 mg/L,	Review and optimize treatment process at WTP. All essential water quality parameters shall to be tested regularly and ensured supplied water meets the drinking water quality standards as per BIS: 10500-2012.
Backwash wastewater/ settled clarified sludge management	Wastewater from Pulsator, backwash water from filter bed are recycled by recycle mechanism. Information on sludge was not available	information on sludge generation, handling, and disposal is not available	Document sludge handling, storage, and disposal practices.
WTP - 300 MLD Puzhal			
Compliance with environmental regulatory framework	No information was available on CTO for use of DG sets	Operating DG set without CTO from TNPCB is a non-compliance of the requirements of the Air (Prevention and Control of Pollution) Act, 1981. The WTP has a chlorine storage of 18 tonnes (threshold quantity greater than 10 tonnes but less than 25 tonnes); however, statutory clearance/approval	Corrective action: CMWSSB to obtain CTO from TNPCB The WTP shall obtain the required statutory clearance/approval for chlorine storage (18 tonnes) from the Tamil Nadu Pollution Control Board (TNPCB) and the Directorate of Industrial Safety and Health (DISH), and maintain valid documentation at the

Items	Preliminary audit observations	Issues of concern/non-compliances	Proposed corrective actions*
		from the TNPCB and the Directorate of Industrial Safety and Health (DISH) was not available	facility to ensure regulatory compliance.
Treatment efficiency – treated water to meet Indian drinking water standards (10500-2012)	As per the external laboratory report dated 28 Nov 2025, treated water meets all drinking water standards.	None	All essential water quality parameters shall to be tested regularly and ensured supplied water meets the drinking water quality standards as per BIS: 10500-2012.
Backwash wastewater/ settled clarified sludge management	Wastewater from clarifier are recycled by recycle mechanism, information on sludge disposal was not available	Information on sludge generation, handling, and disposal is not available	Prepare a sludge management plan Document sludge handling, storage, and disposal practices.
WTP - 270 MLD Kilpauk			
Compliance with environmental regulatory framework	No information was available on CTO for use of DG sets	Operating DG set without CTO from TNPCB is a non-compliance of the requirements of the Air (Prevention and Control of Pollution) Act, 1981.	Corrective action: CMWSSB to obtain CTO from TNPCB
Treatment efficiency – treated water to meet Indian drinking water standards (BIS: 10500-2012)	The in-house laboratory is testing only a limited number of parameters, and complete test results for all required parameters are not available.	Laboratory report does not include all required parameters as per BIS: 10500-2012	All essential water quality parameters shall to be tested regularly and ensured supplied water meets the drinking water quality standards as per BIS: 10500-2012.
Backwash wastewater/ settled clarified sludge management	Wastewater from the backwash is collected in reclamation well and then pumped to collection and sedimentation tank for sludge treatment, information on sludge disposal was not available	Information on sludge generation, handling, and disposal is not available	Prepare a sludge management plan Document sludge handling, storage, and disposal practices
WTP – 180 MLD - Veeranam -Vadakuthu			
Compliance with environmental regulatory framework	No information was available on CTO for use of DG sets	Operating DG set without CTO from TNPCB is a non-	Corrective action: CMWSSB to obtain CTO from TNPCB

Items	Preliminary audit observations	Issues of concern/non-compliances	Proposed corrective actions*
		compliance of the requirements of the Air (Prevention and Control of Pollution) Act, 1981.	
Treatment efficiency – treated water to meet Indian drinking water standards (BIS: 10500-2012)	As per the external laboratory report dated 02 Dec 2025, treated water meets all drinking water standards.	None	All essential water quality parameters shall to be tested regularly and ensured supplied water meets the drinking water quality standards as per BIS: 10500-2012.
Backwash wastewater/ settled clarified sludge management	Wastewater from the backwash is collected in reclamation well and then pumped to collection and sedimentation tank for sludge treatment, information on sludge disposal was not available	Information on sludge generation, handling, and disposal is not available	Prepare a sludge management plan Document sludge handling, storage, and disposal practices.
DSP -110 MLD – Nemmeli			
Compliance with environmental regulatory framework	The DSP has valid required CTO under Air and Water Act. Facility also has valid CRZ clearance	None	-
Treatment efficiency – treated water to meet Indian drinking water standards (BIS: 10500-2012)	As per the available laboratory report dated 25 Nov 2025, treated water is meeting the drinking water parameters for all the tested parameters	Laboratory report does not include bacteriological parameters	All essential water quality parameters shall to be tested regularly and ensured supplied water meets the drinking water quality standards as per BIS: 10500-2012.
Backwash wastewater / settled clarified sludge management and brine disposal system	Low-TSS sludge from backwash water and clarifier underflow is collected in a sludge tank, where supernatant is recycled and dilute sludge is mixed with RO reject and discharged to the sea through a diffuser, ensuring adequate dilution.	As per CRZ clearance sludge should be disposed in secure landfill only as per PCB norms	Ensure that all sludge is disposed of only at a PCB-authorized secure landfill in strict compliance with CRZ clearance conditions and applicable PCB norms.
DSP -150 MLD – Nemmeli			

Items	Preliminary audit observations	Issues of concern/non-compliances	Proposed corrective actions*
Compliance with environmental regulatory framework	The DSP has valid required CTO under Air and water act. Facility also has valid CRZ clearance	None	None
Treatment efficiency – treated water to meet Indian drinking water standards (BIS: 10500-2012)	As per the available laboratory report dated 12 Nov 2025, treated water is meeting the drinking water parameters for all the tested parameters	None	All essential water quality parameters shall to be tested regularly and ensured supplied water meets the drinking water quality standards as per BIS: 10500-2012.
Backwash wastewater/ settled clarified sludge management and brine disposal system	Sludge is treated in sludge treatment – decanted centrifuge system	Information on sludge disposal was not available	Prepare a sludge management plan • Document sludge handling, storage, and disposal practices. • Establish authorized disposal or reuse arrangements and maintain records with periodic reviews.

*Since the associated facilities are not in the scope of this project, audit findings will be shared with CMWSSB.

219. Prior to commencement of the O&M phase, CMWSSB shall ensure that all DG sets obtain valid CTOs and PESO licenses for diesel storage exceeding 1,000 liters (L) and comply with National Green Tribunal (NGT) requirements through retrofitting of approved emission-control devices or conversion to cleaner fuels. All facilities shall implement mandatory personal protective equipment (PPE) usage, adequate spill-control measures, safety signage, and rectify electrical safety deficiencies. Comprehensive emergency response plans for chemical spills, chlorine leaks, fire incidents, marine discharge failures, and operational disruptions shall be updated, displayed, and periodically drilled. CMWSSB is responsible for implementation and TNUIFSL is responsible for ensuring that all statutory permits and approvals are obtained before the O&M phase. Details of environmental audits of existing WTPs and DSPs is presented in Appendix 11.

220. **Potential impacts on surface water, groundwater, and competing use of water resources**. The proposed Project aims to strengthen the city's bulk transmission and distribution network by laying new feeder mains and ring mains to improve system reliability and reduce non-revenue water losses, without introducing any new abstraction from surface or groundwater sources. Therefore, no impacts on water resources are envisaged.

221. The project will, however, interface with existing reservoirs, transmission mains, and treatment systems that depend on regional water bodies such as Chembarambakkam, Red Hills, and Poondi reservoirs, as well as seawater-based sources. Potential impacts include temporary disruption of existing municipal water supply lines during tie-in or interconnection works, the risk

of accidental contamination or wastage of treated water during dewatering, flushing, or testing of pipelines, localized groundwater drawdown in areas where trench or pit dewatering is required due to high water tables, and additional pressure on stormwater drains and urban water bodies if construction wastewater or runoff is not effectively managed.

222. The project design and construction methodology have been developed to ensure that water resource use remains fully sustainable and that there is no competition or adverse effect on existing users or sources. The following mitigation measures will be implemented:

223. Following mitigation measures/actions shall be implemented:
- The project will not introduce any new surface or groundwater abstraction.
- All water supply improvements will rely solely on the existing CMWSSB treatment and transmission system, ensuring no added pressure on current reservoirs.
- Carry out detailed utility mapping to locate and protect existing water supply pipelines during excavation.
- All tie-ins or interconnections with existing mains will be done during planned shutdowns and supervised by CMWSSB engineers to avoid service interruptions.
- If temporary dewatering is needed, the pumped water will be collected, filtered for sediment, and safely discharged into stormwater drains or reused for construction.
- No groundwater extraction will be allowed for construction or curing; only municipal or recycled water sources will be used.
- Pipeline testing and flushing water will be discharged into designated drains equipped with silt traps to prevent erosion and pollution.
- No untreated wastewater or chemical-laden testing water will be allowed to enter surface drains, lakes, or marshlands.
- Maintain continuous coordination with CMWSSB for any operational adjustments needed during the integration of new mains.
- Ensure no interference with reservoir systems, intake operations, or distribution balancing.

D. Anticipated impacts and mitigation measures during the pre-construction phase

224. **Legal compliance.** Environmental non-compliance poses direct project-execution risks. Failure to secure mandatory clearances, permits, or NOCs can lead to regulatory action, suspension of work, legal penalties, or forced redesign of sensitive stretches. In extreme cases, authorities may mandate stoppage until corrective submissions are completed, resulting in time and cost overruns.

225. Following mitigation measures/actions shall be implemented:
- Obtain all required statutory consents, clearances, permits, and NOCs (including CTE/CTO from TNPCB) prior to initiation of construction.
- Ensure that all contractor-obtained approvals for construction activities are secured before mobilization.
- Refer to Table 11 for applicable environmental compliances.

226. **Baseline environmental conditions (air, noise, water, soil).** Absence of reliable baseline data limits the ability to characterize existing environmental conditions and establish objective benchmarks for impact prediction. Without accurate information on ambient air and noise levels, water quality, drainage performance, ecological sensitivities, or existing utility conditions, it becomes difficult to quantify incremental project impacts or assess whether mitigation measures are functioning as intended. Data gaps may also weaken regulatory submissions, delay

approvals, and force re-assessment during construction, increasing uncertainty and compliance risk.

227. Following mitigation measures / actions shall be implemented:
- Undertake pre-construction baseline monitoring of ambient air quality, noise levels, groundwater/surface-water quality, and soil quality using National Accreditation Board for Testing and Calibration Laboratories (NABL)-accredited laboratories.
- Select monitoring stations to represent sensitive receptors (schools, hospitals, residential zones) and key project influence areas to ensure that baseline conditions are accurately captured for subsequent comparison and verification.

228. **Sourcing of construction materials.** Unsustainable extraction of construction materials can accelerate riverbed degradation, alter natural flow regimes, deplete groundwater–surface water interactions, increase bank erosion, and reduce habitat availability for aquatic species. It may also trigger illegal mining activities, contribute to loss of riparian vegetation, and elevate sediment loads downstream, potentially affecting water quality and community water-use reliability.

229. Following mitigation measures/actions shall be implemented:
- Materials shall be sourced only from licensed quarries/sand mines permitted by the Department of Geology and Mines.
- No new quarries shall be proposed under this project.
- All batches must carry Environmental Clearance (EC) documentation.
- PMU/PIU will conduct periodic audits of sourcing and chain-of-custody records.

230. **Design of construction camps, storage yards, and disposal sites:** Improper sitting and layout of temporary construction facilities can lead to removal of urban vegetation and loss of green buffers, localized soil contamination from fuel, lubricants, and construction waste, and deterioration of surface-water and drain quality from uncontrolled runoff and wastewater discharge. Inadequate separation from residential areas may generate nuisance through noise, dust, traffic, and visual intrusion, adversely affecting community wellbeing. Poor ecological consideration within dense urban settings may also disturb urban fauna, disrupt micro-habitats, and create unmanaged waste zones.

231. Following mitigation measures/actions shall be implemented:

(i) Site selection controls
- Camps or storage areas shall not be in CRZ areas, marshlands, riverbanks, or within 50 m of waterways.
- Sites require prior PIU approval.

(ii) Camp layout and facilities
- Designs will include sanitation blocks, drainage, waste segregation zones, oil storage with bunds, and emergency access routes.
- Open garbage pits will be strictly prohibited.

(iii) Social requirements
- Encourage hiring of local labour to reduce camp establishment.
- Plan for full restoration of camp areas after use.

232. **Stakeholder communication and disclosure in design stage.** Insufficient consultation with affected communities, institutions, and other stakeholders may result in misinformation about project scope and timelines, lack of public support, and elevated complaints and objections. This can delay statutory clearances and construction mobilization, and may prevent accurate

identification of sensitive receptors, vulnerable households, utility constraints, or site-specific risks. Consequently, design solutions may be sub-optimal, socially unacceptable, or environmentally non-compliant.

233. Following mitigation measures/actions shall be implemented:

(i) Stakeholder identification and consultation –
- Maintain a stakeholder register including vulnerable groups.
- Conduct consultations to disclose scope, timelines, and expected impacts.

(ii) Communication tools
- Install Project Information Boards across key locations (in Tamil and English).
- Use GCC/CMWSSB platforms for periodic updates.

(iii) Grievance redress mechanism (GRM)
- Establish a multi-tier GRM with a 7–15 day response window.
- Display GRM contact information at all design-stage public points.

(iv) Documentation
- Maintain consultation minutes, attendance sheets, and disclosure materials for PIU/ADB review.

234. **Contractor training and capacity building requirements:** If contractors and field personnel lack sufficient knowledge of the environmental management plan (EMP), occupational health and safety (OHS) protocols, and the grievance redress mechanism (GRM), non-compliance risks increase during design finalization and pre-construction mobilization. This may result in incorrect interpretation of environmental safeguards, poor integration of safety controls into method statements, and weak documentation. In addition, gaps in OHS awareness can lead to unsafe planning of work fronts, elevated accident risk, and inadequate emergency preparedness. Weak understanding of the GRM may also limit early issue resolution and increase the likelihood of stakeholder dissatisfaction and work disruption.

235. Following mitigation measures/actions shall be implemented:

(i) Mandatory Induction Training
- PIU shall train contractor management on EMP, OHS, GRM, and SPS 2009 requirements.

(ii) Periodic Refreshers
- Monthly toolbox and risk refresher training, with attendance reports submitted to PIU.

(iii) Monitoring and Capacity Support
- PIU safeguard officers will provide joint inspections and on-site corrective guidance.

236. **Occupational health and safety (OHS) and labour standards integration in design:** For large-diameter pipeline works in dense urban corridors, insufficient incorporation of occupational health and safety (OHS) and labour standards at the design stage can significantly elevate worker and public safety risks. Key consequences include unsafe trench geometry, inadequate shoring provisions, and higher probability of trench collapse. Inaccurate utility mapping and absence of protection plans increase the likelihood of third-party utility strikes, service disruptions, and associated hazards. Elevated noise exposure and restricted workspaces may lead to non-compliance with statutory limits and poor ergonomic conditions. Failure to integrate labour standards requirements can also result in excessive working hours, inadequate welfare facilities, and other labour-related violations, increasing regulatory exposure and work stoppages.

237. Following mitigation measures/actions shall be implemented:

(i) Labour Standards
- Minimum age: 18 years.
- Compliance with Building and Other Construction (BOCW) Act 1996 and other labour acts.

(ii) Design of Health and Safety Plan
- Risk assessment for trenches, metro tunnels, deep excavations, confined spaces.
- Emergency response procedures including coordination with CMRL and utility agencies.

(iii) PPE Specifications
- PPE to conform to BIS/IS standards, covering helmets, reflective jackets, gloves, boots, and respiratory protection.

(iv) Accident Liability and Compensation Mechanisms
- Design/tender documents must require contractors to comply with the Employee's Compensation Act 1923.

(v) Public Safety Considerations
- Designs shall include barricading plans, lighting arrangements, pedestrian access, and traffic management at all work fronts.

(vi) Utility and Metro Coordination
- Design must include joint utility verification, NOCs from CMRL/Highways, and vibration control criteria near metro structures.

238. **Environmental documentation and compliance requirements.** Weak environmental documentation during project preparation, including incomplete baseline information or insufficient incorporation of stakeholder inputs, can directly affect the approval and compliance pathway. Deficiencies may trigger clarification requests from regulators, cause delays in obtaining statutory clearances, and require revisions to technical reports, designs, and EMPs. Poor-quality inputs may compromise impact identification, resulting in inappropriate or ineffective mitigation measures and higher residual impacts. Inadequate documentation also weakens auditability and traceability, increasing exposure to non-compliance findings during monitoring or external review.

239. Following mitigation measures/actions shall be implemented:

(i) Baseline Data Collection
- Collect air, noise, water, soil, and traffic baseline data prior to construction.

(ii) Consultations
- Engage GCC, TNPCB, CMWSSB, and local communities; integrate feedback of stakeholders into design.

(iii) Environmental Planning and Monitoring
- Contractor to prepare site environmental management plan (SEMP) based on design and baseline conditions.
- Maintain a Compliance Register of all approvals.

(iv) Reporting and Capacity Development
- Monthly environmental reports to PIU.
- Training for contractor environmental staff on monitoring and compliance.

E. Anticipated impacts and mitigation measures during the construction phase

240. **Contractor training and capacity building.** Construction activities are highly dependent on the awareness and competence of the contractor's workforce. Inadequate training, poor orientation, or limited understanding of environmental, health, and social obligations can directly

compromise compliance with statutory and contractual requirements and weaken the enforcement of approved environmental management measures. Insufficient worker capability increases the likelihood of unsafe work practices and occupational accidents, generates poor waste-handling practices that can degrade soil, water, and air quality, and heightens the risk of social conflicts with surrounding communities when inappropriate behaviour, lack of communication, or disregard for community sensitivities occurs.

241. Following mitigation measures/actions shall be implemented:

(i) Mandatory induction training
- All Contractor personnel—engineers, supervisors, machinery operators, and labourers—shall undergo environmental and OHS induction training before mobilization.

(ii) Induction shall include:
- EMP requirements, pollution prevention, and waste management.
- OHS protocols, PPE use, first aid, and emergency response.
- Community Health and Safety (CHS) protocols, respectful conduct, and behavioral expectations.
- Roles and procedures under the GRM.
- Overview of applicable Indian laws including:
  – Factories Act, 1948
  – Building and Other Construction Workers Act, 1996
  – Water Act, 1974; Air Act, 1981
  – Solid Waste Management Rules, 2016
  – Construction and Demolition Waste Management Rules, 2025
- Contractor's internal code of conduct regarding waste disposal, restrictions in sensitive areas, and zero tolerance for harassment or misconduct.

(iii) Continuous capacity building
- Refresher training shall be conducted every six months and for all newly inducted workers.
- Specialized training shall be provided for high-risk tasks such as confined space entry, working near metro tunnels, excavation, lifting operations, hazardous materials handling, and emergency response.
- The PIU/Environmental Specialist shall periodically observe or review training sessions to ensure consistency and adequacy.

(iv) Documentation and recordkeeping
- The Contractor shall maintain detailed, verifiable records of all training events including attendance, topics covered, trainer credentials, and participant evaluation.
- Training records shall be available for inspection by PIU, ADB, or statutory authorities during audits or monitoring visits.

242. **Establishment of construction camps, yards, and ancillary facilities.** The establishment of construction camps, storage yards, and ancillary facilities is required given the wide, linear work front of the RMS Project across Chennai. If these temporary facilities are not properly sited, designed, or managed, they can generate localized environmental and social impacts, including vegetation clearance and topsoil loss, pollution arising from poor runoff control, inadequate solid-waste handling, and untreated wastewater, as well as nuisance impacts from dust generation, noise, and increased vehicle movements. Improper sanitation and poorly controlled worker activities may also result in public health concerns and trigger tensions or conflicts with adjacent communities.

243. Following mitigation measures/actions shall be implemented:

(i) Site selection and establishment

- All camp and yard locations shall be selected in consultation with GCC/local bodies and approved by the PIU. Preference shall be given to government or institutional land.
- No camps shall be located near water bodies (Cooum, Adyar), stormwater drains, metro tunnels/pillars, or other sensitive zones.
- Camps shall be sited in already cleared areas. If unavoidable, tree cutting shall follow the approved compensatory plantation plan.
- Camps shall be equipped with potable water, sanitation systems (septic tank–soak pit), drainage, power supply, and waste collection prior to occupancy.

(ii) Camp management and environmental controls

- Ensure full compliance with the Guidance Note on Workers' Accommodation: Processes and Standards (August 2009[5]) for planning, establishing, and operating all worker accommodation facilities. (Appendix 19)
- Provide safe drinking water, conduct periodic drinking water testing
- Provide safe, ventilated, and structurally sound living spaces, ensuring adequate room size, lighting, and fire-safety measures.
- Maintain clean and adequate sanitation facilities, including toilets and bathing areas.
- Prevent discharge of wastewater into drains; ensure septic tanks are regularly de-sludge by authorized agencies.
- Provide segregated solid waste bins for biodegradable, recyclable, and non-recyclable waste; hand over waste only to authorized GCC collection services or recyclers.
- Prohibit burning of waste within or near worker camps.
- Install lined drains, silt traps, and stormwater management systems to prevent flooding, soil erosion, and uncontrolled runoff.
- Ensure safe, hygienic food services, including clean cooking areas, proper food storage, and regular inspections.
- Conduct regular health check-ups, maintain vector-control measures (e.g., mosquito control), and organize hygiene awareness programs for workers.
- Provide appropriate security arrangements while ensuring workers' privacy and dignity.
- Display codes of conduct at visible locations; prohibit alcohol, drugs, violence, and any unlawful activities inside worker camps.
- Ensure a functional internal GRM is accessible to all workers.
- Continue monitoring and supervision to ensure worker health, safety, hygiene, and dignity throughout the project lifecycle.
- Details of Helplines, rules and regulations, contact details of nearest hospital, ambulance, fire department, police station, etc. in the camps. Multilingual instructions and posters to be provided in the camp.

244. **Construction activities in urban areas:** Construction of the RMS along narrow urban roads and densely populated neighborhoods in Chennai will lead to temporary disturbances to traffic flow, pedestrian movement, and access to local businesses. Dust, noise, and vibration from excavation and pipe-laying machinery may affect nearby residents, schools, hospitals, religious places, and other sensitive locations. Improper storage of construction materials may create safety hazards and localized flooding during rainfall. The proposed alignment involves pipelines of diameters ranging from 1,181 mm to 1,976 mm, laid primarily along existing road ROW. Earthwork widths vary between 1.8 m and 3.0 m, with excavation depths between 2.7 m and 3.5 m. Barricading will be provided on both sides, generally 3–4 m wide, to ensure safe movement of pedestrians and vehicles during construction. Major roads impacted include Ambattur Redhills

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[5] https://www.ifc.org/en/insights-reports/2000/publications-gpn-workersaccommodation

Road, Grand Northern Trunk Road, Chennai-Srikakulam Highway, Madhavaram Milk Colony Road, Kamaraj Salai, East Coast Road (ECR), Velachery-Tambaram Main Road, NH-55/Mount Poonamallee Road, Porur Link Road, and Bengaluru-Chennai Highway, among others. Construction on double-lane and single-lane roads will be carefully phased to maintain at least partial traffic flow, with temporary diversions, signage, and flag persons deployed to minimize inconvenience and ensure safety. Appendix 9 provides the details of road width, pipe dia and details of barricading.

245. Following mitigation measures / actions shall be implemented:

(i) Traffic and pedestrian safety
- Enforce speed limits of 20 km/h for construction vehicles in built-up areas.
- Deploy trained flagmen at busy intersections and near schools/hospitals.
- Provide hard barricades, reflective signage, lighting, and temporary diversions for safe pedestrian and vehicular movement.

(ii) Noise management
- Restrict noisy work to daytime (08:00–18:00).
- Use well-maintained equipment with silencers; blasting is prohibited.
- Conduct monitoring at sensitive locations to ensure compliance with CPCB norms.

(iii) Dust and air quality control
- Regular water sprinkling at active sites and unpaved access roads.
- Cover stockpiles; ensure trucks carrying loose material are tarpaulin-covered.
- Maintain machinery to minimize exhaust emissions.

246. **Immediate Road Restoration after Refilling the Trench.** Excavation and subsequent refilling of trenches disturb the topsoil, which, under the influence of wind, vehicular movement, pedestrian traffic, and other roadside activities, can result in significant dust generation. There is a high potential for dust emissions from refilled trenches, particularly in the period between refilling and final road reinstatement. Conventionally, road restoration is often delayed allowing the refilled material to stabilize naturally; however, under dry and windy conditions and on roads carrying heavy traffic, such delays can lead to excessive dust generation and create unhealthy and unsafe conditions for road users and nearby residents. Furthermore, once trench refilling is completed, barricades and dust screens are typically removed, leaving no physical measures in place to control dust emissions. Routine dust suppression measures, such as surface wetting, are likely to be ineffective under these site conditions. Therefore, it is essential to restore or relay the road surface immediately after trench refilling, or to adopt suitable soil consolidation and stabilization techniques that enable immediate road restoration and effectively arrest dust generation. Following mitigation measures/actions shall be implemented:

(i) **Immediate Surface Reinstatement**
- The Contractor shall ensure that the road is made motorable immediately after trench refilling through proper compaction and provision of temporary or permanent surface layers.
- Refilled trenches shall not be left exposed or unpaved for extended periods.
- Temporary restoration shall ensure safe passage of vehicles and pedestrians and effective dust suppression.

(ii) **Soil Consolidation and Stabilization**
- Refilled trenches shall be compacted in layers using mechanical compaction and/or water-assisted compaction to achieve adequate bearing capacity.

- Where required, approved stabilization techniques (lean concrete, granular sub-base, or stabilizers) shall be adopted to prevent settlement and dust generation.

(iii) **Temporary Surface Treatment (where immediate permanent paving is not feasible)**

- Provide temporary bituminous surfacing, precast concrete slabs, or lean concrete topping to seal the surface.
- Temporary surfaces shall be traffic-worthy and smoothly integrated with the existing pavement until final reinstatement.

(iv) **Dust Control Measures**

- Controlled water sprinkling or approved dust suppressants shall be applied during refilling and compaction.
- Moisture content of backfilled material shall be maintained until the surface is sealed.

(v) **Traffic and Public Safety**

- Barricades, warning signage, reflective markers, and night lighting shall remain in place until the road is restored to a safe condition.
- Safe pedestrian access to residences and commercial establishments shall be always maintained.

247. Permanent Road Reinstatement – Responsibility and Standards

(a) Non-Municipal Asphalt Roads and Other Paved Areas

- Other than Municipal Roads, all asphalt roads and other paved areas shall have permanent reinstatement carried out by the Contractor, in accordance with the requirements and standards of the road-owning authority concerned.
- Permanent reinstatement shall comprise provision of surfacing at least equal to that existing immediately before the Contractor first entered the site.
- The Contractor shall carry out permanent reinstatement as per a programme agreed with the Engineer.
- Permanent reinstatement of private lands shall similarly restore surfaces to a condition at least equal to the pre-construction condition.
- All permanent reinstatement shall include the Contractor’s return to site at regular intervals during the maintenance period to carry out remedial works as necessary, ensuring that the final condition of the surface is not inferior to the original.
- All costs related to permanent reinstatement, maintenance-period visits, remedial works, and any charges payable to the concerned authority shall be deemed included in the contract price.

(b) Highways

- Restoration and reinstatement of Highways carriageway and footpath surfaces shall be carried out by the Highways Department.
- CMWSSB shall bear the cost of such restoration as per the approved estimates and standards of the Highways Department.

(c) Municipal Roads

- Reinstatement of Municipal Roads, including asphalt roads, WBM roads, and sidewalks, shall be carried out by the Municipal Roads Department of the Greater Chennai Corporation (GCC) or by the Highways Department, as applicable.
- CMWSSB shall bear the cost of such reinstatement.

248. Monitoring, Supervision, and Accountability

- The Contractor shall be fully responsible for:

- Immediate motorable restoration after trench refilling.
- Preventing dust nuisance, settlement, and unsafe conditions.
- Rectifying any defects arising from inadequate restoration at no additional cost.

- The concerned line agency (Highways Department/GCC/other road-owning authority) shall:
  - Undertake final permanent restoration where applicable.
  - Ensure compliance with approved technical standards.
- CMWSSB shall:
  - Monitor and supervise trench refilling, temporary motorable restoration, and coordination with line agencies.
  - Ensure timely completion of final road reinstatement.
  - Enforce compliance with environmental, safety, and EMP requirements.

249. **Night works.** Due to the requirement for significant nighttime construction under the Chennai project, potential impacts may arise related to noise, lighting, worker safety, traffic management, and community disturbance; therefore, proper planning and implementation of mitigation measures are essential to ensure compliance with regulatory norms and to minimize inconvenience to surrounding residents, businesses, and road users. As high daytime traffic congestion, critical safety considerations, and directives from local authorities necessitate that a substantial portion of construction activities be carried out during night hours, these works shall be undertaken with strict safeguards to ensure public safety, worker welfare, and minimal disturbance to nearby communities.

250. Following mitigation measures/actions shall be implemented:

- Night works shall be executed only as per approved schedules and after obtaining permissions from PMU/PIU/CMSC (Appendix 17).
- Advance written intimation must be provided to district administration, police/traffic police, utility agencies, Resident Welfare Associations (RWAs), and business associations.
- The Contractor shall prepare a detailed Night Works Plan and obtain PIU approval prior to commencement.
- As per Noise pollution (regulation and control) rules, 2000 (under the Environment Protection Act, 1986), as amended, night-time noise limits must be complied with:
  - Industrial: 70 dB
  - Commercial: 55 dB
  - Residential: 45 dB
  - Silence Zones: 40 dB
- Ensure adequate illumination levels at work sites:
  - General work areas: 54 lux
  - Medium-accuracy work: 108 lux
  - High-accuracy activities: 216 lux
- Prefer grid electricity for lighting; where not feasible, use super-silent DG sets.
- Avoid or minimize high-noise activities (cutting, hammering, crushing) during night hours.
- Workers shall receive adequate daytime rest, be medically fit, and trained for night operations.
- Vulnerable workers (elderly, pregnant women, and women with small children) shall not be deployed on night shifts.
- All personnel must use PPE, including helmets, gloves, and retro-reflective safety vests.

- Provide full traffic management arrangements including barricades, reflective signage, caution boards, channelizers, and warning lights.
- Avoid use of vehicle horns; all construction machinery must have effective silencers/mufflers.
- Night works should conclude at least one hour before morning peak traffic begins.
- PIU/CMSC and site safety personnel shall conduct hourly monitoring of noise levels, illumination, barricading, and safety compliance.
- Maintain photographic/video documentation, monitoring registers, and daily checklists.
- Ensure emergency preparedness: first-aid kits, rescue equipment, fire extinguishers, and emergency vehicles must always remain operational.
- Avoid night works near hospitals, examination centers, and during major festivals/tourist peak periods, unless special permissions are obtained.
- Ensure minimal nuisance to surrounding residents by controlling noise, lights, and movement of construction vehicles.
- Maintain a safe, peaceful, and well-managed work environment to avoid community grievances.

251. **Impacts on physical cultural resources (PCRs)**. Chennai's dense and historic urban fabric includes numerous temples, churches, mosques, heritage buildings, and culturally important structures along or near the Ring Main alignment. While no direct impact on known PCRs is expected, indirect impacts may arise from vibration, dust, noise, and temporary access restrictions during excavation or pipe-laying works in their vicinity.

252. Following mitigation measures/actions shall be implemented:

- All PCRs within 50 m shall be identified, mapped, and communicated to site teams.
- Work schedules near religious institutions shall be discussed with temple committees/community groups to avoid festival timings.
- Intensified sprinkling and restricted working hours near pcrs.
- Chance finds protocol to be implemented if cultural artifacts are discovered:
- Stop work immediately and secure the site.
- Notify PIU and Department of Archaeology, Tamil Nadu.
- Resume construction only after obtaining formal clearance.

253. **Construction near sensitive areas**: The subproject components are located entirely within existing road ROWs and previously converted urban areas, with no works proposed inside or adjacent to any KBAs. The nearest major KBA is Pulicat lake (24 km from nearest proposed component), an ecologically critical wetland that supports numerous migratory waterbirds and forms part of the East Coast KBA network. Based on project sitting and the urbanized context of all work locations, no direct or indirect impacts on Pulicat Lake or any other KBA are anticipated, as there will be no land take, no discharge, no hydrological alteration, and no material transport in the direction of these sensitive habitats. The Environment Chapter identifies approximately 80 threatened species reported at the regional level (IUCN Red List), primarily associated with wetlands, coastal lakes, estuaries, and forested habitats; however, these species do not occur within the modified and built-up footprint of the proposed project areas. Therefore, significant impacts on threatened flora or fauna are ruled out. Nevertheless, to ensure precaution, the project will implement standard biodiversity safeguards, including: (i) avoiding night construction near vegetated patches to minimize disturbance; (ii) strict control of dust, noise, and lighting; (iii) ensuring no construction waste or wastewater enters any waterways; (iv) prohibiting any resource extraction, vegetation clearance, or temporary camps near ecologically sensitive areas; and (v) proper transport and containment of materials to avoid accidental run-off. With these measures,

the project will not result in any adverse impact on KBAs, Pulicat Lake, or regionally threatened species.

254. The Ring Main and Transmission Main alignments pass just outside Guindy National Park and 62 m from Pallikaranai Ramsar Site, but no direct impacts are expected since all construction is confined to existing road ROW. Indirect, temporary impacts such as noise, dust, vibration, and potential disturbance to wildlife or wetlands may occur during excavation, material transport, and other construction activities. These impacts can be effectively managed with proper mitigation measures.

255. Following mitigation measures/actions shall be implemented:

(i) Obtain all mandatory clearances (CRZ, TNPCB, Forest Department, Wetland Authority) before construction.
(ii) Follow conditions from Guindy NP, Pallikaranai Wetland Division, and CZMA authorities.
(iii) Restrict all activities strictly within existing road ROW; no diversion, storage, or encroachment into parks, wetlands, or CRZ zones.
(iv) Use HDD or micro-tunneling near sensitive habitats to avoid deep excavation.
(v) Minimize vibration, noise, and dust near wildlife zones; avoid nighttime construction near Guindy NP and Pallikaranai marsh.
(vi) Avoid construction during early morning and evening peak wildlife activity; install wildlife-safe lighting and prohibit loud machinery or horn use near the park.
(vii) Store fuels and hazardous materials at least 200 m away from wetlands; prevent any discharge of wastewater, silt, or construction runoff into marshes.
(viii) Install silt traps, sedimentation tanks, and barricades along marsh edges.
(ix) Follow CRZ norms: no stockpiling in CRZ, use pre-assembled pipe sections, and implement spill-prevention during transport near coastlines or backwaters.
(x) Use low-noise equipment, provide temporary noise barriers, sprinkle water to suppress dust, and cover trucks carrying soil or waste.
(xi) Prevent water pollution by using drip trays for parked machinery and maintaining spill response kits at sensitive locations.
(xii) Remove all construction waste daily from sensitive areas; no dumping, burning, or temporary storage near wetlands, marshes, or CRZ zones; dispose of spoils only at approved locations (refer to spoils management plan in Appendix 13).
(xiii) Conduct mandatory environmental induction training for workers; prohibit alcohol use, smoking, open flames, and littering near sensitive habitats.
(xiv) Do not establish worker camps or park machinery within 100 m of ecological or sensitive zones.

256. **Transport of construction materials.** Pipe-laying along major arterial and sub-arterial corridors will inevitably affect traffic operations by reducing carriageway width, introducing lane closures, and slowing vehicular movement. These constraints can lead to congestion and prolonged travel times, elevate accident risks for pedestrians and two-wheelers navigating construction zones, and increase air and noise pollution from idling vehicles and diversion routes. Heavy vehicle movement and repeated haulage may also damage road surfaces that are not engineered for such loading. In extreme cases, poor traffic management can restrict access for emergency vehicles and compromise response times.

257. Following mitigation measures/actions shall be implemented:

(i) Traffic management planning

- Prepare a detailed traffic management plan (TMP) in coordination with Chennai traffic police.
- Use designated haul routes, avoiding residential lanes where possible.
- Schedule material movement during off-peak hours.
- Maintain coordination with authorities for diversions, temporary closures, and road safety measures.

(ii) Vehicle and driver management

- Ensure all trucks comply with Regional Transport Office (RTO) norms, pollution standards (Bharat Stage VI norms), and are well-maintained.
- Provide safety orientation to drivers, especially regarding school/hospital zones.
- Enforce strict speed limits along haul routes.

(iii) Storage and loading/unloading controls

- Use only pre-approved laydown areas for material storage.
- Deploy traffic marshals during loading/unloading.

(iv) Community safety and awareness

- Provide advance notice of heavy vehicle movement and diversions to residents and businesses.
- Install barricades, reflective safety signs, and temporary walkways.
- Ensure emergency vehicle access is never obstructed.

(v) Monitoring and compliance

- Assign trained traffic supervisors for continuous monitoring.
- Maintain daily vehicle movement logs and incident reports for review of PIU.
- Implement corrective measures promptly in case of violations or congestion.

(vi) Environmental safeguards during transport

- Ensure tarpaulin covering of trucks carrying loose materials, sprinkle water on dusty haul roads.
- Maintain functional silencers to minimize noise.
- Securely transport fuels/chemicals to avoid spillage.

258. **Access disruption to shops, houses, and institutions.** Access disruption to shops, residences, and public institutions is a typical construction-stage impact for linear pipeline works undertaken within urban ROW. Temporary blockages arising from trenching, material stocking, barricading, or traffic diversions may limit the ability of residents to reach their homes and may reduce customer footfall for commercial establishments. For institutions such as schools, hospitals, and government offices, even short-duration access constraints can interfere with normal operations, delay student or patient movement, and create safety risks at entry and exit points. Emergency services are particularly vulnerable; impeded access can slow ambulances, fire, or police response times. While these disruptions are temporary and localized, they can generate livelihood impacts, trigger grievances, and heighten community dissatisfaction if not managed through advance notice, well-designed pedestrian and vehicular access arrangements, and continuous contractor coordination with affected stakeholders.

259. Following mitigation measures/actions shall be implemented:

(i) Provision of safe pedestrian access

- Ensure uninterrupted pedestrian movement throughout construction zones.
- Install temporary walkways using metal planks, handrails, and toe guards.

- Provide barricaded passages to prevent unsafe crossings and exposure to open trenches.

(ii) Temporary access ramps for vulnerable users

- Provide temporary ramps at all affected entry points.
- Facilitate movement for elderly persons, persons with disabilities, school children, and delivery/service providers.
- Ensure gradients and surfaces are safe, non-slip, and stable.

(iii) Coordination with sensitive receptors and emergency services

- Coordinate construction schedules with hospitals, schools, and commercial establishments.
- Avoid peak operational and patient-movement hours when working near hospitals.
- Maintain access for students, staff, and emergency medical transport.
- Liaise with ambulance operators, fire services, and local police to ensure priority passage in case of emergencies.

(iv) Minimization of full-access closures

- Avoid full closure of access to residences, shops, or institutions wherever possible.
- If closure is technically unavoidable, restrict it to the shortest duration.
- Provide alternative temporary access, including provisions for emergency-vehicle entry, and prioritize reinstatement activities.

(v) Advance public notification

- Issue prior notice (48–72 hours) to residents, shop owners, and institutions before starting work at their frontage.
- Communicate duration, expected impacts, access arrangements (including emergency-vehicle passage), and contractor contact points.
- Use multiple communication modes, including verbal intimation, leaflets, and signboards.

260. **Working near old and structurally vulnerable buildings.** Construction activities along sections passing close to old, dilapidated, or structurally sensitive buildings specially in Pulianthope high road, new Farrance road, Medavakkam tank road, vasu street, Kilpauk, Valluvarkottam High Road and Raja Muthiah road area may induce vibration, ground settlement, and accidental structural distress due to excavation, pipe laying, movement of heavy machinery, and dewatering. Such impacts may result in cracking of walls, damage to foundations, or safety risks to occupants, particularly in congested urban areas with narrow road widths and aged structures. Temporary obstruction of access, increased noise, and dust may further affect residents and businesses housed in old buildings, many of which may lack adequate structural resilience.

261. Following mitigation measures/actions shall be implemented:

a. Pre-construction building condition survey

- Conduct a detailed pre-construction structural condition survey of all old and vulnerable buildings located within the zone of influence of excavation and construction activities.
- Document baseline conditions through photographs, videos, and written records, and obtain acknowledgment from property owners.
- Identify high-risk structures and define site-specific protection measures prior to commencement of works.

(ii) Excavation and vibration control

- Adopt controlled excavation methods and avoid sudden or deep open cuts near old buildings.
- Prohibit use of heavy vibratory equipment near sensitive structures.
- Monitor vibration levels where required and ensure compliance with applicable standards.

(iii) Structural protection and safety

- Provide temporary shoring, bracing, or support systems for trenches near old buildings, as required.
- Maintain safe setback distances for material storage and heavy equipment movement.
- Ensure continuous supervision by qualified engineers during work near vulnerable structures.

(iv) Access, dust, and noise management

- Maintain uninterrupted and safe access to buildings through temporary walkways and signage.
- Implement dust suppression measures such as water sprinkling and covering of materials.
- Restrict high-noise activities to daytime hours and minimize duration near occupied old buildings.

(v) Grievance redress and corrective action

- Establish a mechanism for prompt reporting and resolution of any structural damage or safety concerns.
- Undertake immediate corrective measures in case of observed distress, at the contractor's cost.

262. **Safety risks to road users and workers.** Open trenches, construction equipment movement, and temporary reduction of road width create elevated accident risks for motorists, pedestrians, and workers. Without adequate controls, these conditions may lead to collisions, falls into trenches, sideswipe incidents, and unsafe nighttime driving conditions. Open excavations, machinery operations, and constrained carriageways may increase the likelihood of accidents involving road users and site personnel

263. Following mitigation measures/actions shall be implemented:

(i) Installation of safety barriers and signage

- Install reflective barricades, caution tape, blinking lights, and direction boards in accordance with applicable IRC work-zone safety guidelines.
- Position signage at appropriate distances upstream and downstream of the work zone.

(ii) Minimization of open trench duration

- Keep trenches open only for the minimum construction duration.
- Backfill on the same day wherever feasible, particularly in pedestrian-heavy or narrow-road locations.

(iii) Nighttime illumination

- Ensure adequate lighting during low-visibility hours using solar lights, blinking lamps, or equivalent illumination devices.
- Position lights to highlight trench edges, barricades, and traffic-flow paths.

(iv) Mandatory use of PPE for workers

- Provide all workers with appropriate PPE, including helmets, reflective jackets, gloves, and safety footwear.
- Enforce PPE usage through continuous supervision and toolbox talks.

(v) Traffic-speed management
- Install rumble strips, cones, barriers, and warning boards to reduce vehicular speed in construction zones.
- Coordinate with traffic police as needed in high-risk corridors.

264. **Transportation of material.** Construction of the RMS Project will require transporting large quantities of materials such as pipes, concrete, aggregates, and machinery through Chennai's urban and peri-urban roads. This increased movement may cause traffic congestion, especially on narrow roads and in areas near schools, hospitals, and commercial centers. The presence of heavy trucks and machinery can create safety risks for pedestrians, two-wheelers, and other road users, and open trenches or temporary roadside storage may further increase the chances of accidents. Continuous heavy vehicle movement may also damage local roads, particularly those not designed to carry such loads. Increased dust, diesel emissions, and noise from transport activities may affect nearby communities. If traffic is not properly managed, the movement of construction vehicles may delay or obstruct emergency services such as ambulances and fire engines.

265. Following mitigation measures/actions shall be implemented:

(i) Traffic management planning and regulatory coordination
- A detailed traffic plan will be prepared in coordination with Chennai Traffic Police, identifying approved haul routes and traffic control measures.
- Continuous coordination with local authorities will ensure safe diversions, temporary signals, and road closures.
- Local residents, schools, and businesses will be informed in advance about heavy-vehicle movement and temporary diversions.

(ii) Approved haul routes and vehicle scheduling
- Material transport will use only pre-approved main roads, avoiding narrow residential streets.
- Movement of heavy vehicles will be scheduled during off-peak hours to reduce congestion.
- Speed limits, such as 20 km/h in residential areas, will be strictly enforced with proper signage.

(iii) Vehicle safety, maintenance, and driver training
- All trucks and machinery will be regularly maintained and must meet pollution and safety standards.
- Drivers will be trained in safe driving, mandated speed limits, and heightened awareness near schools, hospitals, and busy pedestrian areas.
- Trained traffic supervisors will monitor haul routes and ensure compliance with safety protocols.

(iv) Pedestrian safety and work-zone controls
- Barricades, safe walkways, and reflective signs will be provided for pedestrian safety near work zones.
- Emergency-vehicle access will always be maintained.
- Loading and unloading zones will have trained flag persons to guide traffic and ensure safety.

(v) Monitoring, reporting, and corrective action
- Daily reports on vehicle movement, traffic issues, and any incidents will be submitted to the PIU.
- Immediate corrective actions will be taken in case of violations or safety concerns.

(vi) Dust, noise, and hazardous material controls
- Trucks carrying loose materials will be fully covered, and haul routes will be sprinkled with water to control dust.
- All vehicles will have functional silencers, and horns will be used only when required for safety.
- Hazardous materials will be transported securely to prevent spills during movement.

266. **Impacts on Surface Water Quality and Stormwater Conveyance.** Construction activities undertaken in proximity to waterbodies, stormwater drains, canal crossings, and minor watercourses have the potential to temporarily affect surface-water quality and natural stormwater conveyance. For the RMS Project, impacts may arise from trench excavation, pipe laying, stockpiling of materials, temporary storage within work corridors, and movement of construction vehicles. These activities disturb soil along trenches, access tracks, and stockpile zones, as well as at the 24 river and canal bridge-crossing locations, creating exposed surfaces susceptible to wash-off during rainfall or runoff events. Resultant sediment-laden discharge can increase turbidity, sedimentation, and siltation in adjoining drains, canals, and rivers.

267. Construction interfaces with flowing water also increase the likelihood of accidental release of fuels, lubricants, hydraulic fluids, curing compounds, or other chemicals during refueling, maintenance, and storage operations. If uncontrolled, such spills can enter storm drains or directly discharge to surface water systems, degrading aquatic quality and creating ecological and public-health risks. In addition, inadequate management of wastewater from worker facilities, fabrication or batching areas, and equipment-washing locations may introduce pathogens, nutrients, and elevated organic loads into the stormwater network.

268. Improper handling of excavation spoils, debris, and packaging waste can obstruct drainage channels and stormwater conveyance systems, resulting in localized flooding, bank instability, and pollution. Partial blockage at bridge-crossing works may also modify natural drainage behavior and temporarily alter hydrologic conditions. In cumulative terms, these interfaces can increase pollutant loads on an already stressed surface-water environment in Chennai. However, the risks are short-term in nature and can be effectively avoided or minimized through systematic planning, sediment and erosion-control measures, hazardous-materials and waste-management protocols, and routine monitoring of water quality.

269. Following mitigation measures/actions shall be implemented:

(i) Work planning and scheduling
- Schedule construction near rivers, canals, drains, and stormwater corridors during dry periods wherever feasible.
- Carry out in-drain and bridge-crossing works in short, controlled segments to limit exposed soil areas.

(ii) Sediment and erosion control
- Install silt fences, sediment traps, and geotextile barriers along trench edges, access tracks, stockpiles, and waterbody interfaces.
- Backfill trenches crossing drains and canals immediately after pipe installation.
- Cover or stabilize stockpiles to prevent wash-off during rainfall events.

(iii) Hazardous materials management
- Store fuels, lubricants, and chemicals in bunded, impervious areas at least 50 meters away from waterbodies or storm drains.
- Maintain spill-prevention kits and emergency response procedures at all

work sites.

- Conduct refueling and equipment maintenance in designated locations to prevent contamination.

(iv) Wastewater and sanitation management

- Collect and treat wastewater from worker facilities, batching yards, or equipment-washing areas before disposal.
- Ensure no untreated greywater or sewage enters drains, canals, or rivers.

(v) Solid waste and debris management

- Remove construction debris, spoil, and packaging waste from work zones daily.
- Store waste only in designated areas away from watercourses and flood-prone zones.
- Excavated spoil generated from pipelines and allied works will be reutilized to the extent feasible within project facilities. Suitable and non-contaminated spoil shall be transported to designated pumping station sites of CMWSSB for controlled filling to raise finished ground levels above recorded flood levels, subject to site-specific geotechnical suitability and engineer approval.
- Surplus spoil, after meeting internal filling requirements, shall be transported and disposed of only at Government-approved Construction and Demolition (C&D) Waste Processing Facilities, namely:
    - Kodungaiyur Dumping Ground
    - Perungudi Dumping Ground
- Transportation shall be undertaken in covered vehicles in compliance with applicable Solid Waste Management and C&D Waste Management Rules, and disposal receipts shall be maintained for environmental monitoring and audit purposes.

(vi) Water quality monitoring

- Conduct regular monitoring of turbidity, pH, and visible sedimentation downstream of works.
- Carry out additional monitoring following rainfall events to assess effectiveness of mitigation measures.
- Regular monitoring of turbidity and pH downstream of works, especially after rainfall, to assess the effectiveness of mitigation (Appendix 18).

(vii) Bridge and river-crossing management

- Use temporary barriers and cofferdams during pipe-carrying bridge construction to minimize sediment entry into flowing water.
- Sequence bridge-crossing works to avoid prolonged interference with natural flow or drainage.
- Maintain access for aquatic organisms where possible.

270. **Soil erosion and sediment control in open areas.** Excavation activities for pipelines, trenches, and access tracks in unpaved, open areas can lead to the removal of topsoil, destabilization of slopes, and increased susceptibility to erosion. Loose soil from these disturbed areas may be transported by wind or rainfall into nearby drains, canals, or waterbodies, contributing to sedimentation and reducing water quality. In addition to impairing surface-water systems, uncontrolled erosion can degrade site productivity, undermine embankments or temporary works, and generate sediment-related impacts downstream, potentially affecting drainage efficiency, aquatic habitats, and surrounding land stability.

271. Following mitigation measures/actions shall be implemented:

(i) Minimization of disturbance
- Limit soil disturbance to the minimum area required for construction.
- Phase works to reduce exposed soil duration.

(ii) Soil stockpiling
- Locate stockpiles away from drains and waterbodies.
- Cover with tarpaulin and install silt fencing around stockpiles.

(iii) Structural controls
- Provide check dams, diversion drains, and silt fences to intercept runoff.
- Ensure proper regrading and compaction of backfilled trenches.

(iv) Progressive rehabilitation
- Stabilize disturbed areas with vegetation once works are complete.
- Use bioengineering techniques (coir logs, erosion mats) on slopes.

272. **Air quality impacts.** Construction activities associated with the RMS Project, including excavation, material transportation, and movement of vehicles, can temporarily degrade local air quality in the project area. Dust is generated from exposed soil, trenching, stockpiles, and unpaved access roads, contributing to elevated concentrations of particulate matter (PM_{10} and $PM_{2.5}$). Exhaust emissions from diesel-powered machinery and vehicles further increase the presence of air pollutants such as nitrogen oxides (NOx), carbon monoxide (CO), and sulfur oxides (SOx). During periods of low wind or calm atmospheric conditions, which are common in Chennai, these pollutants may accumulate locally, exacerbating air-quality impacts. Although temporary, these emissions can affect nearby residents, pedestrians, workers, and sensitive receptors, including schools, hospitals, and commercial areas, if not adequately controlled.

273. Following mitigation measures/actions shall be implemented:

(i) Dust control
- Regular water sprinkling at work sites and on unpaved roads.
- Cover material stockpiles with tarpaulin.
- Cover trucks transporting loose material.
- Enforce vehicle speed limits.
- Provide temporary enclosures in areas or activities generating dust

(ii) Emission control
- Maintain all machinery as per manufacturer guidelines.
- Avoid idling; switch off engines when not in use.

(iii) Work sequencing
- Carry out excavation, laying, and backfilling sequentially to reduce the duration of exposed soil.

(iv) Monitoring
- Conduct daily visual checks for dust and smoke.
- Perform periodic ambient air quality monitoring as per the EMP.

274. **Noise and vibration.** Construction activities in urban and sensitive areas associated with the RMS Project can temporarily elevate ambient noise and generate vibrations. Noise arises from excavation, trenching, material handling, operation of construction machinery, and movement of vehicles, while vibrations result from heavy machinery operation, piling, and vehicular traffic on compacted surfaces. Elevated noise and vibrations can disturb residents in nearby settlements, disrupt learning environments for students and staff in schools, interfere with patient care in hospitals, and cause discomfort to visitors of religious or cultural institutions. Sensitive urban green corridors and associated wildlife may also be affected by both sound and vibration disturbances. Although these impacts are typically short-term and localized, they can

affect human health, productivity, and ecological well-being if not adequately controlled.

275. Following mitigation measures/actions shall be implemented:

(i) Planning and scheduling
- Schedule high-noise and high-vibration activities during daytime hours to minimize disturbance to residents, schools, hospitals, and religious institutions.
- Avoid continuous operation of noisy equipment near sensitive receptors.

(ii) Equipment management
- Use well-maintained, modern construction machinery with noise-reducing features such as silencers and mufflers.
- Ensure that vehicles and machinery comply with applicable noise and vibration standards.
- Limit engine idling of stationary equipment.

(iii) Noise barriers and enclosures
- Install temporary acoustic barriers, screens, or enclosures around noisy construction areas where activities are close to sensitive receptors.
- Use berms, hoardings, or site fencing to reduce sound propagation.

(iv) Vibration control
- Monitor vibration levels at sensitive structures and adjust construction techniques if thresholds are exceeded.
- Use low-vibration machinery for operations near existing buildings or heritage structures.
- Avoid pile driving or heavy compaction near sensitive structures without prior assessment and protective measures.

(v) Public awareness and coordination
- Inform nearby residents, institutions, and hospitals about planned high-noise and vibration activities in advance.
- Establish a grievance redress mechanism for complaints related to noise and vibration.

(vi) Monitoring
- Conduct regular noise and vibration monitoring at sensitive locations to ensure compliance with prescribed standards.
- Implement corrective actions immediately if monitoring indicates exceedances.

276. **Occupational health and safety (OHS) and labour standards.** Construction activities for the RMS Project involve excavation, trenching, operation of heavy machinery, and working in confined spaces or near underground utilities and metro infrastructure and high-traffic corridors. These activities pose significant occupational health and safety risks, including accidents due to trench collapses, falls, machinery operation, and working in confined or narrow spaces. Workers are also exposed to health hazards from dust, noise, vibration, and potential contact with communicable diseases, as well as risks arising from fatigue, extended working hours, and improper work practices. Inadequate sanitation and welfare facilities may aggravate health concerns. Without strict adherence to safety protocols and labour standards, such hazards may result in serious injuries, occupational illnesses, reduced productivity, work stoppages, legal liabilities, and non-compliance with statutory labour and safety regulations.

277. Following mitigation measures/actions shall be implemented:

(i) Planning and budgeting
- Prepare detailed OHS budgets and cost estimates following national/local regulations and good practice standards.
- Develop a comprehensive Occupational Health and Safety Plan (OHSP) for PIU approval prior to commencement of construction activities.

(ii) Hazard identification and risk assessment (HIRA)
- Conduct thorough hazard identification and risk assessments for trenching, confined spaces, and work near metro infrastructure and underground utilities.
- Maintain safe working distances and monitor vibration and other risks near tunnels, pillars, and utilities.
- HIRA will be developed as an integral part of the site-specific environmental management plan (SEMP) during the implementation phase. Prior to the commencement of any physical works,

(iii) Emergency preparedness
- Establish emergency response procedures in coordination with CMRL, local authorities, and emergency services.
- Include specific procedures for accidental utility damage, trench collapses, and machinery-related incidents.

(iv) Personal protective equipment (PPE)
- Provide BIS/IS-standard PPE to all workers, including helmets, gloves, reflective vests, safety boots, and hearing protection.
- Always ensure mandatory and continuous use of PPE on site.

(v) Site safety measures
- Install proper barricading, warning signage, and lighting around all trenches, metro structures, and high-risk areas.
- Maintain safe traffic and pedestrian flow around construction zones.

(vi) Health monitoring and reporting
- Maintain site accident registers and corrective action logs, and submit them to the PIU for review.
- Ensure immediate medical care, proper reporting, and compensation for work-related injuries in accordance with Indian law and labour standards.

(vii) Training and awareness
- Conduct regular safety training and toolbox talks for all workers, emphasizing risks associated with trenches, confined spaces, heavy machinery, and work near utilities.
- Promote a safety culture through ongoing supervision and enforcement of OHS protocols.

(viii) Worker rights and compensation
- Ensure compensation, medical care, and insurance for injuries as per law.

278. **Anticipated impacts and mitigation measures during the post-construction (demobilization) phase.** Improper closure and demobilization of construction sites can lead to several environmental and safety issues. Leftover construction materials, hazardous waste, or debris may cause soil and water pollution. Visual quality of urban and peri-urban areas may be degraded due to scattered materials, temporary structures, or unfinished works. Additionally, open excavations, unstable spoil piles, or inadequately reinstated sites may pose long-term safety hazards to the public and workers, including risk of falls, erosion, or localized flooding.

279. Following mitigation measures/actions shall be implemented:

(i) Waste removal and disposal
- Remove all leftover materials, debris, and temporary structures.
- Dispose hazardous waste as per Hazardous and Other Wastes Rules, 2016.
- Grade and stabilize spoil disposal sites.

(ii) Site rehabilitation and revegetation
- Restore all temporary sites (camps, offices, yards) to original condition.
- Stabilize slopes using bioengineering and native vegetation.

(iii) Final site inspection and handover
- Conduct joint inspection with PIU, CMWSSB and local authorities.
- CMWSSB and GCC to issue the final project completion certificate after satisfactory restoration of all sites.

280. **Long-Term Management of Project Access Roads and Post-Construction Site Closure.** Temporary access roads and construction sites, if not properly managed during and after the construction phase, can lead to several environmental, safety, and social impacts. Unmanaged access roads may facilitate unauthorized entry, illegal dumping, and unintended vehicular movement, causing erosion, sedimentation, and degradation of sensitive riparian areas, urban green spaces, or other environmentally important sites. Leftover construction materials, spoil, and hazardous substances can result in soil and water pollution, aesthetic degradation, and long-term safety hazards from open excavations, unstable spoil piles, or unfinished structures. Overall, these impacts can compromise environmental quality, public safety, and community acceptance of the project if adequate measures are not implemented.

281. Following mitigation measures/actions shall be implemented:

(i) Management of temporary access roads
- Install lockable gates at all entry points to sensitive zones to prevent unauthorized access.
- Place clear signage indicating “Restricted Access – Authorized Personnel Only.”
- Assign custody and management of keys to the designated PIU/CMWSSB unit.
- Coordinate with local authorities and the Forest Department for monitoring and control where required.
- Maintain access roads at minimum necessary width and stabilize verges with native vegetation to prevent erosion and sedimentation.

(ii) Post-construction site clearance and restoration
- Remove all leftover construction materials, debris, and temporary structures.
- Dispose of hazardous wastes at authorized facilities following environmental regulations.
- Grade, compact, and stabilize spoil sites to prevent erosion and runoff.
- Restore temporary construction sites, including replanting vegetation as per the compensatory plantation plan.
- Stabilize slopes and embankments using bioengineering techniques such as brush layering, turfing, or vegetative mats.
- Conduct a joint final inspection of restored sites with the PIU, contractor, and local authorities.
- Condition project acceptance on satisfactory restoration of sites, access roads, and environmentally sensitive areas.

F. Operation phase – impacts and mitigation measures

282. **Water Loss, Leakage, and Pipeline Integrity.** During operation, pipe bursts, leakages, or maintenance activities can result in loss of treated water, localized flooding, erosion of nearby utilities, and potential risks to road users. Temporary traffic congestion may occur when pipeline sections require repair or maintenance.

283. Following mitigation measures/actions shall be implemented:

- Conduct routine inspection and pressure monitoring through SCADA systems.
- Implement preventive maintenance and timely repair of valves, joints, and fittings.
- Maintain adequate buffer pressure to avoid surge and reduce the risk of pipe bursts.
- Develop and implement an Emergency Response Plan (ERP) for immediate isolation of affected pipeline sections.
- Schedule maintenance and repair activities during off-peak hours to minimize traffic disruption.
- Coordinate with traffic police for temporary diversions and deploy barricades, signage, and safety cones to protect the public.
- Restore road surfaces promptly after maintenance or repair works.

284. **Energy Consumption and Carbon Footprint**: Pumping stations in the pipeline network consume significant electricity, contributing to greenhouse gas (GHG) emissions and increasing the operational environmental footprint.

285. Following mitigation measures/actions shall be implemented:

- Use energy-efficient pumps and motors to reduce electricity consumption.
- Optimize pumping schedules to avoid peak-hour energy loads.
- Implement power factor correction and conduct periodic energy audits to improve efficiency.
- Explore integration of renewable energy sources such as solar power to offset conventional electricity use.

286. **Water quality and safety.** Operation of pipelines carries the risk of water contamination, biofilm formation, or fluctuations in residual chlorine levels, potentially compromising public health and water safety. Unauthorized public access to WDSs, including Overhead Tanks (OHTs) and Underground Tanks (UGTs), can also lead to accidents, injury, or interference with the water supply infrastructure.

287. Following mitigation measures/actions shall be implemented:

- Conduct continuous water quality monitoring for turbidity, residual chlorine, pH, and other relevant parameters.
- Perform regular flushing of transmission mains to prevent stagnation and biofilm buildup.
- Maintain adequate pipeline pressure to prevent ingress from contaminants.
- Apply chemical dosing at the Water Treatment Plant (WTP) as per Central Public Health and Environmental Engineering Organisation (CPHEEO) standards.
- Restrict access to OHT/UGT premises with secure fencing, locked gates, and warning signage.
- Deploying security personnel and surveillance measures where necessary.

- Conduct routine inspections to ensure integrity of barriers, signage, and access control.

288. **Occupational health & safety (OHS) risks:** Operation and maintenance activities, including pump operation, confined space entry, and valve chamber works, expose workers to mechanical, electrical, and chemical hazards, as well as hazardous atmospheres, increasing the likelihood of accidents and injuries.

289. Following mitigation measures/actions shall be implemented:

- Provide task-specific PPE, including helmets, gloves, gas detectors, safety harnesses, and electrical protection gear.
- Implement Confined Space Entry Procedures with gas testing prior to entry.
- Apply Lock-out/Tag-out (LOTO) procedures for all electrical systems during maintenance.
- Conduct periodic safety training, emergency drills, and awareness sessions to ensure workers are prepared for hazardous situations.

VII. INFORMATION DISCLOSURE, CONSULTATION AND PARTICIPATION

A. Overview

290. The active participation of stakeholders including local community, NGOs/CBOs, etc., in all stages of project preparation and implementation is essential for successful implementation of the project. It will ensure that the subprojects are designed, constructed, and operated with utmost consideration to local needs, ensures community acceptance, and will bring maximum benefits to the people. Public consultation and information disclosure is required by SPS 2009.

291. Most of the key stakeholders have been identified and consulted during preparation of this IEE. Additional stakeholders identified during project implementation will be included in the consultation process in accordance with the meaningful consultation requirements of ADB SPS (2009).

292. **Primary stakeholders** are defined as individuals, groups, or institutions that are directly or potentially affected (positively or adversely) by the subproject. These include:

(i) Residents, shopkeepers, and business owners located in or near sites where facilities will be constructed under the RMS;
(ii) Vulnerable groups within the project influence area; and
(iii) Government and utility agencies directly responsible for service delivery and infrastructure within the project area.

293. **Secondary stakeholders** are defined as those who are not directly affected but have an interest in, or influence over, the project's implementation and outcomes. These include:

(i) Local non-governmental organizations (NGOs) and community-based organizations (CBOs);
(ii) Community representatives and civil society groups;
(iii) The broader beneficiary community;
(iv) State-level government agencies;
(v) TNUIFSL;

(vi) GoTN; and
(vii) ADB.

294. Consultation with both primary and secondary stakeholders will continue throughout project implementation and operation to ensure transparency, participation, and effective grievance redress.

B. Public Consultation

295. The public consultation and disclosure program is a continuous process throughout the project implementation, including project planning, design and construction.

1. Consultation during Project Preparation

296. The subproject proposal is formulated by CMWSSB in consultation with the public representative bodies in the project area to suit their requirements.

297. Focus-group discussions with local residents of the area and other stakeholders were conducted to learn their views and concerns. A socio-economic household survey has been conducted in the project area, covering sample households, to understand the household characteristics, health status, and the infrastructure service levels, and the demand for infrastructure services. The public and the people residing along the project activity areas were consulted through a project area-level consultation workshop, which was conducted in Chennai on 29.07.2025. The minutes of the public consultation meeting are presented in Appendix 12.

2. Stakeholders Consultation

298. A city-wide stakeholders' consultation was conducted on 20 July 2023 at the CMWSSB Office, Greater Chennai Corporation, with participation from people's representatives and senior officials from multiple departments. The objective was to disseminate project-related technical information, explain the implementation schedule, highlight expected benefits, and discuss potential temporary adverse impacts during construction. Key themes such as environmental and social safeguards, gender inclusion, and community participation were presented. A total of 58 participants attended the consultation, including 41 male and 17 female participants. Stakeholders occupying senior and decision-making positions discussed the project's technical aspects, shared insights, and provided valuable recommendations. The summary of safeguard documents was also disclosed in local languages to ensure transparency and accessibility. All feedback received has been duly recorded and appended to this report. Minutes of the Meeting are provided in Appendix 13.

299. **Public Consultation on the Ring Main Project.** A public consultation meeting was conducted on 29 July 2025 to engage the public and local community groups on the proposed Ring Main Project. The consultation provided an open platform to disseminate project information and obtain direct feedback from citizens.

300. The Consultant team explained the project's objectives and conceptual framework, noting that the project is designed considering multiple planning horizons. Prior to construction, the PIU will carry out further consultations with affected communities, including advance notifications on road cutting, temporary shifting of stalls/vendors, safety measures, and traffic diversions.

301. **Focus Group Discussions (FGDs) – Site-Specific Consultations.** To ensure inclusive

community participation, FGDs were conducted from 25 July 2025 to 29 July 2025 at six locations across the proposed UGT construction areas. Dissemination of project-related technical information was undertaken, and safeguard summaries were disclosed in English and Tamil languages.

302. Both male and female community members participated and shared location-specific concerns regarding access disruptions, dust, traffic diversions, and safety during construction. Table 26 shows the breakdown of participants during the FGDs.

Table 26: Details of location-wise participants

S. No	Location	Male	Female	Total
1	Puzhal	18	1	19
2	Tiruvottiyur	7	1	8
3	Thoraipakkam	7	2	9
4	Sholinganallur	7	3	10
5	Puzhuthivakkam	13	0	13
6	Semmencherry, Ezhilma Nagar	2	13	15
Total		**54**	**20**	**74**

303. **FGDs with Commercial Stalls and Street Vendors.** A separate round of FGDs was held with street vendors, stall keepers, auto drivers, local commuters, and other mobile livelihoods likely to experience temporary disruptions. A total of 118 participants attended (69 males, 49 females). Table 27 presents the breakdown of participants. The Team explained the project's objectives and potential construction-phase impacts. Participants shared feedback regarding access issues, temporary relocation, air dust pollution, business interruptions, and safety near work sites.

304. These discussions helped identify localized challenges and informed the preparation of impact-mitigation measures.

Table 27: Location-wise participants for vendor/stall FGDs

S. No	Location	Zone	Ward	Male	Female	Total	Target Group
1	Purasaivakkam	6	77	0	5	5	Street vendors
2	Kilpauk	6	100	0	4	4	Street vendors
3	Medavakkam Tank Road	6	75	3	2	5	Street vendor
4	Valluvarkottam	9	113	2	6	8	Tea stall
5	Thiruvanmiyur	15	179	2	2	4	Street Vendors
6	Kottivakkam	15	181	4	2	6	Juice stall
7	Vanagaram	11	150	5	0	5	Tea stall
8	Ayanambakkam	7	79	6	0	6	Tea stall
9	Ambattur	7	86	–	–	–	Cycle stall
10	Nammalvarpet	10	72	3	0	3	Auto drivers
11	Madhya Kailash	13	170	4	0	4	Auto drivers
12	Pallikaranai	14	189	8	0	8	Auto driver
13	Medavakkam	15	191	9	0	9	Bus stop
14	Velachery	13	176	4	3	7	Bus stop
15	Porur	11	153	6	2	8	Bus stop
16	Tondiarpet	4	47	0	6	6	Bus stop
17	Surapattu	3	32	0	3	3	Petrol bunk
18	Puzhal	2	24	8	3	11	Residents
19	Madhavaram	2	27	0	5	5	Clinic
20	Kotturpuram	13	122	0	6	6	Residents
Total				**69**	**49**	**118**	

C. Information Disclosure

305. Executive summary of the IEE will be translated into Tamil and will be made available at the offices of PMU, PIU, and CMWSSB and will be displayed on their notice boards. Hard copies of the IEE will be accessible to citizens to disclose the document and at the same time create wider public awareness. An electronic version of the IEE in English and an Executive Summary in Tamil will be placed in the official website of the TNUIFSL and CMWSSB after approval of the IEE by ADB. Stakeholders will also be made aware of grievance register and redress mechanism.

306. Public information campaigns to explain the project details to a wider population will be conducted by the project team including officials from the contractor(s), PIU, and the design consultants. Public disclosure meetings will be conducted at key project stages to inform the public of the progress and future plans. Prior to start of construction, the PIU will issue Notification on the start date of implementation in local newspapers. A board showing the details of the project will be displayed at the construction sites to inform the public on the project.

307. Local communities will be continuously consulted regarding location of construction camps, access and hauling routes and other likely disturbances during construction. The road closure, together with the proposed detours will be communicated via advertising, pamphlets, road signage, etc.

VIII. GRIEVANCE REDRESS MECHANISM

308. A common grievance redress mechanism (GRM) will be in place to redress environmental and social issues or any other project related grievances. The GRM described below has been developed in consultation with stakeholders. Public awareness campaign will be conducted to ensure that awareness on the project and its grievance redress procedures is generated. The campaign will ensure that the poor, vulnerable and others are made aware of grievance redress procedures and entitlements per project entitlement matrix, and PMU and concerned PIUs will ensure that their grievances are addressed.

309. Affected persons will have the flexibility of conveying grievances/suggestions by dropping grievance redress/suggestion forms in complaints/suggestion boxes or through telephone hotlines at accessible locations, by e-mail, by post, or by writing in complaints register in CMWSSB or PIU or implementing agency offices. PIU safeguards officer will be responsible for the timely grievance redress on safeguards and gender issues and for registration of grievances, related disclosure, and communication with the aggrieved party.

310. GRM provides an accessible, inclusive, gender-sensitive and culturally appropriate platform for receiving and facilitating resolution of affected persons' grievances related to the project. A multi-tier grievance redress mechanism (GRM) is conceived with redress arrangements at project level and beyond project level for grievances that are/cannot be redressed at project level. Safeguards officer and Social and Gender Officer will be responsible for creating awareness among affected communities and help them through the process of grievance redress, recording and registering grievances of non-literate affected persons.

311. GRM aims to provide a time-bound and transparent mechanism to voice and resolve social and environmental concerns linked to the project. All grievances – major or minor, will be registered. Documentation of the name of the complainant, date of receipt of the complaint, address/contact details of the person, location of the problem area, and how the problem was

resolved will be undertaken. PIU will also be responsible for follow-through for each grievance, periodic information dissemination to complainants on the status of their grievance and recording their feedback (satisfaction/dissatisfaction and suggestions).

312. In the case of grievances that are immediate and urgent in the perception of the complainant, the matter will first be addressed at the field level. The contractor, together with the field-level staff of the CMSC and PIU, will attempt to resolve the issue on site. If the grievance cannot be resolved at this stage, it will be escalated to the PIU level for timely resolution.

313. Should the field unit of the PIU (PIU-I level) be unable to resolve the complaint within the stipulated timeframe, the grievance will be forwarded to the PIU-II level, where senior engineering and administrative officers will review the matter and take corrective action. If the issue remains unresolved at the PIU level—particularly grievances involving environmental pollution, compensation, compliance matters—it will be forwarded to the Project Grievance Committee chaired by senior leadership of CMWSSB and TNUIFSL and other committee members for further examination and redress.

314. For grievances requiring inter-departmental coordination beyond CMWSSB and TNUIFSL, or those involving broader policy, regulatory, or multi-agency issues, the matter will be escalated to the State Steering Committee, which serves as the apex body for final decision-making and resolution.

315. Grievance redress committee (GRC) will meet every month (if there are pending, registered grievances, as required), determine the merit of each grievance, and resolve grievances within specified time upon receiving the complaint-failing which the grievance will be addressed by the state-level steering committee. The Steering Committee will resolve escalated/unresolved grievances received.

316. Composition of the Grievance Redress Committee. GRC will be headed by the MD, TNUIFSL with MD, CMWSSB as co-chair, and other members include: chief engineer, CMWSSB, Safeguards Officers of PIU and PMU, a representative of TNPCB, Mayor/deputy Mayor (or elected ULB representative designated by the Mayor), one person of repute (as required). GRC must have at least one women member.

317. State Level Steering Committee will be the highest tier for grievance management and will include CMA as chair MDs of CMWSSB and TNUIFSL and other committee members.

318. Areas of Jurisdiction. The areas of jurisdiction of the GRC, headed by the District Magistrate will be: (i) all locations or sites within the district where subproject facilities are proposed, or (ii) their areas of influence within the district. The Steering Committee will have jurisdictional authority across the state (i.e., areas of influence of subproject facilities beyond district boundaries, if any).

319. The multi-tier GRM (Figure 35) for the project is outlined below, each tier having time- bound schedules and with responsible persons identified to address grievances and seek appropriate persons' advice at each stage, as required. The GRC will continue to function throughout the project duration. The implementing agencies/CMWSSBs shall issue notifications to establish the respective PIU level grievance redress cells, with details of composition, process of grievance redress to be followed, and time limit for grievance redress at each level.

320. **First-Level Redress – Field Level** (Resolution within 3 days); Affected persons may submit complaints verbally or in writing to the field-level team. Grievances will be reviewed by the

Assistant Executive Engineer of the PIU responsible for field implementation, supported by the Contractor's field and supervisory personnel (including safeguards) and the CMSC field staff. The team will verify the concern on-site, identify corrective actions, and communicate the response to the complainant. If resolved, the case will be closed at this stage; if unresolved within three days, the complainant may escalate the issue.

321. **Second-Level Redress – PIU Level (Resolution within 7 days).** Unresolved grievances will be referred to the PIU and handled by the Executive Engineer/zone in-charge of the PIU, assisted by the Deputy Construction Manager and the Environmental Specialist of the CMSC and the PIU safeguards officer. The PIU team will review previous actions, undertake additional technical or safeguards review if required, and determine corrective or preventive measures. If accepted by the complainant, the matter will be closed. If dissatisfaction persists after seven days, the case moves to the next tier.

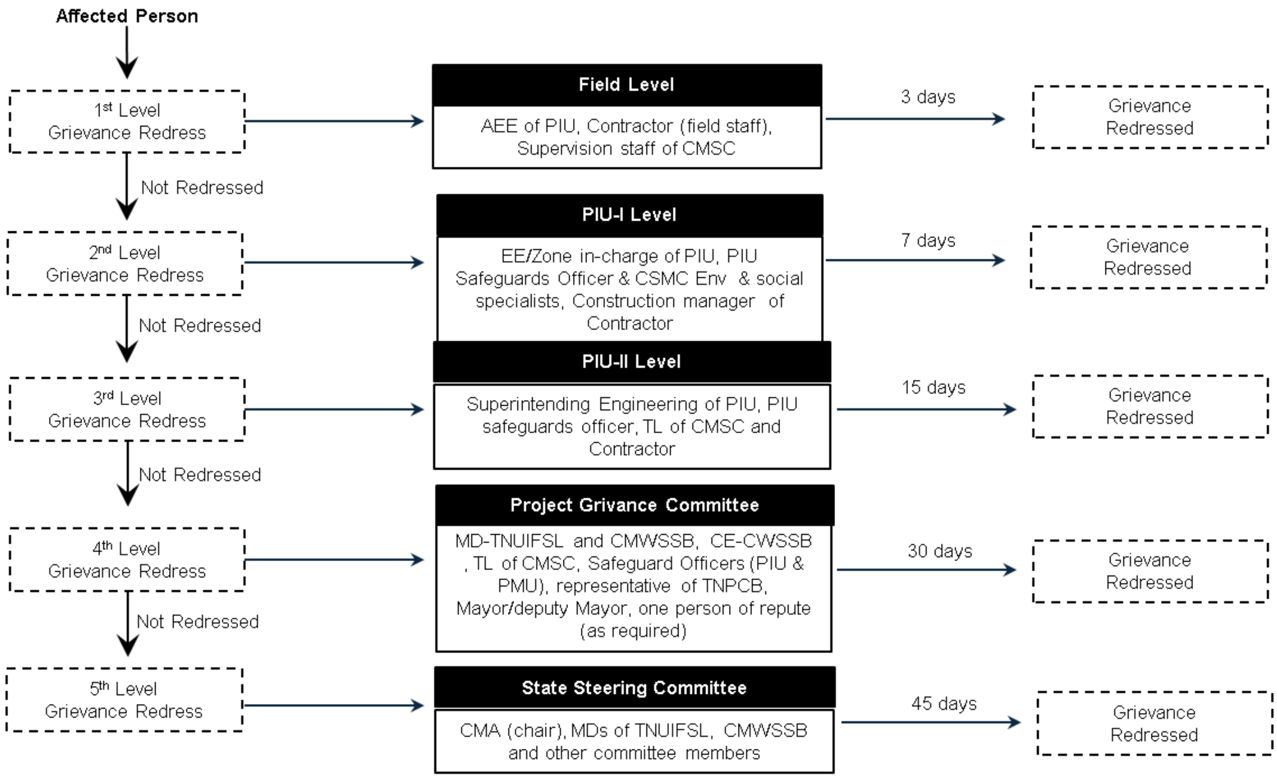
322. **Third-Level Redress – PIU (Head of Planning) Level (Resolution within 15 days).** At this level, the grievance will be examined by the Superintending Engineer (Planning)/PIU Project Manager, with support from PIU safeguards officer and the Team Leaders of the CMSC and construction manager of contractor. This review will assess contractual responsibilities, environmental and social safeguards, traffic and safety impacts, and community-interface concerns, and will deliver an authoritative decision within fifteen days. If unresolved, escalation continues.

323. **Fourth-Level Redress – Grievance Redress Committee (Resolution within 30 days)**: Complex or persistent grievances will be referred to the project-level GRC. The GRC will conduct hearings, review documentation, and recommend corrective actions. A decision must be issued within thirty days. If a complainant remains dissatisfied, the matter may be referred to the State level.

324. **Fifth-Level Redress – State-Level Steering Committee (Resolution within 45 days):** Unresolved grievances will be escalated to the State-level Steering Committee. The Committee will be chaired by the Secretary to Government, Municipal Administration, with participation from other government agencies as required. The Committee will conduct high-level review of the grievance and deliver a decision within forty-five days. This constitutes the final platform for administrative redress.

325. **Judicial Recourse – Courts of Law:** The complainant may pursue judicial remedy to the grievance at any stage of the redress process, in accordance with applicable law. Access to the GRM does not preclude or restrict the legal rights of affected persons at any stage.

Figure 36: Grievance Redress Process



APs= Affected Persons, CMA= Commissionerate of Municipal Administration, CMWSSB= Chennai Metro Water Supply and Sewerage Board, CMSC= construction management and supervision consultant, GCC = Greater Chennai Corporation, IA= implementing agency, MD= managing director, PIU= program implementation unit, PMU= program management unit, TNUIFSL= Tamil Nadu Urban Infrastructure Financial Services Ltd.

326. The project GRM notwithstanding, an aggrieved person shall have access to the country's legal system at any stage and accessing the country's legal system can run parallel to accessing the GRM and is not dependent on the negative outcome of the GRM.

327. **ADB Accountability Mechanism**. If the established GRM is not in a position to resolve the issue, the affected person can use the ADB Accountability Mechanism through directly contacting (in writing) the Complaint Receiving Officer at ADB headquarters or the ADB India Resident Mission (INRM). Before submitting a complaint to the Accountability Mechanism, it is necessary that affected persons make a good faith effort to solve the problem by working with the concerned ADB operations department and/or INRM. Only after doing that, and if they are still dissatisfied, will the ADB Accountability Mechanism consider the compliant eligible for review. The complaint can be submitted in any of the official languages of ADB's developing member countries. The ADB Accountability Mechanism information will be included in the project-relevant information disclosure leaflet to be distributed to the affected communities, as part of the project GRM.

328. **Recordkeeping.** Records of all grievances received, including contact details of complainant, complaint received date, nature of grievance, agreed corrective actions and the date these were in effect and outcome will be kept by CMWSSB/PIU (with the support of CMSC) and submitted to PMU.

329. **Information Dissemination Methods of the GRM.** The PIU, assisted by the CMSC will be responsible for information dissemination to affected persons and the public in the project area

on GRM. Public awareness campaign will be conducted to ensure that awareness on the project and its grievance redress procedures is generated. The campaign will ensure that the poor, vulnerable and others are made aware of grievance redress procedures and entitlements per this policy framework including contact details of officials/members of GRC, where/how to register grievance, various stages of grievance redress process, time likely to be taken for redress of minor and major grievances, etc. Grievances received and responses provided will be documented and reported back to the affected persons. The number of grievances recorded and resolved, and the outcomes will be displayed/disclosed in the PIU, offices, CMWSSB notice boards and on the web, as well as reported in the semi-annual environmental and social monitoring reports to be submitted to ADB. A sample grievance registration form is given as Appendix 16.

330. **Periodic Review and Documentation of Lessons Learned.** The PMU will periodically review the functioning of the GRM and record information on the effectiveness of the mechanism, especially on the PIU's ability to prevent and address grievances.

331. **Costs.** All costs involved in resolving the complaints (meetings, consultations, communication and reporting/ information dissemination) will be borne by the respective PIU.

IX. ENVIRONMENTAL MANAGEMENT PLAN

332. An Environmental Management Plan (EMP) has been developed to provide mitigation measures to reduce all negative impacts to acceptable levels.

333. The EMP will guide the environmentally-sound construction of the subproject and ensure efficient lines of communication between PMU, PIU, consultants and the contractors. The EMP will (i) ensure that the activities are undertaken in a responsible non-detrimental manner; (i) provide a pro-active, feasible and practical working tool to enable the measurement and monitoring of environmental performance on site; (ii) guide and control the implementation of findings and recommendations of the environmental assessment conducted for the subproject; (iii) detail specific actions deemed necessary to assist in mitigating the environmental impact of the subproject; and (iv) ensure that safety recommendations are complied with. The EMP includes a monitoring program to measure the environmental condition and effectiveness of the implementation of the mitigation measures. It will include observations on- and off-site, document checks, and interviews with workers and beneficiaries.

334. The contractor will be required to submit to PIU, for review and approval, a site environmental management plan (SEMP) including (i) proposed sites/locations for construction work camps, storage areas, hauling roads, lay down areas, disposal areas for solid and hazardous wastes; (ii) specific mitigation measures following the approved EMP; and (iii) monitoring program as per EMP. No works are allowed to commence prior to the approval of SEMP by the PIU.

335. A copy of the EMP/approved SEMP will be always kept onsite during the construction period. The EMP is included in the bid and contract documents. Non-compliance with, or any deviation from, the conditions set out in this document constitutes a failure in compliance.

336. For civil works, the contractor will be required to (i) carry out all the mitigation and monitoring measures set forth in the approved EMP; and (ii) implement any corrective or preventative actions set out in safeguards monitoring reports that the employer will prepare from time to time to monitor implementation of this IEE and SEMP. The contractor shall allocate a budget for compliance with

these SEMP measures, requirements and actions.

337. The following tables show the potential environmental impacts, proposed mitigation measures and responsible agencies for implementation.

Table 28: Anticipated impacts and mitigation measures for location and design phase

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and Source of Funds
Design Stage – Ring Main and Transmission Main	Potential long-term leakage, pipeline failure, ecological disturbance, heritage/cultural impacts, climate risks, and utility conflicts if design does not adequately address sensitive locations.	Integrate all CRZ clearance conditions and statutory requirements into design drawings. • Incorporate approved EMP and method statements into design documentation. • Conduct detailed utility mapping, GPS survey, and clash analysis to avoid conflicts with existing utilities. • Integrate sediment control, erosion control, and environmentally sensitive layout principles into design (silt fencing, controlled excavation zones, drainage paths, etc.). • Incorporate climate-resilient features (corrosion-resistant materials, thermal expansion allowances, flood-resilient structures, SCADA protection, elevated electrical assets). • Any conditions specified by CRZ authority should be strictly complied with.	Design Consultant and CMWSSB	TNUIFSL	• Approved design drawings with integrated EMP and CRZ conditions. • Utility clash analysis and DGPS-verified alignment. • All statutory permissions/NOCs included in design package. • Climate- and disaster-resilient design features documented.	Included in project design and EMP budget
		• Ensure that no new groundwater or surface water abstraction is included in the design. • Validate hydraulic design to ensure uninterrupted reservoir operations and distribution flows. • Approve tie-in locations based on operational feasibility and service continuity. • Review and confirm that dewatering discharge plans and testing water disposal concepts meet regulatory norms.	CMWSSB; TNUIFSL			
		• Obtain NOC from Forest Department for works within 10 km of Guindy National Park (ESZ clearance). • Obtain CRZ clearance from DoECC as per CRZ Notification,	Design consultant, CMWSSB			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and Source of Funds
		2019. • Ensure CRZ, and environmental compliance are incorporated before finalizing designs. • Ensure EMP and safeguards requirements are fully integrated in bid documents and technical specifications. • Coordinate with regulatory bodies (DoECC, ASI, TN Forest Dept, MOEFCC) to confirm all requirements are reflected in design.				
Sensitive areas (Guindy national park, Pallikaranai marshland, CRZ zones) in project area	• Risk of encroachment into sensitive areas (Guindy ESZ, Pallikaranai Marsh, CRZ, ASI buffer) if alignment is incorrect. • Delays in approvals due to incomplete or inaccurate IEE/CRZ submissions. • Potential disturbance to wetlands, biodiversity, and protected habitats if boundaries are not properly identified in design.	• Prepare IEE/EMP, CRZ maps, and statutory submissions with verified coordinates before initiating works. • Maintain coordination with TN Forest Department, DoECC, GCC, and ASI for location verification and compliance. • Obtain No Objection Certificate (NOC) from the TN Forest Department for works within 10 km of Guindy National Park and secure CRZ clearance in accordance with CRZ Notification 2019, complying with issued conditions	TNUIFSL, Design Consultant and CMWSSB	TNUIFSL	• Submission of updated IEE/EMP and CRZ maps with verified coordinates. • Copies of NOC from TN Forest Department and CRZ clearance issued. • Records of coordination meetings with regulatory authorities. • GIS-based alignment validation reports completed. • Alignment drawings with all sensitive area boundaries clearly	Included in project design and EMP budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and Source of Funds
	• Inadequate planning for trenchless or non-monsoon work may increase risk of ecological disturbance during execution. • Lack of early ecological input can result in mitigation measures that are impractical or ineffective during construction.	• Verify GIS-based alignment and ground-truth surveys prior to construction. • Endorse alignment drawings with marked Guindy ESZ, Pallikaranai Marsh boundaries, CRZ zones, and ASI buffers. • Ensure avoidance of any deviation near sensitive locations unless re-screened and cleared by relevant authorities. • Assist in marking sensitive boundaries (Guindy ESZ, Pallikaranai Marsh, CRZ, ASI buffers) on alignment drawings. • Provide technical inputs for statutory submissions and CRZ map updates.	Design consultant, CMWSSB		marked and endorsed. • Documented approvals for any deviations near sensitive locations. • Technical inputs/comments incorporated in alignment drawings and statutory submissions. • Confirmation that sensitive boundaries (ESZ, Ramsar, CRZ, ASI) are correctly represented in design drawings.	
River and canal crossings (Adyar river, Cooum river, Buckingham canal).	Bridge foundation works can increase turbidity by disturbing bed sediments, reduce water clarity, and affect aquatic habitat. Concrete, fuel, and wash-water spills may pollute the channel. Temporary works may alter tidal hydraulics in sensitive stretches of the Adyar, Cooum, and Buckingham Canal, affecting flushing and habitat conditions.	• Ensure CRZ Clearance from DoECC, GoTN is obtained prior to any work within the influence zone. • Approve final designs incorporating EMP and method statements in compliance with CRZ clearance conditions.	TNUIFSL, Design Consultant;	CMWSSB/ TNUIFSL	CRZ clearance obtained; CRZ compliance reflected in approved design; documented design review reports.	Included in project and EMP budget
		• Verify that designs comply with CRZ clearance conditions and approved EMP/method statements. • Confirm placement of temporary structures (ramps, platforms, staging) outside 100 m buffer from tidal waterbodies. • Review hydraulic/tidal flow studies to validate bridge elevation and free-board requirements.	Design Consultant; CMWSSB			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and Source of Funds
		• Ensure no heavy machinery, fuel, or material storage is planned within 50 m of CRZ boundary. • Incorporate silt fencing, controlled excavation, sediment barriers, and slope protection using native vegetation in designs. • Design temporary bunds/cofferdams to avoid altering tidal flow and ensure removal post-work. • Use pre-cast bridge components to minimize in-stream work and avoid bed disturbance. • Ensure zero discharge of spoil, debris, or wastewater into water bodies. • Design abutments to avoid constricting natural river/canal width. • Plan works near water bodies only during non-monsoon periods. • Include provisions for inspection of bridge supports during operation to prevent leakage or structural settlement.	Design consultant			
Climate Change and Extreme Weather (Design Integration)	Risks of flooding, storm surges, pipeline flotation, erosion, deterioration of structures, and disruptions to SCADA and power systems.	• Incorporate climate-resilient materials and structural designs. • Include elevated SCADA rooms, waterproof enclosures, and backup power systems. • Design flood-resilient pipe supports, foundations, and encasements. • Integrate predictive analytics, over-pressure protection, and automatic isolation valves in SCADA design.	Design Consultant; CMWSSB; TNUIFSL	TNUIFSL; CMWSSB	Design drawings demonstrating climate-proofing; elevation and protection of critical components; verification during design review.	Included in project and design budget
		• Provide operational data for sizing and risk assessment (historic flooding, river water levels, pressure transients). • Approve	CMWSSB			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and Source of Funds
		resilience features.				
		• Ensure that climate adaptation criteria are fully integrated before issuing design approval.	Design Consultant; CMWSSB			
Trenching, excavation, material movement, and heavy-equipment operation within proximity of ASI-protected fort-wall remnants (approx. 400 m from corridor).	No direct intersection with notified assets, but potential indirect impacts including: • Uncontrolled excavation causing accidental encroachment toward regulated buffer zones. • Excessive vibration from heavy equipment accelerating deterioration of old masonry. • Structural instability due to movement/compaction loads.	• Superimpose finalized pipeline alignment on ASI heritage maps and verify horizontal offsets prior to mobilization • Conduct pre- and post-construction structural condition surveys for heritage structures within influence distance • Implement a formal Chance-Find Procedure (CFP) and stop-work protocol if archaeological materials are discovered • Restrict equipment lay-down, fuel storage, and worker congregation away from mapped heritage control zones	Design Consultant; CMWSSB	TNUIFSL	• Verified overlay of alignment on ASI heritage maps. • Records of vibration-minimizing construction method statements. • Pre- and post-condition survey reports.	design budget
Impacts due to utility congestion near metro corridors and underground services	• Damage to underground water lines, sewers, storm drains, gas mains, OFC networks, and power cables. • Risks to Metro infrastructure: • Localized congestion, dust, and public safety hazards: • Delays due to rework or emergency restoration	• Conduct GPR surveys and collect utility details from Metro Rail, GCC, telecom agencies, and CMWSSB. • Prepare a detailed Utility Relocation Plan with ownership and depth. • Identify all Metro-related crossings and secure written NOCs.	Design consultants, Consultant's survey team	TNUIFSL	• CMRL-approved method statements • Clearance certificates • No structural flags • Traffic permits	design budget
		• Hold joint coordination meetings with each utility agency. • All works within 25 m of Metro tunnels/pillars require prior CMRL approval and supervision.	CMWSSB; TNUIFSL			
		• Use manual or low-vibration excavation near sensitive utilities. • Maintain minimum offsets from foundations and chambers. • Ensure proper barricading,	Contractor/ CMSC			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and Source of Funds
		signage, and public safety arrangements.				
Loss of Trees and Impacts on Urban Green Cover	Approx. 75 roadside trees (common LC species) may be affected due to proximity to trenching, working width, and equipment movement. • Loss of shade/green cover • Disturbance to urban fauna • Dust & noise during felling/pruning • Temporary loss of micro-climate benefits	• Undertake micro-alignment adjustments in design to avoid mature trees. • Restrict design working width to essential needs only. • Integrate trenchless technology in narrow, tree-dense areas and all CRZ/marsh buffer zones. • Map all identified trees accurately using chainage and GIS layers. • Incorporate tree protection plan in drawings and BOQs.	Design Consultant	Monitoring: PIU–CMWSSB Environmental Specialist; PMU–TNUIFSL	• Alignment finalization report showing tree avoidance. • Approved tree inventory with GPS coordinates. • BOQs include plantation and tree protection line items. • Approval/NOC from GCC/Horticulture Dept. • Tree Protection Plan included in tender documents.	Included in design and PMU/PIU administrative budgets
		• Obtain statutory approvals (GCC/Horticulture) for any pruning/felling. • Specify compensatory plantation ratio (minimum 1:10) using native species. • Define plantation locations (road corridors, waterbody buffers, public lands). • Include tree management protocol in bid documents. • Require contractor to maintain tree impact register and quarterly reports.	CMWSSB & TNUIFSL			
Flood-Prone and Waterlogging Zones	Alignment passes through flood-prone corridors (Velachery, Madipakkam, Sholinganallur, Saidapet). • Impediment to local drainage • Soil instability affecting pipeline bedding • Increased urban flooding risk • Construction delays	• Conduct local hydrology and drainage assessments. • Avoid trenching segments during monsoon by construction phasing in the design-stage EMP. • Provide temporary cross-drainage provisions in design drawings. • Specify granular sub-base (GSB) bedding and reinforced side-support where high groundwater or weak soils exist. • Ensure design does not block existing stormwater drains or	Design Consultant	PIU–CMWSSB; Environmental Specialist PMU–TNUIFSL	• Approved drainage assessment report. • Design drawings showing cross-drainage provisions. • BOQs include bedding reinforcement, temporary drainage, and dewatering systems. • Monsoon construction restriction plan integrated in tender documents.	Included in design and PMU/PIU administrative budgets

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and Source of Funds
	and safety risks	culverts.				
		• Validate drainage assessment and confirm integration with GCC stormwater master plan. • Ensure BOQ includes dewatering pumps, silt traps, and temporary drainage channels. • Include mandatory requirement to restore road drainage immediately after pipe laying. • Approve construction sequencing and monsoon-restriction plan before tendering.	CMWSSB & TNUIFSL			
Surface Water, Groundwater and Competing Water Use (Design Controls)	Unplanned interruption of water supply; accidental contamination during tie-ins; improper handling of dewatered/testing water if not addressed in design.	Prepare tie-in design ensuring controlled, planned shutdowns. • Integrate dewatering filtration and discharge systems into design. • Ensure construction water is not sourced from groundwater (design basis).	Design Consultant	TNUIFSL; CMWSSB	No proposed groundwater extraction; tie-in plans approved; silt-trap and discharge arrangements reflected in design.	Included in project and EMP budget
		Review tie-in design and approve sequencing based on network hydraulics. • Confirm no changes in water allocation or reservoir operations.	Design Consultant; CMWSSB			
		Ensure environmental safeguards related to water use are incorporated.				

Table 29: Anticipated impacts and mitigation measures during pre-construction phase

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
Regulatory Clearances and Permissions	Delays or non-compliance due to missing permits; risk of legal violations if clearances not obtained before mobilization.	Submit applications for all statutory permits (road cutting, tree pruning, traffic approval, utility permissions, CTE, CTO, tree cutting, CRZ, working near sensitive areas) before starting construction.	Contractor; CWMSSB (As applicable)	CMSC ES, TNUIFSL	Approvals, permits, NOCs available and verified; compliance report from CMSC; no work started without permits.	Contractor cost; Included in project management budget
		Issue approvals for road cutting, traffic diversion, waste disposal, and work permits.	GCC/TNPCB/ Forest department DOECC, MOEFCC			
		Verify completeness of permits; provide utility clearance letters; coordinate with GCC where required.	CMWSSB			
		Ensure pre-construction compliance with ADB safeguard and state regulatory requirements.	TNUIFSL			
		Review and certify all permit documents before allowing construction to start.	CMSC Environmental Specialist			
Baseline environmental conditions (air, noise, water, soil)	Lack of baseline data may affect proper assessment of project impacts	Conduct environmental monitoring through NABL-approved laboratories before construction	Contractor, CMWSSB	CMSC ES, TNUIFSL	Environmental Monitoring Reports on air, noise, soil, and water quality	Contractor budget
Sourcing of construction materials	Unsustainable sourcing of sand, gravel, and aggregates may degrade riverbeds, cause illegal quarrying, or violate environmental norms.	• Procure materials only from licensed quarries and sand mines approved by the Department of Geology and Mines, Tamil Nadu. • No new quarrying for the project. • Maintain proof of source and EC for all materials. • Conduct regular audits by PMU/PIU.	Contractor, CMWSSB	CMSC ES, TNUIFSL	Audit reports, material source certificates, and environmental clearances verified.	Contractor budget
Utility Shifting and Site Readiness	Damage to utilities; service interruptions; delays due to improper surveys or uncoordinated shifting.	Conduct joint utility verification and prepare utility-shifting plan. • Schedule relocation in coordination with utility owners	Contractor,	CMSC TL, TNUIFSL	Utility verification completed; signed joint inspection report; approved shifting plan; no accidental damage.	Contractor budget
		Coordinate with utility agencies (GCC, TNEB, telecoms).	CMSC; CMWSSB			
		Provide approvals for shifting of municipal assets (streetlights, drains).	GCC			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		Prepare hazard identification and risk mitigation plan for utility works	Contractor EHS Expert			
		Validate plan ensuring there are no safety or environmental risks.	CMWSSB, CMSC Environmental Specialist			
Temporary Traffic Management	Traffic congestion, safety risks, and community disruption during pre-construction planning.	Prepare traffic management plan (TMP) with diversion routes, barricading layout, pedestrian access, signages.	Construction Contractor; CMSC ES	Traffic Management specialist of CMSC, PIU; CMSC	Approved TMP; signages and diversion plans ready before mobilization; records of coordination meetings.	Contractor budget
		Approve TMP and define timing for works in high-traffic zones.	GCC/Traffic Police			
		• Review TMP against safety, accessibility, and environmental requirements.	CMSC Environmental Specialist			
		• Integrate road safety measures, lighting, and emergency procedures into TMP.	Contractor EHS Expert			
		• Ensure TMP approval is obtained before construction start.	PIU and PMU			
Environmental Management Plan Integration	EMP not implemented due to lack of awareness, training, or documentation.	• Prepare Site Environmental Management Plan (SEMP) aligned with approved EMP.	Contractor; CMSC ES; PIU; PMU; Contractor EHS Expert	Health and Safety Specialist of CMSC, PMU, PIU; CMSC ES	Approved SEMP; site-specific plans; induction training records; EHS method statements approved.	Contractor budget
		• Review and approve SEMP, site-specific plans (SSP), emergency response plan.	CMSC Environmental Specialist			
		• Verify that SEMP is approved before issuing site handover.	CMWSSB			
		• Prepare worksite risk registers and method statements incorporating EHS controls. • Conduct preconstruction environmental monitoring and site photography/ videography	Contractor EHS Expert			
		• Ensure EMP compliance is integrated into contract management system.	PMU – TNUIFSL			
Construction Camps, Storage, and Disposal Areas	Poorly managed camps and storage areas may cause vegetation loss, soil contamination, water	• Avoid camps/storage in wetlands, parks, CRZ, or urban green spaces; maintain ≥ 50 m from rivers, canals, drains. • Provide sanitation facilities with septic systems.	Construction Contractor	PMU – TNUIFSL	Layout approval, waste disposal records, sanitation and fuel storage inspection reports.	Contractor budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
	pollution, nuisance to residents, and disturbance to urban fauna.	• Segregate and dispose of waste at government-approved sites. • Store fuels/oils in bunded, leak-proof areas. • Prefer hiring local workers; remove and rehabilitate temporary facilities post-construction.				
		• Approve layout for offices, equipment, fuel storage, waste management, and drainage.	PIU-			
Labour Accommodation and Worker Facilities (Pre-Construction Planning)	Poor camp siting; sanitation or waste mismanagement; health risks.	Identify sites for labour camps; submit layout plan with sanitation, drainage, waste management.	Contractor	EHS-CMSC; CMSC -ES; PIU	Approved camp layout; sanitation and waste systems documented; EHS plan approved.	Contractor budget
		Prepare camp EHS plan; ensure compliance with IFC/WB standards & national labour laws.	Contractor EHS Expert			
		Inspect proposed camp sites; verify compliance with EMP and local norms.	CMSC Environmental Specialist			
		Issue approvals for temporary camp establishment if required.	GCC			
		Maintain oversight of environmental and safety conditions related to labour accommodation	PIU			
Stakeholder Consultations and GRM Setup	Unaddressed community concerns; access problems; misunderstanding of project impacts.	Conduct consultations with affected persons, shop owners, residents. • Publicize GRM process, complaint numbers, contact persons. • Document consultations and ensure concerns are integrated into SEMP	CMSC, PIU	PMU; PIU	Consultation records; GRM public information displayed; issues integrated into pre-construction planning.	Included in project/consulting budget
		Participate in consultations; incorporate site-specific issues in work planning.	Contractor, PIU			
		Provide oversight to ensure ADB requirements are followed.	CMSC; TNUIFSL			
Pre-Construction Training and EHS Mobilization	Unsafe construction start due to untrained workers, improper PPE, and unprepared emergency systems.	Mobilize EHS team, procure PPE, establish first-aid facilities. emergency response.	Construction contractor	PIU; CMSC ES, CMSC EHS and PMU	Training records; PPE issued; EHS team mobilized; emergency numbers and contacts displayed.	Contractor budget
		• Conduct induction training on OHS, traffic safety • Verify workforce training records before issuing construction clearance.	Contractors EHS Expert; CMSC Environmental Specialist			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		Ensure contractor staffing and resources comply with contract requirements.	PIU & PMU			

Table 30: Anticipated impacts and mitigation measures during construction phase

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
Contractor Training and Capacity Building	Poor awareness of EMP, OHS, pollution control, GRM, community relations, emergency response.	Conduct induction, OHS, EMP, emergency response, pollution control, and GRM training; specialized sessions for high-risk tasks; maintain training records. Overview of applicable Indian laws including: – Factories Act, 1948 – Building and Other Construction Workers Act, 1996 – Water Act, 1974; Air Act, 1981 – Solid Waste Management Rules, 2016 – C&D Waste Management Rules, 2016 Contractor's internal code of conduct regarding waste disposal, restrictions in sensitive areas, and zero tolerance for harassment or misconduct.	Contractor's EHS Expert; Environmental Specialist-CMSC	TL- CMSC /PIU/ PMU	Training attendance sheets, refresher training logs, PIU/CMSC verification notes.	Included in project/contract budget
		Review training content; verify records; conduct periodic audits.	CMSC Environmental Specialist/PIU			
		• The Contractor shall maintain detailed, verifiable records of all training events including attendance, topics covered, trainer credentials, and participant evaluation. • Training records shall be available for inspection by PIU, ADB, or statutory authorities during audits or monitoring visits.	Contractor's EHS Expert;			
Construction Camps, Yards and Ancillary Areas	Vegetation loss, soil disturbance, contamination, nuisance from	• Ensure full compliance with the Guidance Note on Workers' Accommodation: Processes and Standards (August 2009[6]) for planning,	Contractor	Environmental Specialist-CMSC; Health and Safety	Camp approvals, sanitation logs, waste disposal records, site	Included in project/contract budget

[6] https://www.ifc.org/en/insights-reports/2000/publications-gpn-workersaccommodation

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
	dust/noise, waste mismanagement.	establishing, and operating all worker accommodation facilities. • Provide safe drinking water, conduct periodic testing of drinking water • Provide safe, ventilated, and structurally sound living spaces, ensuring adequate room size, lighting, and fire-safety measures. • Maintain clean and adequate sanitation facilities, including toilets and bathing areas. • Prevent discharge of wastewater into drains; ensure septic tanks are regularly de-sludge by authorized agencies. • Provide segregated solid waste bins for biodegradable, recyclable, and non-recyclable waste; hand over waste only to authorized GCC collection services or recyclers. • Prohibit burning of waste within or near worker camps. • Install lined drains, silt traps, and stormwater management systems to prevent flooding, soil erosion, and uncontrolled runoff. • Ensure safe, hygienic food services, including clean cooking areas, proper food storage, and regular inspections. • Conduct regular health check-ups, maintain vector-control measures (e.g., mosquito control), and organize hygiene awareness programs for workers. • Provide appropriate security arrangements while ensuring workers' privacy and dignity. • Display codes of conduct at visible locations; prohibit alcohol, drugs, violence, and any unlawful activities inside worker camps.		Specialist of CMSC; PIU	restoration certificates.	

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		• Ensure a functional internal grievance redress mechanism is accessible to all workers. • Continue monitoring and supervision to ensure worker health, safety, hygiene, and dignity throughout the project lifecycle.				
		• Coordinate waste collection and disposal	GCC			
		• Approved locations; monitor camp compliance.	CMSC/PIU			
Construction in Urban Areas	Traffic disruption, noise, vibration, dust, access issues, safety risks.	• Enforce speed limits of 20 km/h for construction vehicles in built-up areas. • Deploy trained flagmen at busy intersections and near schools/hospitals. • Provide barricades, reflective signage, lighting, and temporary diversions for safe pedestrian and vehicular movement. • Restrict noisy work to daytime (08:00–18:00). • Use well-maintained equipment with silencers; blasting is prohibited. • Regular water sprinkling at active sites and unpaved access roads. • Cover stockpiles; ensure trucks carrying loose material are tarpaulin-covered. • Maintain machinery to minimize exhaust emissions.	Contractor	CMSC; PIU	Daily inspection logs, dust/noise monitoring reports, TMP implementation records.	Included in project/contract budget
		Approve diversions; support traffic control.	Traffic Police/ GCC			
		• Daily monitoring of TMP compliance. • Conduct monitoring at sensitive locations to ensure compliance with CPCB norms	CMSC/PIU			
Physical Cultural Resources (PCRs)	Vibration, noise, dust impacts; chance finds; access limitations.	• All PCRs within 50 m shall be identified, mapped, and communicated to site teams. • Work schedules near religious institutions shall be discussed with temple committees/community groups to avoid festival timings.	Contractor	Environmental Specialist-CMSC; PMU, Department of Archaeology	PCR mapping records, chance find reports, vibration logs.	Included in project/ contract budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		• Intensified sprinkling and restricted working hours near PCRs. • Vibration monitoring shall be carried out as per DGMS/CMRL guidelines near old or sensitive structures. • Chance finds protocol to be implemented if cultural artifacts are discovered: • Stop work immediately and secure the site.				
		• Respond to chance find notifications.	TN Department of Archaeology			
		• Notify PIU and TN Department of Archaeology. • Resume construction only after obtaining formal clearance. • Monitor PCR handling and compliance.	CMSC/PIU			
Traffic Congestion and Mobility Restrictions	Lane closures, pedestrian inconvenience, emergency vehicle delays.	Implement TMP; diversions; signage; flagmen; avoid peak-hours.	Contractor	PIU; Traffic Police; Environmental Specialist-CMSC; Traffic Management Specialist; PMU, GCC	TMP approvals, incident reports, traffic deviation logs.	Included in project/contract budget
		TMP approval and supervision.	Traffic Police			
		• Always maintain emergency access lanes. • Coordinate with emergency services to provide real-time information about road closures. • Immediately pause work to give priority passage during emergency situations.	Contractor; Traffic Police			
		Coordinate with public transport authority.	GCC			
		Oversight and compliance monitoring.	CMSC/PIU:			
Transport of construction materials through arterial and sub-arterial corridors during pipe-laying works	Traffic congestion, reduced carriageway width, lane closures, increased travel time, elevated accident risk for pedestrians/two-wheelers, restricted emergency access, increased air and noise emissions	• Prepare a detailed Traffic Management Plan (TMP) in coordination with Chennai Traffic Police and Chennai Metropolitan Development Authority (CMDA). • Use designated haul routes, avoiding residential lanes wherever practicable. • Schedule material movement during off-peak hours. • Maintain continuous coordination with authorities for traffic diversions, temporary closures, and road safety installations.	Contractor (Traffic Manager and Site Engineer)	• Approved TMP on record before mobilization. • Weekly verification by PIU/CMSC of lane closure and diversion compliance	PIU Engineer; CMSC Transport and Safety Specialist; Chennai Traffic Police (approvals)	Included in project/ contract budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
	from idling vehicles and diversion routes, road surface damage from repeated haulage	• Ensure all trucks comply with RTO requirements, emission standards (BS-VI), and valid permits • Provide safety orientation to all drivers, including protocols in school and hospital zones. • Enforce strict speed limits on designated haul routes	Contractor (Logistics Manager; Drivers)			
		• Use only pre-approved lay-down areas for storage. • Deploy trained traffic marshals during loading/unloading. • Prohibit storage of materials on public roads or pedestrian pathways.	Contractor (Storekeeper and Traffic Marshals			
		• Provide advance notice to residents and businesses on heavy-vehicle movement and traffic diversions. • Install barricades, reflective warning signs, and temporary pedestrian walkways. • Always maintain unobstructed access for emergency vehicles.	Contractor (Community Liaison Officer; EHS Manager)			
		• Assign trained traffic supervisors for continuous field monitoring. • Maintain daily vehicle-movement logs and incident records for PIU review. • Implement corrective actions immediately in case of violations or congestion.	Contractor (Traffic Supervisors)			
Immediate Road Restoration after Trench Refilling	Delayed road restoration may cause excessive dust generation, traffic hazards, unsafe conditions, and public complaints.	• Restore road surface immediately after trench refilling to avoid exposure of loose soil. • Compact refilled soil in layers using mechanical / water-assisted compaction to achieve traffic-worthy condition. • Where permanent paving is delayed, provide temporary bituminous layer, precast slabs, or lean concrete topping. • Maintain barricades, warning signage, and reflective markers until the road is	Contractor (temporary & non-municipal permanent restoration) Highways Dept./ GCC (final restoration of highways and municipal roads)	CMWSSB/PIU /CMSC/ Engineer	• Road made motorable immediately after refilling • No prolonged dust or safety complaints • Permanent restoration completed to original or approved	Included in Contractor's BOQ (where applicable)/ CMWSSB funds for line agency works

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		motorable. **All asphalt roads and other paved areas (other than municipal roads)** shall be permanently reinstated by the Contractor in accordance with the requirements of the concerned authority, with surfacing at least equal to the pre-existing condition. • Permanent reinstatement shall be carried out as per a programme agreed with the Engineer. • Permanent reinstatement of private lands shall restore surfaces to a condition at least equal to that existing prior to entry. • Contractor shall return to site during the maintenance period to carry out any remedial works required to ensure long-term surface integrity. **Highways:** Final reinstatement of carriageway and sidewalks shall be carried out by the Highways Department; costs to be borne by CMWSSB. **Municipal Roads:** Final reinstatement of asphalt, WBM roads, and sidewalks shall be carried out by GCC/Highways Department; costs to be borne by CMWSSB. • Contractor shall ensure the road is made motorable immediately after refilling. • All costs for permanent reinstatement, remedial visits, and statutory charges shall be deemed included in the contract price, unless executed by line agencies.			condition • Maintenance-period defects rectified • Restoration completion certificates from concerned authority	
Nighttime Construction Activities	Noise, lighting disturbance, fatigue risks, safety hazards, community grievances.	• Night-time noise limits must be complied with: o Industrial: 70 dB o Commercial: 55 dB o Residential: 45 dB o Silence Zones: 40 dB • Ensure adequate illumination levels at work sites: o General work areas: 54 lux o Medium-accuracy work: 108 lux o High-accuracy activities: 216 lux	Contractor/EHS Expert	PIU; Environmental Specialist-CMSC; Traffic Police	Noise/illumination logs, PPE audits, barricade inspections, grievance records.	Included in project/contract budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		• Prefer grid electricity for lighting; where not feasible, use super-silent DG sets. • Avoid or minimize high-noise activities (cutting, hammering, crushing) during night hours. • Workers shall receive adequate daytime rest, be medically fit, and trained for night operations. • Vulnerable workers (elderly, pregnant women, and women with small children) shall not be deployed on night shifts. • All personnel must use PPE, including helmets, gloves, and retro-reflective safety vests. • Provide full traffic management arrangements including barricades, reflective signage, caution boards, channelizers, and warning lights. • Avoid use of vehicle horns; all construction machinery must have effective silencers/mufflers. • Night works should conclude at least one hour before morning peak traffic begins. • PIU/CMSC and site safety personnel shall conduct hourly monitoring of noise levels, illumination, barricading, and safety compliance. • Maintain photographic/video documentation, monitoring registers, and daily checklists. • Ensure emergency preparedness: first-aid kits, rescue equipment, fire extinguishers, and emergency vehicles must remain operational at all times. • Avoid night works near hospitals, examination centers, and during major festivals/tourist peak periods, unless special permissions are obtained. • Ensure minimal nuisance to surrounding residents by controlling noise, lights, and movement of construction vehicles.				
		Approve night work plans; conduct	CMSC			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		compliance audits; verify monitoring logs.	Environmental Specialist / PIU			
		Night traffic enforcement	Traffic Police:			
Surface Water Quality Protection	Sedimentation, turbidity, contamination, flooding, drain blockage.	• Schedule in-drain construction during dry months. • Conduct works in short, controlled segments to limit exposed soil. • Install silt fences, sediment traps, and geotextile barriers along drains and watercourses. • immediately backfill trenches crossing drains after pipe installation. • Store fuel and chemicals at least 50 m away from drains in bunded, impervious areas. • Maintain spill kits and emergency response protocols. • Daily removal of cement bags, debris, and construction materials from areas adjacent to drains.	Contractor:	Contractor; CMSC; PIU	Water quality monitoring reports, spill logs, inspection records.	contract budget
		• Water quality checks; compliance audits.	CMSC/PIU			
Construction Near Sensitive Areas (CRZ, Wetlands, NP, Ramsar)	• Noise, dust, vibration, habitat disturbance.	• Obtain all mandatory clearances (CRZ, TNPCB, Forest Department, Wetland Authority) before construction. • Follow conditions from Guindy NP, Pallikaranai Wetland Division, and CZMA authorities. • Restrict all activities strictly within existing road ROW; no diversion, storage, or encroachment into parks, wetlands, or CRZ zones. • Use HDD or micro-tunneling near sensitive habitats to avoid deep excavation. • Minimize vibration, noise, and dust near wildlife zones; avoid nighttime construction near Guindy NP and Pallikaranai marsh. • Avoid construction during early morning and evening peak wildlife activity; install wildlife-safe lighting and prohibit loud	Contractor	Environmental Specialist-CMSC; PMU	Clearance approvals, compliance logs, training records.	contract budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		machinery or horn use near the park. • Store fuels and hazardous materials at least 200 m away from wetlands; prevent any discharge of wastewater, silt, or construction runoff into marshes. • Install silt traps, sedimentation tanks, and barricades along marsh edges. • Follow CRZ norms: no stockpiling in CRZ, use pre-assembled pipe sections, and implement spill-prevention during transport near coastlines or backwaters. • Use low-noise equipment, provide temporary noise barriers, sprinkle water to suppress dust, and cover trucks carrying soil or waste. • Prevent water pollution by using drip trays for parked machinery and maintaining spill response kits at sensitive locations. • Remove all construction waste daily from sensitive areas; no dumping, burning, or temporary storage near wetlands, marshes, or CRZ zones; dispose spoil only at approved locations.				
		Provide oversight into sensitive zones.	Forest & Wetland Authorities			
		Verify adherence to clearance conditions.	CMSC/PIU			
Construction in Waterbodies/Stormwater Corridors	Sediment-laden runoff, contamination, flow obstruction, localized flooding.	• Install silt fences, sandbags, and temporary bunds downslope of excavation and stockpile areas. • Stabilize exposed earth surfaces using tarpaulins, mulch, or temporary covers. • Backfill trenches immediately after pipe laying to minimize exposed soil duration. • Schedule major earthworks during dry months where feasible. *Hazardous materials and chemical management* • Store fuels, oils, and chemicals in bunded, impervious areas at least 50 m	Contractor	Contractor; CMSC; PIU	Inspection logs, debris removal records, silt trap maintenance logs.	Included in project/contract budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		from drains or waterbodies. • Conduct refueling and maintenance only within designated contained areas. • Maintain spill kits at work sites and train workers in spill response procedures. *Wastewater and solid waste management* • Provide septic tank–soak pit systems or portable toilets; discharge of sanitary wastewater into drains is prohibited. • Segregate, collect, and dispose of construction waste and debris at authorized facilities. • Wash vehicles and equipment only in designated areas with proper containment. *Stormwater and drainage management* • Construct temporary diversion drains to route clean water around construction zones. • Provide silt traps or retention basins to remove sediment before discharge into municipal drains. *Monitoring and compliance* • Conduct regular monitoring of turbidity, pH, and oil/grease in waterbodies upstream and downstream of work sites.				
		PIU to carry out weekly inspections of erosion control measures and ensure immediate corrective actions for non-compliance.	CMSC/PIU			
High Groundwater Levels/ Dewatering	• Ground instability, trench collapse, flooding of excavations, reduced productivity, safety risks to workers • Turbid discharge and siltation in drains	• Identify high-water-table zones using geotechnical investigation and utility surveys. • Approve contractor's Dewatering and Drainage Management Plan. • Ensure provision for silt traps, sedimentation tanks, and controlled discharge in EMP.	PIU/CMSC Environmental Specialist	PIU; PMU Environmental Specialist	Indicator of Compliance: • Dewatering plan approved • Trenches kept stable and dry • No flooding or property damage • Turbidity levels within limits	Included in project/contract budget
		• Deploy well-point dewatering systems or sump-pump arrangements, depending on soil profile.	Construction – Contractor			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
	• Risk of inundation of nearby properties	• Maintain trench width, shoring, timbering, and strutting per IS codes to prevent collapse. • Pump dewatered water through silt traps before discharge to storm drains. • Avoid continuous pumping near building foundations; use staged dewatering. • Keep standby pumps and generators to prevent sudden flooding of trenches.				
		• Verify discharge quality (turbidity, silt load). • Inspect shoring and groundwater control daily.	PIU/CMSC / Environmental Specialist			
Soil Erosion and Sediment Control	Topsoil loss, trench instability, erosion, sedimentation into drains/ waterbodies.	• Limit soil disturbance to the minimum area required for construction. • Phase works to reduce exposed soil duration. *Soil stockpiling* • Locate stockpiles away from drains and waterbodies. • Cover with tarpaulin and install silt fencing around stockpiles. *Structural controls* • Provide check dams, diversion drains, and silt fences to intercept runoff. • Ensure proper regrading and compaction of backfilled trenches. *Progressive rehabilitation* • Stabilize disturbed areas with vegetation once works are complete. • Use bioengineering techniques (coir logs, erosion mats) on slopes.	Contractor	Environmental Specialist-CMSC; PIU	Erosion control logs, inspection sheets.	Included in project/contract budget
		Regular erosion monitoring and enforcement.	CMSC/PIU			
Air Quality Impacts	Dust emissions, equipment exhaust affecting workers and residents.	• Regular water sprinkling at work sites and on unpaved roads. • Cover material stockpiles with tarpaulin. • Cover trucks transporting loose material. • Enforce vehicle speed limits. *Emission control*	Contractor	Environmental Specialist-CMSC; PMU	Dust monitoring logs, equipment maintenance records.	Included in project/contract budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		• Maintain all machinery as per manufacturer guidelines. • Avoid idling; switch off engines when not in use. *Work sequencing* • Carry out excavation, laying, and backfilling sequentially to reduce the duration of exposed soil.				
		Air quality monitoring and compliance review.	CMSC/PIU			
Noise and Vibration	Increased noise; impacts to schools, hospitals, PCRs; structural vibration.	• Restrict noisy activities to 08:00–18:00 hours. • Schedule high-noise and high-vibration activities during daytime hours to minimize disturbance to residents, schools, hospitals, and religious institutions. • Avoid continuous operation of noisy equipment near sensitive receptors • Avoid high-noise works during festivals, examinations, or sensitive events. • Maintain equipment and fit silencers on exhausts. • Avoid blasting; use mechanical cutting for hard material. • Install temporary acoustic barriers, screens, or enclosures around noisy construction areas where activities are close to sensitive receptors. • Use berms, hoardings, or site fencing to reduce sound propagation. • Limit horn use to safety purposes only.	Contractor	Environmental Specialist-CMSC; PMU	Noise/vibration logs, equipment maintenance reports.	Included in project/contract budget
		• Inform nearby residents, institutions, and hospitals about planned high-noise and vibration activities in advance. • Establish a grievance redress mechanism for complaints related to noise and vibration. • Conduct regular noise and vibration monitoring at sensitive locations to	CMSC/PIU			

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
		ensure compliance with prescribed standards. • Implement corrective actions immediately if monitoring indicates exceedances				
Community Health and Safety	Pedestrian risks, open trenches, machinery movement, sanitation issues, community conflicts.	• Enforce 20 km/h speed limit near settlements and schools. • Deploy flagpersons at junctions and high-risk areas. • Park vehicles only in designated areas. • install barricades, reflective tapes, and warning signage at all sites. • Backfill or cover open trenches near public areas within 48 hours. *Workers conduct and public health* • Enforce a worker code of conduct. • Provide sanitation, waste management, and health protocols at camps.	Contractor	Environmental Specialist-CMSC; Health and Safety Specialist CMSC; PMU	Incident reports, GRM logs, inspection records.	Included in project/contract budget
		• Provide advance notifications of work schedules.	Contractor, GCC			
		• Ensure GRM access for complaints and prompt resolution.	CMSC/PIU			
Access disruption to shops, houses, and institutions	Temporary loss of access may affect residents, businesses, schools, hospitals, and emergency services.	• Ensure safe, continuous pedestrian access through temporary walkways with hand rials, metal planks, and barricaded passages. • Provide temporary access ramps for elderly persons, differently abled persons, and delivery services. • Coordinate works with hospitals, schools, and commercial areas to minimize inconvenience. • Avoid full closure of entry points; where unavoidable, ensure short-duration, notified closures. • Give 48–72 hours advance notice before work in front of shops or residences.	Contractor	Environmental Specialist-CMSC; PIU		
Utility-Rich Corridors/Utility Risk Management	• Electrocution, gas leaks, water main bursts, fiber-optic	• Prepare and approve joint utility verification map. • Conduct physical validation using GPR,	PIU/CMSC/Utility Departments	Environmental Specialist-CMSC; PMU	• Zero electrocution/gas leak incidents	Included in project/contract budget

Field	Anticipated impact	Mitigation measures	Responsible for implementation	Responsible for monitoring	Indicators of compliance	Cost and source of funds
	damage • Service interruptions to residents and businesses • Safety hazards for workers and public	trial pits. • Obtain NOCs and shutdown/relocation schedules.		Utility Departments	• Updated utility map • Utility strike log maintained • Approvals/NOCs available onsite	
		• Follow approved Utility Risk Mitigation plan.• Manual digging near utilities; mechanical excavation allowed only outside buffer zones. • Use insulated tools, gas detectors, and safety observers during trenching. • Avoid exposing live electricity lines; maintain safe distance per CEIG requirements. • Report any utility strike immediately; stop work until reinstatement. • Provide protective encasement or supports for exposed utilities. • Coordinate with utility owners before crossing, connecting, or supporting lines.	Construction – Contractor			
		• Daily supervision in utility-dense stretches • Utility strike register and incident reporting	PIU/CMSC / Utility engineers			
Occupational Health and Safety (OHS)	Worker injuries, machinery hazards, fatigue, communicable diseases.	HIRA; PPE; sanitation; emergency response; first aid; drills; safe excavation; toolbox talks.	Contractor/ EHS Expert	Contractor; EHS Expert; CMSC; PIU	HIRA reports, PPE records, OHS inspection logs, drill reports.	Included in project/contract budget
		OHS audits; incident investigations; safety compliance checks.	CMSC/PIU			

Table 31: Anticipated impacts and mitigation measures during post construction phase

Field	Anticipated impact	Mitigation Measures	Responsible for implementation	Responsible for monitoring	Indicator of compliance	Cost and source of funds
Site Closure and Demobilization	Soil, stormwater, and waterbody contamination from leftover materials; visual blight; safety hazards from open excavations, spoil piles, or leftover temporary structures.	Prepare and implement a site demobilization and restoration plan for all completed stretches. • Remove all construction materials, debris, packaging, wood/steel scraps, and temporary structures. • Collect and dispose hazardous waste (oil, grease, paints, solvents, contaminated soil) only at authorized TSDF facilities per Hazardous and Other Wastes Rules, 2016. • Backfill, grade, compact, and stabilize all excavated and disturbed areas. • Restore all temporary facilities (camp sites, site offices, storage yards, access roads) to pre-construction condition. • Replant trees and shrubs as per compensatory plantation plan	Contractor	PMU/CMSC GCC; CMWSSB	• Final inspection report approved by PIU.• Photographic documentation of restored areas.• Records of waste disposal at authorized facilities.• Certification of site restoration and plantation completion.	Included in project/contract budget

Field	Anticipated impact	Mitigation Measures	Responsible for implementation	Responsible for monitoring	Indicator of compliance	Cost and source of funds
		using native species. • Stabilize embankments/slopes using bioengineering (coir logs, turfing, vegetation).				
		• Conduct joint final inspection with GCC, PIU, and relevant authorities. • Provide completion certificate only after full restoration is verified	CMSC / Local Authorities			
Long-Term Management of Project Access Roads	Unauthorized access leading to dumping, encroachment, illegal vehicle movement; erosion and sedimentation; degradation of riparian/urban green zones.	• Restore temporary access roads; remove debris and reinstate drains and cross-drainage structures. • Install lockable gates and fencing as per design.	Contractor	PMU / CMSC GCC; CMWSSB	• Gate installation and access control logs. • Signage installation records. • Periodic PIU maintenance inspection reports. • Vegetation stabilization / erosion control documentation.	Included in project/contract budget; O&M budget for long-term maintenance
		• Convert temporary access paths into controlled-use service tracks only for periodic O&M. • Keep gates locked; assign key control and access logs. • Install signage: "Restricted Access – Authorized Personnel Only"; include warning of fines for illegal dumping. • Coordinate with GCC, Forest Department, and Police for enforcement in sensitive areas. • Stabilize service tracks and adjoining shoulders using native vegetation; maintain minimum required width to prevent encroachment.	PIU / CMWSSB			

Table 32: Anticipated impacts and mitigation measures during Operation phase

Activity/ Source of Impact	Potential Impact/ Risk	Mitigation/Management Measures	Responsibility (Implementation)	Responsibility (Supervision/ Enforcement)	Cost Implication	Monitoring and Reporting Requirements
Routine operation of transmission and ring mains; pressure variations and aging pipeline assets	Water leakage, pipe bursts, loss of treated water, localized flooding and erosion, damage to nearby utilities, public-safety risks to road users, reactive maintenance and traffic disruption	• Conduct routine inspection and pressure monitoring via SCADA. • Implement preventive maintenance and timely repair of valves, joints, fittings. • Maintain buffer pressure and surge-control systems to reduce burst risk. • Develop and operationalize an Emergency Response Plan (ERP) for immediate	O&M Division of CMWSSB; O&M Contractor	CMWSSB Operations Engineers; TNUIFSL (compliance oversight)	Included in O&M budget (SCADA, preventive maintenance, ERP drills)	• Monthly SCADA pressure logs. • Number/frequency of bursts and repair response time. • Records of road restoration timelines and traffic coordination.

Activity/ Source of Impact	Potential Impact/ Risk	Mitigation/Management Measures	Responsibility (Implementation)	Responsibility (Supervision/ Enforcement)	Cost Implication	Monitoring and Reporting Requirements
		isolation of affected pipeline sections. • Schedule maintenance during off-peak hours. • Coordinate with Traffic Police for diversions and deploy barricades/signage/cones. • Restore road surfaces promptly after corrective works.				
Continuous pumping operations at pumping stations	High electricity consumption, increased GHG footprint, avoidable operational cost	• Install and operate high-efficiency pumps and motors. • Optimize pumping schedules to avoid grid peak hours. • Implement power-factor correction and conduct periodic energy audits. • Explore/implement solar or other renewable power integration.	CMWSSB O&M Electrical/Mechanical Division; Energy Manager	CMWSSB; TNUIFSL (periodic reporting)	Included in O&M cost; renewable energy subject to capex allocation	• Monthly energy-consumption trend. • Energy audit reports. • PF-correction records and renewable-energy share.
Operation of pressurized transmission grid; stagnation in long stretches; storage at OHT/UGT	Water-quality deterioration, biofilm formation, low residual chlorine, public-health risk, possible contamination intrusions due to low pressure; unauthorized access to elevated/underground tanks leading to injury or tampering	• Continuous monitoring of turbidity, residual chlorine, pH, and other key indicators. • Routine flushing of transmission mains to avoid stagnation. • Maintain adequate operating pressure to prevent ingress. • Apply chemical dosing at WTP as per CPHEEO standards. • Restrict access to OHT/UGT via fencing, locked gates, signage. • Deploy security/ surveillance as required.	CMWSSB Water Quality Division and O&M Units; Security Contractor	CMWSSB O&M Engineers; TNUIFSL (oversight through periodic audits)	Included in O&M and WTP-chemical budget; security cost where applicable	• Daily chlorine/turbidity logs. • Storage security inspection reports. • Grievance and incident records.

Activity/ Source of Impact	Potential Impact/ Risk	Mitigation/Management Measures	Responsibility (Implementation)	Responsibility (Supervision/ Enforcement)	Cost Implication	Monitoring and Reporting Requirements
		• Conduct routine inspection of barriers, signage, and access-control assets.				
Operation and maintenance workforce—confined spaces, pump rooms, electrical systems, chemical dosing points	Mechanical, electrical, and chemical exposure; confined-space hazards; electrocution; accidents during valve-chamber entry; inadequate preparedness for emergency scenarios	• Provide task-specific PPE (helmets, gloves, gas detectors, harnesses, electrical protection). • Implement Confined-Space Entry Procedures with atmospheric testing prior to entry. \• Apply Lock-out/Tag-out (LOTO) for all electrical maintenance. • Conduct periodic safety training, drills, and awareness sessions.	CMWSSB O&M Division; Contractor-supplied Maintenance Crews	CMWSSB Safety Officer; CMSC/Third-party Safety Auditor (if applicable)	O&M safety budget (PPE, training, gas detectors)	• PPE inventory and issuance records. • Confined-space entry permits. • LOTO registers. • Safety-training attendance sheets and drill reports.

Table 33: Pre-construction and construction stage environmental monitoring plan

Sample	Monitoring Location	Responsibility	Parameter to monitor	Frequency	Cost (INR) and USD
Ambient air quality	9 Sampling locations	Construction Contractor in Coordination with PIU	PM_{10}, $PM_{2.5}$, SO_2, NO_2 and CO	Once before the start of construction. Three seasons in a year for 4 years during construction	109 samples x 5500 per sample = ₹599,500 (USD 6674.46)
Noise level	9 Sampling locations	Construction Contractor in Coordination with PIU	Noise level: Daytime and nighttime noise, dB(A)	Once before the start of construction. Three seasons in a year for 4 years during construction	109 samples x 1500 per sample = ₹163,500 (USD 1820.31)
Surface water quality	9 Sampling locations	Construction Contractor in Coordination with PIU	River water quality – standard parameters	Once before the start of construction. Three seasons in a year for 4 years	109 samples x 5000 = ₹554,500 (USD 6173.46)

Sample	Monitoring Location	Responsibility	Parameter to monitor	Frequency	Cost (INR) and USD
				during construction	
Construction disturbances, nuisances, public and worker safety	All work sites	Supervising staff and safeguards specialists of PIU	Implementation of construction stage EMP including dust control, noise control, traffic management, and safety measures. Site inspection checklist to review implementation is appended	Weekly during construction	Staff and consultant costs are part of the incremental administration costs

A. Institutional arrangements and implementation schedule

338. The Program will adopt a multi-tier institutional framework that integrates state-level oversight, project-level execution, and independent construction-phase supervision. The approach draws on lessons learned from TNUIFSL and other urban utility programs, particularly the need for early budgeting of environmental, health, and safety (EHS) provisions, systematic monitoring, and enforceable contractor accountability.

339. **Project Executing and Implementing Agencies.** The Municipal Administration and Water Supply Department (MAWS), acting through **TNUIFSL**, will function as the Executing Agency. A Program Steering Committee chaired by the Principal Secretary (MAWS) will provide policy guidance, approve key environmental decisions, and review compliance performance.

340. A strengthened Project Management Unit (PMU) will be established within TNUIFSL, headed by the Chairperson and Managing Director, TNUIFSL, acting as the Project Director, for overall management, planning, implementation, monitoring, reporting, and coordination of the Chennai climate resilient water security and sewerage project (CCRWSSP). As part of the PMU structure, a dedicated Environmental and Social Safeguards Cell/Unit will be established within TNUIFSL to oversee compliance with ADB Safeguard Policy Statement (2009) and applicable national environmental and social regulations. The Consultant Management and Supervision Consultant (CMSC) will support TNUIFSL in establishing and operationalizing this safeguards cell, including staffing, systems, and procedures.

341. The PMU, through the Safeguards Cell, will be responsible for overall project planning, management, coordination, supervision, and progress monitoring with specific accountability for environmental and social safeguard compliance. The Safeguards Cell, comprising an Environmental Officer and an additional mobilized environmental safeguards personnel, will implement a due-diligence and compliance regime and undertake the following environmental safeguards responsibilities:

(i) Ensure compliance with Government and ADB environmental safeguard requirements across all project components, including regulatory compliance and statutory clearances.
(ii) Review updated or final IEEs, environmental screening checklists, categorization forms, EMPs, and contractor SEMP submissions, and submit required documents to ADB for approval.
(iii) Coordinate with design engineers and PIUs to incorporate avoidance, minimization, and mitigation measures in detailed engineering designs, and confirm that updated IEEs and EMPs reflect in final designs.
(iv) Ensure that EMP requirements, including Health and Safety provisions and associated costs, are incorporated in bidding documents, civil works contracts, and contractor pricing.
(v) Support continuous assessment of environmental and OHS cost provisions and ensure their adequacy during implementation.
(vi) With support from the CMSC and PIUs, review contractor SEMPs, monitor performance indicators, verify compliance records, and approve corrective or preventive actions as required.
(vii) Provide oversight to PIUs and contractors on implementation of SEMPs, EMPs, and Health and Safety Plans, and ensure full compliance during construction.
(viii) Establish and maintain a system for environmental safeguards monitoring, including tracking performance indicators prescribed in the EMP monitoring plan.

(ix) Facilitate and confirm compliance with environmental regulatory requirements, including permits, consents, and clearances.
(x) With support from CMSC environmental experts, review and evaluate the effectiveness of mitigation measures, environmental performance, and Health and Safety implementation, and recommend corrective measures.
(xi) Prepare and submit Semi-Annual Environmental Monitoring Reports (SEMRs) to ADB.
(xii) Establish the grievance redress mechanism (GRM), identify Grievance Redress Committee (GRC) members, and build the capacity of GRC members, PIUs, consultants, and contractors in addressing environmental safeguard issues.
(xiii) Ensure timely receipt, review, resolution, and disclosure of grievances through the GRM, and maintain oversight of GRM performance.
(xiv) Confirm compliance with mitigation measures and corrective or preventive actions identified in IEEs, EMPs, SEMRs, or safeguard audits.
(xv) Ensure timely disclosure of final IEEs, EMPs, SEMRs, and corrective action plans on the project website in a publicly accessible manner.
(xvi) Conduct periodic review of safeguard-related loan covenants and ensure compliance during implementation.
(xvii) Maintain an enforceable due-diligence regime linking environmental and OHS compliance to contractual enforcement, performance securities, and payment certification.

342. **Project implementation unit (PIU):** CMWSSB will be the IA and will establish a dedicated PIU for the project headed by a Superintending Engineer serving as the full-time Project Manager. CMWSSB will also establish a dedicated Safeguards and Safety Cell, led by an Executive Engineer and supported by one Environmental Engineer and one Health and Safety Engineer, to provide technical oversight to the Project Managers. PIU will include an Assistant Environmental Officer (AEO) responsible for environmental compliance, who will serve as part of, and report functionally to, Safeguards and Safety Cell.

343. The PIUs will comprise dedicated personnel tasked with continuous oversight of construction activities and contractor performance. The PIUs will be supported by a Construction Management and Supervision Consultant (CMSC) recruited by TNUIFSL. For the institutional capacity-building, public awareness, and sector reform components, TNUIFSL will recruit a separate consultant to support CMWSSB in implementing these activities.

344. The environment and safety cell will (i) provide technical oversight to PIUs, including field supervision of EMP/SEMP implementation; (ii) verify contractor staffing and resources prior to site handover; (iii) ensure compliance with statutory clearances, permits, and CRZ approvals; (iv) monitor OHS, Community Health and Safety (CHS), traffic management, and public-interface risks on a daily basis; and (v) submit quarterly reports to the PMU and coordinate with regulatory stakeholders.Each PIU will be following responsibilities:

(i) Ensure compliance with Government and ADB environmental safeguard requirements at the site level, including establishment of procedures for systematic compliance monitoring.
(ii) Review final designs to confirm that IEEs and EMPs remain current, reflect detailed engineering designs, and incorporate any changes in alignment, location, or scope; confirm to the PMU that the environmental category remains unchanged.
(iii) Liaise with local regulatory agencies for clearances and approvals and confirm to the PMU, prior to contract award, that all statutory clearances are in place.

(iv) Coordinate acquisition of rights-of-way in accordance with regulatory and safeguard requirements.
(v) Review and approve contractor Site Environmental Management Plans (SEMPs) and associated Health and Safety Plans.
(vi) Oversee day-to-day implementation of SEMPs and Health and Safety Plans by contractors; conduct routine inspections and verify compliance with Government regulations and EMP requirements.
(vii) Ensure contractors comply with (a) environmental and OHS mitigation measures in the IEEs and EMPs; (b) corrective and preventive actions; (c) budget allocations for environmental and social measures; (d) notification requirements for unanticipated impacts; and (e) reinstatement of community assets, access routes, and agricultural land to pre-project condition.
(viii) Verify pre-construction documentation on road conditions, utilities, and community infrastructure and ensure reinstatement post-construction.
(ix) Ensure provision of OHS induction and periodic training to all workers and enforce a no-work-without-PPE policy.
(x) Review contractors' monthly reports on SEMP implementation, maintain non-compliance records, and recommend contractual remedies as required.
(xi) Prepare and submit quarterly environmental and OHS monitoring reports to the PMU.
(xii) Notify the PMU of unanticipated impacts and recommend corrective action plans.
(xiii) Recommend issuance of construction completion certification following satisfactory environmental clean-up.
(xiv) Ensure continuing public consultation and community awareness during implementation.
(xv) Coordinate the grievance redress process, ensure timely action on complaints, and maintain GRM records.
(xvi) Support all other environmental safeguard-related functions assigned by the PMU.

345. **Program Steering Committee:** The PSC will be chaired by the Principal Secretary, MAWS, and will include:

- Managing Director, TNUIFSL (Convener)
- Managing Director, CMWSSB

346. **Cost of EHS and Traffic-Management Provisions**. Bid and contract documents will incorporate dedicated, measurable, and priced EHS and traffic-management provisions in the Bill of Quantities. These provisions will include trench protection and shoring, pedestrian access platforms and safe crossings, safety signage and lighting, dust and noise suppression, temporary drainage and stormwater diversion, and emergency access and community-protection barriers. These mandatory cost items are intended to eliminate cost avoidance, ensure early budget allocation, and prevent subsequent claims for variations related to environmental and safety compliance.

347. **The Construction Management and Supervision Consultant (CMSC)** will provide comprehensive environmental safeguards, occupational health and safety (OHS), and community safety support to TNUIFSL and CMWSSB. The CMSC team will include dedicated safeguards and safety personnel with clearly defined intermittent and full-time inputs, as follows:

(i) **One Environmental Safeguard Specialist (**full time**)** responsible for overall environmental safeguards oversight, EMP/IEE compliance, and coordination with the PMU and ADB;

(ii) **Two Environmental Safeguard Assistants (full-time)** deployed for day-to-day site monitoring, documentation, contractor coordination, and reporting support at package and PIU levels; (iii) One Chief Safety Expert (intermittent) to provide strategic leadership on occupational and community health and safety across all construction packages; (iv) Two Safety Engineers (civil/construction – full-time) for continuous site-level safety supervision and contractor performance monitoring; and (v) One Safety Engineer (Mechanical & Electrical – full-time) to oversee safety aspects related to pumping stations, E&M works, DG sets, lifting equipment, and electrical installations.The PMU Environmental Officer and PIU Assistant Environmental Officers will be supported by the CMSC Environmental Safeguards Specialist and the full-time Environmental Safeguards Assistants. In parallel, the Chief Safety Expert, Safety Engineers, and Safety Inspectors will support PIUs and contractors in implementing OHS and community safety requirements, ensuring consistent application of safety standards across all project components.

348. Tasks of the Environment Specialist (CMSC)

(i) Environmental Assessment and Regulatory Compliance
- Support PMU in environmental screening and updating ADB REA checklists.
- Integrate environmental management requirements into engineering design, bid documents, and technical specifications.
- Identify statutory environmental approvals and assist PMU/PIU in obtaining clearances and permissions.
- Review contractor SEMPs and update/finalize IEEs and EMPs based on EPC detailed designs.
- Incorporate regulatory and consent-related conditions into implementation plans.

(ii) Stakeholder Engagement and Disclosure
- Lead or support PIUs in structured public consultations and ensure feedback integration into EMPs.
- Assist PIUs in disclosure of safeguard information to affected communities and stakeholders.

(iii) Monitoring, Verification, and Corrective Actions
- Review and clear contractor SEMPs, Health and Safety Plans, and associated method statements from a safeguard perspective.
- Monitor EMP/SEMP implementation and verify field-level compliance.
- Conduct scheduled and unscheduled field inspections to confirm mitigation implementation.
- Identify non-compliance and unanticipated impacts, recommend corrective actions, and report to PMU.
- Update IEEs as required based on design changes or new impacts.
- Assist PIUs in reviewing monthly contractor monitoring reports and in preparing quarterly and semi-annual environmental monitoring reports.

(iv) Capacity Building, GRM, and Gender and Inclusion Measures
- Identify capacity-building needs and conduct training programs for PIU and contractors on environmental safeguards.
- Support PIUs in operating the GRM, ensuring timely documentation, escalation, and closure of grievances.
- Support implementation of gender equality and social inclusion-related actions embedded in the EMP.

349. Tasks of Chief Safety Expert

(i) Occupational and Community Health and Safety Oversight
- Monitor contractor implementation of approved Health and Safety Plans, emergency response procedures, and community safety controls.
- Ensure safe execution of construction activities involving utilities, traffic interfaces, and pedestrian-intensive corridors.
- Verify compliance with occupational safety standards, PPE use, and safe working practices.

(ii) Field inspection, audits, and reporting
- Conduct regular safety audits and daily/weekly inspections of construction work fronts.
- Identify hazards, unsafe acts, and unsafe conditions and issue corrective directions to contractors.
- Record and report OHS non-compliances, near misses, and accidents to PIUs and PMU, and track closure of corrective actions.

(iii) Support to EMP/SEMP Implementation
- Assist contractors and PIUs in implementing construction-related EMP/SEMP mitigation measures.
- Coordinate safe handling of materials, equipment, spoil, and hazardous substances.
- Verify implementation of traffic-management-related mitigation measures where community safety is involved.

(iv) Training, Awareness, and GRM Interface
- Deliver routine OHS toolbox talks, inductions, and awareness programs for workers and subcontractors.
- Support PIUs in building a safety culture at each worksite and in documenting good practices.
- Coordinate with the PIU for the registration and follow-up of construction-related safety grievances under the GRM.

350. In addition, CMSC will engage other technical experts on an intermittent basis to ensure smooth planning and implementation of works. This will include:

(i) **One Traffic Management Expert (intermittent)** to support PIUs and contractors in planning, reviewing, and monitoring Traffic Management Plans (TMPs), traffic diversions, pedestrian safety, and coordination with traffic authorities; and

.. Tasks of TMS will be –

a. Identify and assess traffic- and road-safety-related risks arising from construction activities;

b. Review, strengthen, and advise on contractors' Traffic Management Plans (TMPs) in line with national regulations and international best practices;

c. Guide planning and implementation of traffic diversions, access management, barricading, signage, lighting, and pedestrian safety measures;

d. Coordinate with PIUs, contractors, and local traffic police/authorities for approvals and on-ground implementation of traffic arrangements;

e. Monitor effectiveness of TMP implementation at site and recommend corrective measures where required;

f. Provide training and capacity building to traffic supervisors, flagmen, and site staff on safe traffic control and road safety practices;

g. Support PIUs in addressing traffic-related complaints and grievances under the GRM and ensure timely resolution and documentation; and
h. Assist PIUs and PMU in any other traffic management and road safety–related tasks as required during the construction phase.

351. **Contractor**. The approved IEEs and EMPs will be included in the bidding and contract documents and will be verified by the PIUs and PMU. The PMU and PIUs will ensure that bidding and contract documents include specific provisions requiring contractors to comply with: (i) all applicable laws and regulations relating to environment, health and safety; (ii) reinstate pathways, other local infrastructure, and agricultural land to at least to their pre-project condition upon the completion of construction; and (iii) all applicable labor laws on recruitment and working conditions including ADB's Core Labor Standards on (a) prohibition of child labor as defined in national legislation, international treaties for construction and maintenance activities;(b) equal pay for equal work of equal value regardless of gender, ethnicity, or caste; (c) no discrimination in respect of employment and occupation; (d) allow freedom of association and effectively recognize the right to collective bargaining, and (e) elimination of forced labor; and with (ii) the requirement to disseminate information on sexually transmitted diseases, including HIV/AIDS, to employees and local communities surrounding the project sites.

352. The contractor will be required to appoint a full-time Environment, one Health and Safety (EHS) supervisors supported by field site safety stewards to implement the EMP. Prior to start of construction, Contractor will be required to prepare and submit to PMU and PIU, for review and approval, site-specific EMP (SEMP) which includes (i) proposed sites/locations for construction work camps, storage areas, hauling roads, lay down areas, disposal areas for solid and hazardous wastes; (ii) specific mitigation measures following the EMP in approved draft and final EMP; (iii) monitoring program per EMP; and (iv) budget for SEMP and EMP implementation. No work can commence until the SEMP is approved by the PMU/PIU. Contractors will carry out all the environmental mitigation and monitoring measures outlined in EMP, approved SEMP and their contracts. The contractor will be required to undertake day-to-day monitoring of the implementation of the SEMP and will submit reports to the PIU monthly. Contractor to develop IT based and real-time integrated compliance monitoring and reporting system.

353. A copy of the EMP/approved SEMP will be always kept on-site during the construction phase. Non-compliance with, or any deviation from, the conditions set out in the EMP/SEMP constitutes a failure in compliance and will require corrective actions. The contractors will be required to conduct environmental awareness and orientation of workers prior to deployment to work sites.

354. Key responsibilities of the EHS supervisor are:

(i) Prepare SEMP and submit to PMU/PIU for approval prior to start of construction;
(ii) Ensure implementation of SEMP and report to PIU/CMSC on any new or unanticipated impacts; seek guidance from the PMU/PIU/CMSC to address the new or unanticipated impact in accordance with ADB SPS 2009;
(iii) Ensure that necessary pre-construction and construction permits are obtained;
(iv) Conduct orientation and daily briefing sessions to workers on environment, health and safety;
(v) Ensure that appropriate and adequate worker facilities are provided at the workplace and labor camps as per the contractual provisions;
(vi) Carry out site inspections on a regular basis and prepare site-inspection checklists/reports;

(vii) Record EHS incidents and undertake remedial actions;
(viii) Conduct environmental monitoring (air, noise, etc.,) as per the monitoring plan
(ix) Prepare monthly EMP monitoring reports and submit to PIU;
(x) Work closely with PIU assistant environmental officer (AEO) and CMSC Executive Engineer (EE) to ensure communities are aware of project-related impacts, mitigation measures, and GRM; and
(xi) Coordinate with the PIU and CMSC on any grievances received and ensure that these are addressed in an effective and timely manner.

Figure 37: Institutional Arrangement for Environmental Safeguards

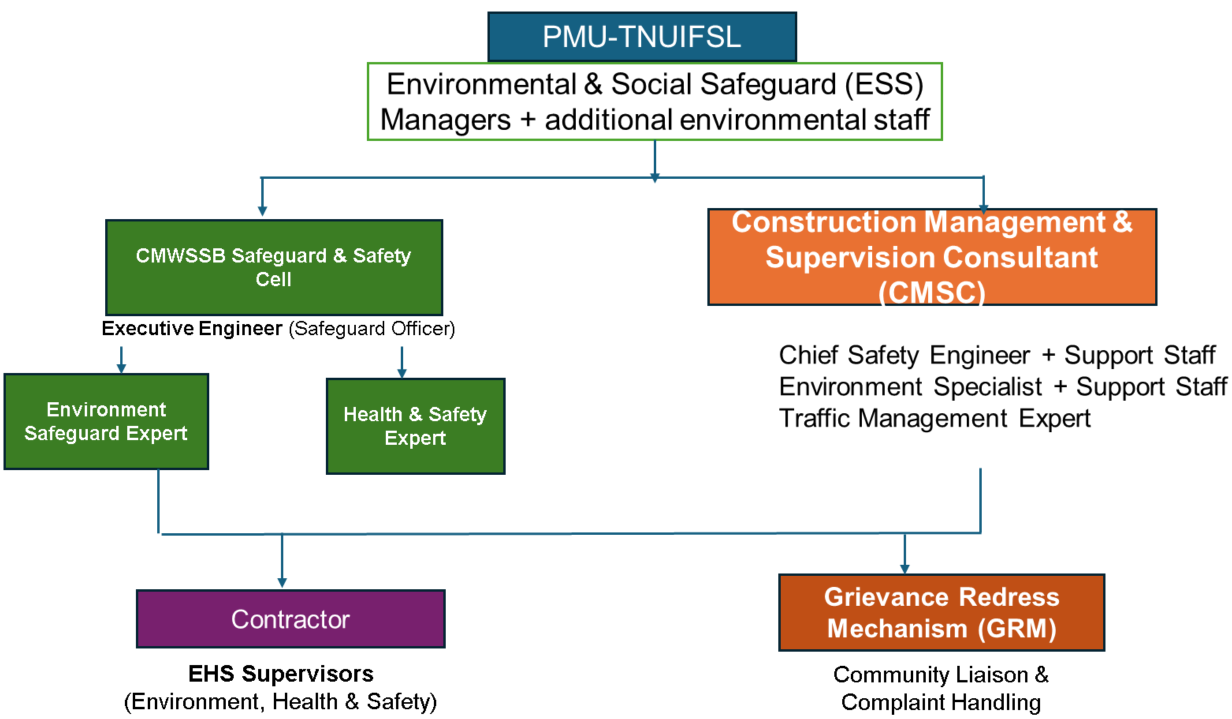


Table 34: Activity and Responsibility – Safeguard Implementation

Project Phase	Activity	Details	Responsible Agency
Pre-construction phase	Project Categorization	Conduct REA for each subproject using the REA checklists	TNUIFSL
		Review the REA and assign project category (A/B/C) based on ADB SPS 2009	PMU
	Conduct environmental assessment	Prepare IEE based on environment category B project categorization and requirements of SPS 2009	Project Consultants
	Project clearances	Fulfilling GoTN/GoI requirements such as clearances from other government agencies	CMWSSB

Project Phase	Activity	Details	Responsible Agency
	Review of IEE	Reviewing IEE Reports to ensure compliance with the requirements of SPS 2009 Submit IEE to ADB for review and clearance	PMU
	Disclosure of IEE	Disclose IEE reports to ADB project websites IEE reports will be made available to the public through the project site office	CMWSSB, ADB
	Incorporation of mitigation measures into Project design	Incorporation of necessary mitigation measures identified in IEE in Project design and in contract documents.	Project Consultants
	Review of design documents	Review of design and contractual documents for compliance of mitigation measures	PMU
Construction Phase	Implementation of mitigation measures	Implementation of necessary mitigation measures	Contractor
	Environmental Monitoring	Environmental monitoring as specified in the monitoring plan during construction stage; monitoring of implementation of mitigation measures	Contractor/ CMSC
	Preparation of progress reports	Preparation of monthly progress reports to be submitted to PMU including a section on implementation of the mitigation measures	CMSC
	Review of progress reports	PMU to review the progress reports, consolidate and send to ADB review Prepare environmental monitoring reports for submission, review, and disclosure by ADB	PMU
Operation phase	Implementation of EMP (O&M Stage)	Execution of mitigation measures related to noise, odor, sludge management, and treated effluent quality as defined in the IEE.	CMWSSB / Operator
	Occupational Health & Safety (OHS)	Ensuring staff use PPE, follow safety protocols for chemical handling (e.g., Chlorine), and undergo regular safety drills.	CMWSSB / Operator
	Grievance Redressal	Addressing community complaints regarding leaks, odors, or service disruptions through the established GRC.	PMU / CMWSSB
	Emergency Response Management	Implementing protocols for natural disasters, pipe bursts, or hazardous chemical spills.	CMWSSB

B. Capacity Development

355. The Project Management Unit (PMU) for the Chennai climate resilient water security and sewerage project (CCRWSSP), established under the Municipal Administration and Water Supply Department (MAWS) through TNUIFSL, consists of experienced staff with basic understanding of environmental and social safeguards, including familiarity with ADB's SPS 2009. However, new staff, PIU engineers, safeguard officers, and personnel assigned to community safety and traffic management functions will require structured training to effectively manage safeguard implementation and construction-stage risks.

356. The CMSC team—comprising of Environmental, Social, Gender, Community Safety, and Traffic Management Experts—will be responsible for building the capacity of PMU and PIU safeguard teams, including site engineers and contractor EHS staff. All trainings will be aligned with the EMP, SEMPs, and the requirements of SPS 2009. Training modules will be updated as detailed designs.

357. The PMU, PIUs, CMSC, and field-level project staff will be trained in the following key areas:

(i) Orientation on ADB Safeguards and Project-Specific Requirements-Training will cover the SPS 2009 requirements for environment, involuntary resettlement, and indigenous peoples; meaningful consultation processes; GRM operations; and responsibilities for safeguard implementation at PMU/PIU/contractor levels.
(ii) Environmental and Social Impact Assessment, and EMP Implementation -Training modules will include environmental impact assessment, risk identification, design-stage safeguards integration, mitigation measures, monitoring requirements, and best practices in construction management for water supply transmission mains, pumping stations, and distribution networks.
(iii) Preparation and Review of Safeguard Documents - Participants will be trained in the preparation and updating of IEE, EMP, DDRs, and safeguards-related reports based on detailed engineering designs, including data collection, monitoring formats, and documentation standards.
(iv) Compensation Disbursement and Social Compliance (if applicable) - Where required, training will include compensation disbursement procedures, livelihood restoration support, and social compliance requirements.
(v) Monitoring, Reporting, and Documentation - Participants will be trained in environmental and social monitoring tools, reporting formats, compliance checklists, contractor performance evaluation, and preparation of monthly, quarterly, and semi-annual/annual EMRs for submission to ADB.

358. A detailed capacity development program, including training modules, schedule, and responsibilities will be developed and customized by the CMSC environmental and social specialists based on the existing skill levels of PMU/PIU staff. Participatory learning methods—such as group exercises, role-playing, field demonstrations, and on-the-job training—will be emphasized to enhance practical understanding.

359. A pre- and post-training assessment will be conducted to evaluate the effectiveness of the capacity-building program and to identify additional training needs during project implementation.

360. Table 35 presents the outline of capacity building program to ensure smooth implementation of the EMP. The estimated cost per year will be is INR 200,000/ USD 2226.68 (excluding trainings of contractors which will be part of EMP implementation cost during construction) to be covered by the project's capacity building program. The detailed cost and specific modules will be

customized for the available skills set after assessing the capabilities of the target participants and the requirements of the project by the environmental specialist of CMSC. The capacity building program will consider participatory learning methods to the extent possible, including learning by doing, role playing, group exercises, on-the-job training, etc. to ensure effectiveness. A post-training assessment that can be compared to the pre-training assessment may be administered to measure the effectiveness of the program.

Table 35: Indicative Training Needs for Environmental Safeguards

Description	Target Participants and Venue	Estimate (INR and USD)	Source of Funds
1. Introduction and Sensitization to Environmental Issues (1 day) ✓ ADB Safeguards Policy Statement 2009 ✓ Government of India and Tamil Nadu applicable safeguard laws, regulations and policies including but not limited to core labor standards, OH and S, etc. ✓ Incorporation of EMP into the project design and contracts ✓ Monitoring, reporting and corrective action planning	All staff and consultants involved in the project at PMU/PIU, Chennai	INR 50,000 (USD 556.67) (LS)	PMU
2. Preparing and implementing SEMR (1/2 day - once at the beginning and at a frequency of once in six months during implementation) ✓ site-specific mitigation & monitoring measures ✓ Roles and responsibilities ✓ Public relations, ✓ Consultations ✓ Grievance redress ✓ Monitoring and corrective action planning ✓ Reporting and disclosure ✓ -Construction site standard operating procedures (SOP) ✓ -Chance find (archaeological) protocol ✓ AC pipe protocol ✓ Traffic management plan ✓ Waste management plan ✓ - Site clean-up & restoration	All staff and consultants involved in the project All contractors immediately after mobilization of the contractor at PIUs	INR 100,000 (USD 1113.34) (LS)	PMU
3. Contractors Orientation to Workers (1/2 day) ✓ Environment, health and safety in project construction (O H and S, core labor laws, spoils management, etc.)	Once before the start of work, and thereafter regular briefing every month once. Daily briefing on safety prior to start of work All workers (including unskilled laborers)	INR 50,000 (USD 556.67) (LS)	Contract or s cost
Total for one year		INR 200,000 (USD 2226.68)	
Total cost for 4 years		800,000 (USD 8906.7)	

Summary of Capacity Building cost for EMP Implementation
Contractor Cost - INR 200,000 (USD 2226.68)
PMU Cost - INR 600,000 (USD 6680.03)
Total - INR 800,000 (USD 8906.70)

C. Monitoring and Reporting

361. Immediately after mobilization and before commencement of construction activities, the contractor will submit a Pre-Construction Compliance Report to the PIU confirming that all pre-construction mitigation measures specified in the EMP have been implemented. This includes mobilization of key safeguard staff such as the EHS Manager, Community Safety Supervisor, and Traffic Supervisor. The PIU, with support from the CMSC, will review and approve the contractor's site environmental management plan (SEMP) before works begin.

362. During construction, the contractor will conduct regular internal monitoring of environmental, health and safety, community safety, and traffic management measures. Findings will be documented in a Monthly EMP Implementation Report and submitted to the PIU. The CMSC will verify compliance, conduct field inspections, and provide corrective action recommendations where required.

363. The PIU and CMSC will jointly prepare a Quarterly Safeguards Monitoring Report summarizing EMP implementation status, incidents, non-compliances, corrective actions, community safety and traffic-related observations, and monitoring results. This report will be submitted to the PMU for review and necessary actions.

364. Based on the monthly and quarterly monitoring results, the PMU—assisted by the CMSC team—will prepare and submit the semi-annual environmental monitoring report (SEMR) to ADB. After ADB's concurrence, the SEMR will be disclosed on the TNUIFSL/ CMWSSB websites for public access. Based on the periodic monitoring results during the operation phase, the **CMWSSB (assisted by the PMU)** will prepare and submit annual environmental monitoring reports **(SEMR)** to ADB. These reports will document compliance with operation phase EMP measures and legal compliance with all applicable national and state regulations. After ADB's concurrence, the SEMR will be disclosed on the **TNUIFSL/CMWSSB** and **ADB** websites for public access.

365. ADB will periodically review the project's performance against the environmental and social safeguard commitments stated in the loan covenants. The scale and frequency of ADB's supervision missions will be commensurate with the project's environmental risks and impacts. Safeguards monitoring will remain an integral part of the project performance management system throughout implementation.

D. EMP Implementation Cost

366. Most mitigation measures included in the EMP are considered part of good engineering and construction practice and are already standard requirements for contractors. As such, no major additional costs are anticipated for implementing routine environmental, health and safety, community safety, and traffic management measures.

367. Key safeguard-related requirements—such as compliance with national labor laws, worker health and safety provisions, PPE supply, insurance, equipment certifications, labor welfare facilities, and emergency preparedness—are part of the contractual obligations of contractors and therefore do not require a separate EMP budget. Costs associated with

establishing safety controls, signage, barricading, community safety measures, and traffic diversions are already embedded within the civil works contract.

368. Mitigation measures to be implemented directly by the PIU, CMWSSB, or other government agencies will be carried out as part of their regular project management responsibilities and do not require duplication under the EMP.

369. Training, awareness programs, and capacity-building activities for PMU, PIU, CMSC, and contractor staff are already budgeted under the project's capacity development component; therefore, no separate EMP cost allocation is required for these activities.

Table 36: Cost Estimates to Implement the EMP

	Particulars	Stages	Unit	Total Number	Rate ₹	Cost ₹	Cost in USD	Costs Covered By
A.	**Implementation staff**							
1	EHS Engineer	Construction	per month	48	50,000	2,400,000	26,720.11	Civil works contractor
	Subtotal (A)					**2,400,000**	26,720.11	
B.	**Mitigation Measures**							
1	Tree Plantation with 3 years of maintenance, including FRP tree guards 0.5 m diameter x 1.5 m	Construction	Per tree	750	2,850	2,137,500	23,797.60	Project costs (PIU)
2	**Safety**: Providing iron barricades of size 2.5 m width and 1.75 m height including danger flags, danger lights, reflectors, fixing stickers etc. as per standard specifications 314552 RM @ Rs 97/m	Construction				30,511,544	339,696.55	Part of the bid documents by Civil works contractor
3	**Pipe carrying bridges:** Creation of Vettiver (Chrusopogon zizanniodes) for soil stabilization to strengthen the bunds.	Construction	Lump sum	24	120,000	2,880,000	32,064.13	Contractor civil cost
4	Water sprinkling for dust suppression	Construction	Days	1,460	3,600	5,256,000	58,517.03	Contractor civil cost
	Subtotal (B)					**10,273,500**	114,378.76	
C.	**Monitoring Measures**							
1	Ambient Air Quality Monitoring – One baseline round before the start of construction. Thereafter, 9 samples location, monitoring for 4 years, covering three seasons each year	Construction	per sample	109	5,500	599,500	6,674.46	Civil works contractor
2	Ambient Noise monitoring -One baseline round before the start of construction. Thereafter, 9 samples locations, monitoring for 4 years, covering three seasons each year	Construction	Per sample	109	1,500	163,500	1,820.31	Civil work contractor
3	Surface water quality monitoring -One baseline round before the start of	Construction	Per sample	109	5,000	545,000	6,067.69	Civil work contractor

	Particulars	Stages	Unit	Total Number	Rate ₹	Cost ₹	Cost in USD	Costs Covered By
	construction. Thereafter, 9 samples location, monitoring for 4 years, covering three seasons each year							
	Subtotal (C)					**1,308,000.00**	14,562.46	
D.	**Capacity Building**							
1	Training on EMP implementation	Pre-construction				600,000	6,680.03	
2	Contractors' Orientation to Workers on EMP Implementation	Prior to dispatch to the worksite				200,000	2,226.68	
	Subtotal (D)					**800,000**	8,906.70	
E	**CRZ EMP**							
1	EMP cost for mitigation measures for CRZ clearance					**3,030,000.00#**	33,734.13	
F	**GRM and Public Consultations**							
1	Grievance Redressal Mechanism Resolution	Construction	Per year	4	100,000	400,000	4,453.35	
2	Public consultations on Environmental aspects	Construction	Per year	4	100,000	400,000	4,453.35	
	Subtotal (E)					**800,000.00**	8,906.70	
	Total (A+B+C+D +E+F)				**INR**	**18,611,500.00**	207,208.86	

#Indicative cost; the actual cost will depend on the final CRZ report and the conditions stipulated by the CRZ Committee.

X. RECOMMENDATIONS AND CONCLUSION

370. The Initial Environmental Examination (IEE) assessed the environmental impacts of all components proposed under the CMWSSB project. Potential negative impacts were identified related to design, location, construction and operation of the subproject. Negative impacts are assessed to be minimal.

371. The potential adverse environmental impacts of the proposed water supply subproject are mainly related to the construction period, which can be minimized by the mitigating measures and environmentally sound engineering and construction practices.

372. Most impacts are due to construction; this is because construction work is to be carried out within the city including densely populated areas. The important impacts identified are generation of dust and noise from construction activities; disturbance to traffic flows; impacts due to disposal of large quantities of surplus soil; disturbance and inconvenience to local people due to trenching along the road; impact on road-side hawkers and vendors; public safety; interference and damage to other infrastructure facilities.

373. These impacts are mostly temporary in nature and can be effectively avoided or mitigated by observing the proposed mitigation measures. The mitigation measures include careful alignment of pipelines in order to minimize the impacts on land, following existing alignment along roads ROW, minimizing the construction area, wetting of soil and construction area to reduce the dust; immediate transport of excess soil; beneficial use of excess soil; scheduling of activities to reduce the noise impacts; special precaution near sensitive areas like schools and hospitals, and, traffic diversions and public information to reduce the potential impacts. Implementation of proper safety measures during construction activities to ensure worker's as well as public safety.

374. Results of this IEE showed that there are no impacts that are significant or complex in nature, or that need an in-depth study to assess the impact. Thus, the subproject is unlikely to cause significant impacts. The potential impacts that are associated with design, construction, and operation can be mitigated to standard levels without difficulty through proper engineering design and the incorporation or application of recommended mitigation measures and procedures.

375. The residents of Chennai City will be the beneficiaries of this subproject. The Ring Main system will provide good quality drinking water and expected to be efficient in providing reliable and efficient water supply. In addition to improved environmental conditions, the subproject will improve the overall public health in the project area.

376. The project components are located within the area of the Chennai City Corporation (GCC). All subproject sites, including UG sumps, are situated on government-owned land, and all pipeline alignments follow existing public roads. While the broader project area includes sensitive features such as Guindy National Park, the Pallikaranai Ramsar Site, and heritage and cultural structures, no works are proposed inside these areas, and no direct impacts are anticipated. There are no forests or protected monuments located within the actual footprint of the subproject components.

377. The proposed scheme, which does not involve any new water sources or treatment plants but connects existing water treatment facilities for improved management of drinking water across the GCC area. Based on ADB SPS 2009, the project is classified as environment category B requiring an IEE. During implementation, the IEE will be updated by the PIU based on the project

detailed design to reflect any changes or amendments and will be submitted to PMU and subsequently to ADB for review and disclosure to their website.

378. **Conclusions and Recommendations.** The following recommendations are applicable to the Chennai Ring Main and Transmission Main subproject to ensure that no significant environmental or social impacts occur during implementation:

(i) Obtain all statutory clearances (including CRZ, road-cutting permissions, tree-cutting approvals, etc.) at the earliest and ensure that all conditions are incorporated into the detailed design.
(ii) Include the updated IEE, along with the EMP and SEMP requirements, in all bid and contract documents.
(iii) Update or revise this IEE based on detailed project design, changes in alignment (if any), or any unanticipated environmental or social impacts during project implementation.
(iv) Conduct pre-construction surveys and utility mapping to avoid disruption to traffic, sensitive receptors, and underground utilities along Chennai's dense urban corridors.
(v) Finalize designs near Guindy National Park, Pallikaranai Ramsar Site, and heritage structures ensuring no direct impacts, adopting trenchless or other low-impact construction methods where required, and fully incorporating all conditions stipulated in the applicable statutory clearances and permits into the detailed design
(vi) Conduct safeguards induction/orientation for all contractors and subcontractors immediately upon award of contract.
(vii) Ensure reuse and safe disposal of construction and demolition waste as per the Construction and Demolition Waste Management Rules, 2016, including use of authorized processing facilities of Greater Chennai Corporation.
(viii) Ensure the contractor mobilizes qualified Environment, Health, and Safety (EHS) officers before the start of any site activity.
(ix) Ensure timely disclosure of project information and effective functioning of the GRM at PIU and contractor levels.
(x) Ensure contractors and subcontractors actively participate in the first-level GRM and maintain a site-level grievance logbook.
(xi) Strictly supervise EMP implementation through regular site inspections, safety audits, and compliance checks by PIU, CMSC, and the PMU.
(xii) Maintain continuous consultations with affected communities, local bodies, traffic police, and institutions along the alignment.
(xiii) Maintain systematic documentation and submit monthly, quarterly, and semi-annual/annual environmental monitoring reports as required in the IEE.
(xiv) PMU, PIUs, CMSC, and contractors shall remain fully committed to protecting the environment, worker safety, and community health throughout the project lifecycle.